TS5000 Robot Controller

TP1000 Operation Manual

(Compatible Robots: THE Series SCARA Robot)
(Compatible Controllers: TS5000 Series)

Rev. 2
October 1, 2022

Notice

- Ensure that this manual is provided to the actual final user of the industrial robot.
- Be sure to read this manual before operating the industrial robot.
- After reading through this manual, keep it nearby for future reference.

Shibaura Machine

The controller supported by this instruction manual is the TS5000.

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Introduction

Thank you for purchasing this SHIBAURA MACHINE robot system.
This manual describes the operating procedures for the TS5000 Series Robot Controller using the TP1000 Teach Pendant.
Before using this manual, refer to the Safety Manual for your robot, and be sure that you fully understand the implemented safety measures.

This manual consists of the following parts.

Part 1: Basic Operation
This presents basic information about the TP1000 that must be understood when reading this manual.

Part 2: Operating Procedures
This describes the TP1000 operating methods focusing on the operating procedures.

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Usage Scope

This system was designed for supporting automation of a wide range of operations, including transportation operations such as workpiece movement and alignment and assembly operations such as part insertion and screw tightening.
We ask that customers use this system for the purpose of supporting automation of these operations and incorporating into automation equipment.

This system was not designed for use in operations where contact is made with people (operations in medical treatment, nursing care, etc.), for incorporating into equipment used for sustaining human life (life support devices, infant incubators, etc.), for incorporating into equipment used for transporting people (automobile and other transportation equipment, airplanes and other aerospace equipment, roller coasters and other amusement park equipment), for equipment used for protecting human life or the human body (safety devices, etc.), or similar equipment.
This system was also not designed for incorporating into equipment for the purpose of controlling high amounts of heat (nuclear control equipment, combustion furnace control equipment, etc.).
In cases where a fault or malfunction of this product could directly threaten human life or cause an injury, its use in operations or incorporating into equipment is prohibited. If this product is used, the customer assumes all responsibility for ensuring safe operation.

Operating Procedures

This describes the operating conditions for the THE series. Usage under different conditions could expose the user to unexpected risks. Be sure to follow the required conditions.

1. Main Specifications of Product — Refer to the robot manual for your robot.
2. Usage Restrictions — The "Restricted Specifications Sheet", which shows the usage restrictions, are contained in Tables 1 and 2.

Table 1. Restricted Specifications Sheet (1)

1. Usage Purpose and Applications for Product (Usage Restrictions)	
1) Intended use	To enable the robot to perform operation in line with the purposes (item transfer, assembly, inspection, machining, etc.). (1) During teaching and adjustment operation - It is highly likely that operation is performed by turning on the servo power within the movable range of the robot. - A co-worker who supervises the worker who performs teaching and operation confirmation is positioned outside the movable range. (1) Teach the position where the robot will operate. (2) Create the operation sequence (program, etc.). (3) Transfer and check the signals and data exchanged with the external equipment. (4) Check operation at low speed (250 mm/sec or less). (2) During operation - Danger prevention measures for preventing contact with the robot (installation of a fence or enclosure) are taken to keep people away from robots during operation. - When a person approaches an operating robot, measures shall be implemented to turn off the servo power. (1) Start the robot. (2) Perform operation in line with the purpose at the specified speed. (3) Stop the robot. (3) During maintenance and inspection - Work performed within the movable range of the robot when the primary power is off (1) Check the bolts for looseness and rattling, and re-tighten them. (2) Check the cable clamps for any looseness. (3) Check for any damage. (4) Clean off any dust. (5) Inspect and apply rust prevention agent. (6) Check the timing belt for any looseness and cracks. (7) Check the cables and air tubes for wear. (8) Check the movement when each axis is pushed and moved by hand. (9) Replace the consumable parts for the robot and controller. (10) Replace the battery for the memory unit in the controller. - Work performed within the movable range of the robot when the primary power is on (1) Check the greasing state, and apply grease (when the servo power is off). (2) Push each axis by hand and check it for any rattling (when the servo power is on). (3) Check for any abnormal vibrations and sounds during operation (when the servo power is on). (4) Check the teaching points for misalignment (when the servo power is on). (5) Replace the battery for the robot position detector (when the servo power is off).

Table 2. Restricted Specifications Sheet (2)

2) Predictable misuse	- Use for work which applies excessive impact to the robot or to which excessive impact is applied by the robot - Modification of robot body, controller, and optional parts and energization with these covers removed - Operation in states other than the specified installation state (placed on floor, suspended from ceiling, mounted to wall, etc.) - Operation on a frame that does not satisfy the specified strength - Operation in an environment AS5other than the specified environment	
3) Unexpected start	- Malfunction caused by excessive noise - Robot start caused by start signal unexpectedly sent from peripheral equipment to the robot - Unintended start caused by abnormal communication data - Malfunction caused by voltage fluctuation	
2. Replacement of product components (time restrictions)		
1) Mechanical restrictions 2) Electrical restrictions	Replace consumable parts as required or periodically through daily inspection, regular inspection (every 3 and 6 months and every year) and overhaul (every 5 years).	
3 Movable range of product (space restriction)		
1) Operating range	Each axis	Check the specifications manual for your robot.
	Additional axis	Conpliant with the individual specifications.
	—	—
	—	—
2) Interface	- The product shall be installed at a location suitable for the external dimensions and operating range. - Space required for maintenance and inspection shall be provided.	
3) Work environment	- Hazard prevention measures for preventing contact with the robot (installation of a fence or enclosure) shall be implemented. - Space shall be provided to prevent a person from getting caught between the robot and the fence or enclosure.	
4. Life cycle of product		
During the life cycle of assembly, shipment test, transportation, installation, teaching, operation, maintenance and disposal, risk assessment is carried out for the "installation" stage (transportation and installation), "usage" stage (teaching, operation, and maintenance), and disposal.		
5. Person who approaches the machine		
Target person	Knowledge, experience, and requirements	
1) Transportation and install	Knowledge: Person who understands the content of the instruction manual	
	Experience: —	
	Requirement: Qualified person on forklift or crane (as required)	
2) Teaching/adjustment operation worker (including co-worker)	Knowledge: Person who understands the content of the instruction manual	
	Experience: Person who has learned how to operate the robot	
	Requirement: Person who has received special training in operation such as teaching of the industrial robot and who is permitted to perform the work by the employer	
3) Operation worker (operator)	Knowledge: Person who understands the content of the instruction manual	
	Experience: Person who understoods the work regulation	
	Requirement: —	
4) Maintenance and inspection worker	Knowledge: Person who understands the content of the instruction manual	
	Experience: Person who has learned how to operate the robot and person who has learned how to perform maintenance work for the target robot	
	Requirement: Person who has received special training in work such as inspection of the industrial robot and who is permitted to perform the work by the employer	
5) Disposal worker	Knowledge: Person understands the content of the instruction manual and the disposal procedures specified by local government regulations	
	Experience: —	
	Requirement: —	
6. Other		

Warranty

This product is provided by SHIBAURA MACHINE after strict testing and inspection and verification that it satisfies our performance standards.
Toshiba provides the following warranty in the event that a fault occurs.

1. Warranty Period

The warranty period of this product is one of the following periods, whichever is shortest.

(1) Period of 18 months from SHIBAURA MACHINE factory shipping
(2) Period of 12 months from customer installation

2. Warranty Terms and Conditions

(1) The warranty applies to this product. The warranty scope is the specifications and functions described in the product specifications, catalog, instruction manual, and other documents. SHIBAURA MACHINE will not be liable for any secondary or incidental damages arising as a result of faults in this product.
(2) Within the warranty period of this product, SHIBAURA MACHINE will repair free of charge only those faults that arise under operation and use in accordance with the instruction manual provided with this product.
A fee will be charged for all repairs of faults that occur after the warranty period has expired.

3. Disclaimers

The following cases are outside the scope of the warranty.

(1) Faults and damage due to usage that did not follow the instruction manual or due to negligent usage.
(2) Problems due to deterioration over time or usage wear (natural fading of the paint coating, deterioration of consumable parts[1])
(3) Problems due to sensory experiences (such as occurrence of noise that does not affect functionality)
(4) When the product was modified or disassembled without the approval of SHIBAURA MACHINE
(5) Faults and damage due to inadequate inspection and maintenance or improper repairs
(6) Faults and damage due to natural disasters, fires, and other external factors
(7) Internal data such as programs and points created or modified by the customer
(8) When this product was purchased in Japan and taken outside the country

1) Consumable parts refer to the maintenance replacement parts shown in the MAINTENANCE section of the Robot Manual for your robot.

4. Notes

(1) SHIBAURA MACHINE cannot guarantee the basic performance of this product when it is used in an environment outside the specifications.

(2) SHIBAURA MACHINE will not be liable for any injuries, damage, faults, or other problems that arise by failing to follow the warning and caution information provided in this instruction manual.

(3) Be aware that the warning, caution, and other safety information in this instruction manual apply to the safety risks foreseen by SHIBAURA MACHINE, and they do not cover every possible scenario.

Main standards, laws, and regulations pertaining to the safety of industrial robots

The international industrial standard ISO/DIS 12100 "Safety of machinery" describes the risk reduction measures for machines and equipment.
Step 1) Inherently safe design measures: Limitations on force, speed, energy, etc.
Step 2) Safeguarding and/or complementary protective measures: Installation of safety fence, etc.
Added protective measures: Installation of emergency stop devices, etc.
Step 3) Information for use: Warning signage, alarms, instruction manuals, etc.

Based on this, the guidelines known as "Guide 51" were defined by classifying the various standards for the ISO/IEC international standards in a hierarchical structure.
The safety standards for industrial robots are standardized as type C in the individual safety standards for machinery.

- ISO 10218 and ISO 10218-1

Be sure to follow the standards, laws, regulations established in the country where the industrial robot is used.

Cautions on Safety

The instruction manual for the controller contains important safety information for ensuring the safe and correct operation of this device and for preventing hazards to the operators and others and property damage.

To ensure that all work procedures from installation to operation of the controller are performed safely, be sure to carefully read the safety manual for your robot before starting the actual work.

Make sure that you well understand the following details (indications and symbols) before reading this manual. Always observe the information that is noted.

[Explanation of indications]

Indication	Meaning of indication
⚠ DANGER	Indicates an imminently hazardous situation which, if not avoided, will result in death or serious injury.
⚠ WARNING	Indicates a potentially hazardous situation which, if not avoided, could result in death or serious injury.
⚠ CAUTION	Indicates a hazardous situation which, if not avoided, could result in minor or moderate injury[1].
NOTICE	Indicates a situation which could result in physical damage[2] only.

1) Injuries refer to injuries, burns and electric shocks, etc., which do not require hospitalization or long-term medical treatment.
2) Physical damage refers to damages due to destruction of assets or resources.

[Explanation of symbols]

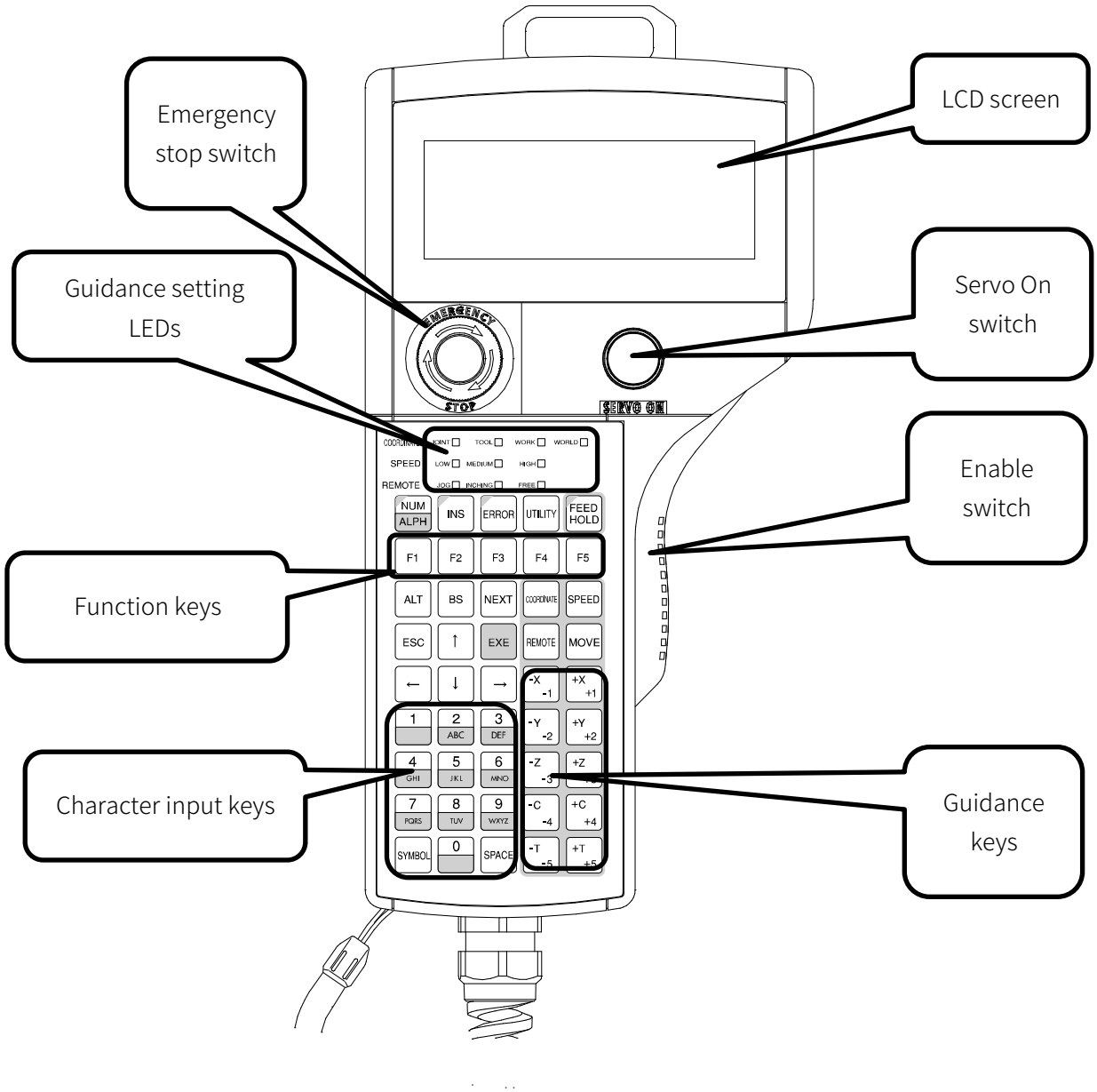
Symbol	Meaning of symbol
	This means that the action is prohibited (must not be done). Details of the actions actually prohibited are indicated with pictures or words in or near the symbol.
	This means that the action is mandatory (must be done). Details of the actions that must be done are indicated with pictures or words in or near the symbol.
	This indicates a danger and precaution. The specific safety information is indicated by the picture or text near or in this symbol.

Basic Operation

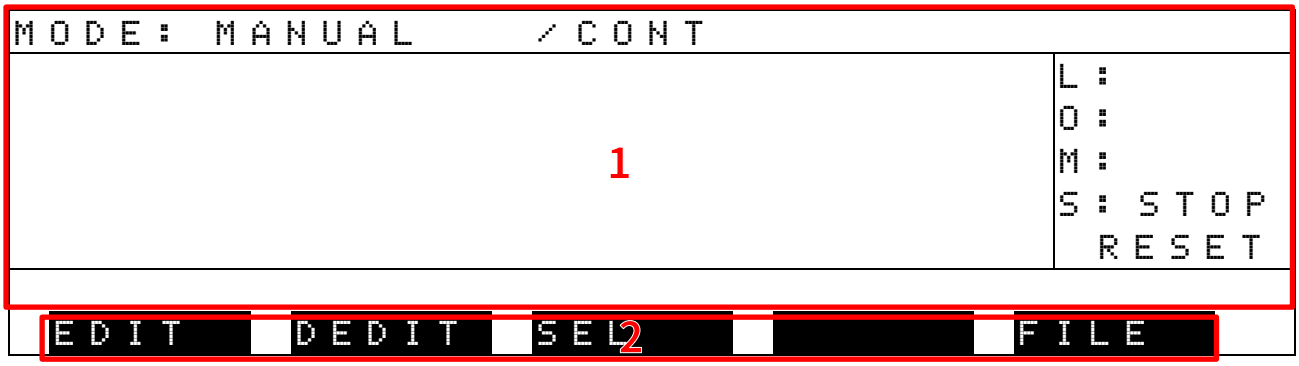
This presents the basic information needed for operating the TP1000.

1. TP1000 Overview

1.1 TP1000 External View

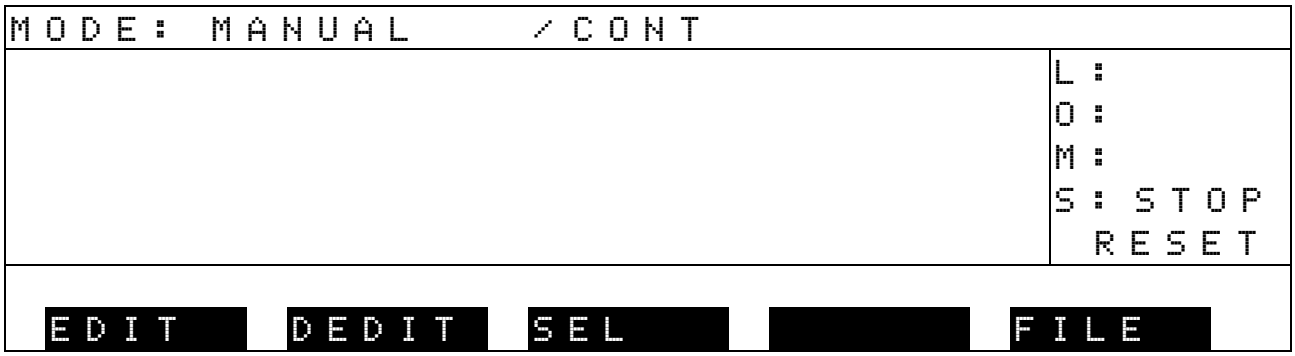


1.2 TP1000 Screen Layout



No.	Name	Description
1	Main screen	This displays various information about the robot. The displayed information varies depending on the screen.
2	Function menu	This displays the name of the operation performed when a function key (F1 to F5) is pressed. From the screen left side, the keys correspond to F1, F2, ..., and F5.

Basic Screen



The valid keys in the basic screen and their actions are shown in the following table.

Key	Description
NEXT key	This switches the function menu.
ALT key + NEXT key	This switches the function menu.

The function menus in the basic screen and their actions are shown in the following table.

Displayed menu name	Description	Section number for detailed explanation
EDIT	This is used to edit the program section.	3.1.1
DEDIT	This is used to edit the data section.	3.2.1
SEL	This is used to select the execution program.	2.3.2
RUN	This executes the program.	2.3.5
FILE	This is used to perform file operations.	3.4
MODE	This is used to set the execution mode.	2.3.3
OVRD	This is used to set the override.	2.3.4 4.2
RESET	This is used to perform the reset operation.	2.3.8
DO	This is used to perform direct execution.	2.4
WATCH	This is used to view and write global variables.	2.3.10 4.4
LDISP	This is used to monitor the execution program.	2.3.9 4.3
LOCK	This is used to enable or disable the switching lock for master mode.	2.1.2 4.1.2
AUTO	This switches master mode to AUTO.	4.1.1
MANU	This switches master mode to MANUAL.	2.1.1

Operating Procedures

This describes the operating procedures for the basic TP1000 operations, which are manual guidance, file editing, test operation, and automatic operation.

1. TP1000 Startup

1.1 Connecting with TS5000

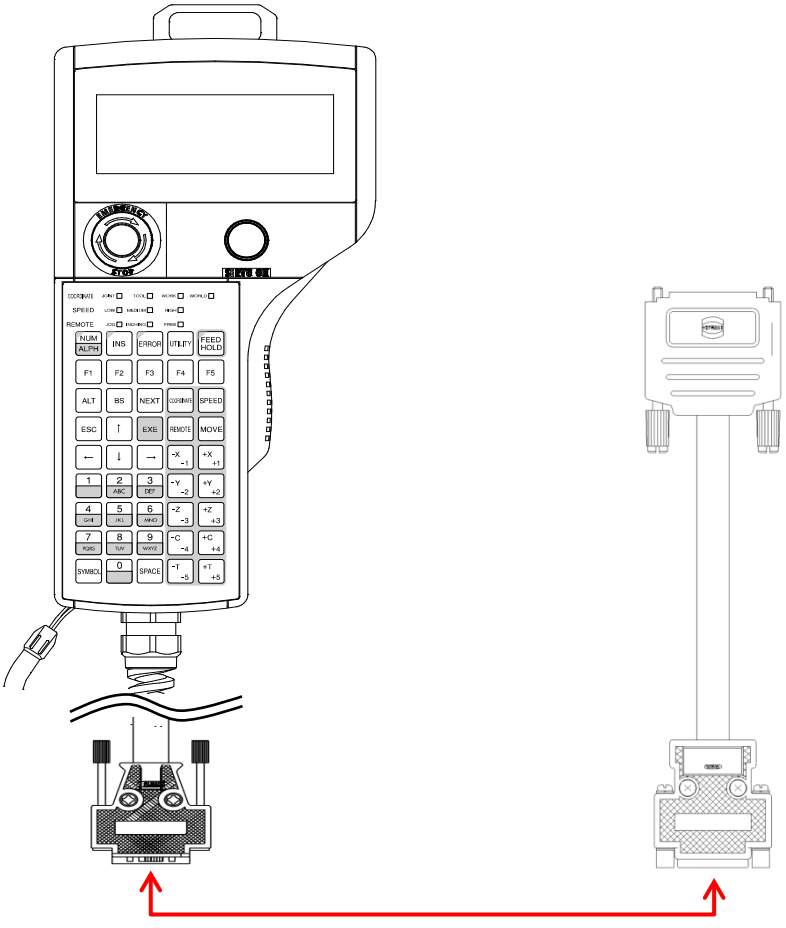
A conversion cable (option) is used to connect the TP1000 to the TS5000.



Conversion cable

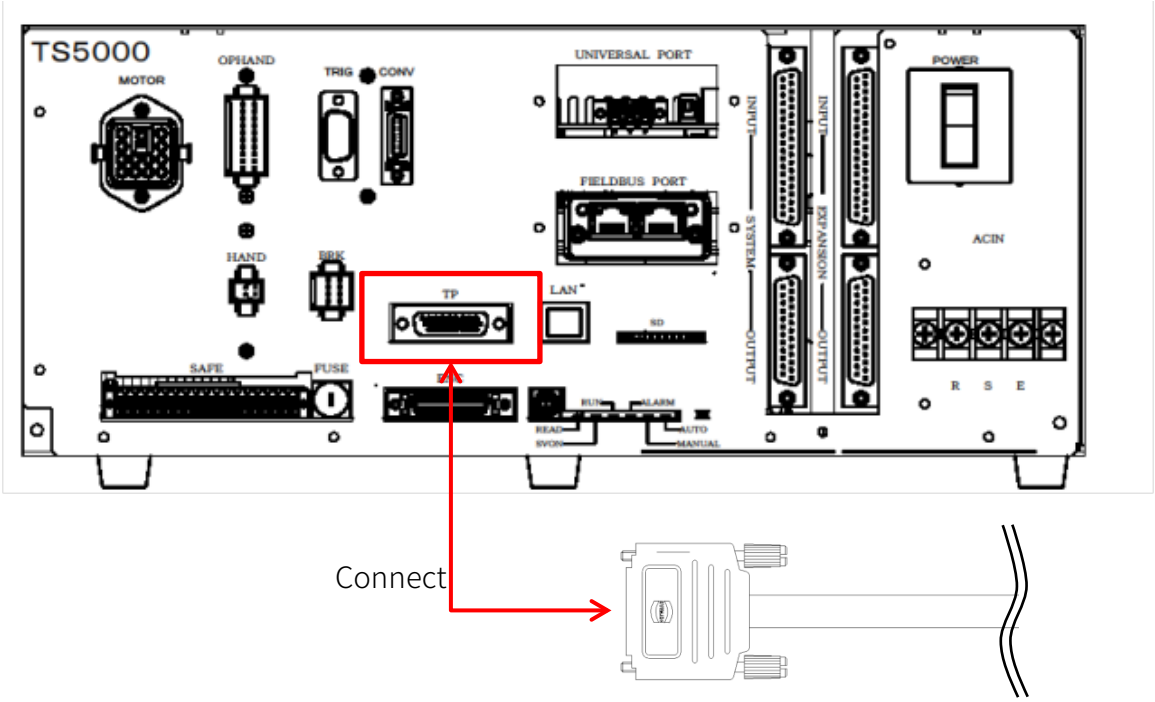
<Connecting the TP1000 and TS5000>

1 Connect the TP1000 and the conversion cable.



Connect

2 Connect the conversion cable to the TP connector of the TS5000.

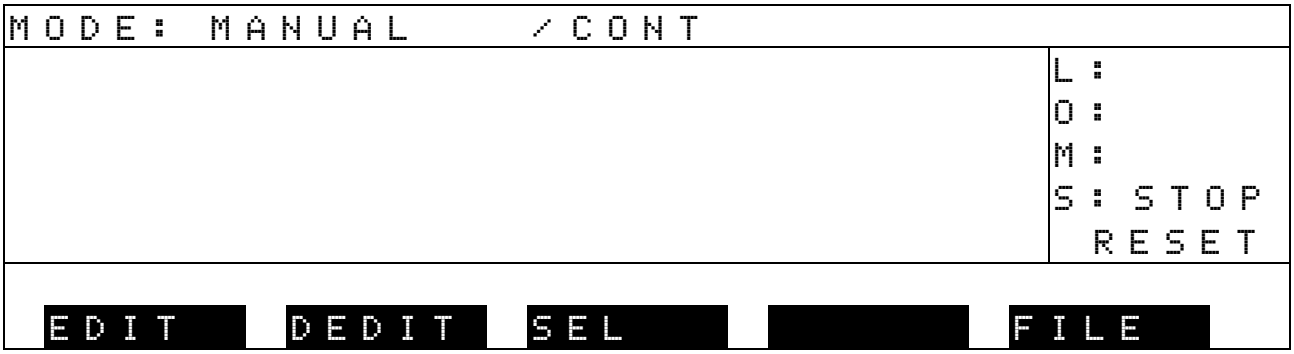


1.2 Starting the TP1000

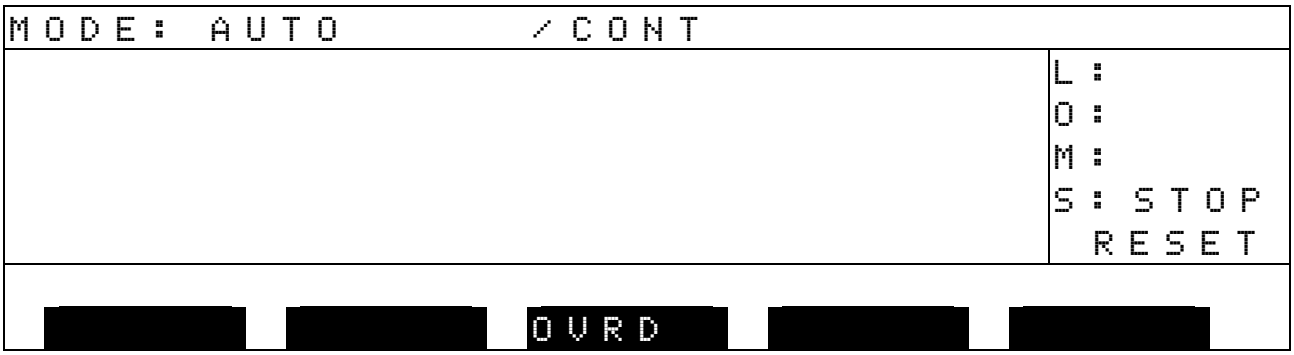
While the TS5000 is connected, turning on the TS5000 power automatically starts up the TP1000.

The displayed screen varies depending on the master mode (MANUAL or AUTO) where the TP1000 was started.
The screens after startup are shown below.

Startup screen (MANUAL mode)



Startup screen (AUTO mode)



2. Operations in MANUAL Mode

This chapter describes the operations performed in MANUAL mode.
MANUAL mode is used when teaching the robot and confirming movements.
During MANUAL mode, the robot tip speed is limited to 250 mm/sec or less.

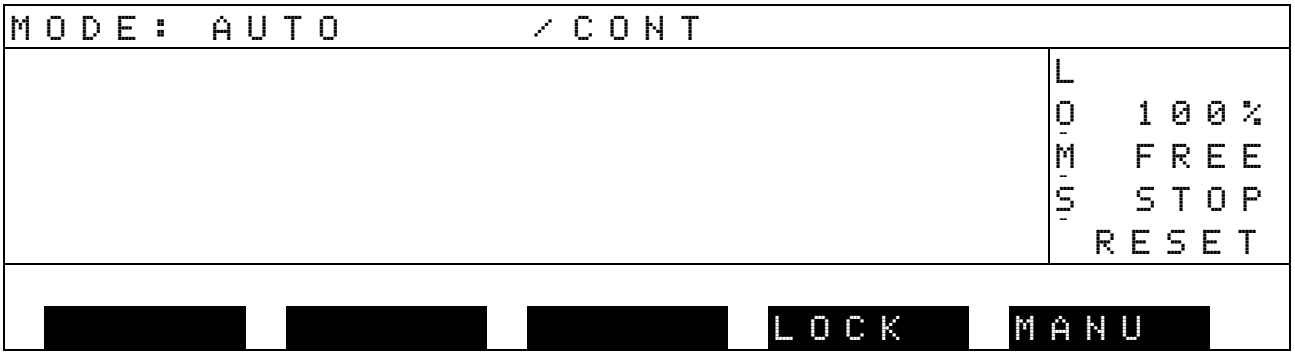
Use the following procedure to switch from AUTO mode to MANUAL mode.

2.1 Master Mode Switching

2.1.1 Master Mode Switching

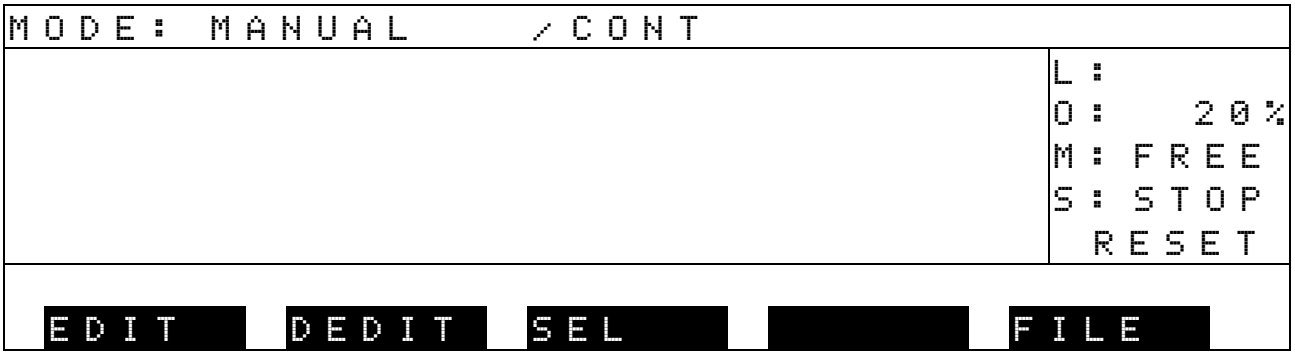
<Switching to MANUAL Mode>

1 Press the NEXT key until "AUTO" appears in the function menu on the basic screen.



2 Press the "AUTO (F5)" key.

The controller master mode is switched, and the mode indicator at the top of the basic screen appears as "MANUAL".



MEMO

Once master mode switching lock is enabled, the message "Master Mode is Locked" is displayed.

If necessary, disable the master mode switching lock by referring to the next section, "Master Mode Switching Lock".

2.1.2 Master Mode Switching Lock

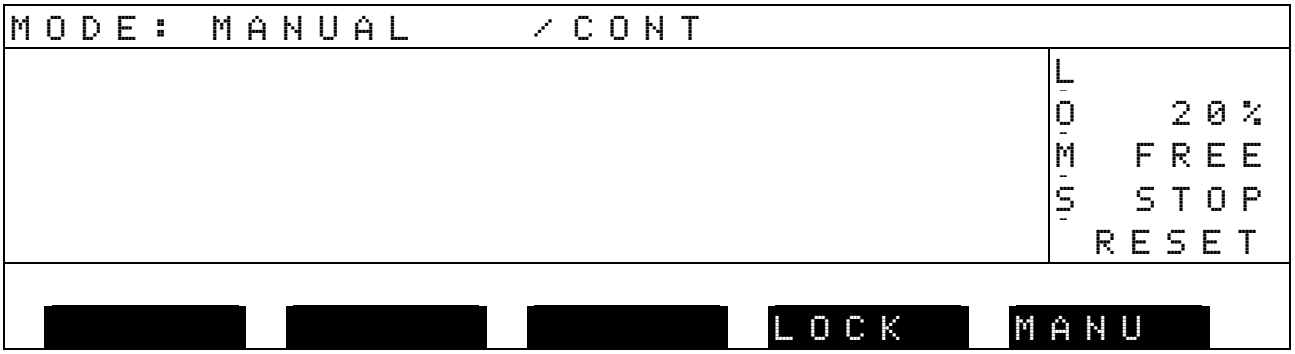
Master mode switching lock can be enabled to prevent switching of the master mode by a third party to a mode unintended by the operator. This section describes how to enable and disable the switching lock for master mode.

MEMO

Master mode switching lock is disabled when the controller power is turned off.

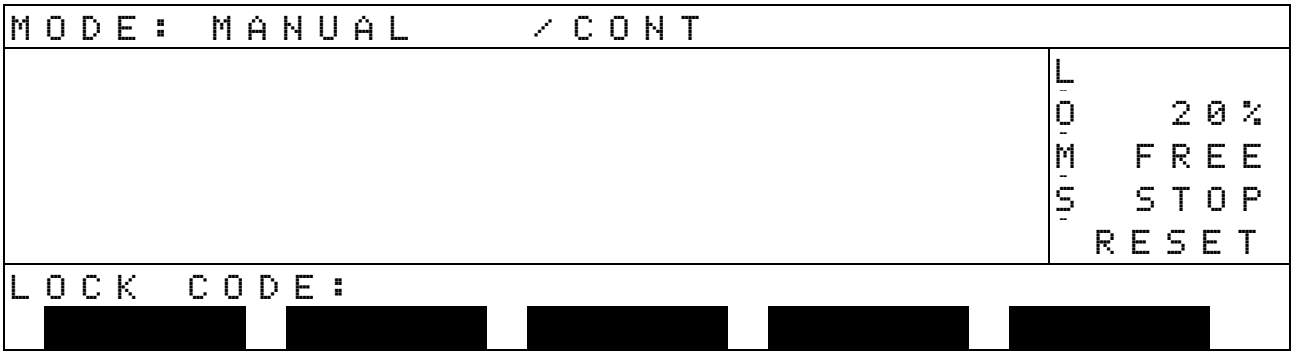
<Master Mode Switching Lock Enable Procedure>

1 Press the NEXT key until "LOCK" appears in the function menu on the basic screen.



2 Press the "LOCK (F4)" key.

The command line changes as shown below.



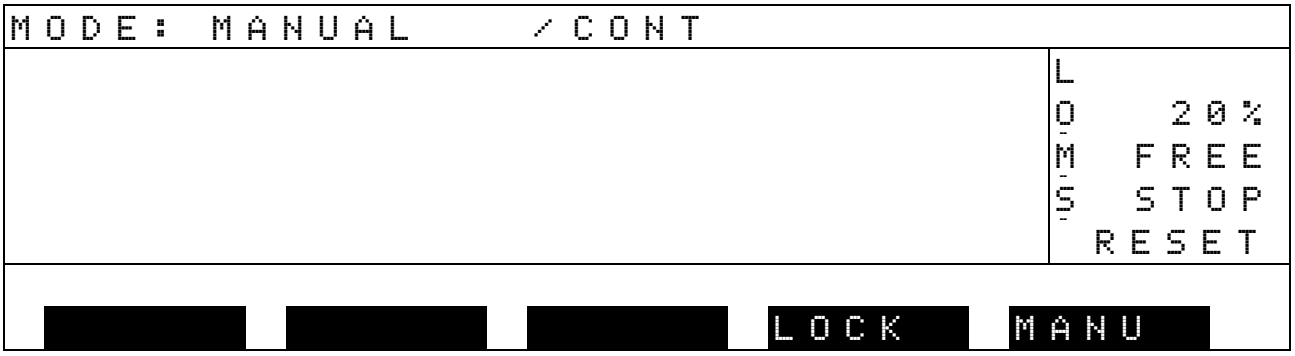
3 Enter your desired lock code, and press the EXE key.

The master mode switching lock is enabled.

> *MEMO*
>
> *Any character string from 1 to 4 characters can be set as the lock code.*

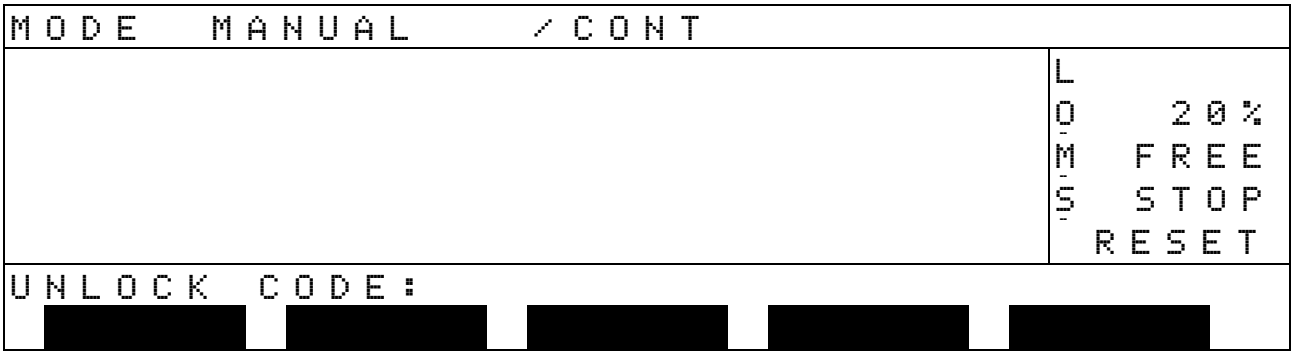
<Master Mode Switching Lock Disable Procedure>

1 Press the NEXT key until "LOCK" appears in the function menu on the basic screen.



2 Press the "LOCK (F4)" key.

The command line changes as shown below.



3 Enter the lock code that was entered when setting, and press the EXE key.

The master mode switching lock is disabled.

> *MEMO*
>
> *If the entered lock code is incorrect, the message "Lock code is missmatch" is displayed.*

2.2 Manual Guidance

This chapter describes manual guidance of the robot where the TP1000 is used.

2.2.1 Overview

In this manual, "manual guidance" refers to the movement of the robot in the axial direction of a coordinate system by using the teach pendant. (Test operation, direct execution, and teaching point movement are not included in manual guidance.)

To perform manual guidance, the guidance type, guidance coordinates, and guidance speed (or guidance amount) must be set.
These are set using the REMOTE, COORDINATE, and SPEED keys, respectively.

2.2.2 Guidance Type Setting

This describes how to set the guidance type.
The guidance types include jog, inching, and free.

• Jog
The robot moves while the guidance key is held down.
The robot stops when the guidance key is released.
Pressing the + guidance key moves in the positive (+) direction of the guidance coordinates, and pressing the - guidance key moves in the negative (-) direction.
Movement is performed at the speed that was set for the guidance speed.
Specifically, setting the guidance type to Jog and performing manual guidance is called "jog operation".

• Inching
When the guidance key is pressed, the robot moves by the movement amount only that was set to the guidance amount and then comes to a stop.
Pressing the + guidance key moves in the positive (+) direction of the guidance coordinates, and pressing the - guidance key moves in the negative (-) direction.

* Even if the operation key is held down, the robot moves by the amount for a single press of the key.
  Even if the key is pressed during robot movement, the robot does not continue to perform movement.

Specifically, setting the guidance type to Inching and performing manual guidance is called "inching operation".

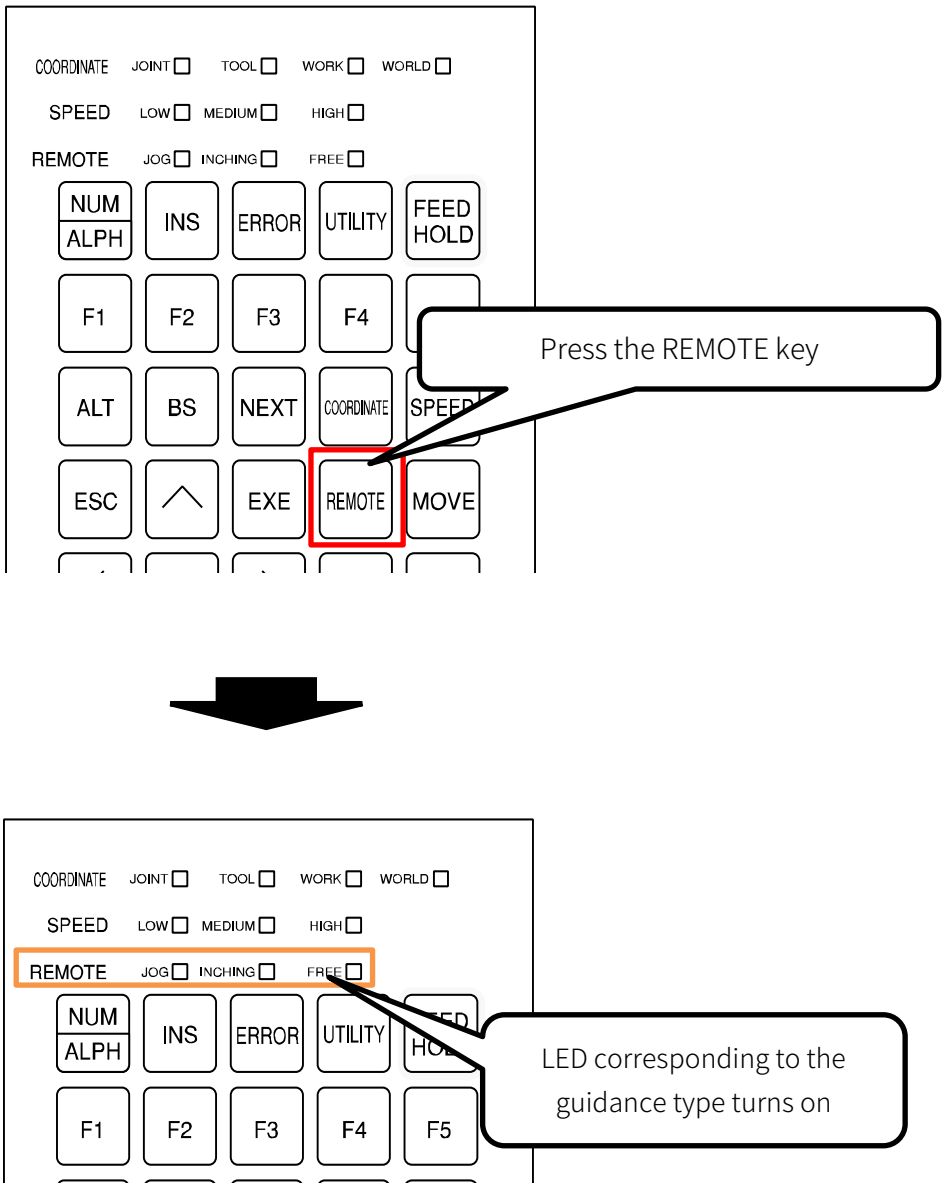
• Free
When the guidance key is pressed, the respective axis is set to servo lock or servo free regardless of the guidance coordinates.
Pressing the + guidance key sets to servo lock.
Pressing the - guidance key sets to servo free.

<Guidance Type Setting Procedure>

1 Press the "REMOTE" key on the key sheet.

Each time that the REMOTE key is pressed, the guidance type cycles in the order of JOG > INCHING > FREE > JOG.

Whenever the guidance type changes, the LED corresponding to the guidance type in the key sheet turns on.



MEMO

When the guidance type is FREE, the LEDs for COORDINATE and SPEED turn off.

2.2.3 Guidance Coordinates Setting

This describes how to set the guidance coordinates.
The guidance coordinates include joint coordinates, world coordinates, work coordinates, and tool coordinates.

• Joint coordinates
The robot moves for each joint axis.

• World coordinates
The robot moves on the Cartesian coordinate system.

• Work coordinates
The robot moves on the work coordinate system that is currently set.

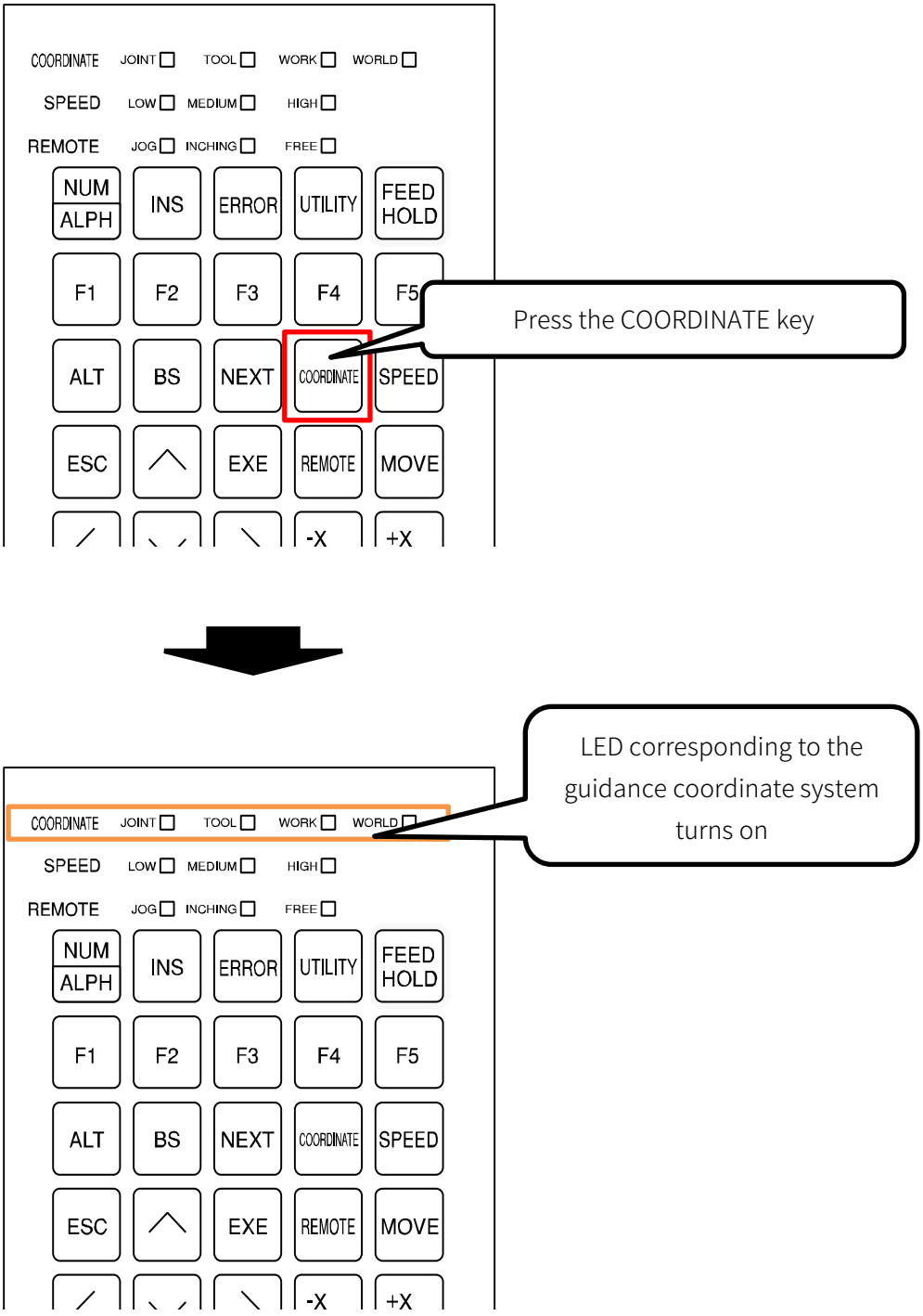
• Tool coordinates
The robot moves on the tool coordinate system that is currently set.

<Guidance Coordinates Setting Procedure>

1 Press the COORDINATE key on the key sheet.

Each time that the COORDINATE key is pressed, the guidance coordinate system cycles in the order of JOINT > TOOL > WORK > WORLD > JOINT.

Whenever the guidance coordinate system changes, the LED corresponding to the guidance coordinate system turns on.



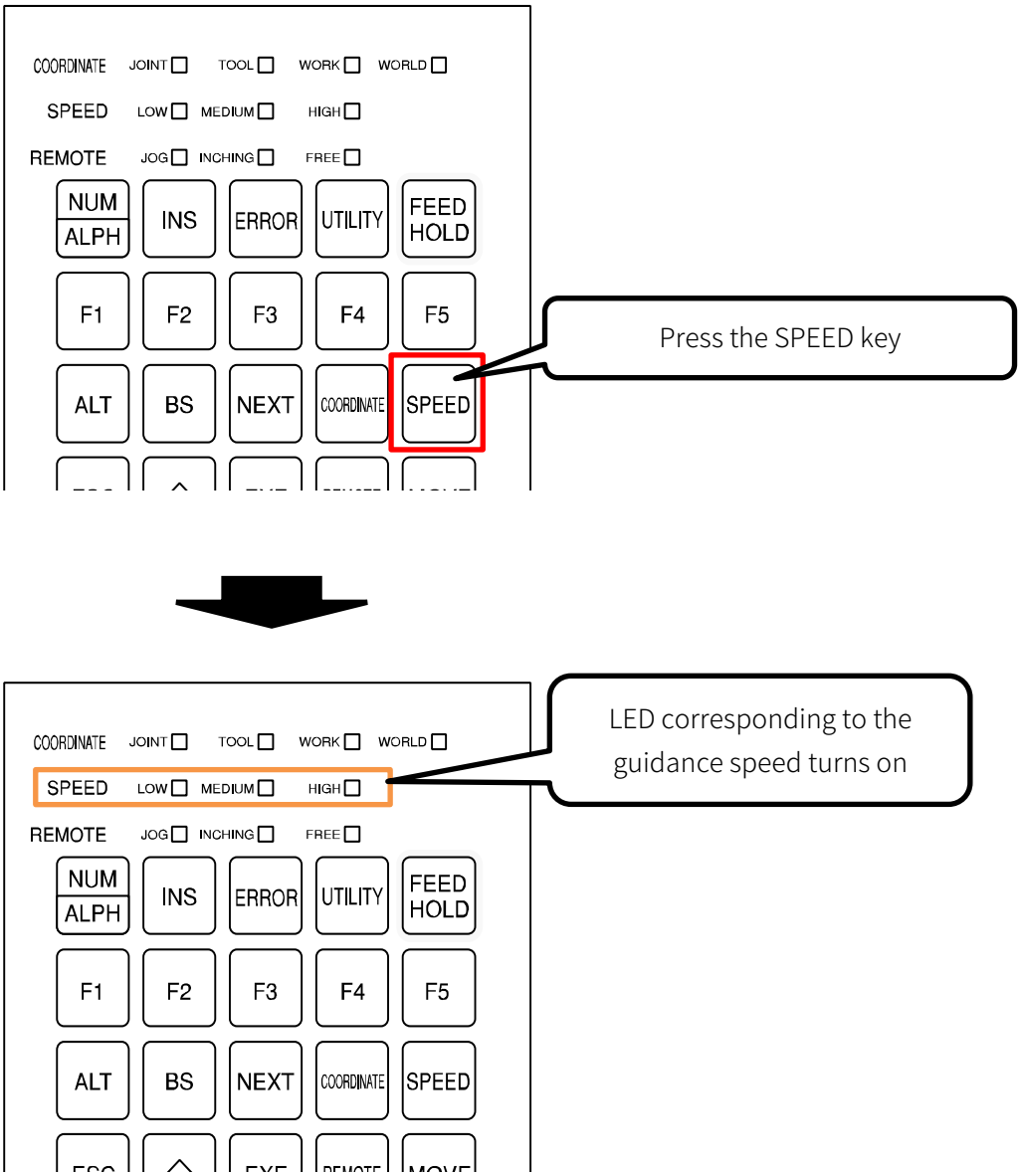
2.2.4 Guidance Speed Setting

This describes how to set the guidance speed.
The guidance speed is the movement speed of the robot when the robot performs movement by jog operation.
The available guidance speeds are LOW (low speed), MID (medium speed), and HIGH (high speed).

<Guidance Speed Setting Procedure>

1 Press the SPEED key on the key sheet.

Each time that the SPEED key is pressed, the guidance speed cycles in the order of LOW > MID > HIGH > LOW.
Whenever the guidance speed changes, the LED corresponding to the guidance speed turns on.



MEMO

The speeds set for LOW, MID, and HIGH can be changed to any selected speed.

For details, refer to the STA1166 Parameter Manual, section U005.

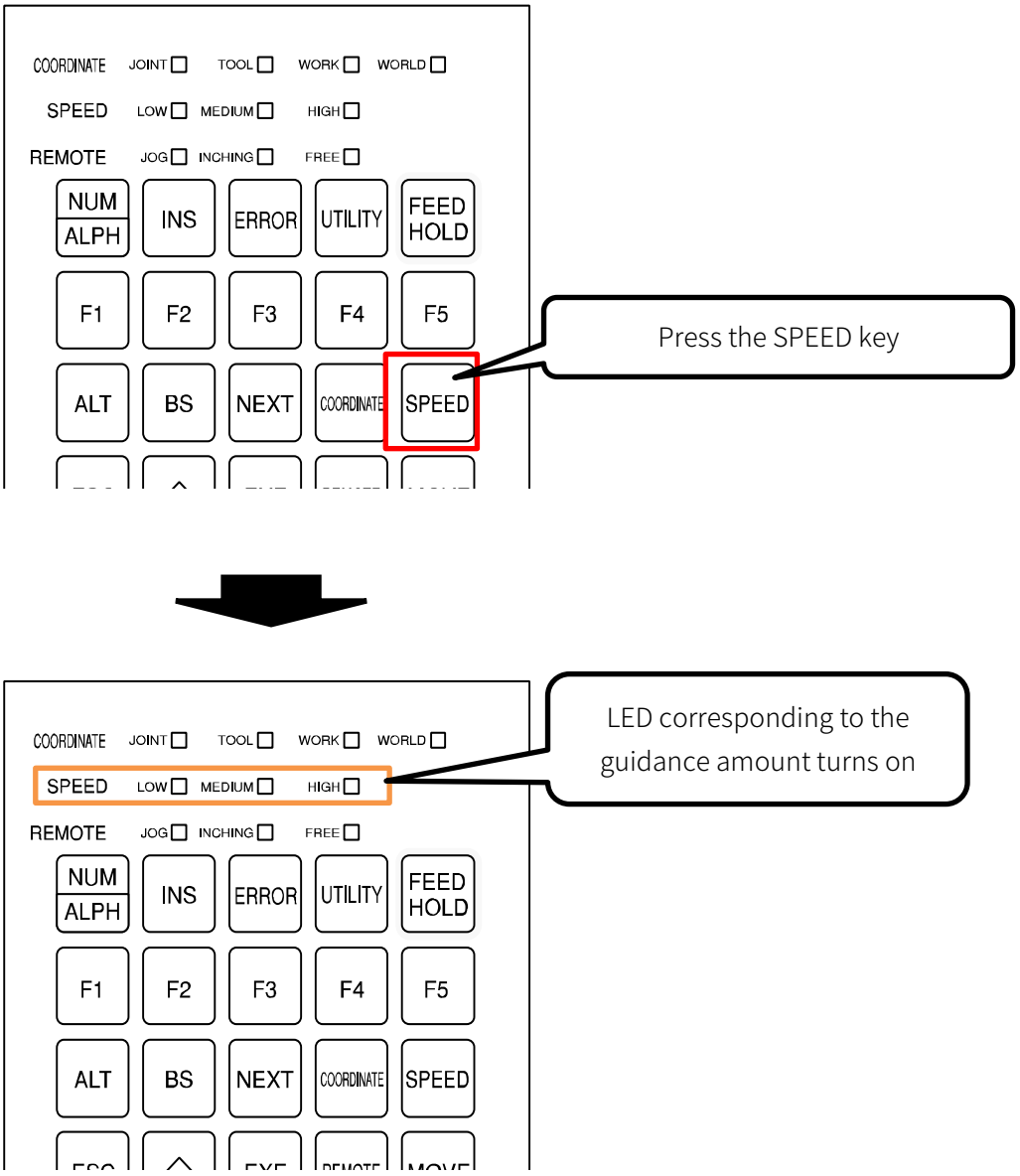
2.2.5 Guidance Amount Setting

This describes how to set the guidance amount.
The guidance amount is the movement amount of the robot when the robot performs movement by inching operation.
The guidance amounts can be set to LOW, MID, and HIGH.

<Guidance Amount Setting Procedure>

1 Press the "SPEED" key on the key sheet.

Each time that the SPEED key is pressed, the guidance amount cycles in the order of LOW > MID > HIGH > LOW.
Whenever the guidance amount changes, the LED corresponding to the guidance amount in the key sheet turns on.



MEMO

The amounts set for LOW, MID, and HIGH can be changed to any selected amount.

For details, refer to the STA1166 Parameter Manual, section U005.

2.2.6 Manual Guidance

This describes how to perform manual guidance.

<Manual Guidance Procedure>

1 Press the Servo On switch while holding the enable switch one level down.

The servo turns on, and the Servo On switch turns on.

2 Press the one of the guidance keys.

The robot moves using the conditions based on the guidance settings (guidance type, coordinates, speed or amount).

MEMO

• The servo cannot be set to ON if an alarm at level 4 or higher has occurred.

• The robot does not operate if two keys are pressed simultaneously.

• If the STOP and BREAK signals are OFF, the servo can be set to ON, but the robot will not operate even if the guidance key is pressed.

⚠ DANGER



MANDATORY

• Be sure to always check that the tool coordinates are set correctly.

Otherwise, a collision could occur with a robot that performed an unexpected motion.

⚠ WARNING



PROHIBITED

When the 3rd axis is set to servo free, do not insert your hands or feet under the 3rd axis.

The 3rd axis could drop and hit your hands or feet.

 **MANDATORY**	• Perform manual guidance in a state where the enable switch or emergency stop switch can always be accessed. Otherwise, a collision could occur due to an unexpected movement by the robot. • Before starting manual guidance, be sure to always check the selected guidance mode, guidance coordinates, guidance speed, and guidance amount. Otherwise, a collision could occur due to an unexpected movement by the robot.

2.3 Test Operation

This section describes how to perform test operation using the TP1000.

2.3.1 Overview

Test operation refers to executing programs in MANUAL mode.
The files used in test operation are called execution programs.

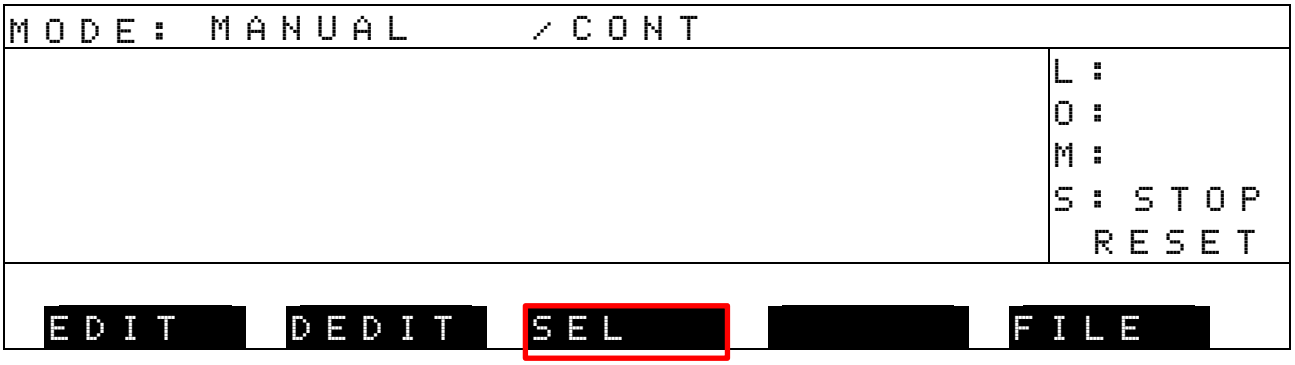
> *MEMO*
>
> *• Test operation requires selection of an execution program and setting of an execution mode and override.*
>
> *These setting procedures are described in detail in the sections "2.3.2 Execution Program Selection (SEL)", "2.3.3 Execution Mode Selection (MODE)", and "2.3.4 Override Settings (OVRD)".*

2.3.2 Execution Program Selection (SEL)

Test operation requires selection of a single program file as the execution program.
This describes how to select the execution program.

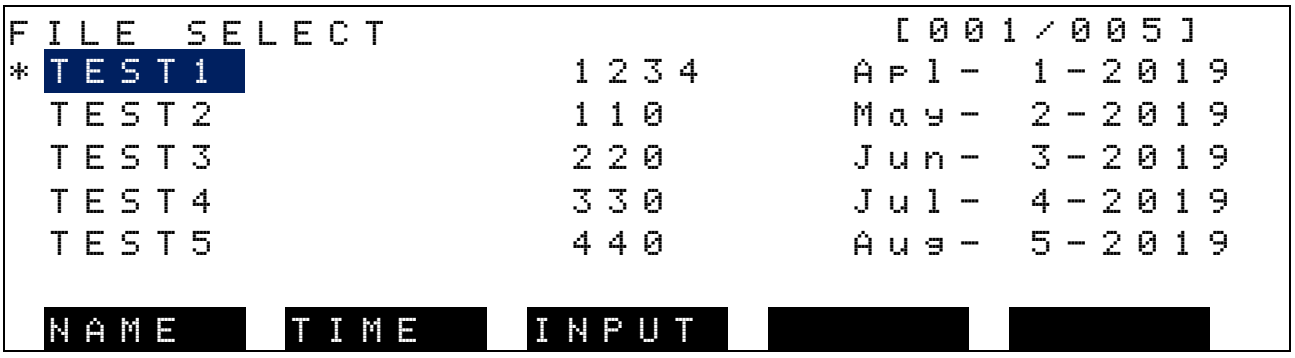
<Execution Program Selection Procedure>

1 Press the NEXT key until "SEL" appears in the function menu on the basic screen.



2 Press the SEL (F3) key.

This opens the File Select screen.



MEMO

The line where the cursor is located appears highlighted.

3 Use the ↑ ↓ keys to the select the program file to be executed, and press the EXE key.

The name of the selected program file appears in the command line.

MEMO

• If there are a large number of stored files, the files are displayed over multiple pages.

To navigate between pages, press the NEXT key or ALT + NEXT key.

• When the INPUT (F3) key is pressed, ":" appears in the command line.

Then, use the character input keys to enter the name of the program file to be selected, and then the same procedure as described above can be used.

4 Press the EXE key.

This analyzes the program file having the selected name and executes a syntax check (this process is called compiling).

When the check is completed, the selected program file is registered as an executable program.

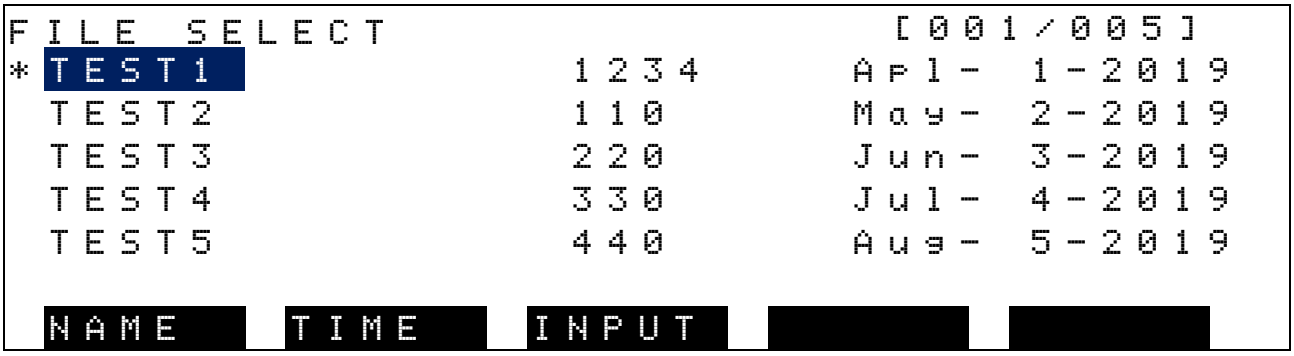
Regardless of the compile results, this returns to the basic screen.

MEMO

• If a syntax error is found, the alarm "1-160 Compile error" occurs.

• When the execution program is selected, the program is reset.

Execution Program Selection Screen



The valid keys in the execution program selection screen and their actions are shown in the following table.

Key	Description
↑ key	This moves the cursor up by one line.
↓ key	This moves the cursor down by one line.
ALT key + ↑ key or ALT key + NEXT key	This moves the cursor to the previous page.
ALT key + ↓ key or NEXT key	This moves the cursor to the next page.
EXE key	This displays the file name where the cursor is located into the command line. Pressing the EXE key again selects the execution program.
ESC key	This returns to the basic screen.

The function menus in the execution program section screen and their actions are shown in the following table.

Displayed menu name	Description	Section number for detailed explanation
NAME	This is used to sort files by name.	
TIME	This is used to sort files by date and time created.	
INPUT	The ":" is displayed in the command line to indicate that the command line can be edited.	2.3.2

2.3.3 Execution Mode Setting (MODE)

This describes how to set the execution mode.
Four modes are available: Continuous operation mode (CONT), cycle operation mode (CYCLE), segment operation mode (SEG), and step operation mode (STEP).
(The uppercase text in parentheses indicates the corresponding mode indicator displayed in the teach pendant.)

• Continuous operation mode
This repeatedly executes the execution program.

• Cycle operation mode
This executes one cycle (from PROGRAM line of main program to END line) of the execution program.

• Segment operation mode
This executes until the execution of the next movement command is completed.

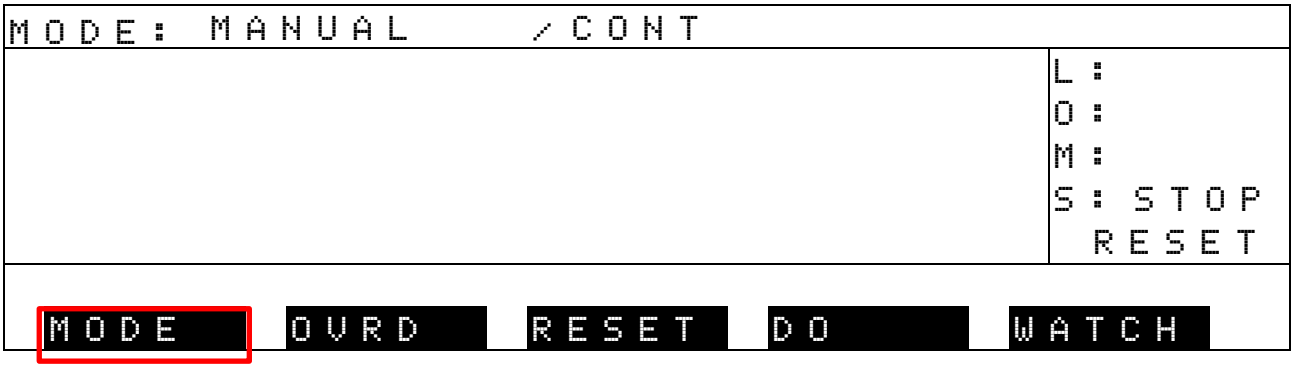
• Step operation mode
This executes the execution program step by step.

MEMO

- *During program execution in continuous operation, cycle operation, and segment operation modes, the execution mode can be switched to continuous operation, cycle operation, and segment operation modes.*
- *During program execution in step operation mode, the execution mode cannot be switched to another execution mode.*

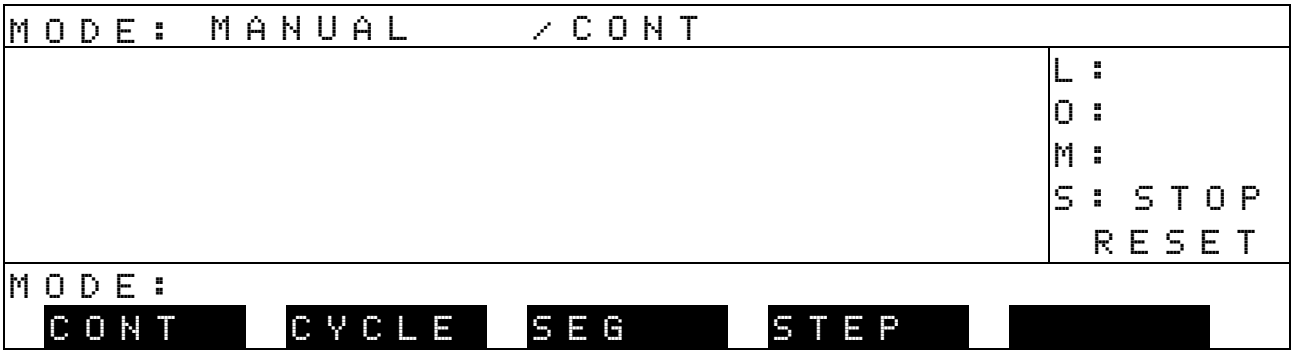
<Execution Mode Setting Procedure>

1 Press the NEXT key until "MODE" appears in the function menu on the basic screen.



2 Press the MODE (F1) key.

"MODE:" is displayed in the command line, and this switches to the function menu.



3 Press the function key corresponding to the mode that you want to select for the execution mode, and press the EXE key. The execution mode is switched to the selected mode.

2.3.4 Override Setting (OVRD)

This describes how to set the override.

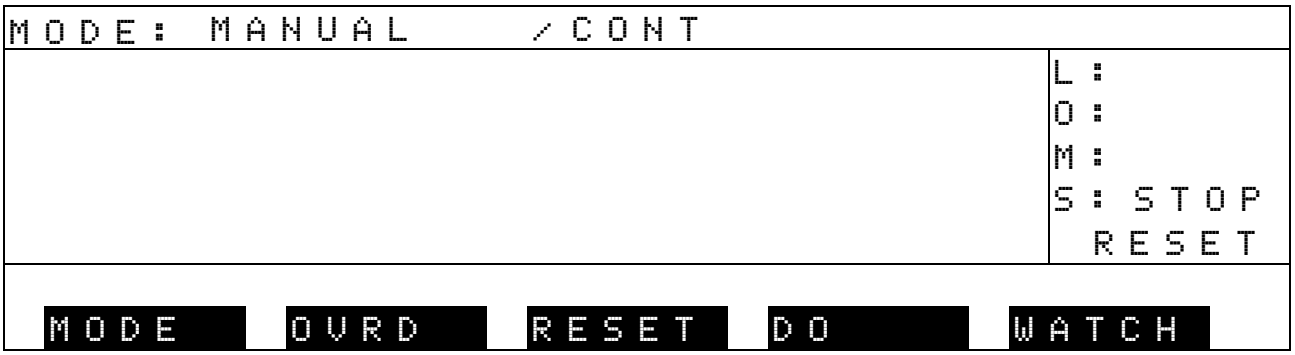
MEMO

• The override can be set even during program execution.

• In MANUAL mode, override is limited to a maximum value of 20%.

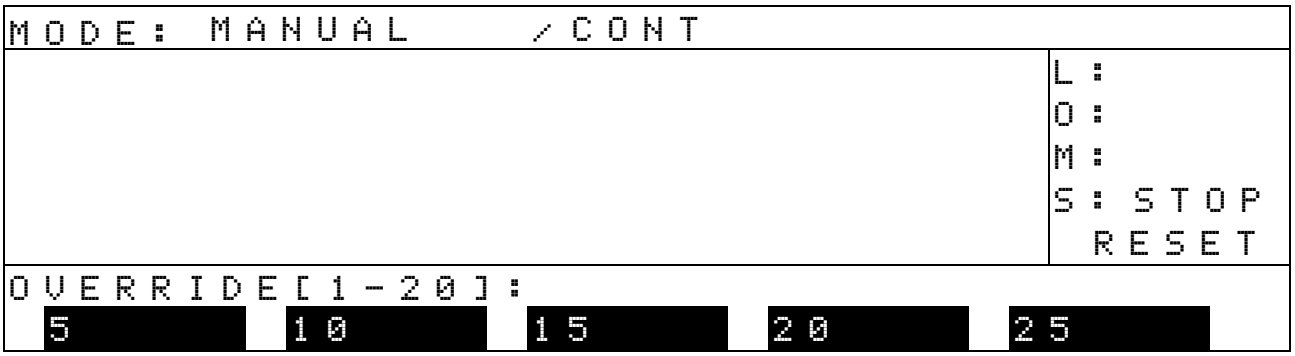
<Override Setting Procedure>

1 Press the NEXT key until "OVRD" appears in the function menu on the basic screen.



2 Press the OVRD (F2) key.

"OVERRIDE[1-20]:" is displayed in the command line, and this switches to the function menu.



3 Enter the value to be set for the override into the command line, and press the EXE key.

A check is conducted that the entered value is within the proper range, and if it is in the range, the override is overwritten by the entered value.

If the value is outside the proper range, "CAN NOT EXECUTE" appears.

MEMO

- *When the function key is pressed when entering to the command line, the number appearing on the function key is entered to the command line.*
- *The override can be set even while a program is running.*

2.3.5 Program Execution (RUN)

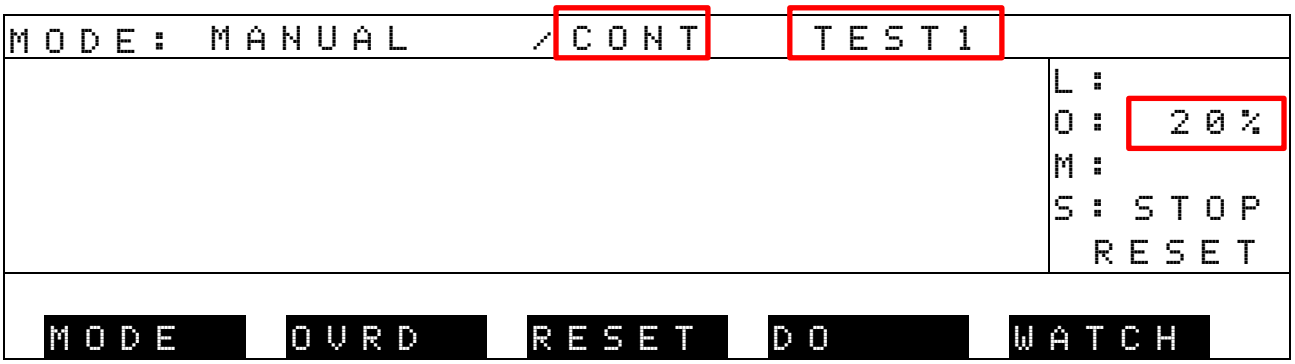
This describes how to run the execution program.

MEMO

• Before running an execution program, check that the selected execution program, execution mode, and override are set correctly.

<Execution Program Run Procedure>

1 Check that the execution program, execution mode, and override are set correctly.



2 Tap the Servo On switch while holding the enable switch one level down.

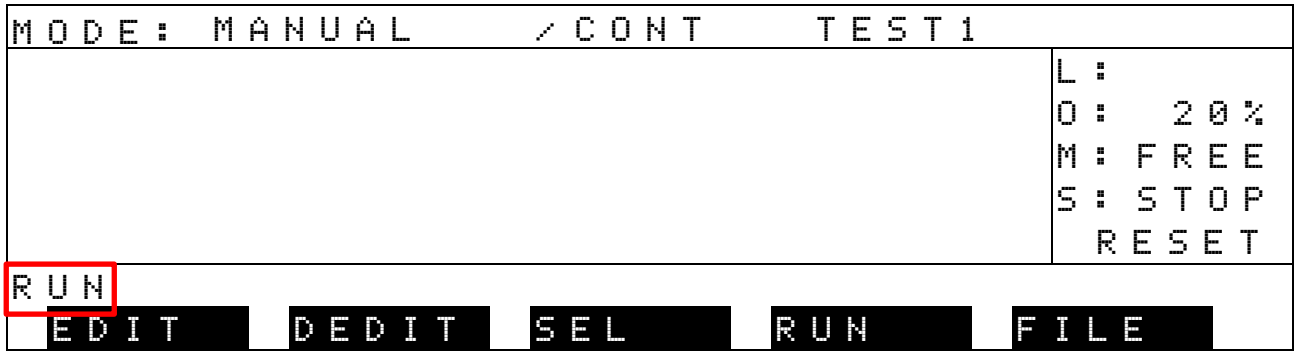
The servo turns on, and the Servo On switch turns on.

3 Press the NEXT key until "RUN" appears in the function menu on the basic screen.



4 Press the RUN (F4) key.

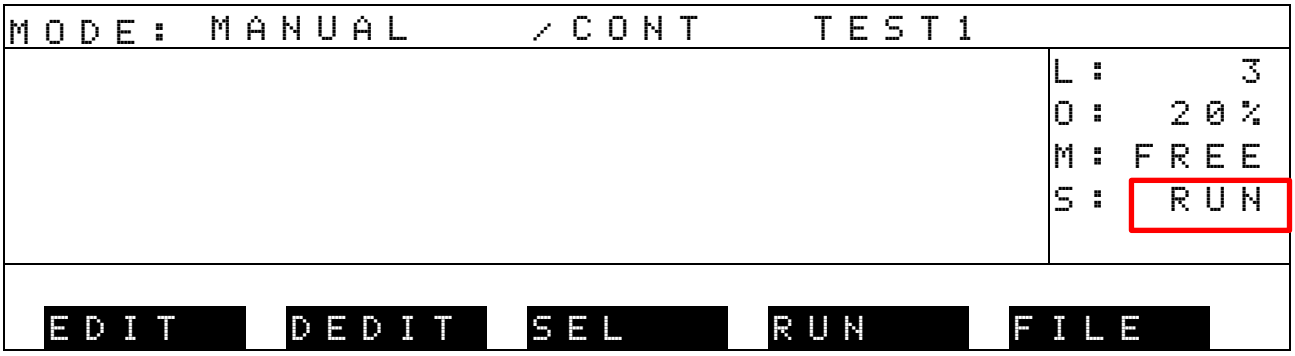
"RUN" appears in the command line.



5 Press the EXE key.

This starts program execution.

The status indicator shows "RUN".



MEMO

In the following cases, the file is not run even if the above operation is performed.

If the file cannot be run, "CAN NOT EXECUTE" appears.

- *When an alarm at level 2 or higher has occurred*
- *When the STOP signal is OFF*
- *When the BREAK signal is OFF*
- *When the signal FEEDHOLD function is enabled and 0 is written to the register G12C*[1]
- *When Servo ON is not set*
- *When an execution program has not been selected*

**1 Instead of """CAN NOT EXECUTE""", the message "feed hold on" appears.*

2.3.6 Program Execution Stopping and Aborting

This describes how to stop program execution.
Program execution can be stopped by a stop or abort operation.

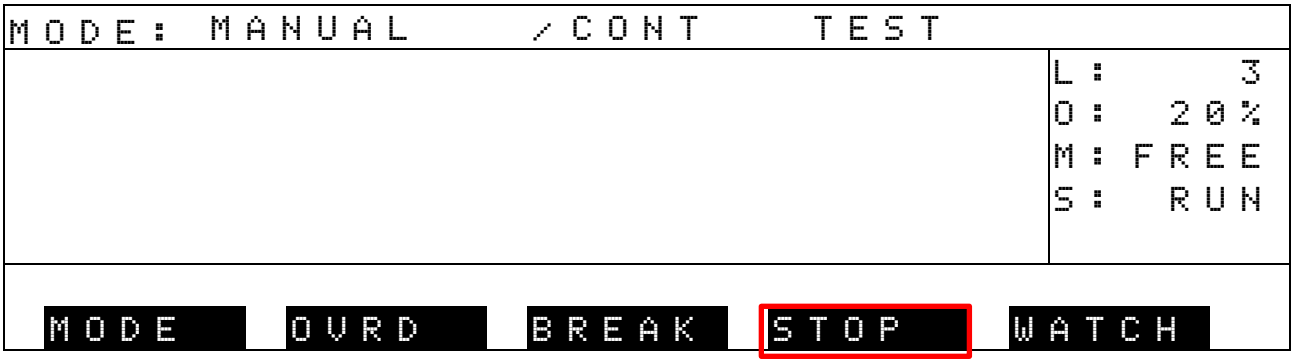
Each procedure is described in detail below.
Stop: Stops program execution once execution of the current command is completed.
Abort: Stops program execution immediately. In this state, the robot decelerates to a stop.

2.3.6.1 Program Execution Stop (STOP)

This describes how to stop program execution.

<Program Execution Stop Procedure>

1 Press the NEXT key until "STOP" appears in the function menu on the basic screen.



2 Press the STOP (F4) key.

This stops execution of the program.
When program execution is stopped, "STOP CONT" appears in the status indicator.



MEMO

Program execution can be stopped by performing the following operations.

• Setting the STOP signal to OFF during program execution

(For details, refer to the "STA1167 Interface Manu".)

• Sending the communication command "SOP1" to the HOST port of the controller

(For details, refer to the "STA1164 Communication Manual".)

• Executing the STOP command in SCOL2 language

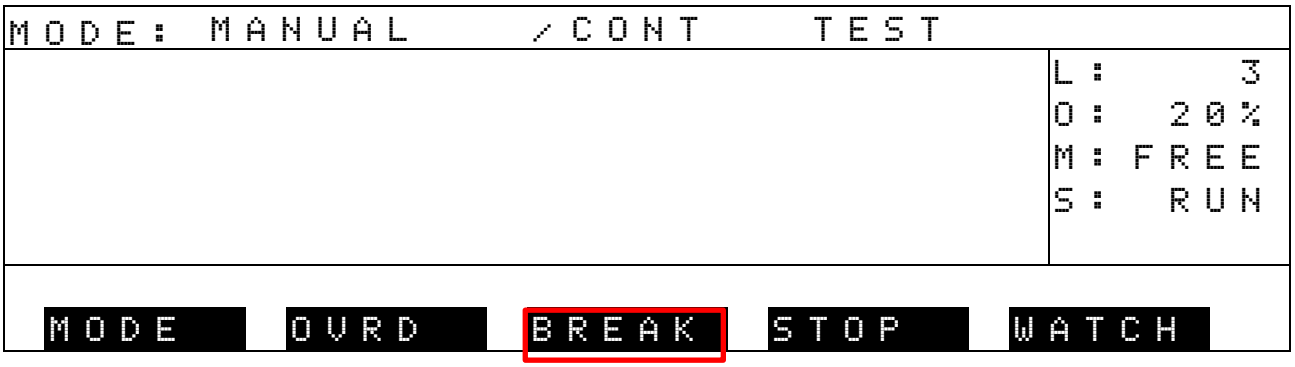
(For details, refer to "STA1163 Robot Language Manual (SCOL2)".)

2.3.6.2 Program Execution Abort (BREAK)

This describes how to abort program execution.

<Program Execution Abort Procedure>

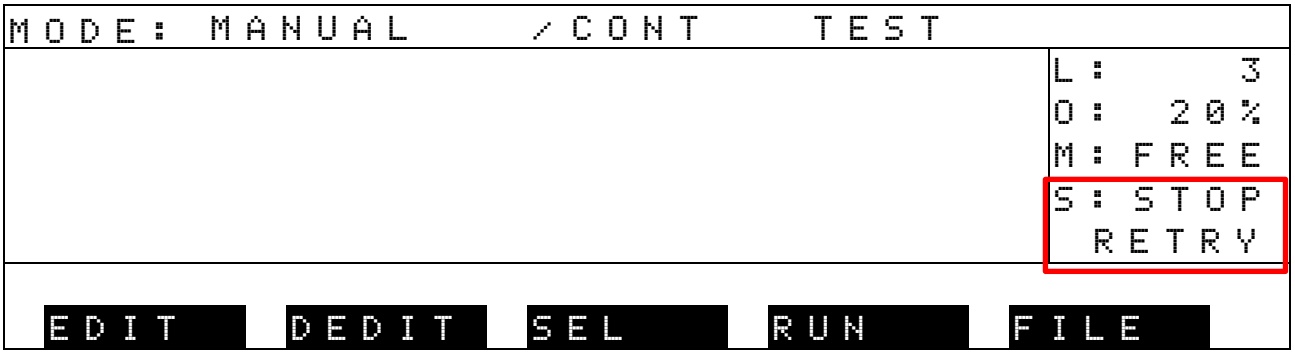
1 Press the NEXT key until "BREAK" appears in the function menu on the basic screen.



2 Press the BREAK (F3) key.

This aborts execution of a program.

When program execution is aborted, "STOP RETRY" appears in the status indicator.



MEMO

After aborting, when program execution is performed, program execution is performed again from the line where execution was aborted.

MEMO

In addition to the above operation, the following operations can be used to abort program execution.

• Setting the BREAK signal to OFF during program execution

(For details, refer to the "STA1167 Interface Manual".)

• Sending the communication command "BRK1" to the HOST port of the controller

(For details, refer to the "STA1164 Communication Manual".)

2.3.7 Program Execution Feedhold (FEEDHOLD)

This describes how to apply or remove a feedhold on program execution.

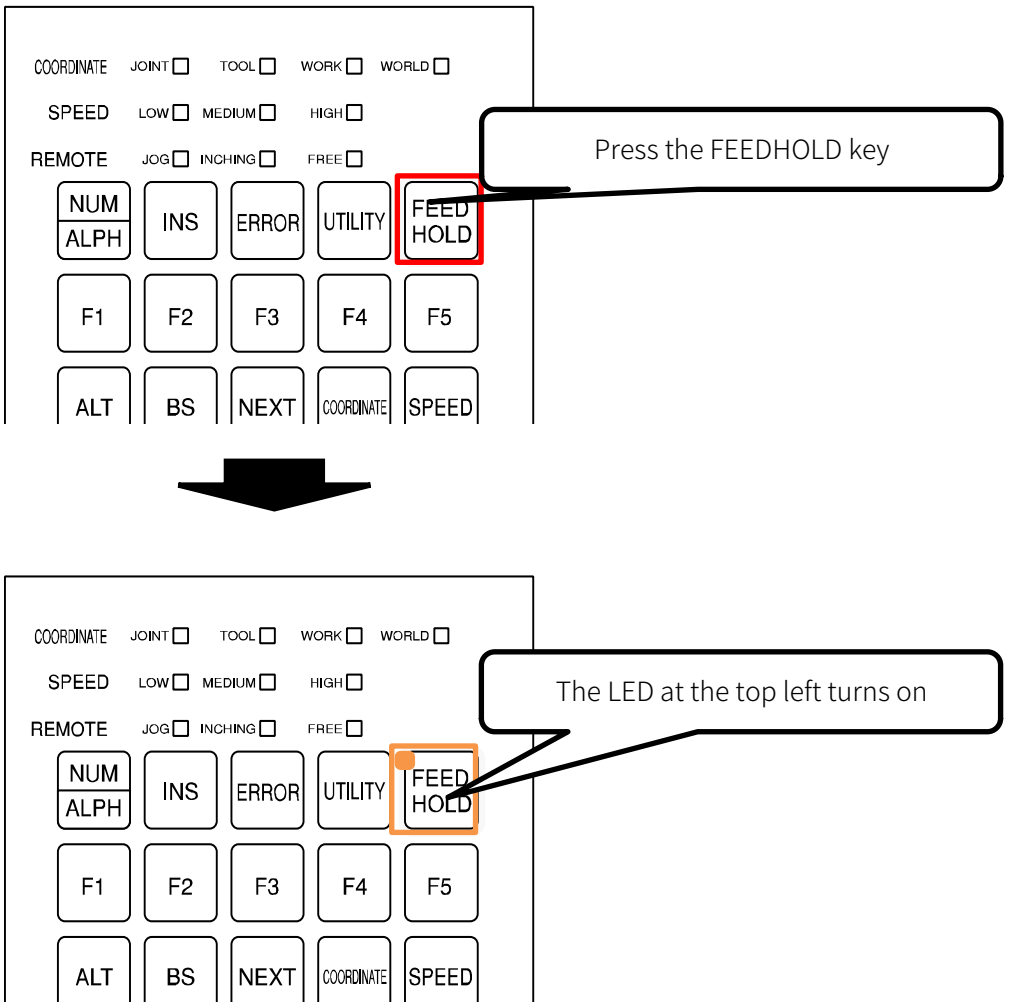
<Applying a Feedhold to Program Execution>

1 Press the FEEDHOLD key on the key sheet.

A feedhold is applied to program execution.

The LED at the top left of the FEEDHOLD key on the key sheet turns on.

While a feedhold is applied to program execution, "RUN" appears in the status indicator.



MEMO

Program execution can still be stopped or aborted during FEEDHOLD.

If this happens, then program execution is immediately stopped.

MEMO

In addition to the above operations, a feedhold can also be applied by performing the following operations.

• Enabling [P006] and setting FEEDHOLD (register G12C) to OFF during program execution

(For details, refer to the "STA1166 Parameter Manual"and "STA1168 TCmini (Internal PLC) Manual".)

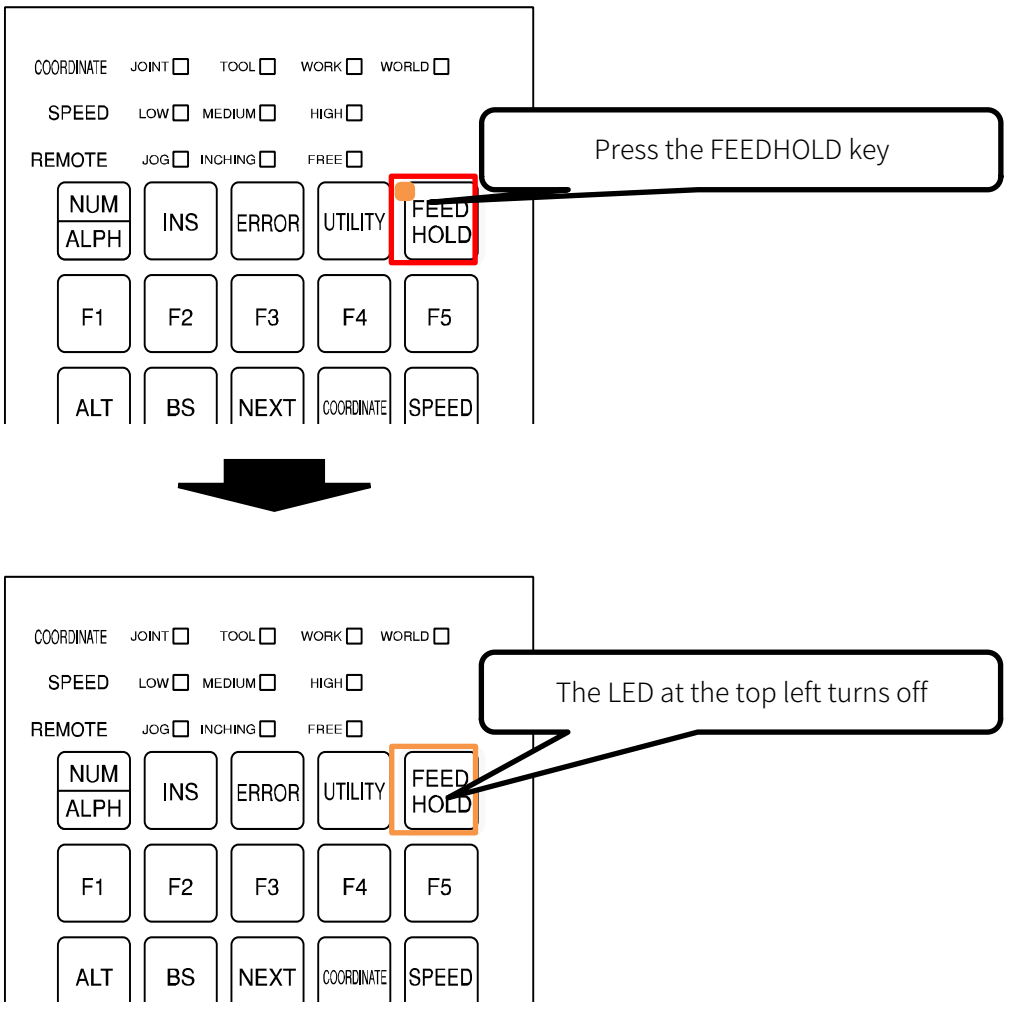
• Sending the communication command "FON1" to the HOST port of the controller

(For details, refer to the "STA1164 Communication Manual".)

<Clearing Feedhold>

1 Press the FEEDHOLD key on the key sheet.

This clears the feedhold on program execution and resumes program execution.
The LED at the top left of the FEEDHOLD key on the key sheet turns off.



MEMO

In addition to the above operations, a feedhold on a file can also be cleared by performing the following operations.

• Enabling [P006] and setting FEEDHOLD (register G12C) to ON during FEEDHOLD.
(For details, refer to the "STA1166 Parameter Manual"and "STA1168 TCmini (Internal PLC) Manual".)

• Sending the communication command "FOF1" to the HOST port of the controller
(For details, refer to the "STA1164 Communication Manual".)

2.3.8 Reset (RESET)

This describes the reset operations.
There are five available reset operations.

• Program reset (PROG)
This resets the execution line and variables of the program that is currently being executed.

• Cycle reset (CYCLE)
This resets the non-global variables in the program being executed and moves the current execution line to the label "RCYCLE".

> *MEMO*
> *If there is no label "RCYCLE", this moves to the start of the program.*

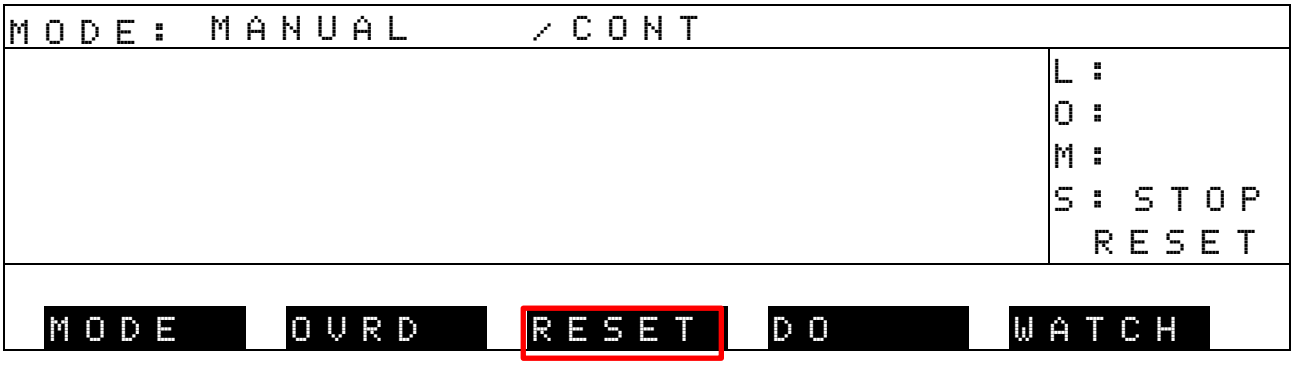
• Step (STEP)
This resets the non-global variables in the program being executed and resets the execution line.

• Signal reset (SIG)
This sets all non-hand output signals to OFF.

• Select reset (SEL)
This resets the execution programs that are currently registered so that no execution programs are registered.

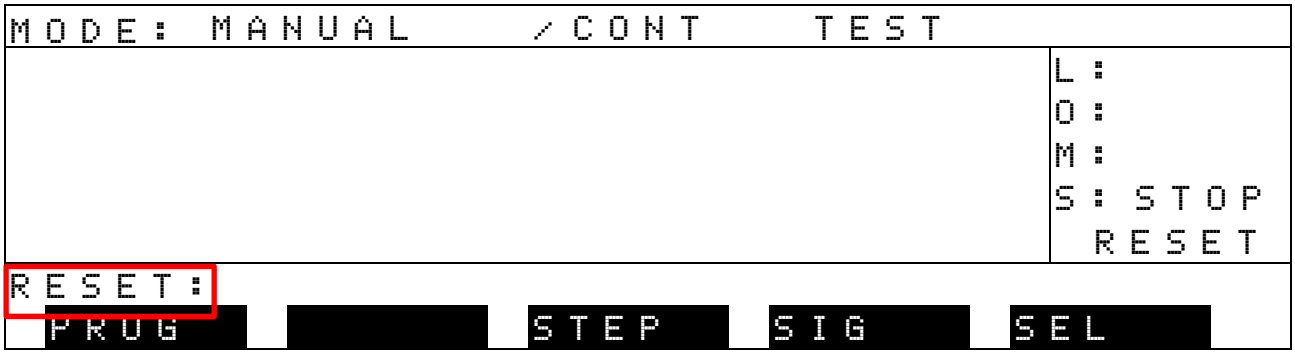
<Reset Procedure>

1 Press the NEXT key until "RESET" appears in the function menu on the basic screen.



2 Press the RESET (F3) key.

"RESET:" is displayed in the command line, and this switches to the function menu.



3 Press the function key corresponding to the item that you want to reset.

The character string assigned to the pressed function key is entered into the command line.
(The character input keys can also be used to enter a character string.)

4 Press the EXE key.

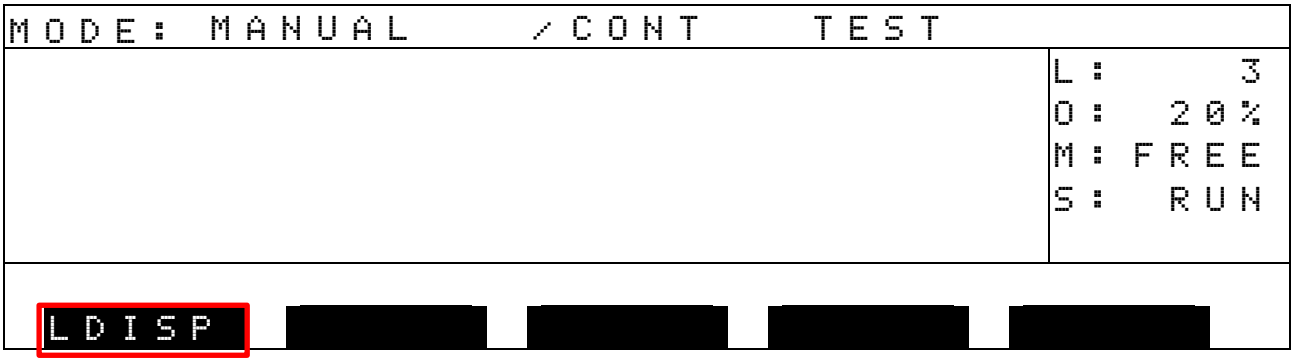
The reset process is performed based on the character string that was entered into the command line.
If the character string is invalid, "CAN NOT EXECUTE" appears.

2.3.9 Execution Program Monitor (LDISP)

This describes how to monitor the content of the execution program.
The content of the execution program is displayed on the LDISP screen.

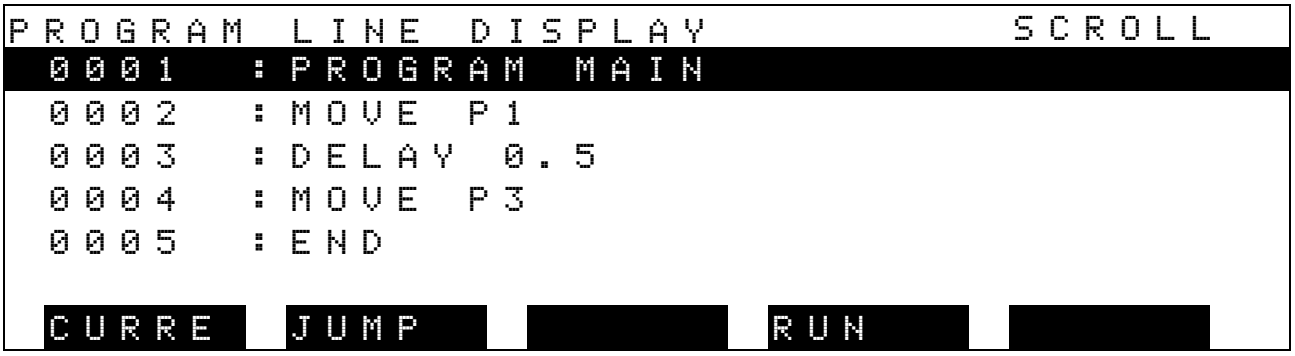
<LDISP Screen Opening Procedure>

1 Press the NEXT key until "LDISP" appears in the function menu on the basic screen.



2 Press the LDISP (F1) key.

The LDISP screen is opened.



The execution program content can be displayed on the LDISP screen using one of two methods.

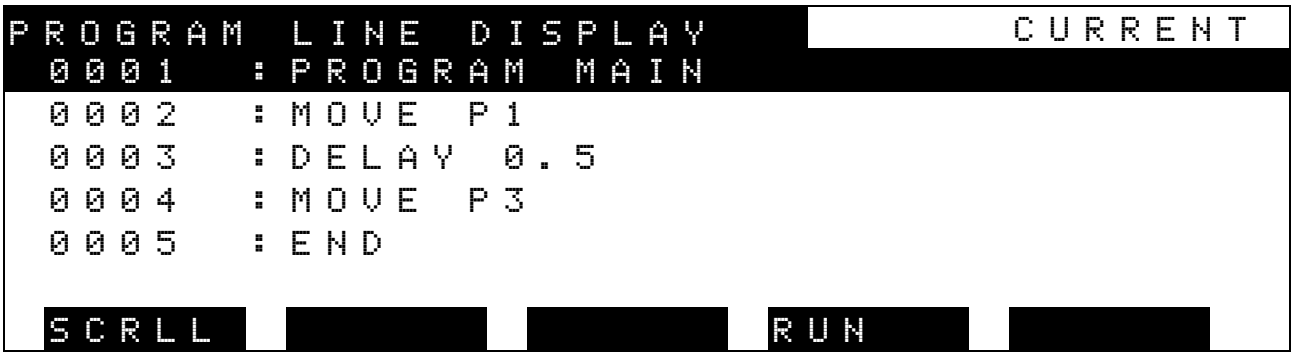
- Displaying in synchronization with the currently-displayed line (Synchronous Display (CURRENT))
- Displaying a selected line (Asynchronous Display (SCROLL))

MEMO

• You can switch between synchronous display and asynchronous display by pressing the SCRLL/CURRE (F1) key.

Execution Program Synchronous Display Screen

In synchronous display, the screen is scrolled based on the line currently being executed.
The currently-displayed execution line appears highlighted as shown in the figure below.



The valid keys in the execution program synchronous display screen and their actions are shown in the following table.

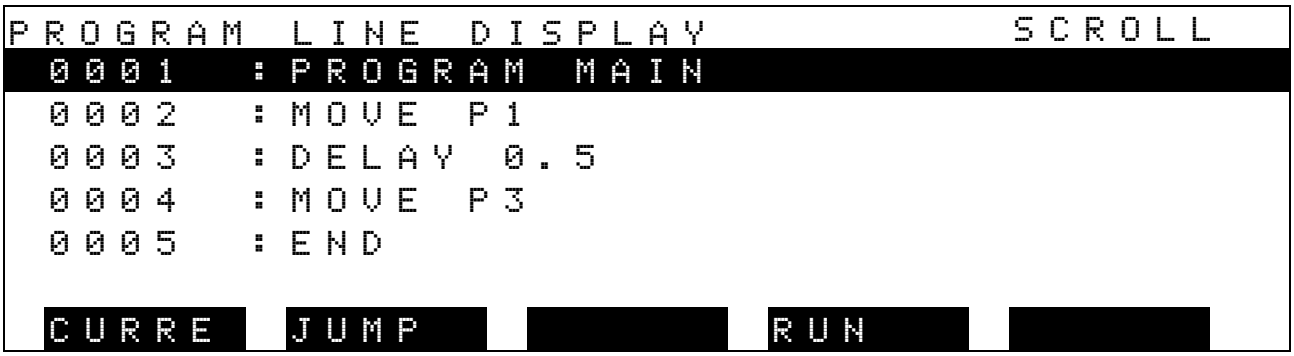
Key	Description
ESC key	This returns to the basic screen.

The function menus in the execution program synchronous display screen and their actions are shown in the following table.

Displayed menu name	Description	Section number for detailed explanation
SCRLL	This opens the asynchronous display screen.	2.8.9
RUN	If program execution is possible, the program is executed.	2.3.5
BREAK	This aborts execution of a program.	2.3.6.2
STOP	This stops execution of the program.	2.3.6.1

Execution Program Asynchronous Display Screen

In asynchronous display, the cursor can be moved and the screen can be scrolled.
The line where the cursor is located appears highlighted as shown in the figure below.



The valid keys in the execution program asynchronous display screen and their actions are shown in the following table.

Key	Description
↑ key	This moves the cursor up by one line.
↓ key	This moves the cursor down by one line.
ESC key	This returns to the basic screen.

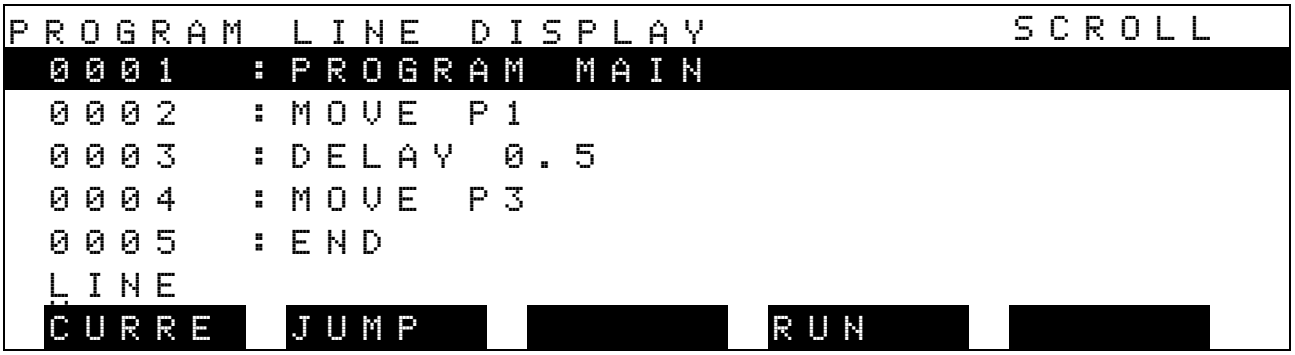
The function menus in the execution program asynchronous display screen and their actions are shown in the following table.

Displayed menu name	Description	Section number for detailed explanation
SCRLL	This opens the asynchronous display screen.	2.2.9
JUMP	This jumps to a specified line.	2.2.9
RUN	If program execution is possible, the program is executed.	2.8.5
BREAK	This aborts execution of a program.	2.3.6.2
STOP	This stops execution of the program.	2.3.6.1

<Jumping to a Selected Line>

1 Press the JUMP (F2) key.

"LINE NO.=" appears in the command line.



2 Enter the line number to be displayed in the command line, and press the EXE key.

The cursor moves to the entered line.

MEMO

If a non-numerical value is entered, this moves to the first line.

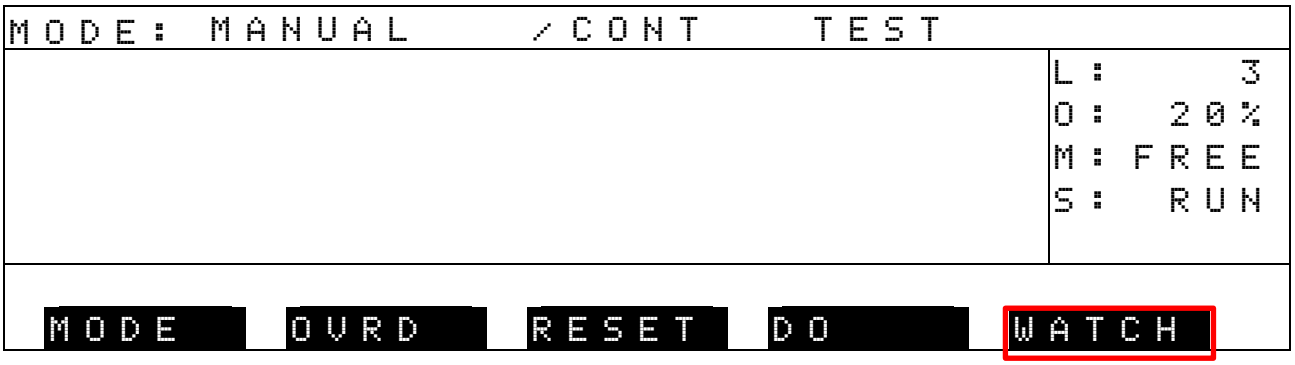
2.3.10 Displaying and Overwriting Global Variables (WATCH)

This describes how to display and overwrite the global variables in the execution program.

A global variable is a variable defined in the area enclosed by GLOBAL-END or DATA-END in a program.
Global variables can be displayed and overwritten on the global variable monitor screen.

<Global Variable Monitor Screen Opening Procedure>

1 Press the NEXT key until "WATCH" appears in the function menu on the basic screen.



MEMO

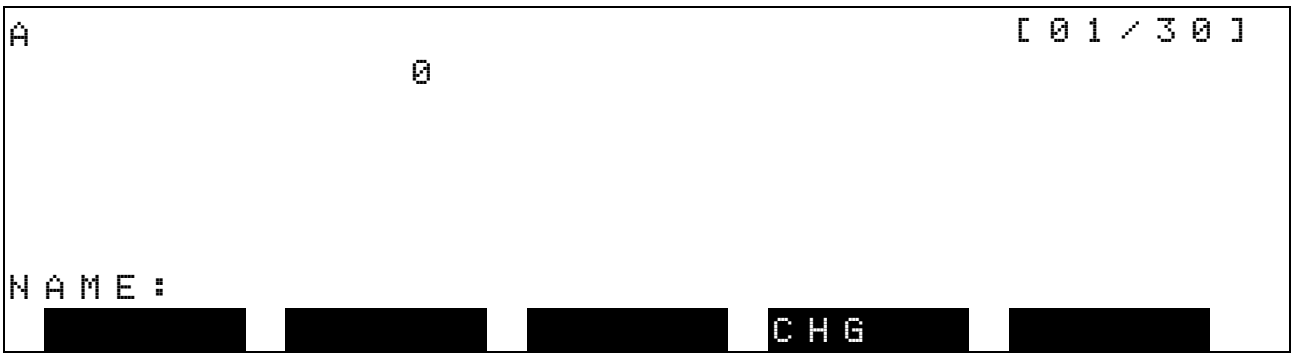
"WATCH" does not appear in the function menu unless the execution program is registered.

2 Press the WATCH (F5) key.

The global variable monitor screen is opened.



Global Variable Monitor Screen



The valid keys in the global variable monitor screen and their actions are shown in the following table.

Key	Description
ESC key	This returns to the basic screen.
↑ key	This displays the previous variable in the watchlist.
↓ key	This displays the next variable in the watchlist.
ALT key + ↑ key or NEXT key	This displays the variable that is five variables before the current variable in the watchlist.
ALT key + ↓ key or ALT key + NEXT key	This displays the variable that is five variables after the current variable in the watchlist.
ALT key + BS key	This removes the displayed variable from the watchlist.

The function menus in the global variable monitor screen and their actions are shown in the following table.

Displayed menu name	Description	Section number for detailed explanation
CHG	This overwrites the value of the displayed variable.	2.8.10

<Registering Variables to Watchlist>

1 Use the character input keys to enter the name of the variable to be registered.

For array variables, include the element number when entering.

MEMO

A maximum of 25 variable names can be registered.

2 Press the EXE key.

The entered variable name is registered to the watchlist, and the current value appears on the global variable monitor screen.

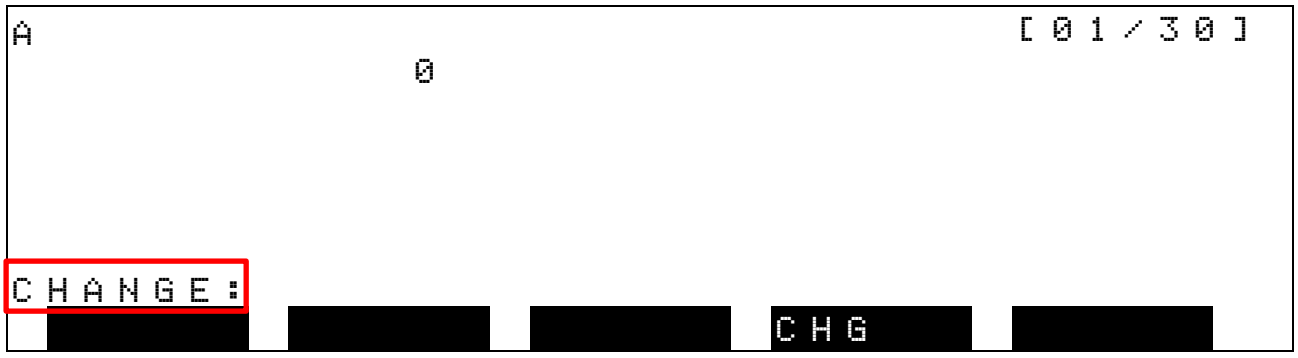
> *MEMO*
>
> *• A maximum of 30 variables can be registered to the watchlist.*
>
> *• Integer, real number, POINT, TRANS, and PAYLOAD data only can be displayed in the watchlist. JOINT and STRING variables are displayed as "----.---".*
>
> *• If there are no global variables for the entered name, "----.---"""appears.*

<Overwriting Variable Values>

1 This shows the variable whose value will be overwritten.

2 Press the CHG (F4) key.

"CHANGE:" appears in the command line.



3 Move the cursor to the element that will be overwritten, and use the character input keys to enter the value.

4 Press the EXE key.

The value of the displayed variable is overwritten by the value in the command line.

MEMO

• To prevent the robot from running out of control due to mistaken input, the POINT and TRANS variables cannot be overwritten with values that are more than ± 3 mm (or deg) separated from the current values for each element.

• The overwritten value is a temporary change, and the value is reset by performing a program reset. To permanently apply the overwritten value, consider either editing the file or using a memory variable. For details on memory variables, see the section "MEMORY" in the Language Manual.

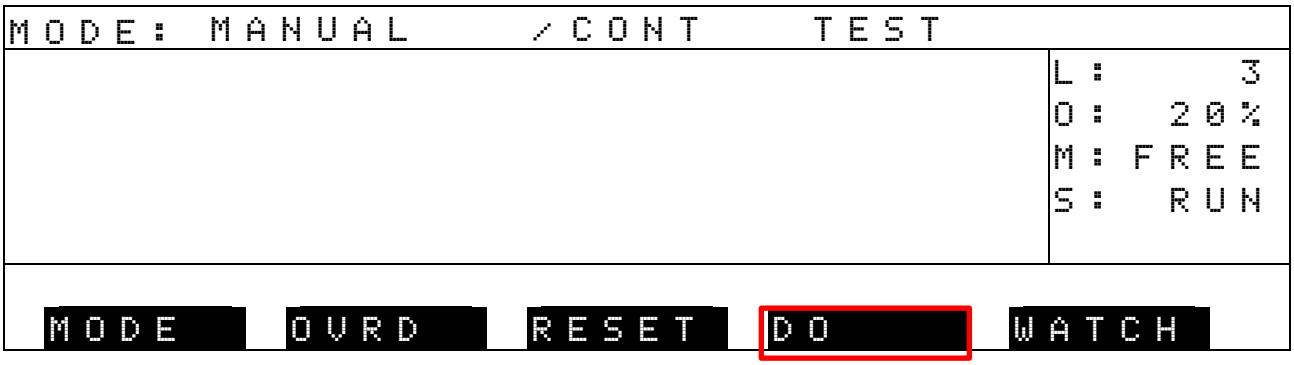
2.4 Direct Execution (DO)

This describes how to perform direct execution.

Direct execution is a function that enables execution of one line of commands in SCOL2 language without using test operation.

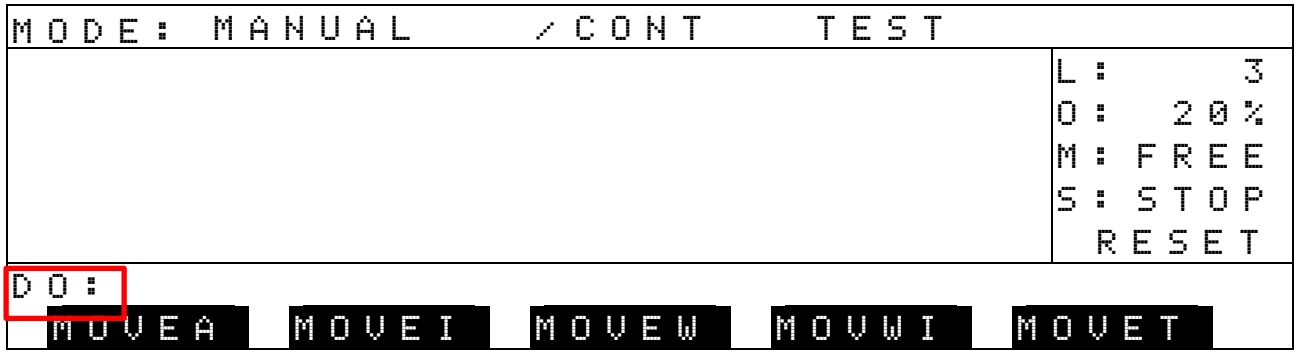
<Direct Execution Procedure>

1 Press the NEXT key until "DO" appears in the function menu on the basic screen.



2 Press the DO (F4) key.

"DO:" appears in the command line.



3 Use the character input keys to enter the command for direct execution.

> *MEMO*
>
> *When the function key is pressed, the character string assigned to the pressed function key is entered into the command line.*

4 Press the EXE key.

The command that was entered into the command line is executed.

When execution is completed or fails, the character string that was entered into the command line is cleared.

3. File Editing and Operations

This chapter describes the file editing and operating procedures.
File editing and operations using the TP1000 can be performed in MANUAL mode only.

The files used by the TS5000 have the following structure.

File Global section
Program section
Data section

Global variables are defined in the global section, movement commands for the robot are contained in the program section, and POINT-type and TRANS-type data are contained in the data section.

3.1 Editing the Program Section

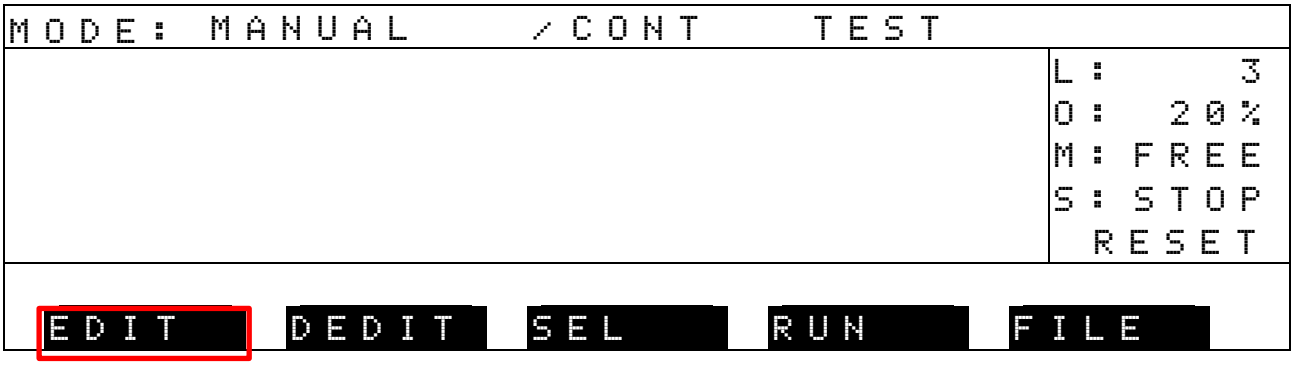
This describes the operations for editing the program section.
The program section editing is described in the "program section edit screen".

3.1.1 Opening the Program Section Edit Screen

This describes how to open the program section edit screen.

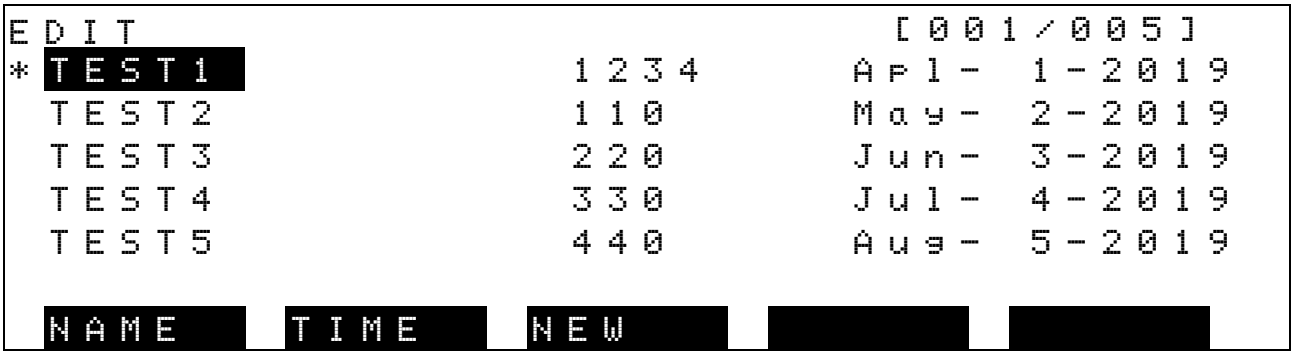
<Program Section Edit Screen Opening Procedure for a New File>

1 Press the NEXT key until "EDIT" appears in the function menu on the basic screen.



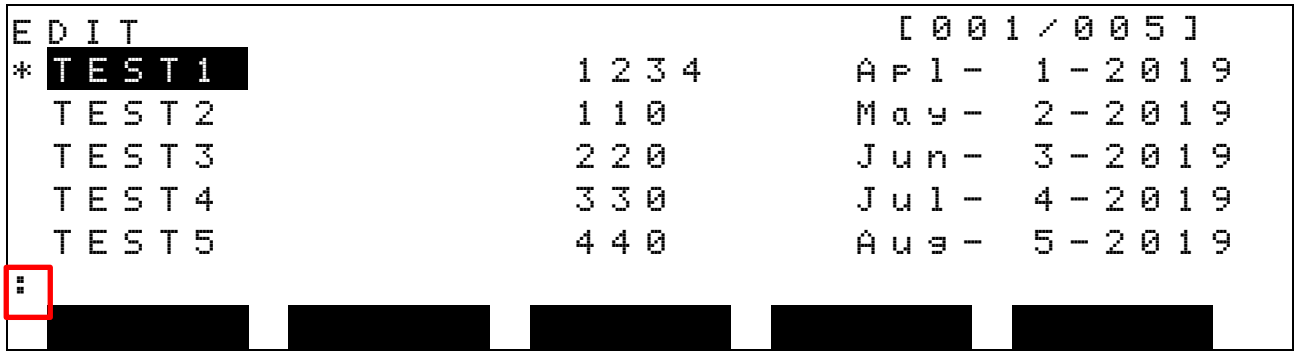
2 Press the EDIT (F1) key.

The select edit file screen is opened.



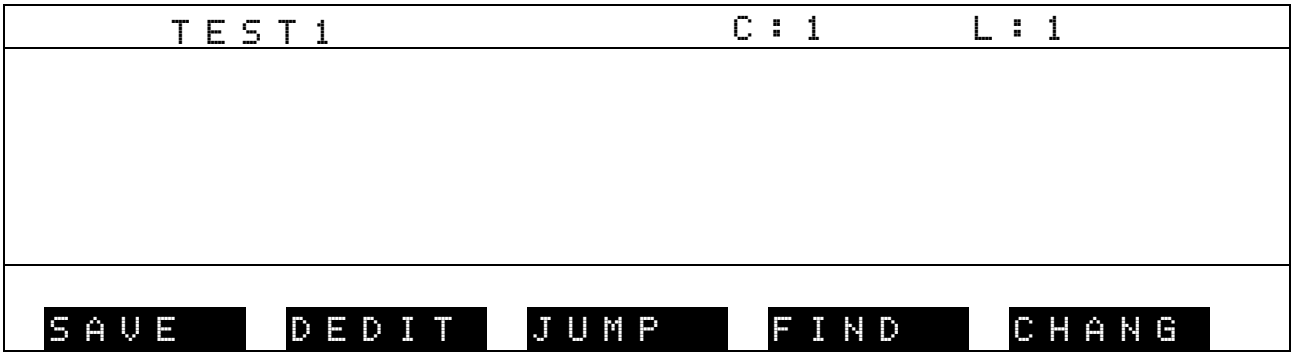
3 Press the NEW (F3) key.

":" appears in the command line.



4 Enter the name of the new file to be created, and press the EXE key.

The program section edit screen is opened.



MEMO

- *The file naming rules are as follows.*

  File name is a maximum of 8 alphanumeric characters.

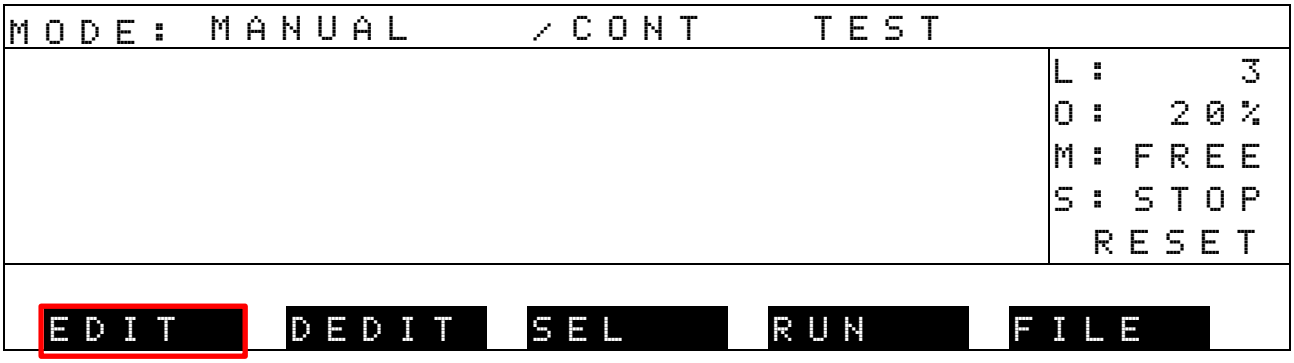
  The extension can use a maximum of 3 characters.
- *If a file name that does not comply with the naming rules is entered, and the EXE key is pressed, "Input error" appears.*

MEMO

- *If the name of an existing file is entered, the new file is not created, and the program section edit screen for the existing file is opened.*

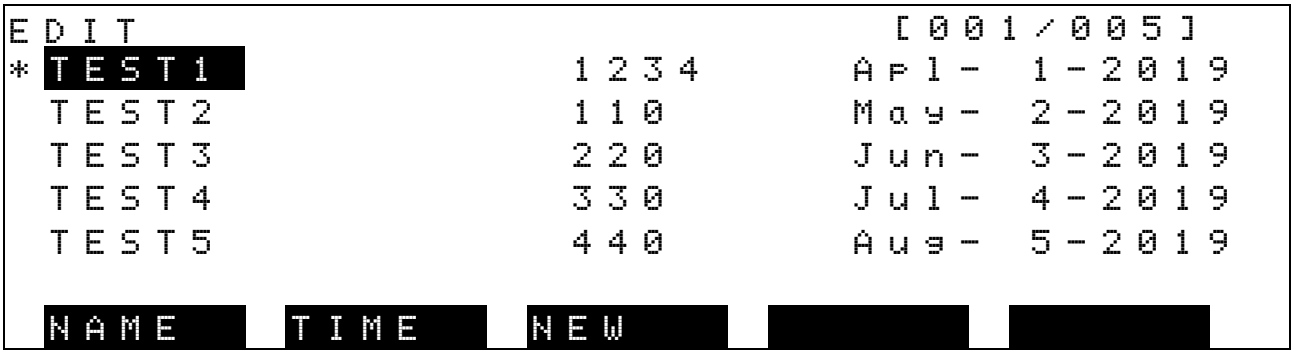
<Program Section Edit Screen Opening Procedure for an Existing File>

1 Press the NEXT key until "EDIT" appears in the function menu on the basic screen.



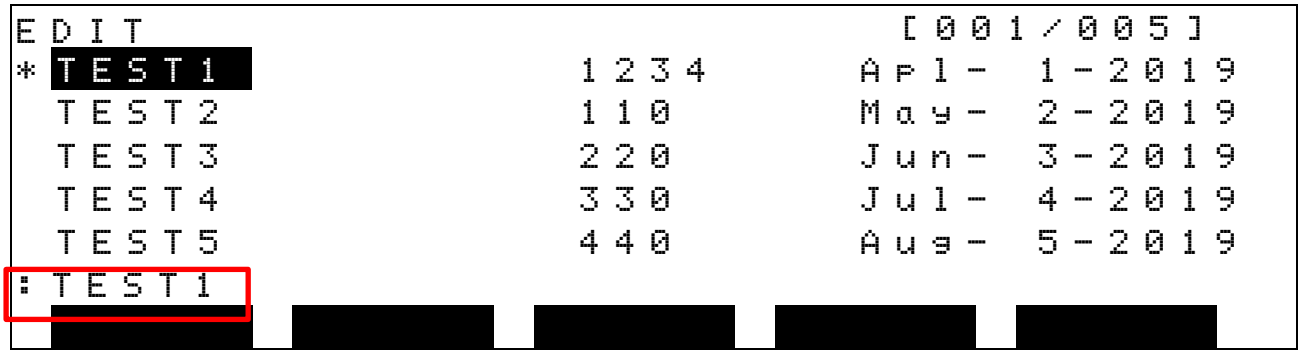
2 Press the EDIT (F1) key.

The select edit file screen is opened.



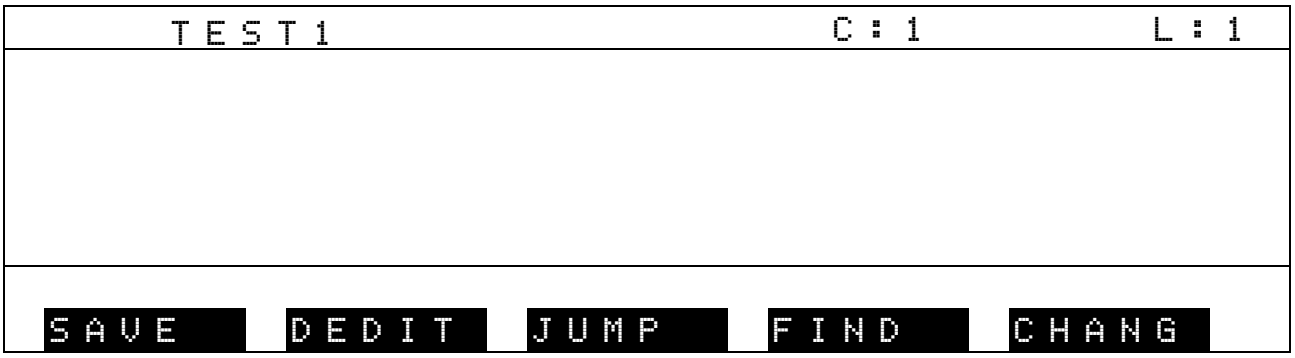
3 Move the cursor to the file to be edited, and press the EXE key.

":<*Selected File Name*>" appears in the command line.



4 Press the EXE key.

The program section edit screen is opened.

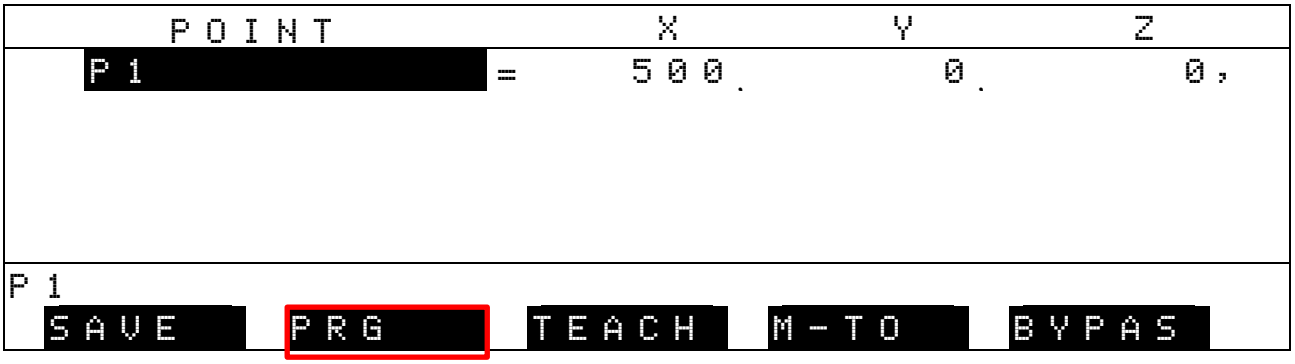


MEMO

If a file with the name shown in the command line does not exist, the file is created as a new file, and the program section edit screen for this new file is opened.

<Opening the Program Section Edit Screen from the Data Section Edit Screen>

1 Press the NEXT key until "PRG" is displayed in the function menu on the data section edit screen.

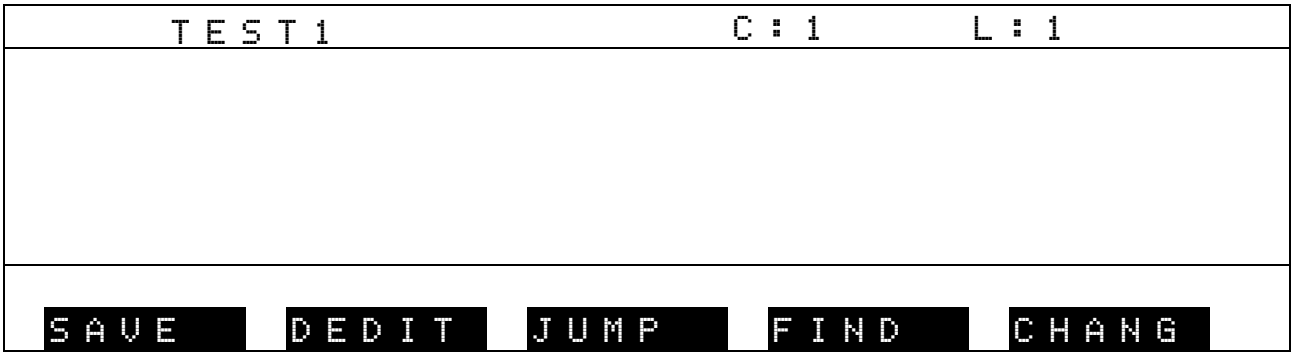


MEMO

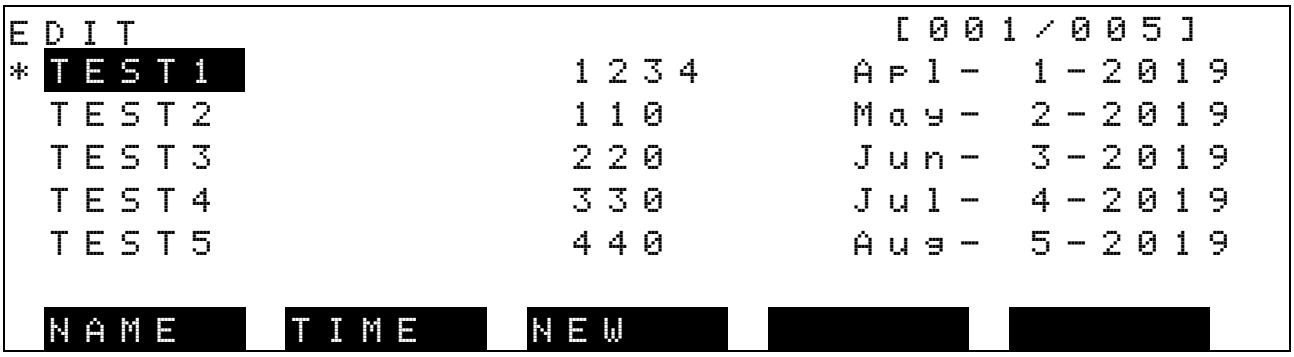
- *If the DEDIT (F2) key was pressed from the basic screen to open the data section edit screen, "PRG" does not appear.*
- *The program section edit screen can also be opened from the POINT-type data list screen, POINT-type data details screen, TRANS-type data list screen, and PAYLOAD-type data edit screen.*

2 Press the PRG (F2) key.

The program section edit screen is opened.



Select Edit File Screen



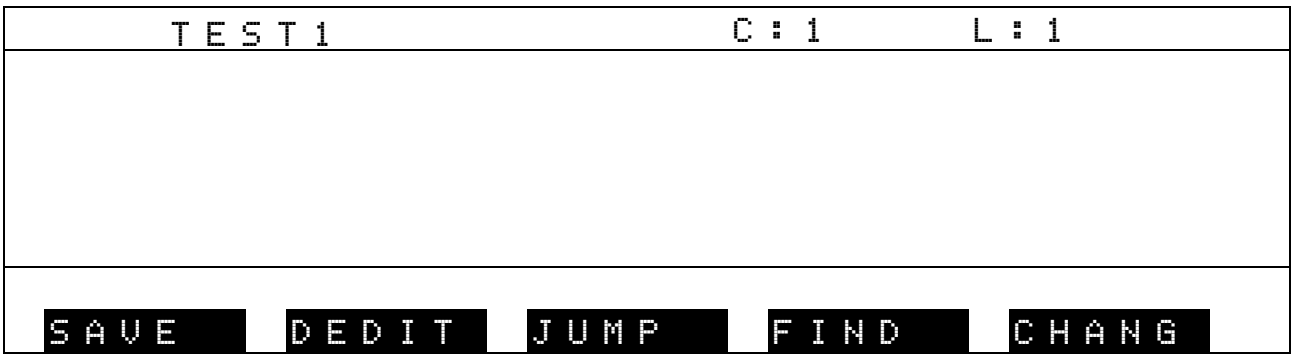
The valid keys in the select edit file screen and their actions are shown in the following table.

Key	Description
↑ key	This moves the cursor up by one line.
↓ key	This moves the cursor down by one line.
ALT key + ↑ key or ALT key + NEXT key	This moves the cursor to the previous page.
ALT key + ↓ key or NEXT key	This moves the cursor to the next page.
ESC key	This returns to the basic screen.

The function menus in the select edit file screen and their actions are shown in the following table.

Displayed menu name	Description	Section number for detailed explanation
NAME	This is used to sort files by name.	
TIME	This is used to sort files by date and time created.	
NEW	The ":" is displayed in the command line to indicate that the command line can be edited. This creates a new file with the name entered in the command line.	

Program Section Edit Screen



The valid keys in the program section edit screen and their actions are shown in the following table.

Key	Description
↑ key	This moves the cursor up by one line.
↓ key	This moves the cursor down by one line.
ALT key + ↑ key	This moves the cursor to the previous page.
ALT key + ↓ key	This moves the cursor to the next page.
← key	This moves the cursor to the previous character. If the cursor is at the start of the line, it moves to the end of the line above it.
→ key	This moves the cursor to the next character. If the cursor is at the end of the line, it moves to the start of the line below it.
NEXT key or ALT key + NEXT key	This switches the function menu.
EXE key	This enters a line feed code to the cursor position.
BS key	This erases the character before the cursor.
ALT key + BS key	This erases the character at the cursor position.
INS key	This switches the character input mode.
Character input keys	This enters the character corresponding to the key at the cursor location.

The function menus in the program section edit screen and their actions are shown in the following table.

Displayed menu name	Description	Section number for detailed explanation
SAVE	This saves and exits the file.	3.3
DEDIT	This opens the data section edit screen.	3.2.1
JUMP	The ":" is displayed in the command line to indicate that the command line can be edited. This creates a new file with the name entered in the command line.	3.1.9
FIND	This searches for a character string.	3.1.7
CHANG	This replaces a character string.	3.1.8
NOSAV	This discards the file.	3.3
COPY	This copies a character string.	3.1.5
CUT	This cuts a character string.	3.1.4
PASTE	This pastes a character string.	3.1.6

3.1.2 Editing the Program Section

When the character input keys are pressed, characters corresponding to the keys are entered at the cursor position.
Create a program by following the syntax rules for SCOL2 language.
For details on the syntax rules for SCOL2 language, see the separate "STA1163 Language Manual".

3.1.3 Switching the Character Input Mode

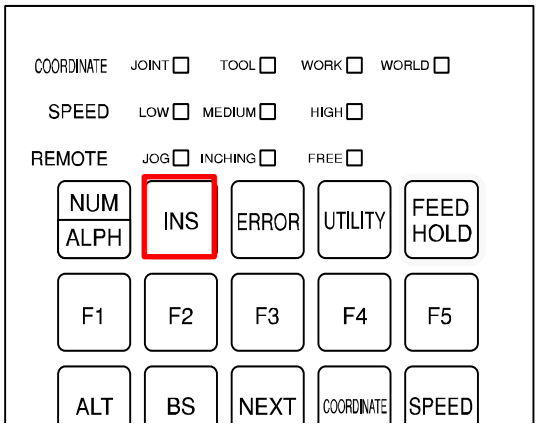
The character input modes include "insert mode", where characters corresponding to the pressed keys are inserted after the cursor, and "overwrite mode", where the characters at the cursor position are replaced by the characters corresponding to the pressed keys.

> *MEMO*
> *When in insert mode, the LED at the top left of the INS key is lit. When in overwrite mode, the LED at the top left of the INS key is off.*

<Switching the Character Input Mode>

1 Press the INS key.

The character input mode is switched.

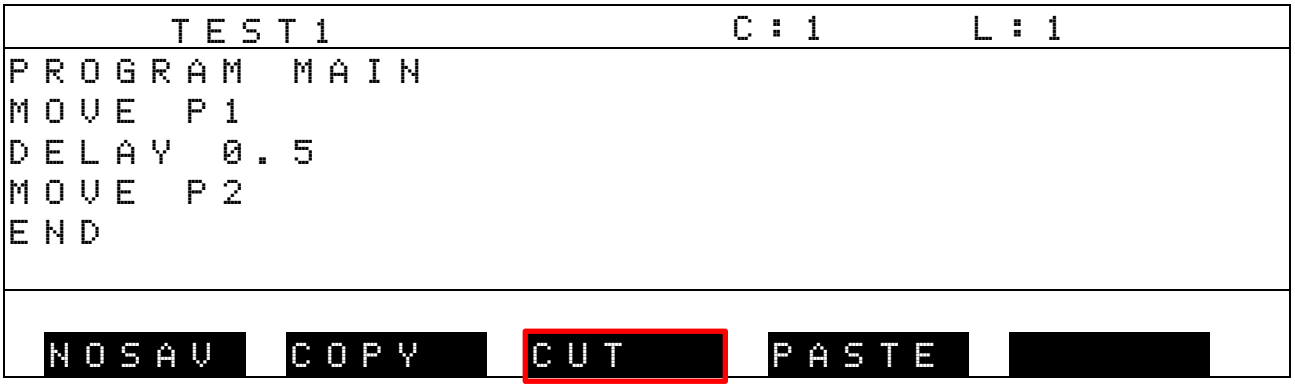


3.1.4 Cutting a Character String (CUT)

This describes how to cut a character string when editing the program section.

<Character String Cutting Procedure>

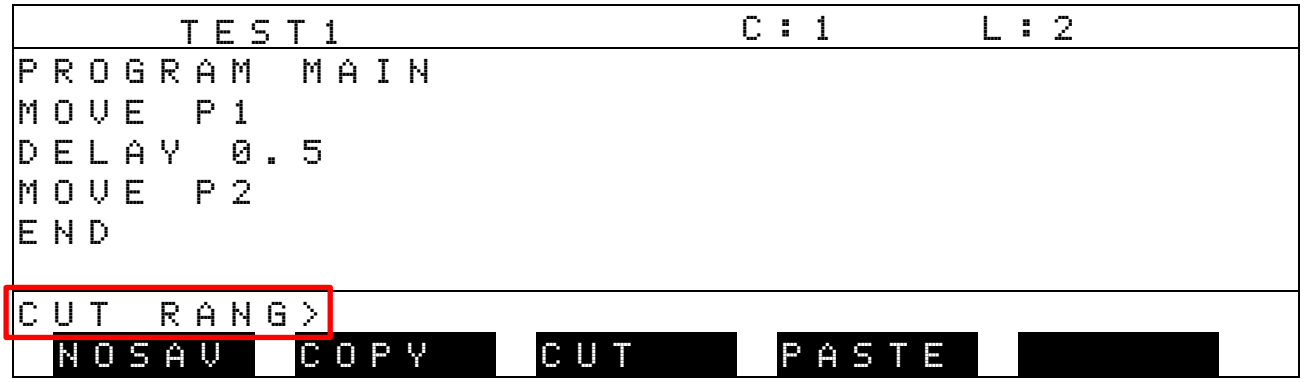
1 Press the NEXT key until "CUT" is displayed in the function menu on the program section edit screen.



2 Move the cursor to the start of the character string to be cut.

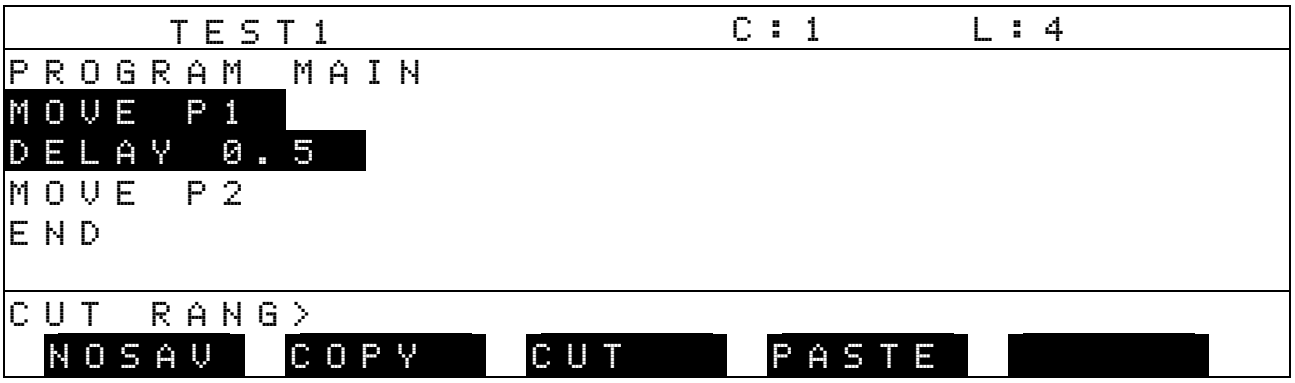
3 Press the CUT (F3) key.

"CUT RANG>" appears in the command line.



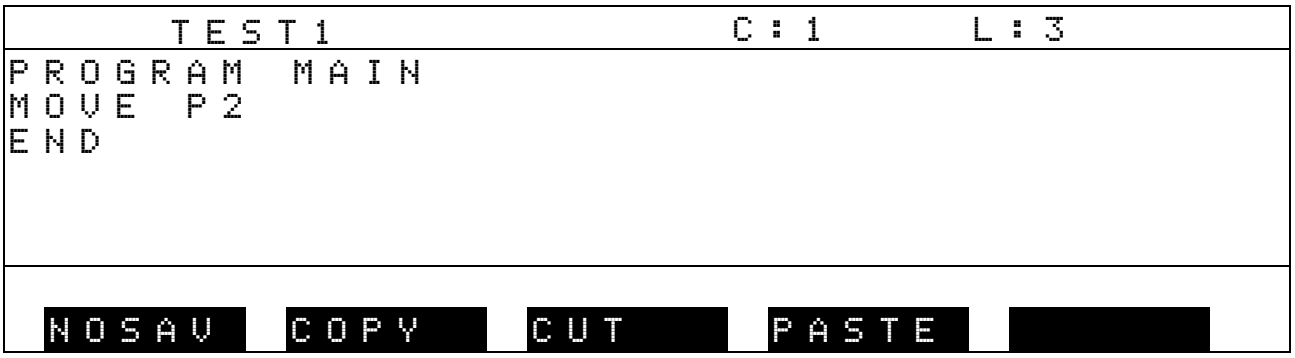
4 Move the cursor to the end of the character string to be cut.

The selected character string appears highlighted.



5 Press the CUT (F3) key.

This selected range of the character string is cut.



MEMO

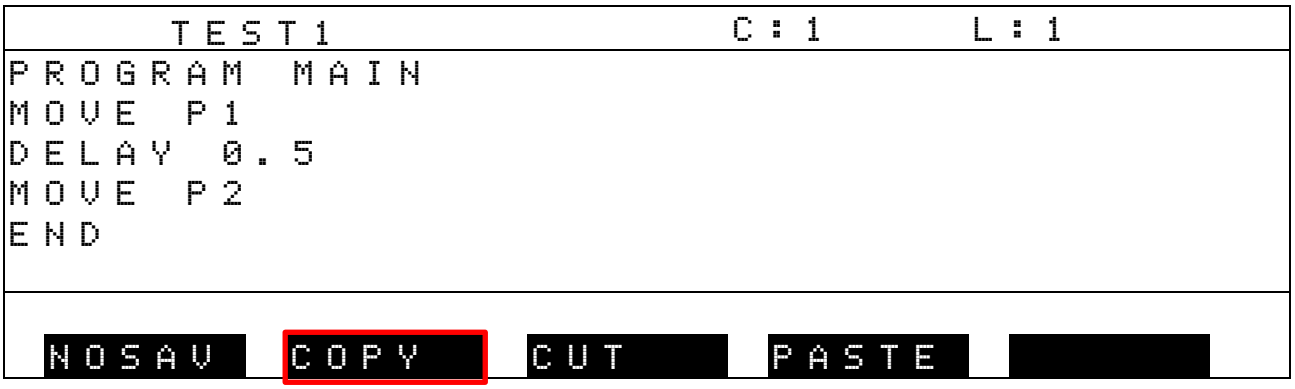
Characters consisting of a maximum of 2248 bytes can be cut. (A line feed is counted as 1 byte.)

3.1.5 Copying a Character String (COPY)

This describes how to copy a character string when editing the program section.

<Character String Copy Procedure>

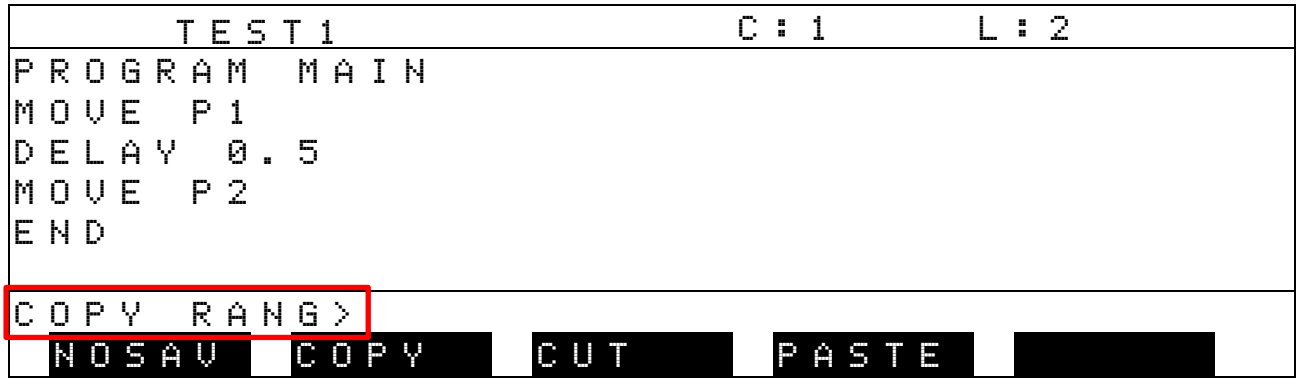
1 Press the NEXT key until "COPY" is displayed in the function menu on the program section edit screen.



2 Move the cursor to the start of the character string to be copied.

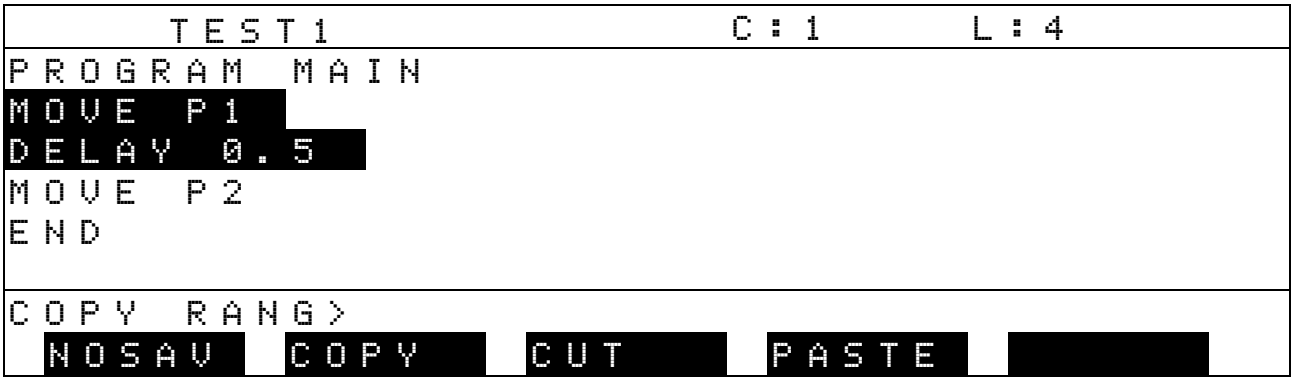
3 Press the COPY (F2) key.

"COPY RANG>" appears in the command line.



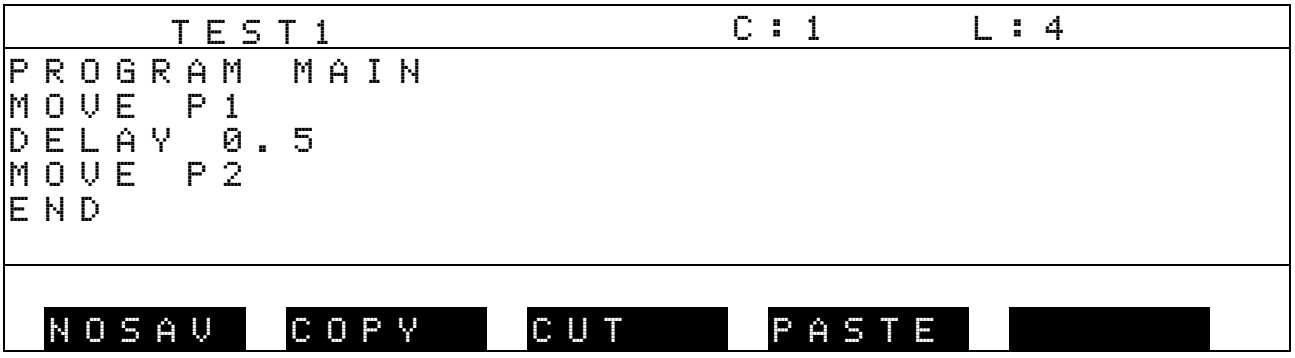
4 Move the cursor to the end of the character string to be copied.

The selected character string appears highlighted.



5 Press the COPY (F2) key.

This selected range of the character string is copied.



MEMO

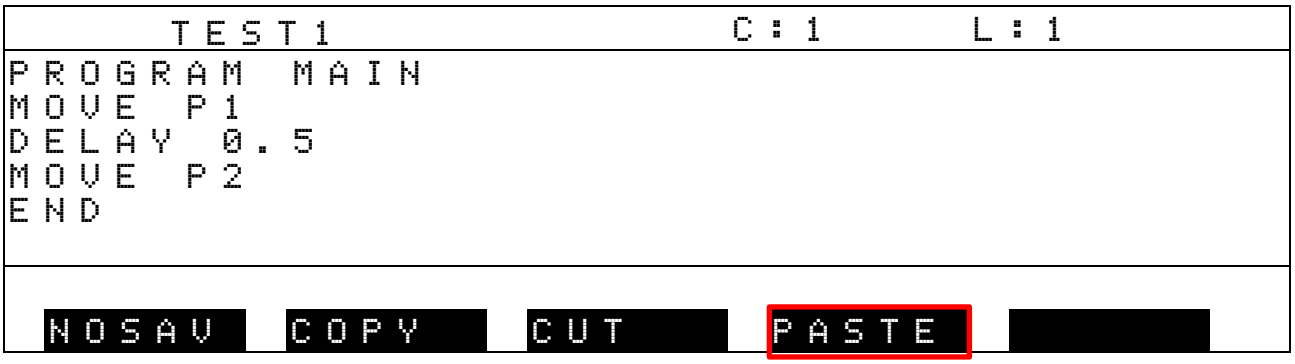
Characters consisting of a maximum of 2248 bytes can be copied. (A line feed is counted as 1 byte.)

3.1.6 Pasting a Character String (PASTE)

This describes how to paste a character string that was cut or copied when editing the program section.

<Character String Paste Procedure>

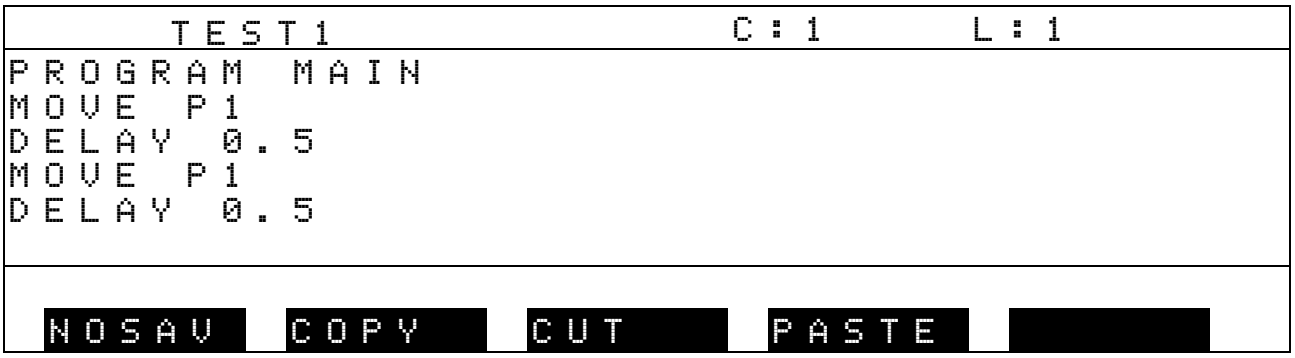
1 Press the NEXT key until "PASTE" is displayed in the function menu on the program section edit screen.



2 Move the cursor to the position where the character string will be pasted.

3 Press the PASTE (F4) key.

The cut or copied character string is inserted at the cursor position.



MEMO

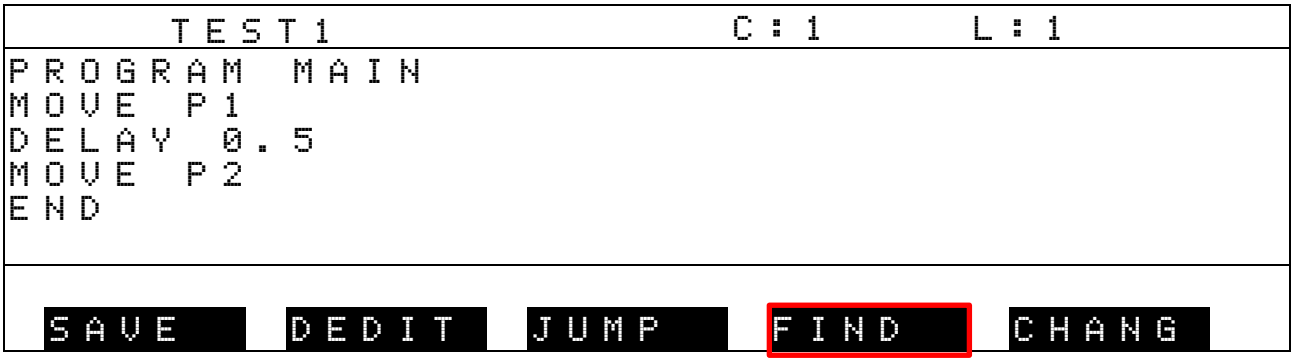
When file editing is completed, the cut or copied character string is cleared.

3.1.7 Searching for a Character String (FIND)

This describes how to search for a character string when editing the program section.

<Character String Search Procedure>

1 Press the NEXT key until "FIND" is displayed in the function menu on the program section edit screen.



2 Press the FIND (F4) key.

"Pattern:" appears in the command line.

3 Enter the character string that you want to search for, and press the EXE key.

A character string search is started from the cursor position, and the cursor moves to the start position of the nearest matching character string.

MEMO

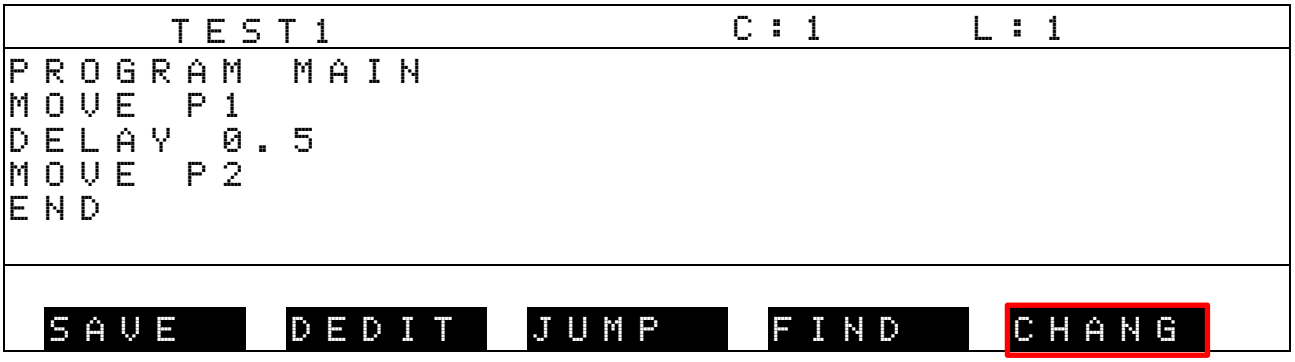
If no matching character string is found, "Not found." appears in the command line.

3.1.8 Replacing a Character String (CHANG)

This describes how to replace a character string when editing the program section.

<Character String Replacement Procedure>

1 Press the NEXT key until "CHANG" is displayed in the function menu on the program section edit screen.



2 Press the CHANG (F5) key.

"Pattern:" appears in the command line.

3 Enter the character string before replacement, and press the EXE key.

"Replacement:" appears in the command line.

4 Enter the character string after replacement, and press the EXE key.

This searches for the character string before replacement from the cursor position, and the cursor moves to the start position of the nearest matching character string.

In the command line, """0: local, 1: global, ESC: exit" appears.



5 Press the 0 key or 1 key.

If the 0 key is pressed, the character string is replaced, and a search is performed for the next matching character string.

If the 1 key is pressed, all matching character strings are replaced.

MEMO

If the ESC key is pressed, this exits replacement of the character string.

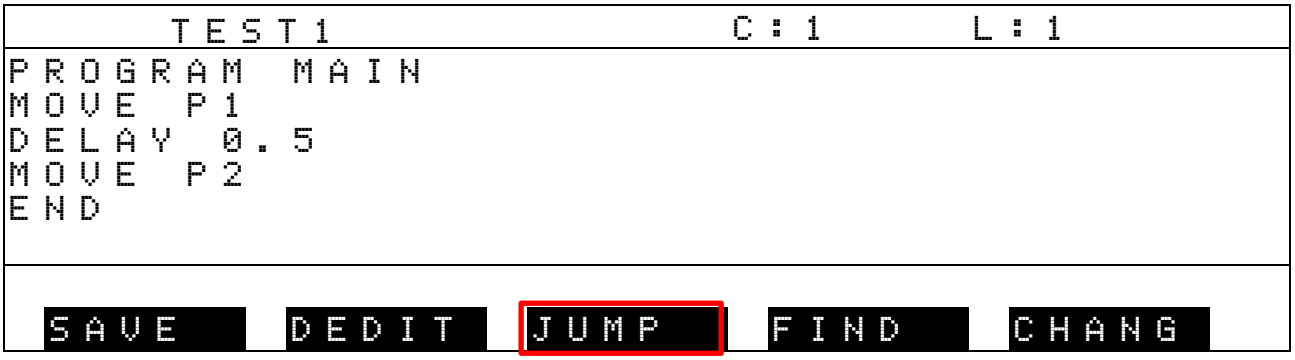
If a key other than 0, 1, or ESC is pressed, the character string is not replaced, and a search is performed for the next matching character string.

3.1.9 Jumping to a Specified Line (JUMP)

This describes how to move the cursor to a specified line when editing the program section.

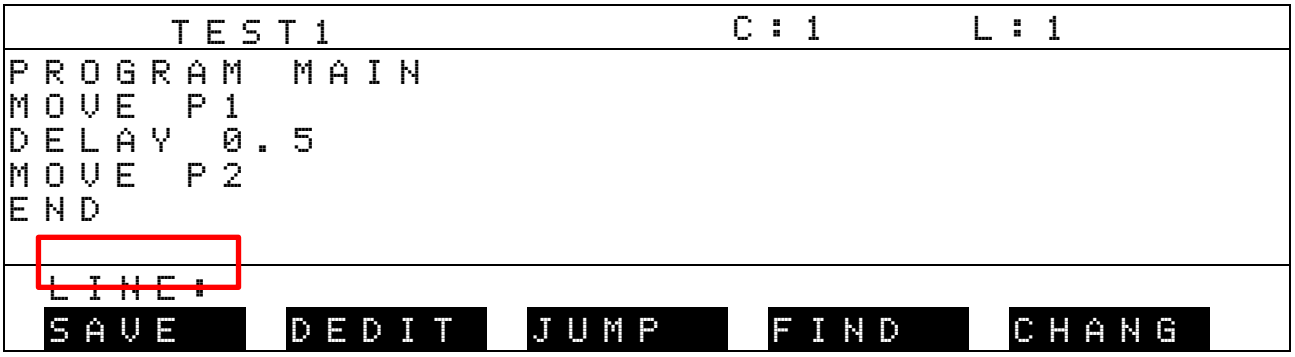
<Procedure for Moving Cursor to Specified Line>

1 Press the NEXT key until "JUMP" is displayed in the function menu on the program section edit screen.



2 Press the JUMP (F3) key.

"LINE:" appears in the command line.



3 Enter the line number to be displayed in the command line, and press the EXE key.

The cursor moves to the entered line.

3.2 Editing the Data Section

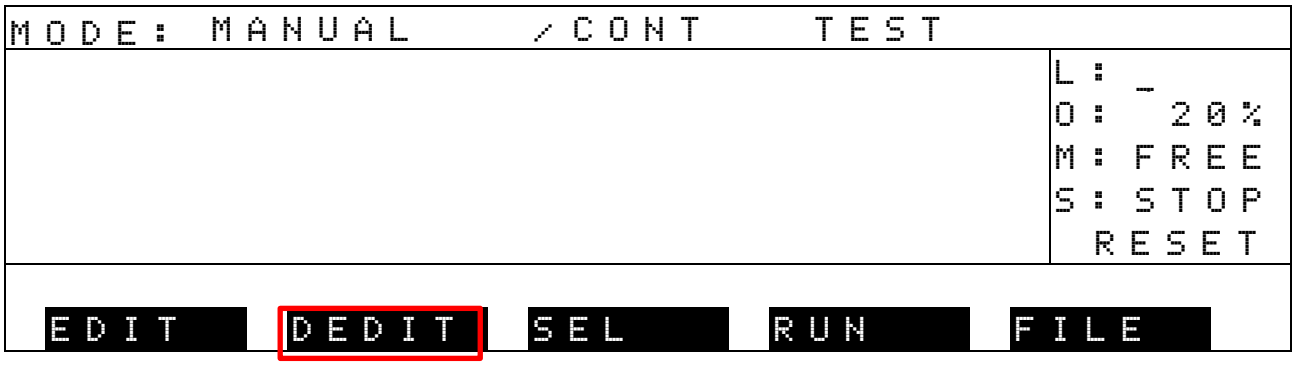
This describes the operations for editing the data section.
The data section contains three types of data -- POINT-type data, TRANS-type data, and PAYLOAD-type data, and this data is edited in the POINT-type data list screen, TRANS-type data edit screen, and PAYLOAD-type data edit screen, respectively.
There is also a TRANS-type data list screen that displays a list of TRANS-type data.
Collectively, these screens are called "data section edit screens".

> *MEMO*
> *The data contained in the data section is handled as global variables.*

3.2.1 Opening the Data Section Edit Screen

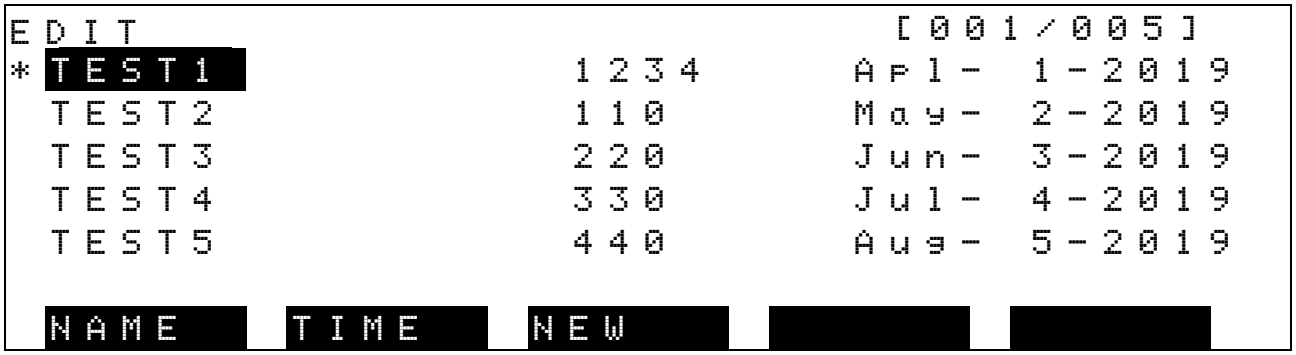
<Data Section Edit Screen Opening Procedure for a New File>

1 Press the NEXT key until "DEDIT" appears in the function menu on the basic screen.



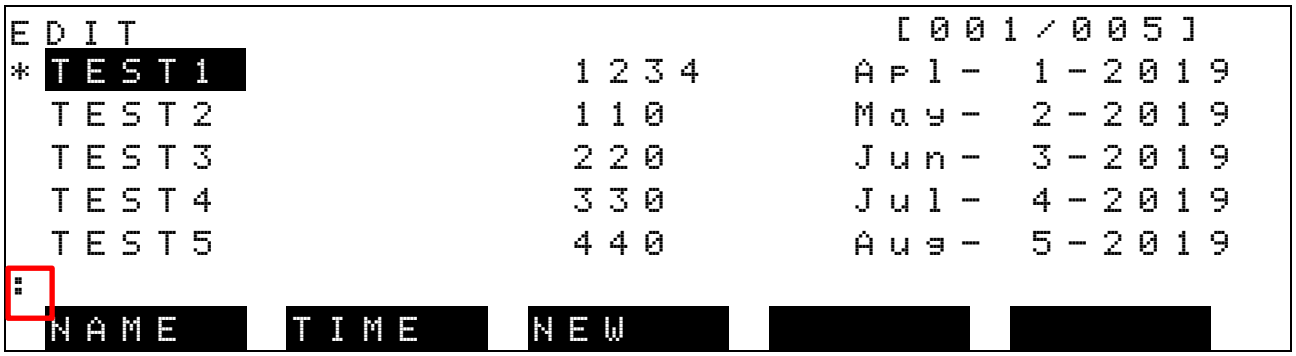
2 Press the DEDIT (F2) key.

The select edit file screen is opened.



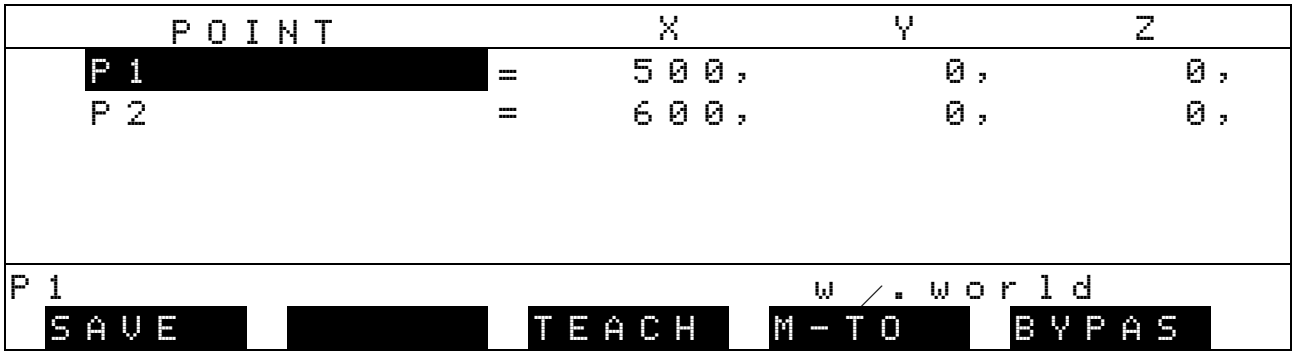
3 Press the NEW (F3) key.

":" appears in the command line.



4 Enter the name of the new file to be created, and press the EXE key.

This opens the data section edit screen (POINT-type data list screen).



MEMO

The file naming rules are as follows.

File name is a maximum of 8 alphanumeric characters.

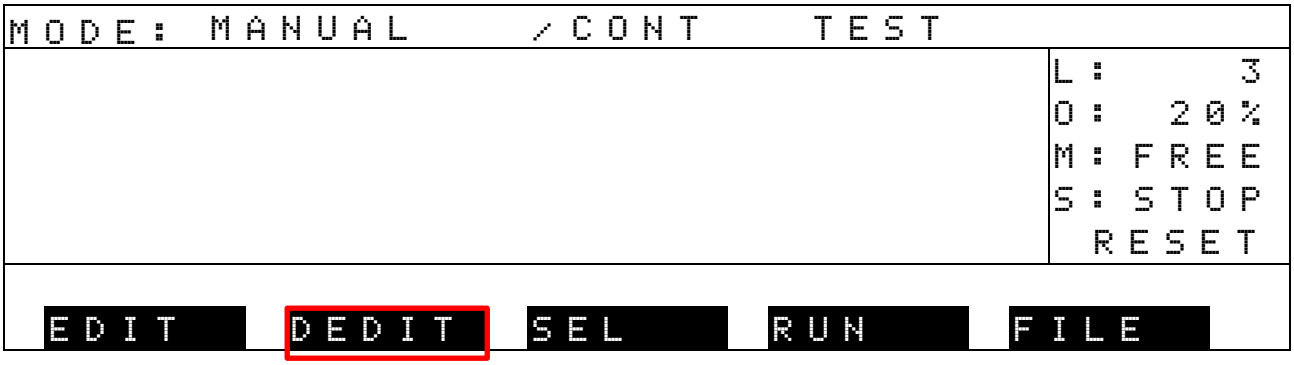
The extension can use a maximum of 3 characters.

MEMO

- *If a file name that does not comply with the naming rules is entered, and the EXE key is pressed, "Input error" appears.*
- *If the name of an existing file is entered, the new file is not created, and the data section edit screen for the existing file is opened.*

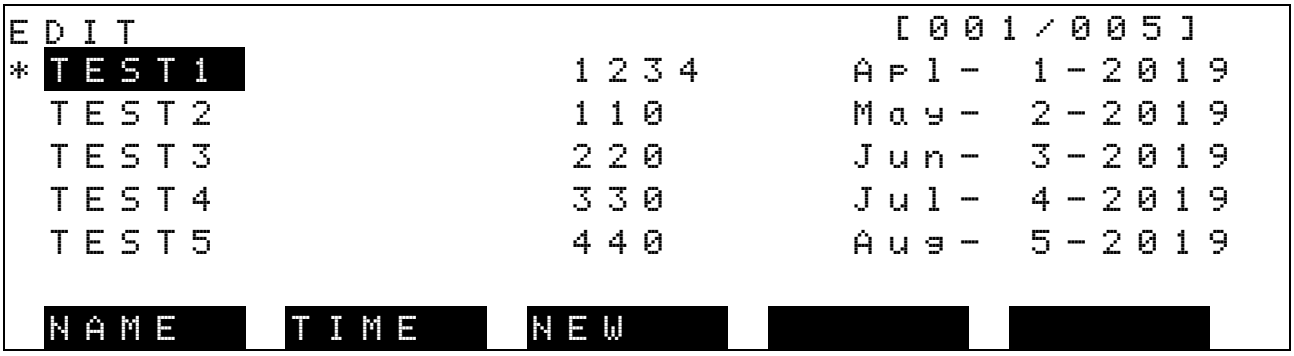
<Data Section Edit Screen Opening Procedure for an Existing File>

1 Press the NEXT key until "DEDIT" appears in the function menu on the basic screen.



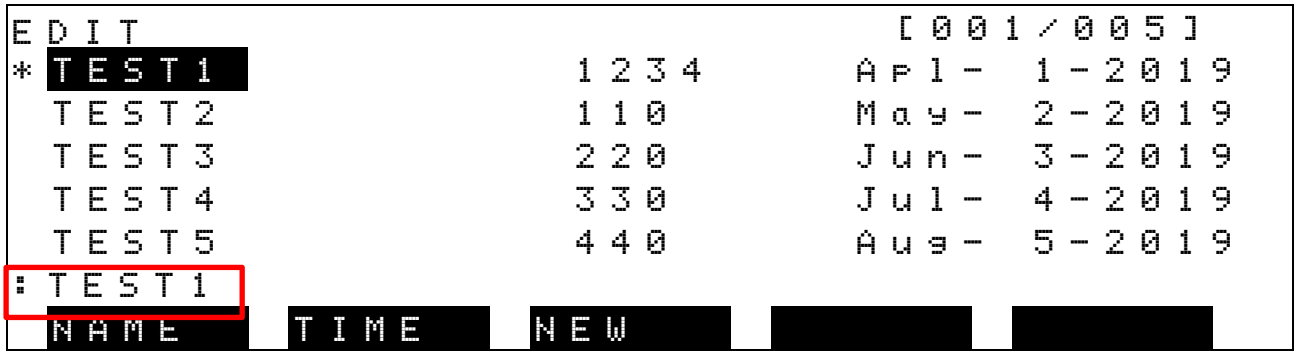
2 Press the DEDIT (F2) key.

The select edit file screen is opened.



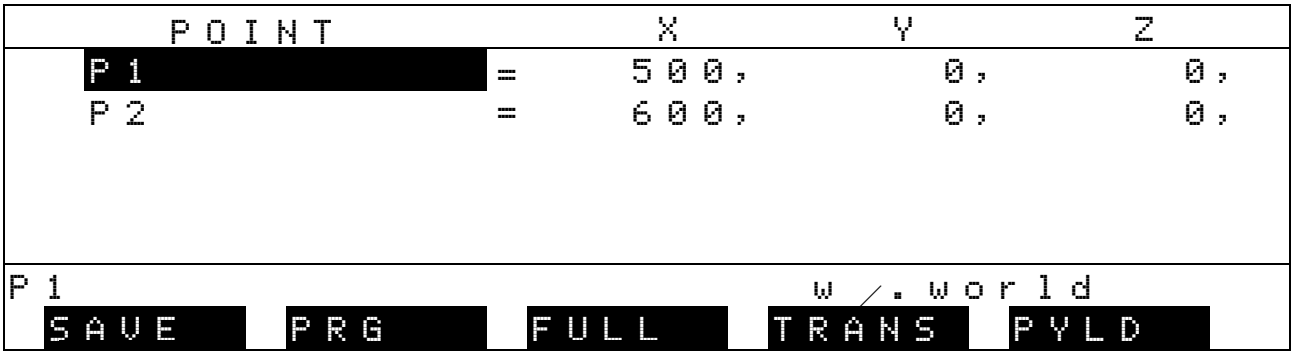
3 Move the cursor to the file to be edited, and press the EXE key.

":<*Selected File Name*>" appears in the command line.



4 Press the EXE key.

This opens the data section edit screen (POINT-type data list screen).



MEMO

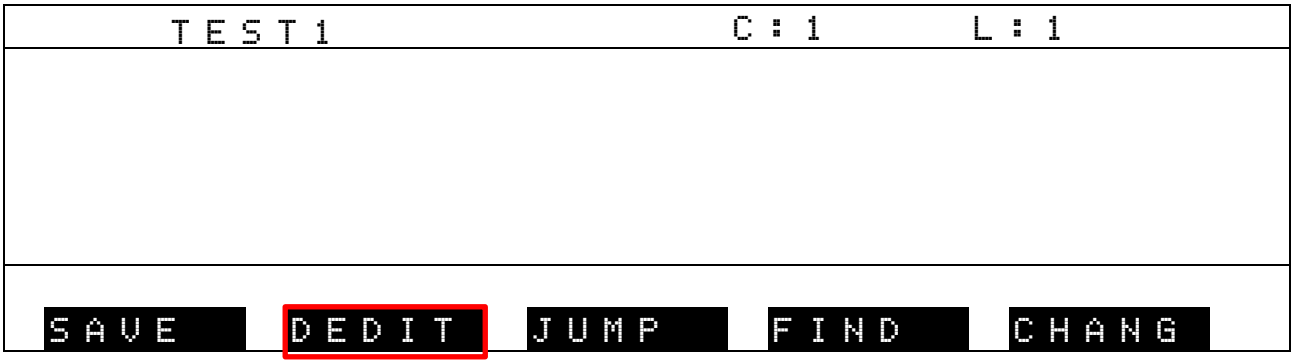
If a file name that does not comply with the naming rules is entered, and the EXE key is pressed, "Input error" appears.

MEMO

If the EXE is pressed when a file with the name shown in the command line does not exist, a new file is created, and the file edit screen is opened.

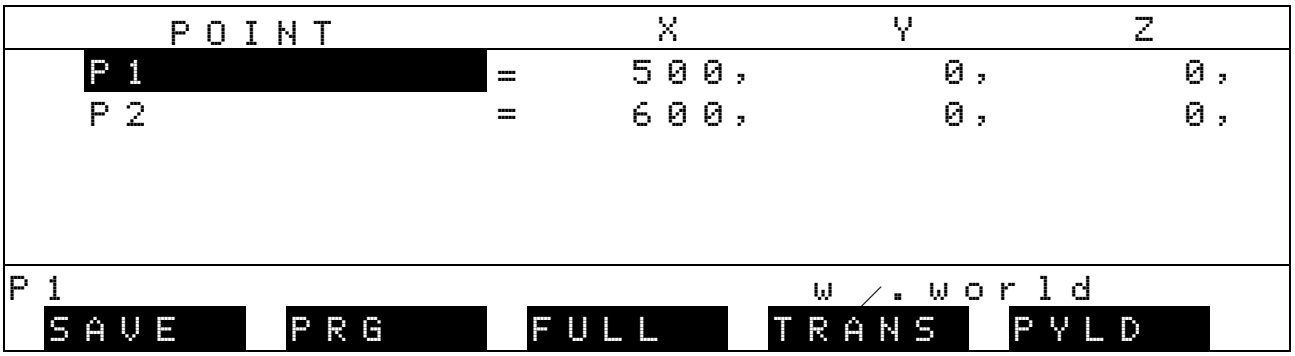
<Opening the Data Section Edit Screen from the Program Section Edit Screen>

1 Press the NEXT key until "DEDIT" is displayed in the function menu on the program section edit screen.



2 Press the DEDIT (F2) key.

The data section edit screen is opened.



MEMO

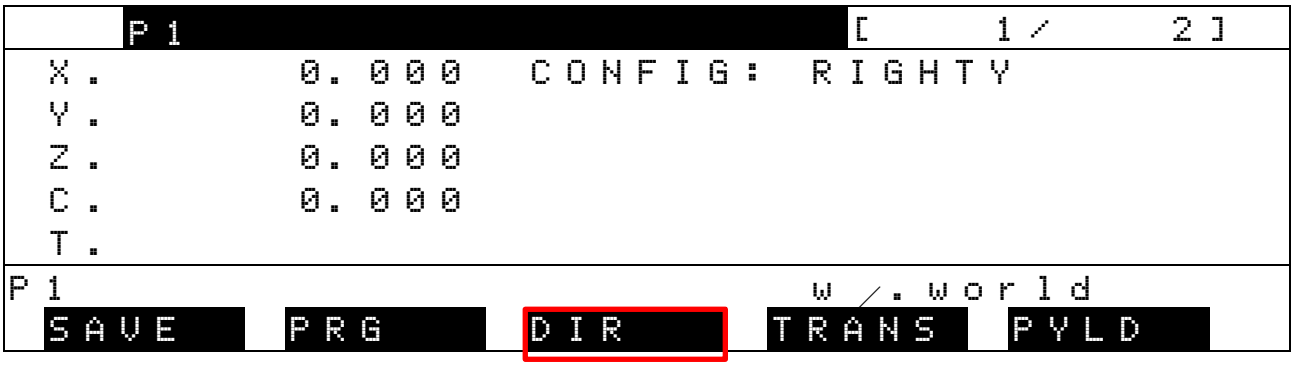
For the data section edit screen, the screen that was displayed last is shown (POINT-type data list screen if the screen is displayed for the first time after starting file editing).

3.2.2 Opening the POINT-type Data List Screen

This describes how to open the POINT-type data edit screen.
The POINT-type data linked to the TRANS-type data that is set in work coordinates is shown in the POINT-type data list screen.

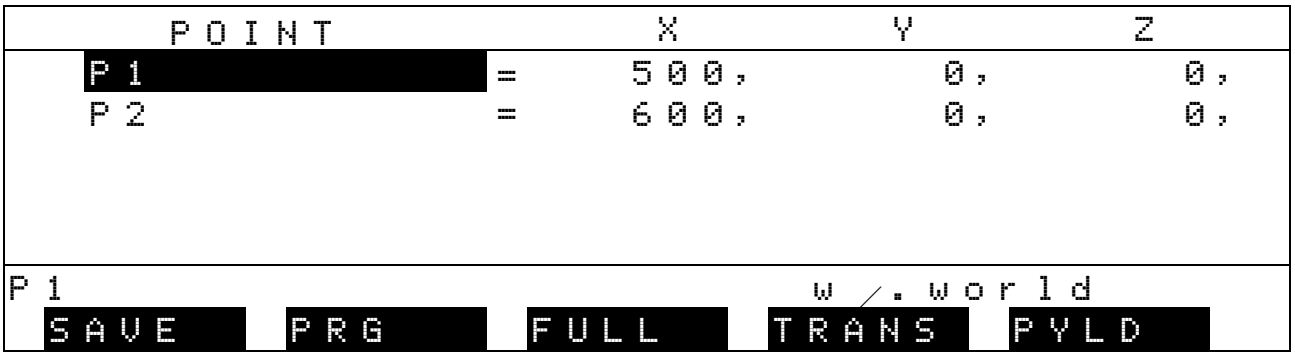
<Opening the POINT-type Data List Screen from the POINT-type Data Details Screen>

1 Press the NEXT key until "DIR" is displayed in the function menu on the POINT-type data details screen.



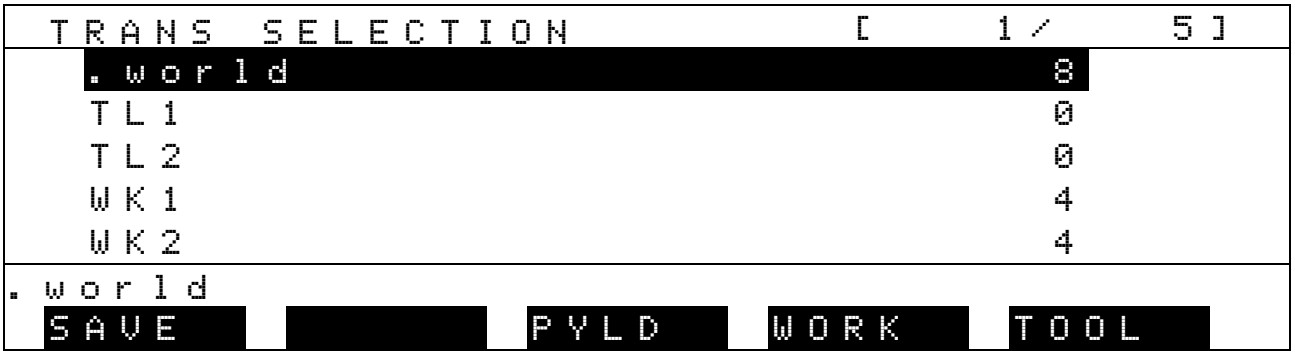
2 Press the FULL (F3) key.

This opens the POINT-type data list screen.



<Opening the POINT-type Data List Screen from the TRANS-type Data List Screen>

1 Move the cursor to the line of the TRANS-type data to be registered in the work coordinates.

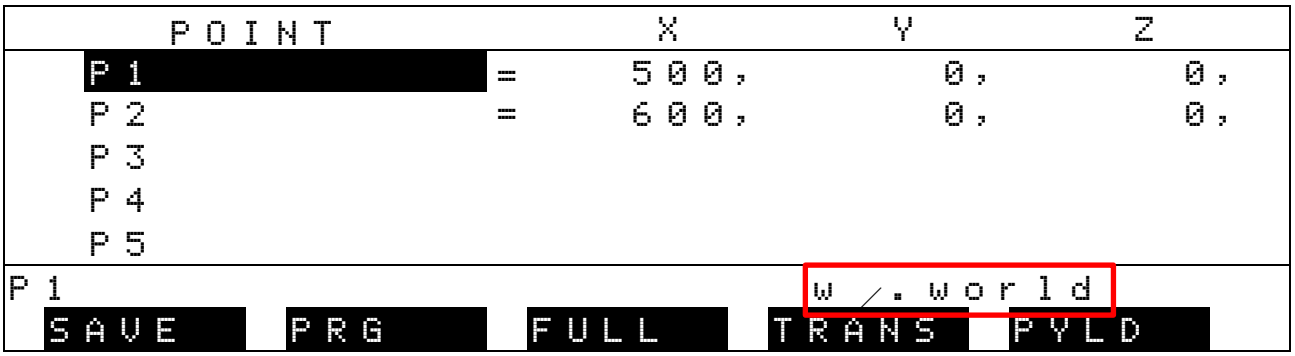


2 Press the NEXT key until "WORK" is displayed on the function menu on the TRANS-type data list screen.



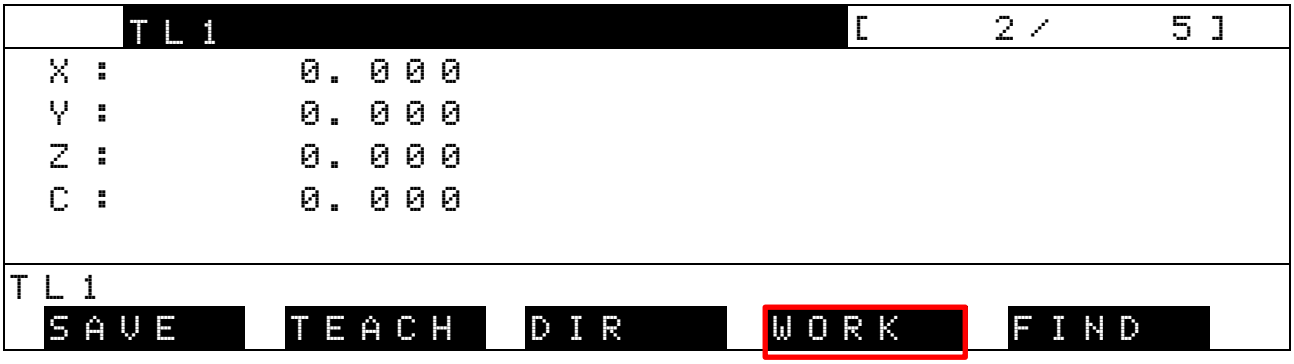
3 Press the WORK (F4) key.

The TRANS-type data in the cursor line is registered to the work coordinates, and the POINT-type data list screen is displayed.



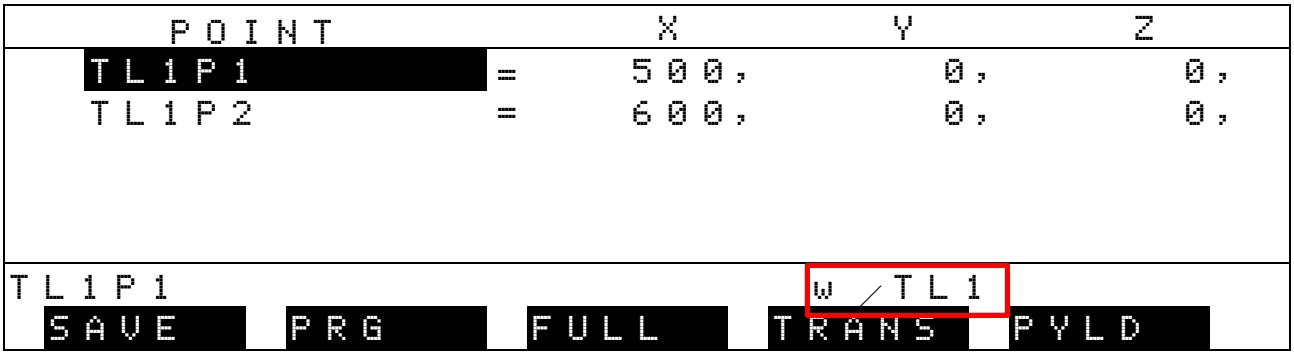
<Opening the POINT-type Data List Screen from the TRANS-type Data Edit Screen>

1 Press the NEXT key until "WORK" is displayed in the function menu on the TRANS-type data edit screen.



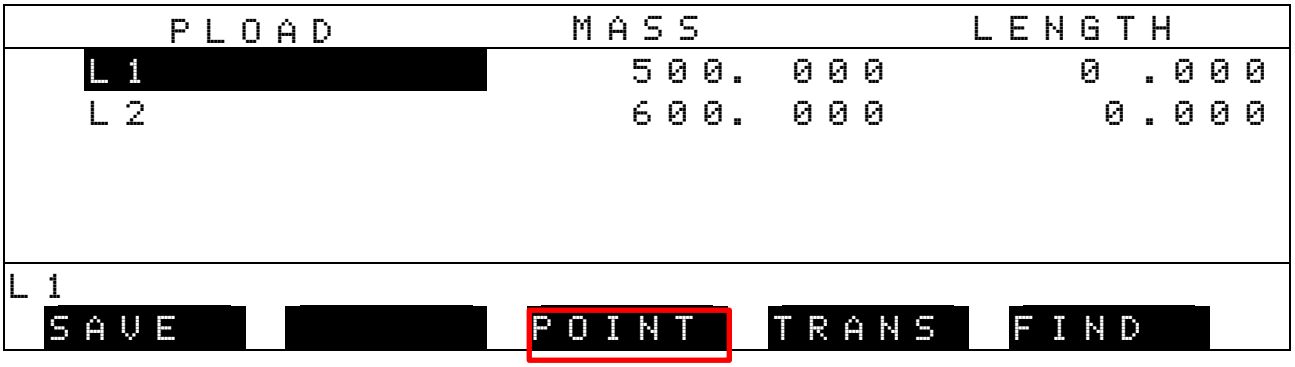
2 Press the WORK (F4) key.

The TRANS-type data appearing in the TRANS-type data edit screen is registered to the work coordinates, and the POINT-type data list screen is displayed.



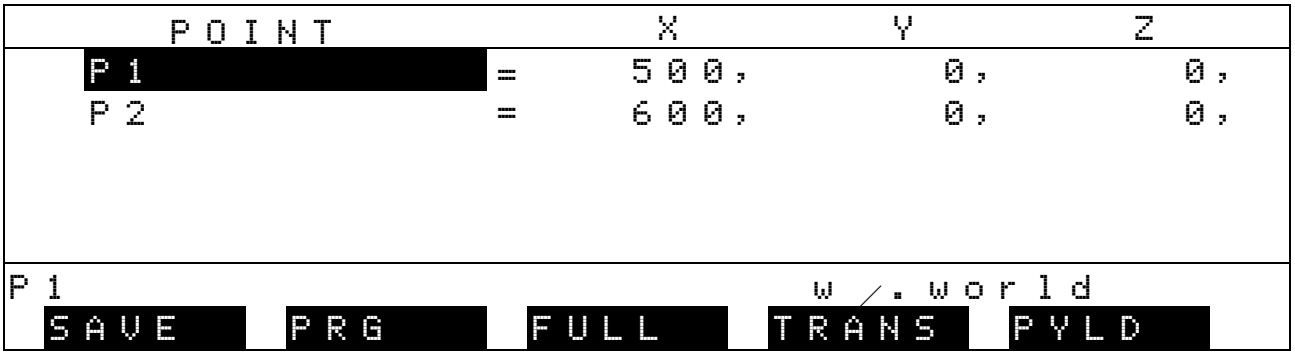
<Opening the POINT-type Data List Screen from the PAYLOAD-type Data Edit Screen>

1 Press the NEXT key until "POINT" is displayed in the function menu on the PAYLOAD-type data edit screen.



2 Press the POINT (F3) key.

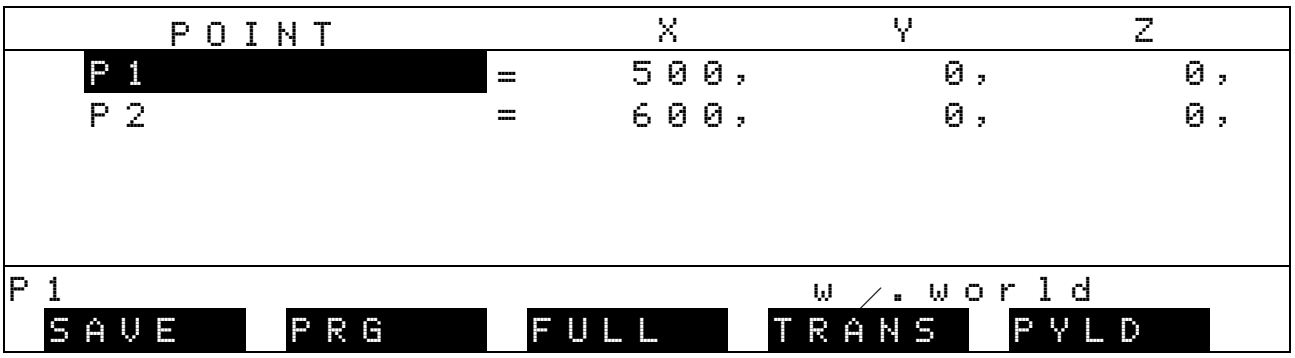
This opens the POINT-type data list screen.



MEMO

When the POINT-type data list screen is opened, the TRANS-type data in the cursor line in the TRANS-type data list screen is registered to the work coordinates.

POINT-type Data List Screen



The valid keys in the POINT-type data list screen and their actions are shown in the following table.

Key	Description
↑ key	This moves the cursor up by one line.
↓ key	This moves the cursor down by one line.
← key	This moves the cursor by one column to the left. *If the C column is displayed, and the cursor is located in the Z column, this changes the screen.
→ key	This moves the cursor by one column to the right. *If the X column is displayed, and the cursor is located in the Z column, this changes the screen.
ALT key + ↑ key	This moves the cursor to the previous page.
ALT key + ↓ key	This moves the cursor to the next page.
ALT key + ← key	The TRANS-type data defined before the TRANS-type data currently registered to the work coordinates is registered to the work coordinates, and the linked POINT-type data list is displayed.
ALT key + → key	The TRANS-type data defined after the TRANS-type data currently registered to the work coordinates is registered to the work coordinates, and the linked POINT-type data list is displayed.
INS key	This adds data to the line after the cursor.
ALT key + BS key	This removes the data in the cursor line.
NEXT key	This switches the function menu.
ALT key + NEXT key	This switches the function menu.
EXE key	This edits the item at the cursor position.

The function menus in the POINT-type data list screen and their actions are shown in the following table.

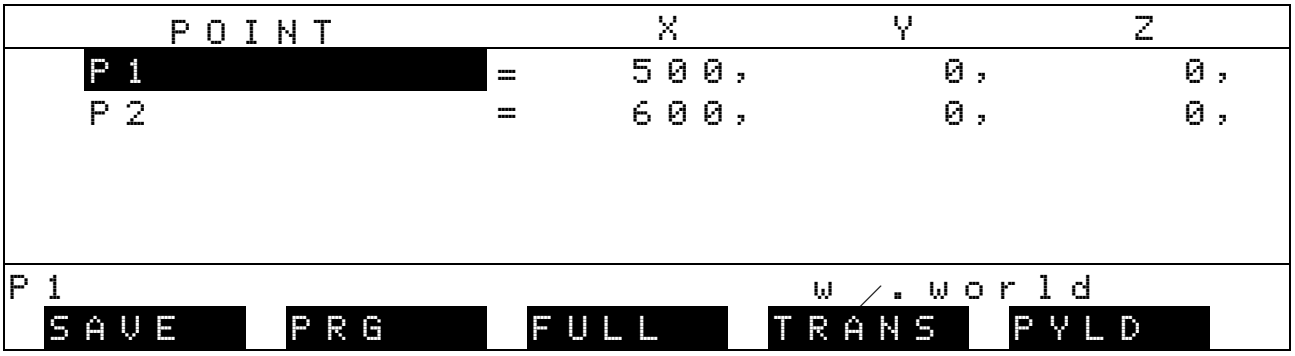
Displayed menu name	Description	Section number for detailed explanation
SAVE	This saves and exits the file.	3.3
NOSAV	This discards the file.	3.3
PRG	This opens the program section edit screen.	3.1.1
FULL	This shows details on the POINT-type data where the cursor is located.	3.2.3
TRANS	This opens the TRANS-type data list screen.	3.2.8
PYLD	This opens the PAYLOAD-type data edit screen.	3.2.12
FIND	This searches for a character string.	3.2.17
TEACH	This teaches the current position.	3.2.5
M-TO	This moves to the teaching point.	3.2.6
BYPAS	This performs teaching point movement using the bypass operation.	3.2.7
CUT	This cuts a character string.	3.2.15
PASTE	This pastes a character string.	3.2.16

3.2.3 Opening the POINT-type Data Details Screen

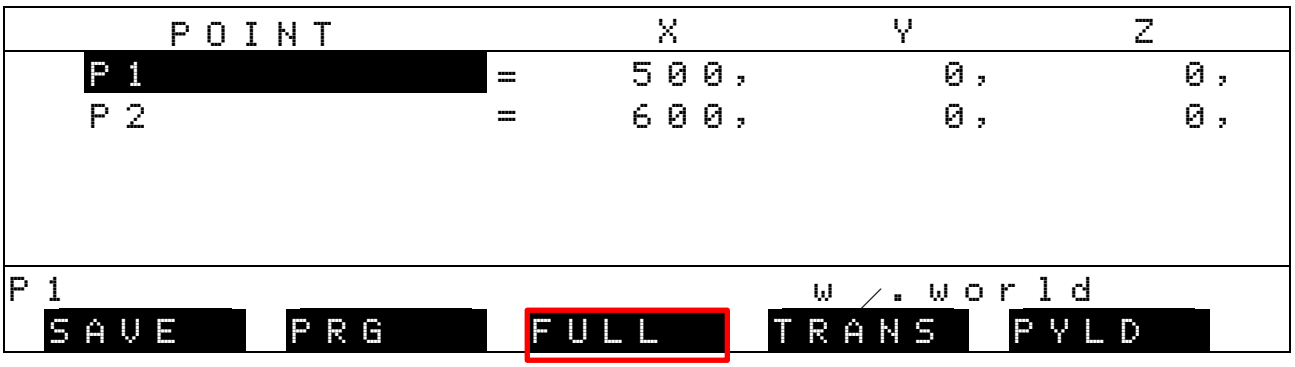
This describes how to open the POINT-type data details.

<POINT-type Data Details Screen Opening Procedure>

1 Move the cursor to the line of the data where the details will be displayed in the POINT-type data list screen.

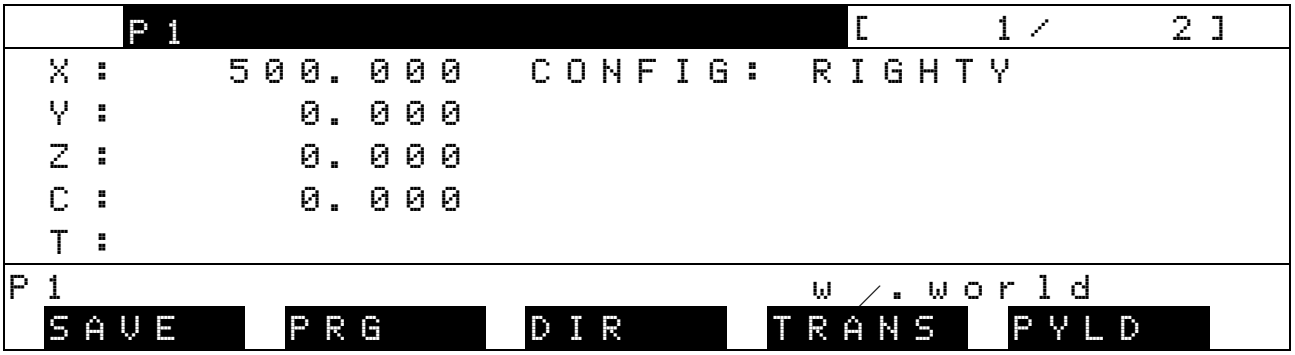


2 Press the NEXT key until "FULL" is displayed in the function menu on the POINT-type data edit screen.



3 Press the FULL (F3) key.

This opens the POINT-type data details screen.



POINT-type Data Details Screen



The valid keys in the POINT-type data details screen and their actions are shown in the following table.

Key	Description
↑ key	This moves the cursor to the position of one element above. If the cursor is at the data name position, this moves to the CONFIG position. If the cursor is at the CONFIG element, this moves to the T position.
↓ key	This moves the cursor to the position of one element below. If the cursor is at the T position, this moves to the CONFIG position. If the cursor is at the CONFIG element, this moves to the data name position.
ALT key + ← key	The TRANS-type data defined before the TRANS-type data currently registered to the work coordinates is registered to the work coordinates, and the linked POINT-type data list is displayed.
ALT key + → key	The TRANS-type data defined after the TRANS-type data currently registered to the work coordinates is registered to the work coordinates, and the linked POINT-type data list is displayed.
INS key	This adds new data to the line following the displayed data.
ALT key + BS key	This removes the data in the cursor line.
NEXT key	This switches the function menu.
ALT key + NEXT key	This switches the function menu.
EXE key	This edits the items at the cursor position.

The function menus in the POINT-type data details screen and their actions are shown in the following table.

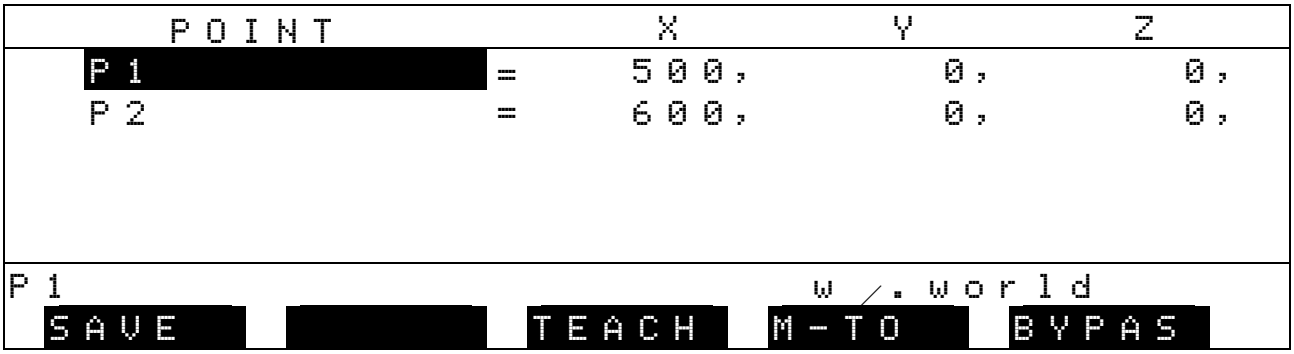
Displayed menu name	Description	Section number for detailed explanation
SAVE	This saves and exits the file.	3.3
NOSAV	This discards the file.	3.3
PRG	This opens the program section edit screen.	3.1.1
DIR	This opens the POINT-type data list screen.	3.2.2
TRANS	This opens the TRANS-type data list screen.	3.2.8
PYLD	This opens the PAYLOAD-type data edit screen.	3.2.12
FIND	This searches for a character string.	3.2.17
TEACH	This teaches the current position.	3.2.5
M-TO	This moves to the teaching point.	3.2.6
BYPAS	This performs teaching point movement using the bypass operation.	3.2.7

3.2.4 Editing POINT-type Data

<Editing Items in Point-type Data>

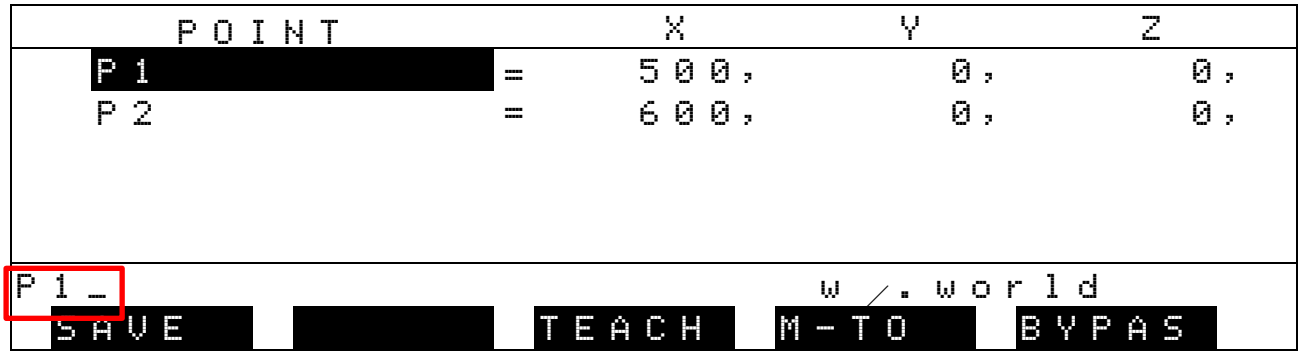
1 Move the cursor to the position of the item to be edited in the POINT-type data list screen or POINT-type data details screen.

The content for the item is displayed in the command line.



2 Press the EXE key.

The cursor is displayed in the command line, and the content can be edited.



3 Edit the content in the command line, and press the EXE key.

This saves the edited changes.

MEMO

- Character strings that are set for the POINT-type data name must have 10 or less alphanumeric characters and begin with a letter.
  If a character string exceeds 10 characters or includes symbols, the editing changes are not accepted, and the message *"ERR: ILLEGAL NAME" is displayed.*
- A value in the range of -9999.999 to 9999.999 can be set for the coordinate value of the POINT-type data.
  If a value outside the range or an invalid character is entered, the editing changes are not accepted, and the message "ERR: ILLEGAL DATA" is displayed.
- *If the entered data name is already being used, the editing changes are not accepted, and the message "ERR: DUPLICATE NAME" is displayed.*

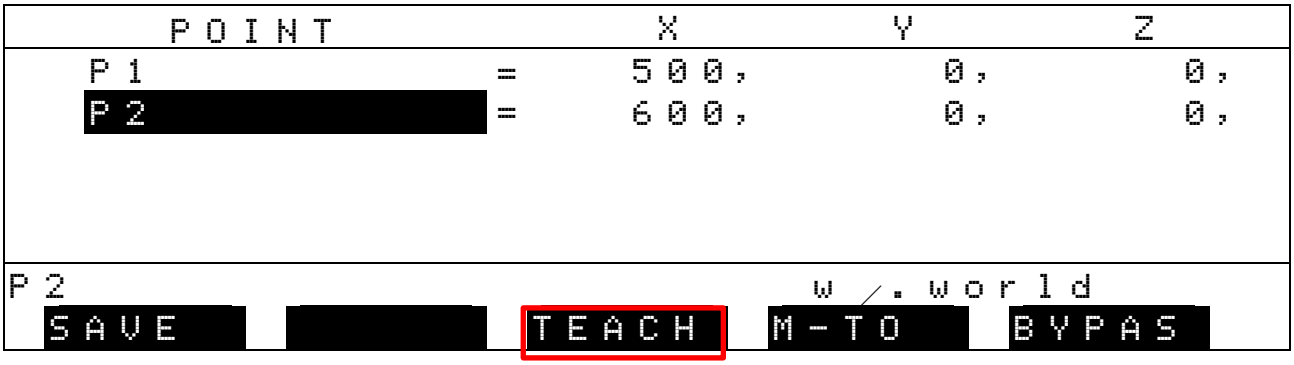
3.2.5 Teaching of current position (TEACH)

This describes how to teach the current position for the robot TCP (tool center position) to the POINT-type data.

<Current Position Teaching Procedure>

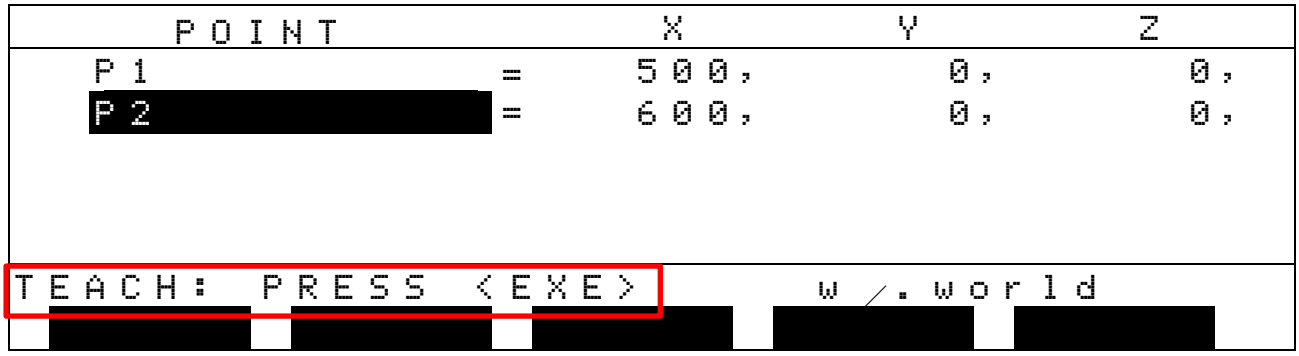
1 Move the cursor to the position of the position-type data where the current position will be taught in the POINT-type data list screen or POINT-type data details screen.

2 Press the NEXT key until "TEACH" is displayed in the function menu.



3 Press the TEACH (F3) key.

"TEACH: PRESS <EXE>" appears in the command line.



4 Press the EXE key.

The coordinate value and configuration of POINT-type data at the cursor position is overwritten by the current position and configuration of the robot.

MEMO

• The current position of the robot is a value that incorporates the tool coordinate and work coordinate settings.

3.2.6 Teaching Point Movement (M-TO)

This describes how to move the robot TCP (tool center position) to the selected POINT-type data position.

In teaching point movement, movement can be performed using PTP movement and CP movement.

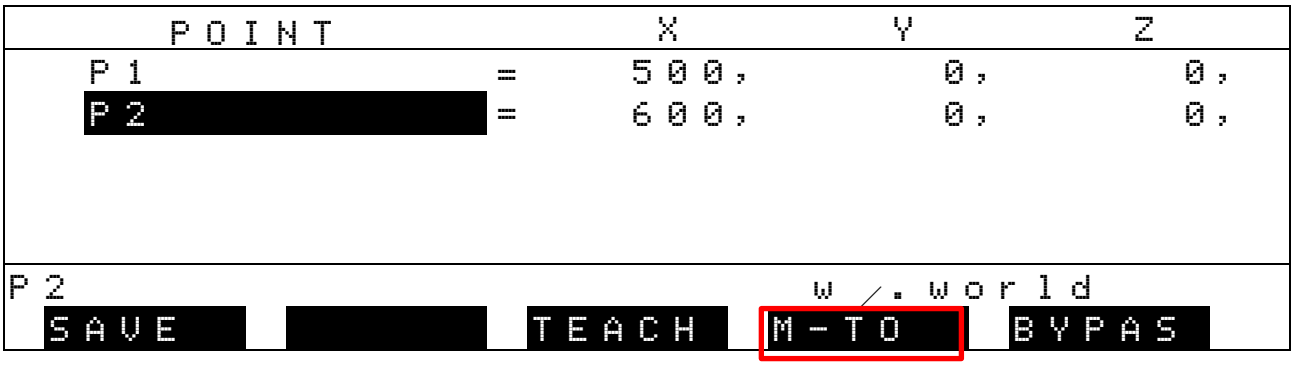
<Moving to Teaching Point Using PTP Movement>

1 Set the guidance coordinates to JOINT.

> *MEMO*
> *For details on the guidance coordinate setting procedure, see the section "2.1.3 Guidance Coordinates Setting".*

2 Move the cursor to the position of the position-type data that will be set as the target position for teaching point movement in the POINT-type data list screen or POINT-type data details screen.

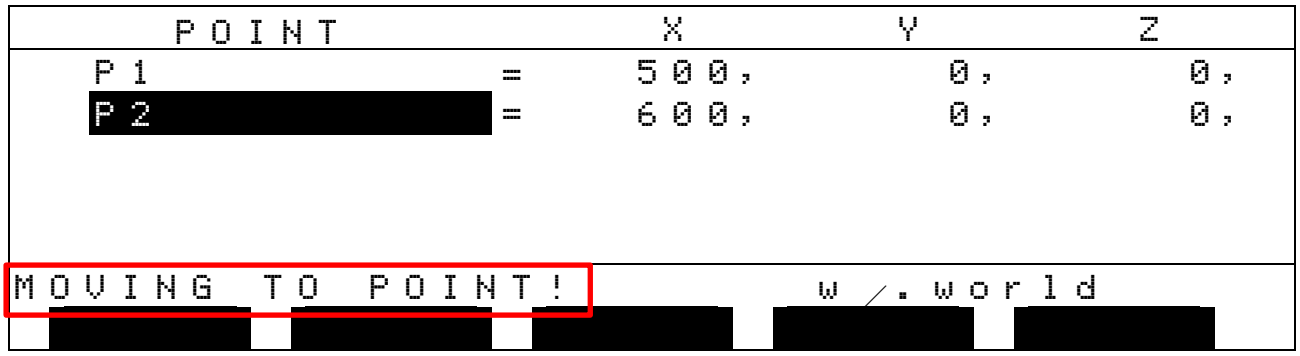
3 Press the NEXT key until "M-TO" is displayed in the function menu.



> *MEMO*
> *"M-TO" does not appear in the function menu when "BYPASS-SW" of [U011] is set to 2.*
> *For details on the setting procedure and the setting descriptions of [U011], see the STA1116 Parameter Manual, section U011.*

4 Press the M-TO (F4) key.

"MOVING TO POINT!" appears in the command line.



5 Press the MOVE key.

While the MOVE key is held down, the robot performs PTP movement with the position of the selected position-type data as the target.

The robot stops movement when the MOVE key is released.

MEMO

The robot does not move even if the MOVE key is pressed in the following cases.

- *When the servo is turned off*
- *When the STOP signal is OFF*
- *When the BREAK signal is OFF*
- *When the position of the selected position-type data is outside the robot operating range*
- *When an alarm at level 2 or higher has occurred*

<Moving to Teaching Point Using CP Movement>

1 Check that [U010] MOVES-TO-SW is set to 1.

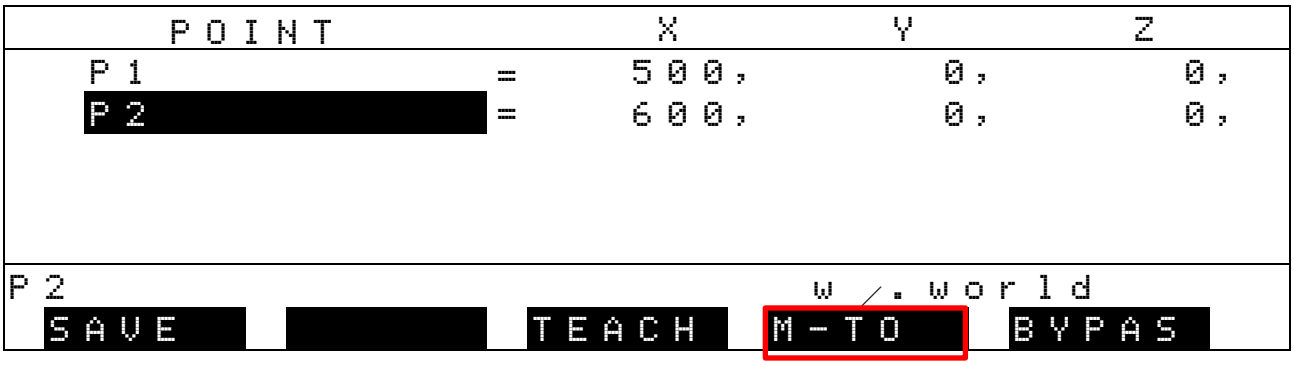
> *MEMO*
> *If [U010] MOVES-TO-SW is set to 0, movement cannot be performed to the teaching point by CP movement.*
> *For details on the setting procedure and the setting descriptions of [U010], see the "STA1116 Parameter Manual", section [U010].*

2 Set the guidance coordinates to TOOL, WORK, or WORLD.

> *MEMO*
> *For details on the guidance coordinate setting procedure, see the section "2.1.3 Guidance Coordinates Setting".*

3 Move the cursor to the position of the position-type data that will be set as the target position for teaching point movement in the POINT-type data list screen or POINT-type data details screen.

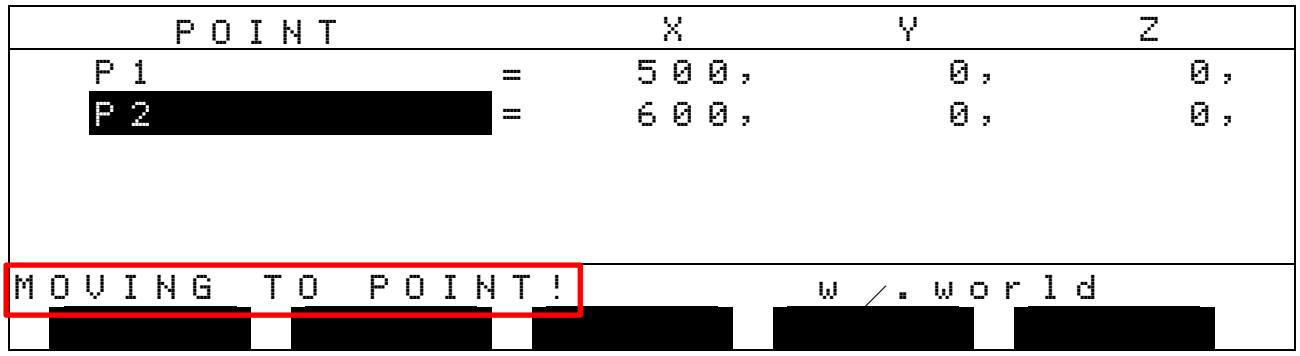
4 Press the NEXT key until "M-TO" is displayed in the function menu.



> *MEMO*
> *"M-TO" does not appear in the function menu when "BYPASS-SW" of [U011] is set to 2.*
> *For details on the setting procedure and the setting descriptions of [U011], see the "STA1116 Parameter Manual", section [U011].*

5 Press the M-TO (F4) key.

"MOVING TO POINT!" appears in the command line.



6 Press the MOVE key.

While the MOVE key is held down, the robot performs CP movement with the position of the selected position-type data as the target.

The robot stops movement when the MOVE key is released.

MEMO

The robot does not move even if the MOVE key is pressed in the following cases.

- *When the servo is turned off*
- *When the STOP signal is OFF*
- *When the BREAK signal is OFF*
- *When the position of the selected position-type data is outside the robot operating range*
- *When an alarm at level 2 or higher has occurred*

3.2.7 Moving to Teaching Point Using Bypass Movement (BYPAS)

This describes how to move the teaching point to the POINT-type data position by bypass movement.
The bypass movement refers to passing through a position offset in the Z-axis direction (via position) to the target position when moving to the target position.
If an obstacle is located between the current position and target position, the bypass movement can be used to avoid the obstacle and perform teaching point movement.

In teaching point movement using bypass movement, movement can be performed using PTP movement and CP movement.

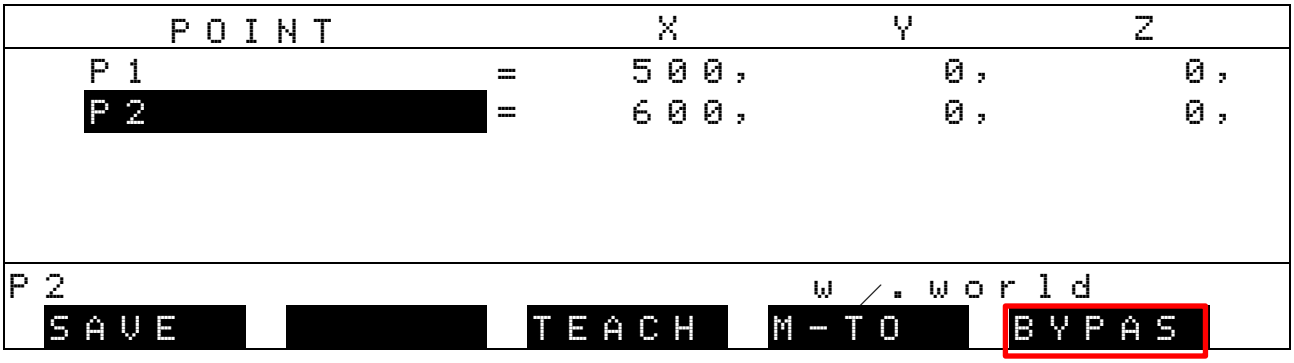
<Bypass Movement Using PTP Movement>

1 Set the guidance coordinates to JOINT.

> *MEMO*
> *For details on the guidance coordinate setting procedure, see the section "2.1.3 Guidance Coordinates Setting".*

2 Move the cursor to the position of the position-type data that will be set as the target position for bypass point movement in the POINT-type data list screen or POINT-type data details screen.

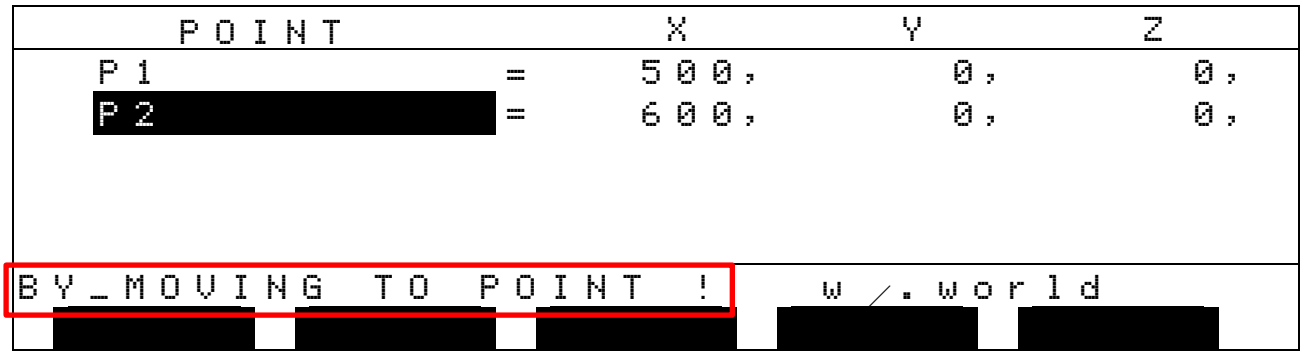
3 Press the NEXT key until "BYPAS" is displayed in the function menu.



> *MEMO*
> *"BYPAS" does not appear in the function menu when "BYPASS-SW" of [U011] is set to 0.*
> *For details on the setting procedure and the setting descriptions of [U011], see the "STA1116 Parameter Manual", section [U011].*

4 Press the BYPAS (F5) key.

"BY_MOVING TO POINT !" appears in the command line.



5 Press the MOVE key.

While the MOVE key is held down, the robot performs bypass movement using PTP movement with the position of the selected position-type data as the target.

The robot stops movement when the MOVE key is released.

MEMO

The robot does not move even if the MOVE key is pressed in the following cases.

- *When the servo is turned off*
- *When the STOP signal is OFF*
- *When the BREAK signal is OFF*
- *When the position of the selected position-type data is outside the robot operating range*
- *When an alarm at level 2 or higher has occurred*

<Bypass Movement Using CP Movement>

1 Check that [U000] is set to 1.

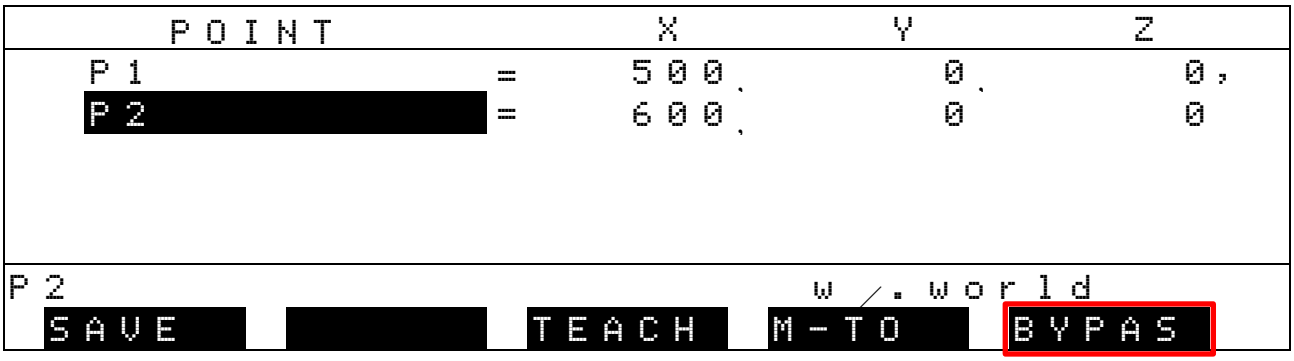
> *MEMO*
> *If [U010] MOVES-TO-SW is set to 0, movement cannot be performed to the teaching point by CP movement.*
> *For details on the setting procedure and the setting descriptions of [U010], see the "STA1116 Parameter Manual", section [U010].*

2 Set the guidance coordinates to TOOL, WORK, or WORLD.

> *MEMO*
> *For details on the guidance coordinate setting procedure, see the section "2.1.3 Guidance Coordinates Setting".*

3 Move the cursor to the position of the position-type data that will be set as the target position for bypass point movement in the POINT-type data list screen or POINT-type data details screen.

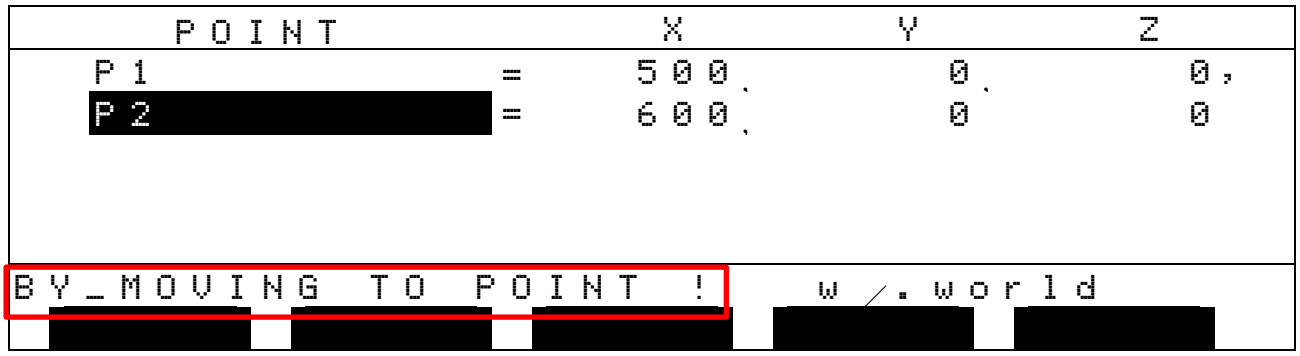
4 Press the NEXT key until "BYPAS" is displayed in the function menu.



> *MEMO*
> *"BYPAS" does not appear in the function menu when "BYPASS-SW" of [U011] is set to 0.*
> *For details on the setting procedure and the setting descriptions of [U011], see the "STA1116 Parameter Manual", section [U011].*

5 Press the BYPAS (F4) key.

"BY_MOVING TO POINT !" appears in the command line.



6 Press the MOVE key.

While the MOVE key is held down, the robot performs bypass movement using CP movement with the position of the selected position-type data as the target.

The robot stops movement when the MOVE key is released.

MEMO

The robot does not move even if the MOVE key is pressed in the following cases.

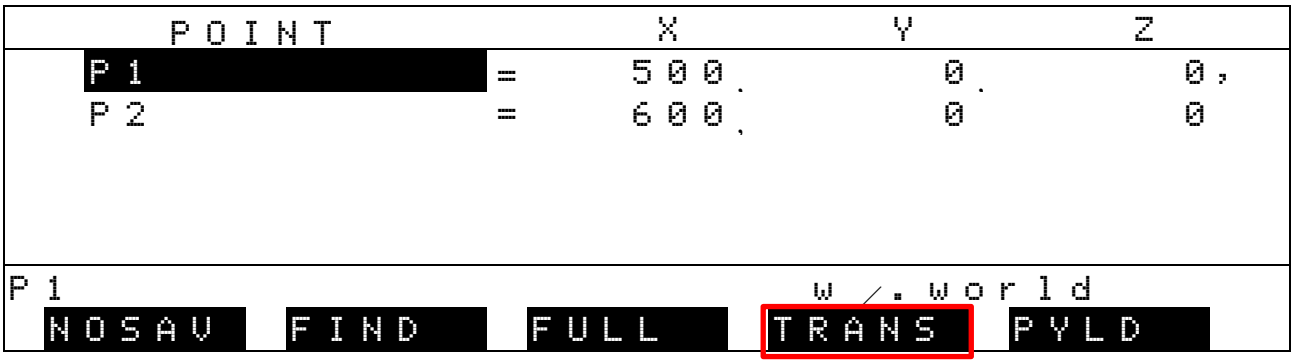
- *When the servo is turned off*
- *When the STOP signal is OFF*
- *When the BREAK signal is OFF*
- *When the position of the selected position-type data is outside the robot operating range*
- *When an alarm at level 2 or higher has occurred*

3.2.8 Opening the TRANS-type Data List Screen

This describes how to open the TRANS-type data list screen.

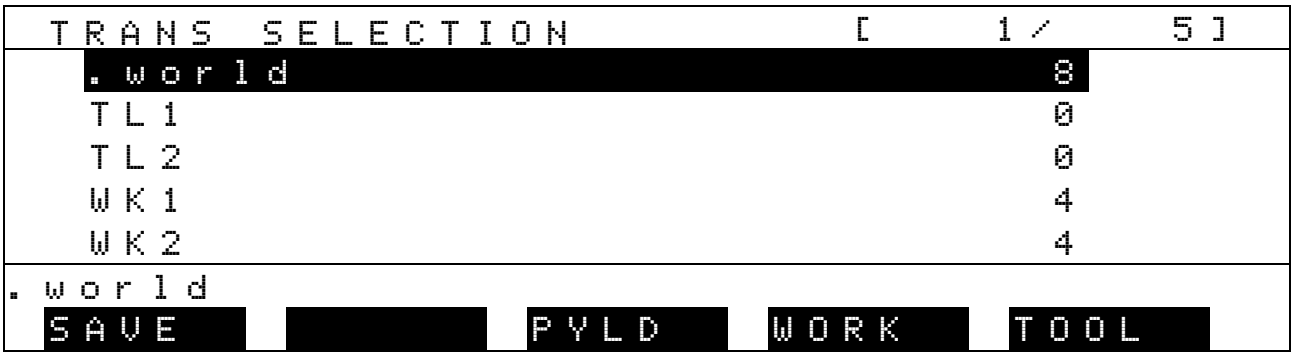
<Opening the TRANS-type Data List Screen from the POINT-type Data List Screen or POINT-type Data Details Screen>

1 Press the NEXT key until "TRANS" is displayed in the function menu on the POINT-type data list screen or POINT-type data details screen.



2 Press the TRANS (F4) key.

This opens the TRANS-type data list screen.



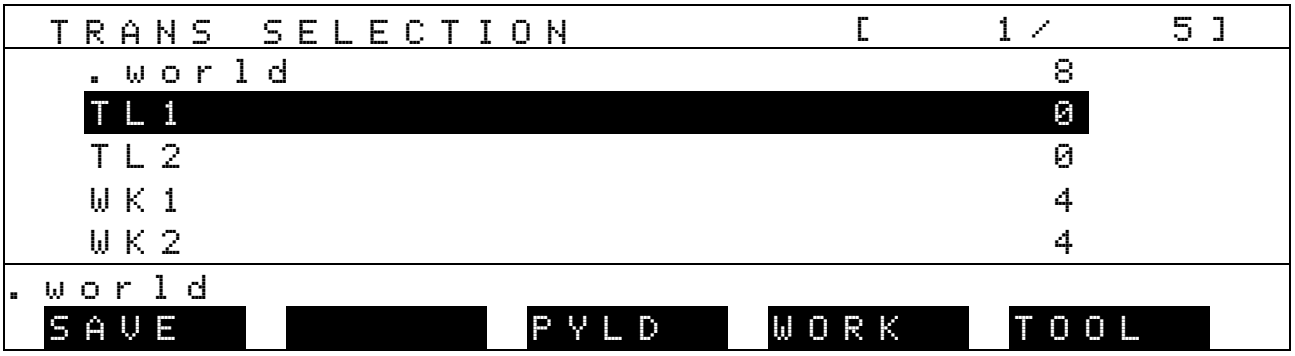
<Opening the TRANS-type Data List Screen from the TRANS-type Data Edit Screen>

1 Press the NEXT key until "DIR" is displayed in the function menu on the TRANS-type data edit screen.



2 Press the DIR (F3) key.

This opens the TRANS-type data list screen.

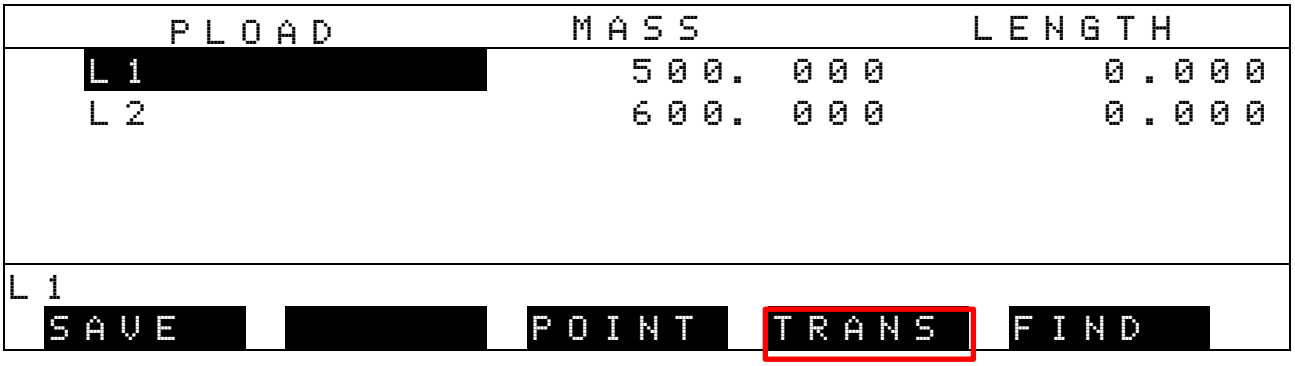


MEMO

The cursor position is the position of the TRANS-type data that was shown in the TRANS-type data edit screen.

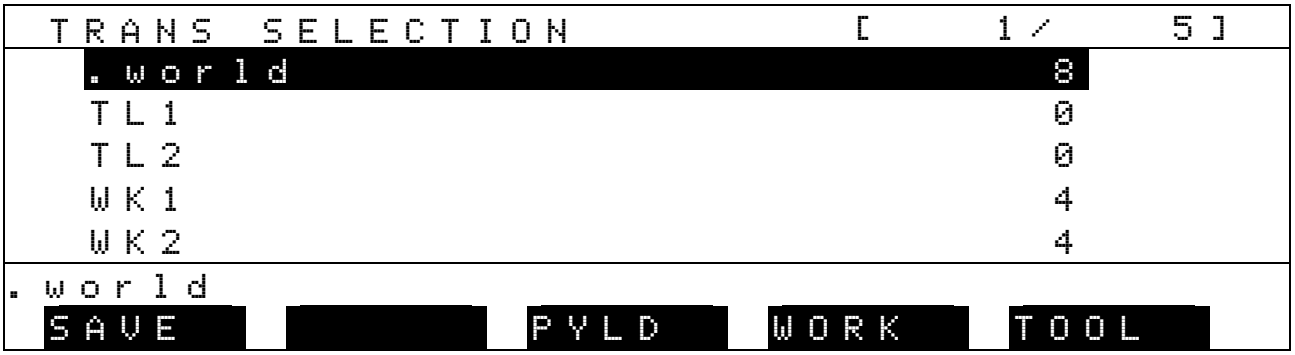
<Opening the TRANS-type Data List Screen from the PAYLOAD-type Data Edit Screen>

1 Press the NEXT key until "TRANS" is displayed in the function menu on the PAYLOAD-type data edit screen.

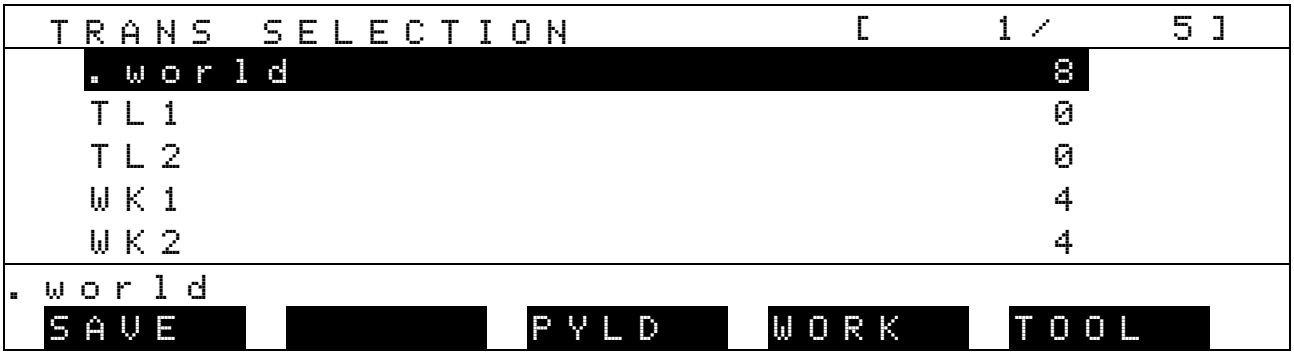


2 Press the TRANS (F4) key.

This opens the TRANS-type data list screen.



TRANS-type Data List Screen



The valid keys in the TRANS-type data list screen and their actions are shown in the following table.

Key	Description
↑ key	This moves the cursor up by one line.
↓ key	This moves the cursor down by one line.
ALT key + ↑ key	This moves the cursor to the previous page.
ALT key + ↓ key	This moves the cursor to the next page.
INS key	This adds data to the line after the cursor.
ALT key + BS key	This removes the data in the cursor line.
NEXT key	This switches the function menu.
ALT key + NEXT key	This switches the function menu.
EXE key	This opens the TRANS-type data edit screen where the cursor is located.

The function menus in the TRANS-type data list screen and their actions are shown in the following table.

Displayed menu name	Description	Section number for detailed explanation
SAVE	This saves and exits the file.	3.3
NOSAV	This discards the file.	3.3
PRG	This opens the program section edit screen.	3.1.1
WORK	This registers the TRANS-type data in the cursor line to the work coordinates and opens the POINT-type data list screen.	3.2.2
TOOL	This registers the TRANS-type data in the cursor line to the tool coordinates. If the data is already registered to the tool coordinates, the registration is cleared.	
PYLD	This opens the PAYLOAD-type data edit screen.	3.2.12
FIND	This searches for data.	3.2.17
CUT	This cuts data.	3.2.15
PASTE	This pastes data.	3.2.16

MEMO

• A "t" is displayed at the start of the line for the TRANS-type data registered to the tool coordinates.

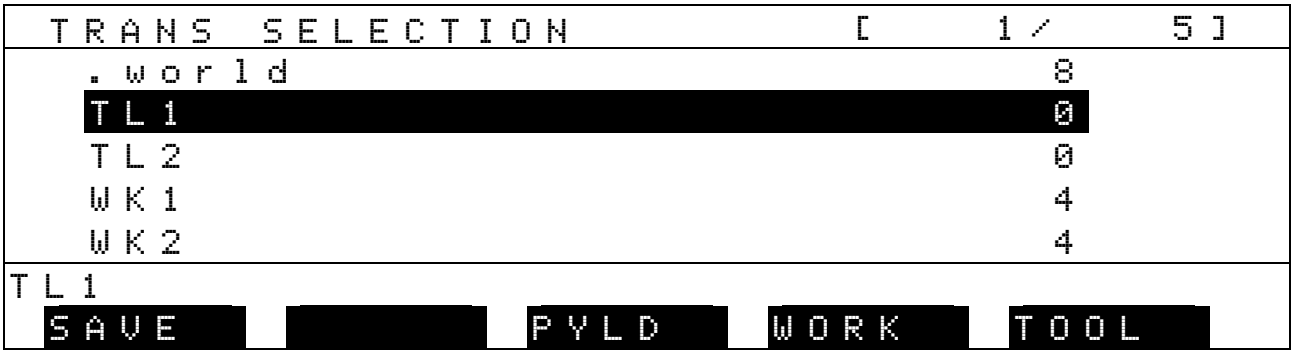
• A "w" is displayed at the start of the line for the TRANS-type data registered to the work coordinates.

3.2.9 Opening the TRANS-type Data Edit Screen

This describes how to open the TRANS-type data edit screen.

<TRANS-type Data Edit Screen Opening Procedure>

1 Move the cursor to the line of the data where editing will be performed in the TRANS-type data list screen.



2 Press the EXE key.

The TRANS-type data edit screen where the cursor is located is opened.



MEMO

".world" cannot be displayed in the TRANS-type data edit screen.

If an operation is performed that will display ".world" in the TRANS-type data edit screen,

"ERR: NO SELECT WORLD" appears in the command line.

TRANS-type Data Edit Screen



The valid keys in the TRANS-type data edit screen and their actions are shown in the following table.

Key	Description
↑ key	This moves the cursor up by one line.
↓ key	This moves the cursor down by one line.
ALT key + ↑ key	This shows the TRANS-type data defined before the currently-displayed TRANS-type data.
ALT key + ↓ key	This shows the TRANS-type data defined after the currently-displayed TRANS-type data.
INS key	This adds new data to the line following the displayed data.
ALT key + BS key	This removes the currently-displayed data.
NEXT key	This switches the function menu.
EXE key	This edits the element at the cursor position.

The function menus in the TRANS-type data list screen and their actions are shown in the following table.

Displayed menu name	Description	Section number for detailed explanation
SAVE	This saves and exits the file.	3.3
NOSAV	This discards the file.	3.3
TEACH	This teaches the TRANS-type data.	3.2.11
DIR	This returns to the TRANS-type data list screen.	3.2.8
WORK	This registers the TRANS-type data in the cursor line to the work coordinates and opens the POINT-type data list screen.	3.2.2
FIND	This searches for a character string.	3.2.17

3.2.10 Editing TRANS-type Data

<TRANS-type Data Editing Procedure>

1 Move the cursor to the position of the item to be edited.



2 Press the EXE key.

The cursor is displayed in the command line, and the content can be edited.



3 Edit the content in the command line, and press the EXE key.

This saves the edited changes.

MEMO

- Character strings that are set for the TRANS-type data name must have 10 or less alphanumeric characters and begin with a letter.
  If a character string exceeds 10 characters or includes symbols, the editing changes are not accepted, and the message *"ERR: ILLEGAL NAME" is displayed.*
- A value in the range of -9999.999 to 9999.999 can be set for the coordinate value of the TRANS-type data.
  If a value outside the range or an invalid character is entered, the editing changes are not accepted, and the message "ERR: ILLEGAL DATA" is displayed.
- *If the entered data name is already being used, the editing changes are not accepted, and the message "ERR: DUPLICATE NAME" is displayed.*

3.2.11 Teaching of TRANS-type Data (TEACH)

This describes how to teach TRANS-type data.

There are three methods for teaching TRANS-type data.

- 1-point teaching
- 2-point teaching
- Tool measurement

3.2.11.1 1-point teaching (1PNT)

This generates the TRANS-type data using the point that was taught as the origin.

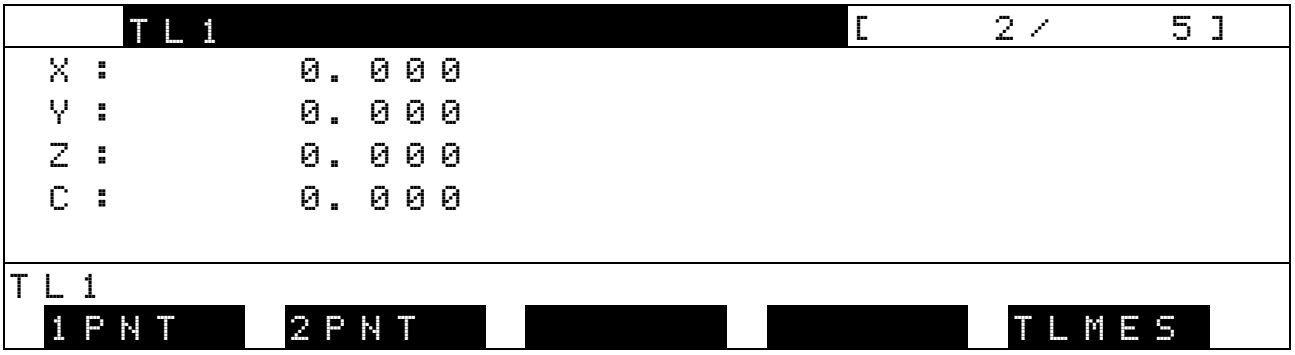
<1-point Teaching Procedure>

1 Press the NEXT key until "TEACH" is displayed in the function menu on the TRANS-type data edit screen.



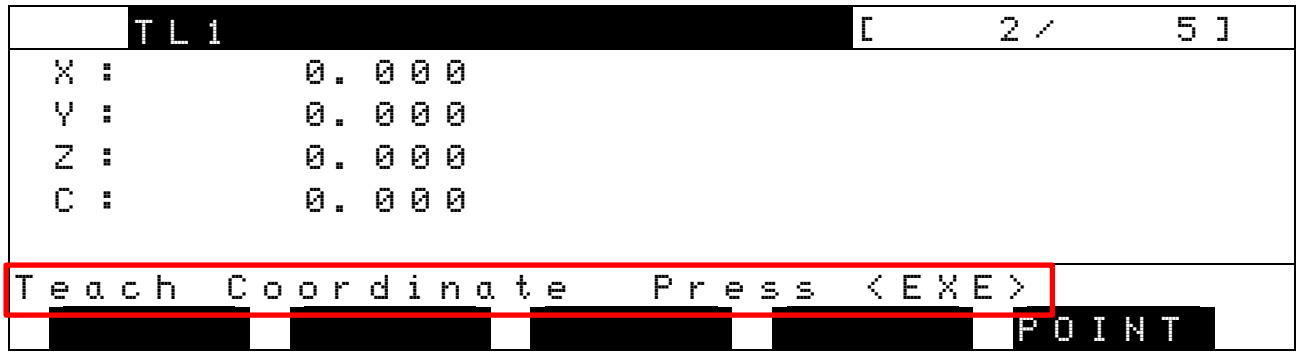
2 Press the TEACH (F2) key.

This switches the function menu.



3 Press the 1PNT (F1) key.

"Teach Coordinate Press <EXE>" appears in the command line.



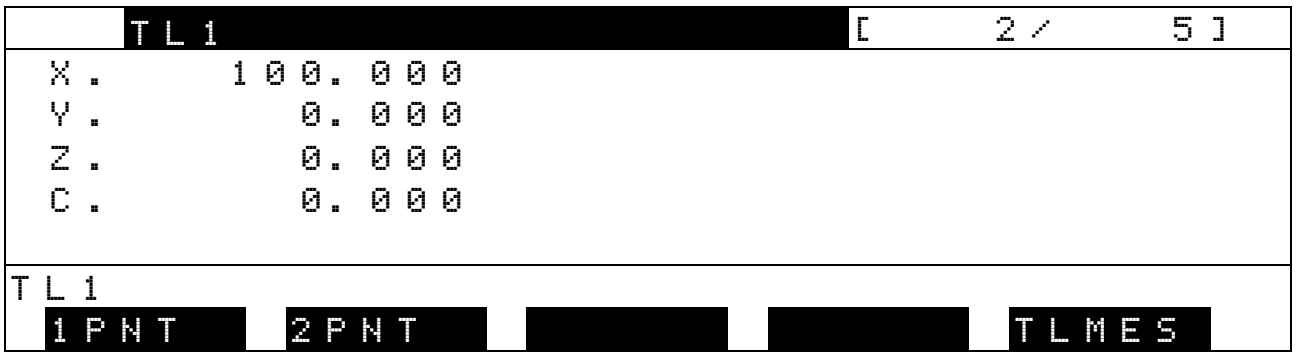
MEMO

• Pressing the ESC key or BS key returns to the previous screen.

• Pressing POINT (F5) opens the POINT-type data selection screen.

4 Move the robot TCP to the origin of the TRANS-type data to be generated, and press the EXE key.

The displayed TRANS-type data value is overwritten by the TRANS-type data value where the robot current position is set as the origin.



3.2.11.2 2-point teaching (2PNT)

Two points (origin and point on coordinate axis) are taught for generating the TRANS-type data.

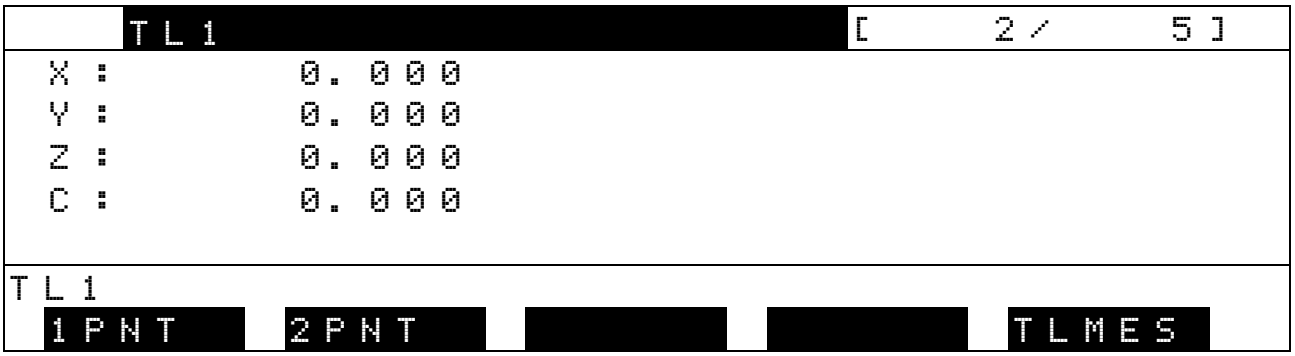
<2-point Teaching Procedure>

1 Press the NEXT key until "TEACH" is displayed in the function menu on the TRANS-type data edit screen.



2 Press the TEACH (F2) key.

This switches the function menu.



3 Press the 2PNT (F2) key.

"Teach Origin Press <EXE>" appears in the command line.



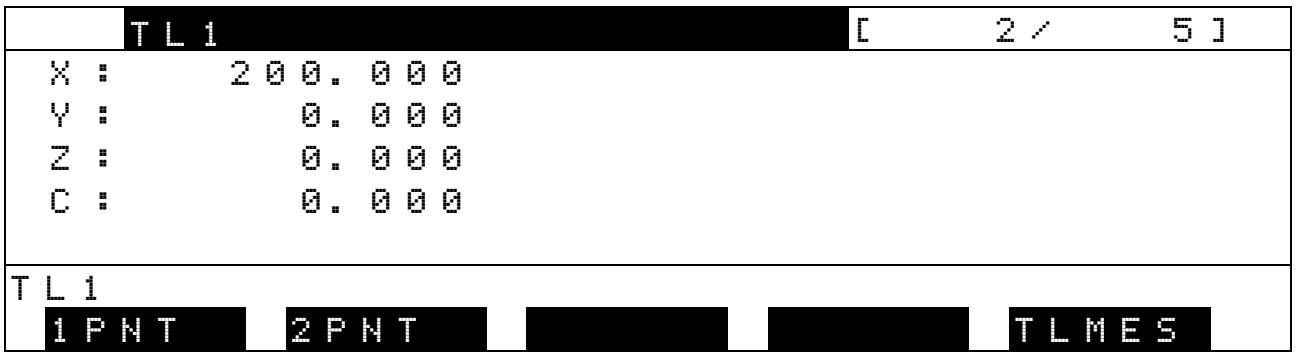
4 Move the robot TCP to the origin of the TRANS-type data to be generated, and press the EXE key.

"Teach +X Direction Press <EXE>" appears in the command line.



5 Move the robot TCP to a point on the coordinate axis of the TRANS-type data to be generated, and press the EXE key.

The value of the currently-displayed TRANS-type data is overwritten by the value of the TRANS-type data generated from the two teaching points.



MEMO

- *Pressing the ESC key during teaching cancels 2-point teaching.*
- *Pressing the BS key during teaching returns to teaching of the previous point.*
- *Pressing POINT (F5) during teaching opens the POINT-type data selection screen.*

3.2.11.3 Tool Measurement (TLMES)

The mechanical interface is moved to four points while the tip of the tool to be measured is fixed at one point, and the TRANS-type data is generated from these teaching points.

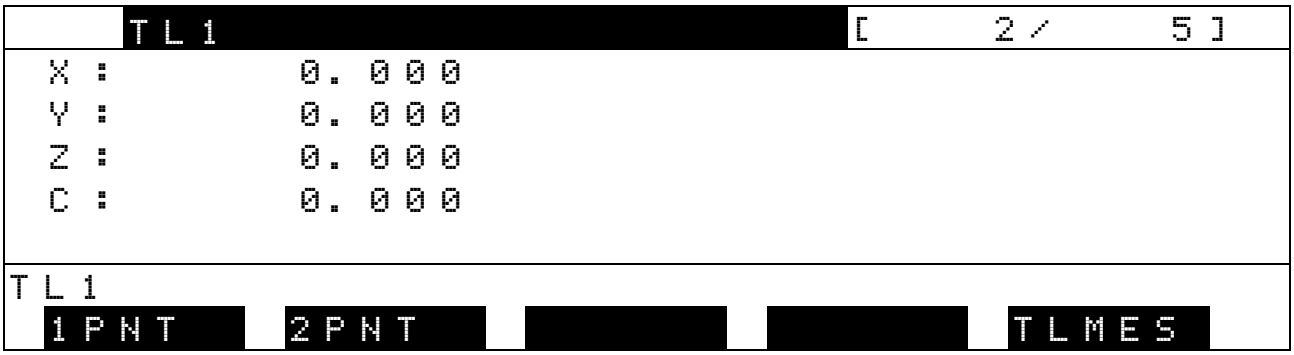
<Tool Measurement Procedure>

1 Press the NEXT key until "TEACH" is displayed in the function menu on the TRANS-type data edit screen.



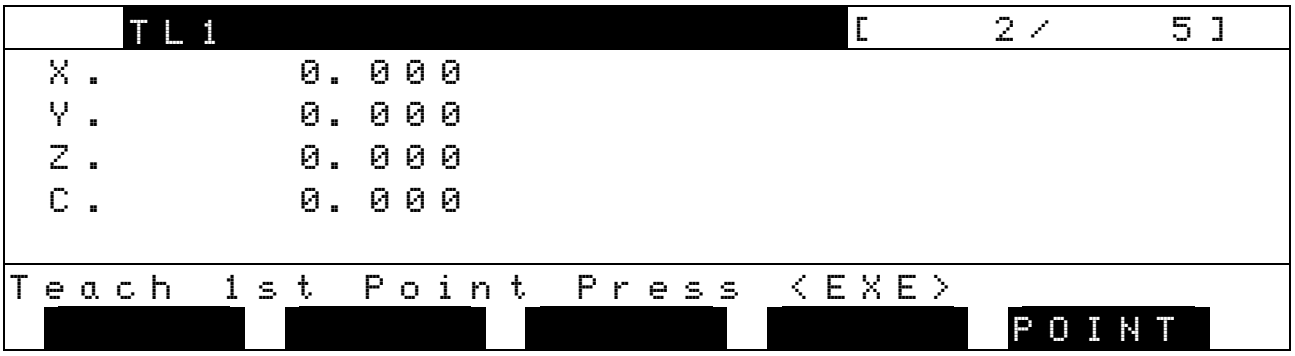
2 Press the TEACH (F2) key.

This switches the function menu.



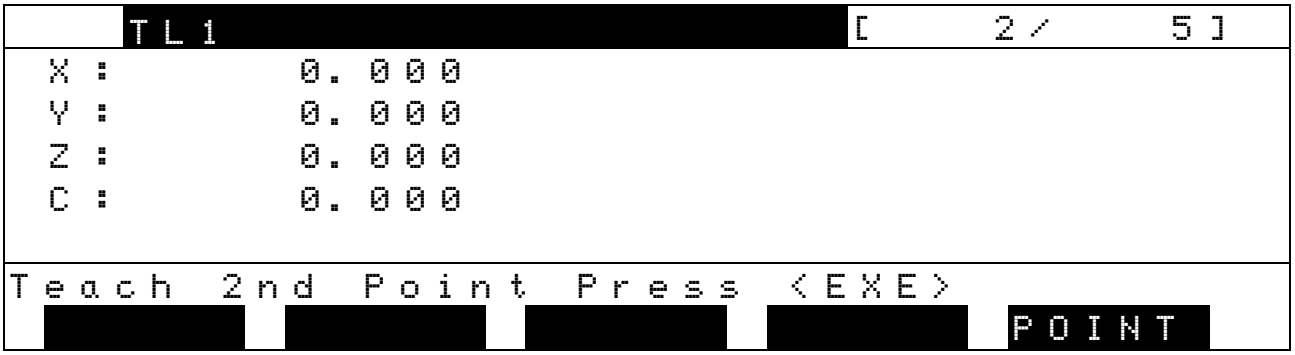
3 Press the TLMES (F5) key.

"Teach 1st Point Press <EXE>" appears in the command line.



4 Fix the tip of the tool to be measured at one point, and press the EXE key.

"Teach 2nd Point Press <EXE>" appears in the command line.

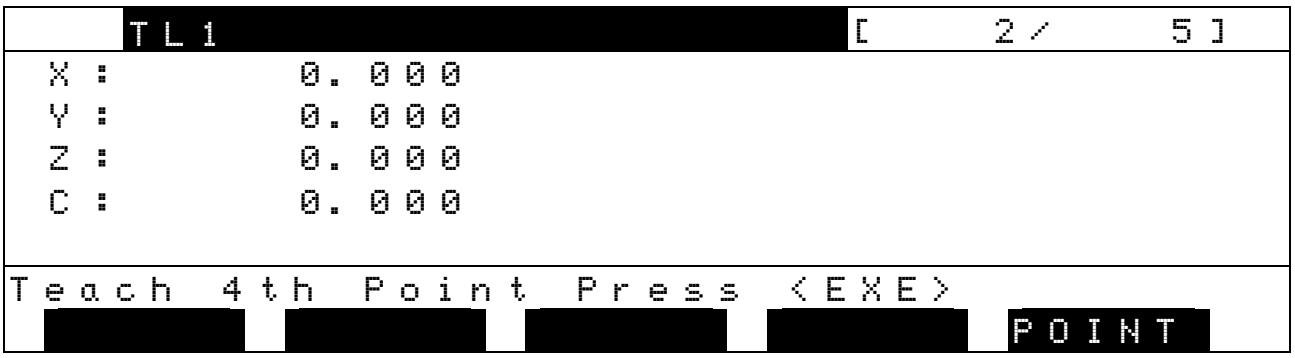


5 Move the mechanical interface position of the robot to the 2nd point while fixing the tip of the tool to be measured at one point, and press the EXE key.

"Teach 3rd Point Press <EXE>" appears in the command line.

6 Move the mechanical interface position of the robot to the 3rd point while fixing the tip of the tool to be measured at one point, and press the EXE key.

"Teach 4th Point Press <EXE>" appears in the command line.



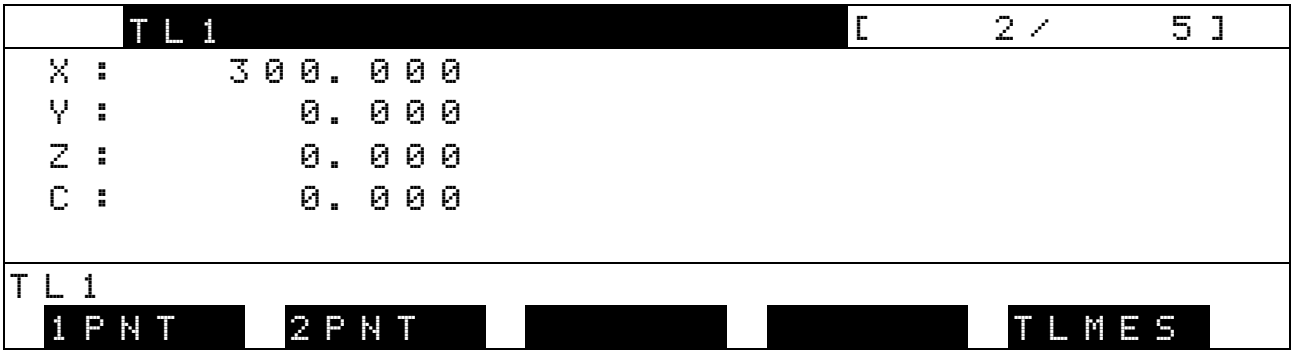
7 Move the mechanical interface position of the robot to the 4th point while fixing the tip of the tool to be measured at one point, and press the EXE key.

"Press <EXE> To Finish" appears in the command line.



8 Press the EXE key.

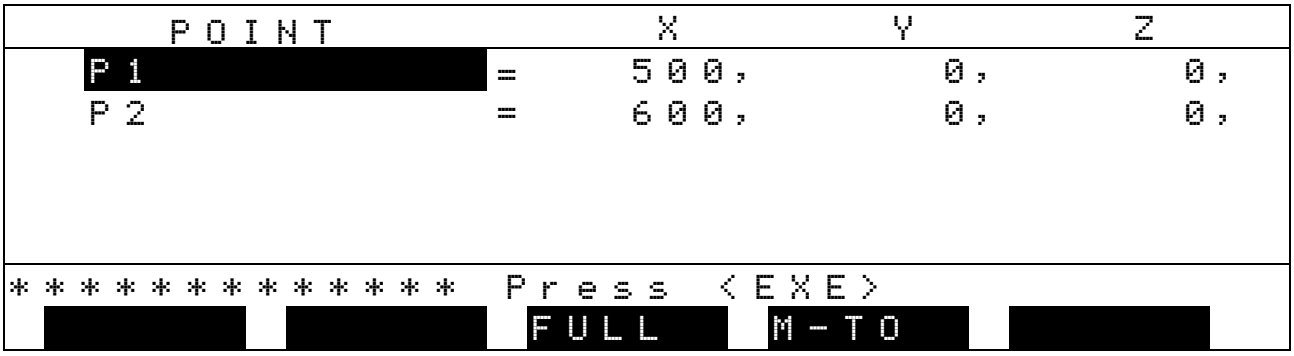
The value of the currently-displayed TRANS-type data is overwritten by the value of the TRANS-type data generated from the four teaching points.



MEMO

- *Pressing the ESC key during teaching cancels tool measurement.*
- *Pressing the BS key during teaching returns to teaching of the previous point.*
- *Pressing POINT (F5) during teaching opens the POINT-type data selection screen.*
- *Calculation will not be performed properly if the tool is measured while tool coordinates are set.*

POINT-type Data Selection Screen



The valid keys in the POINT-type data selection screen and their actions are shown in the following table.

Key	Description
↑ key	This moves the cursor up by one line.
↓ key	This moves the cursor down by one line.
BS key or ESC key	This closes the POINT-type data selection screen.
EXE key	This sets the cursor location as the teaching point and proceeds to the next teaching operation. This also closes the POINT-type data selection screen at the same time.

The function menus in the POINT-type data selection screen and their actions are shown in the following table.

Displayed menu name	Description	Section number for detailed explanation
FULL	This moves to the POINT-type data selection details screen.	This chapter
M-TO	This moves to the teaching point.	3.2.6

MEMO

When the POINT-type data selection screen is being displayed, the registered work coordinates are cleared.

POINT-type Data Selection Details Screen



The valid keys in the POINT-type data selection details screen and their actions are shown in the following table.

Key	Description
↑ key	This switches the displayed POINT-type data to the data one above it.
↓ key	This switches the displayed POINT-type data to the data one below it.
BS key or ESC key	This closes the POINT-type data selection details screen.
EXE key	This sets the cursor location as the teaching point and proceeds to the next teaching operation. This also closes the POINT-type data selection details screen at the same time.

The function menus in the POINT-type data selection details screen and their actions are shown in the following table.

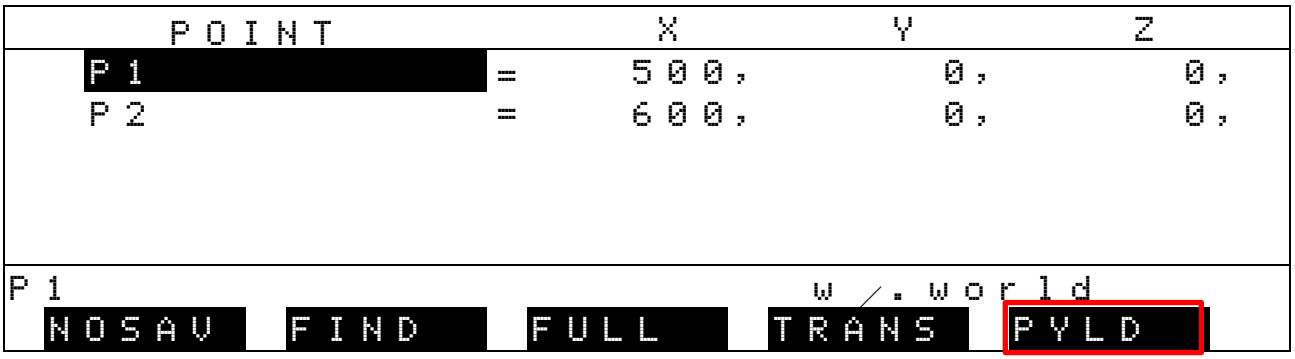
Displayed menu name	Description	Section number for detailed explanation
DIR	This returns to the POINT-type data selection screen.	This chapter
M-TO	This moves to the teaching point.	3.2.6

3.2.12 Opening the PAYLOAD-type Data Edit Screen

This describes how to open the PAYLOAD-type data edit screen.

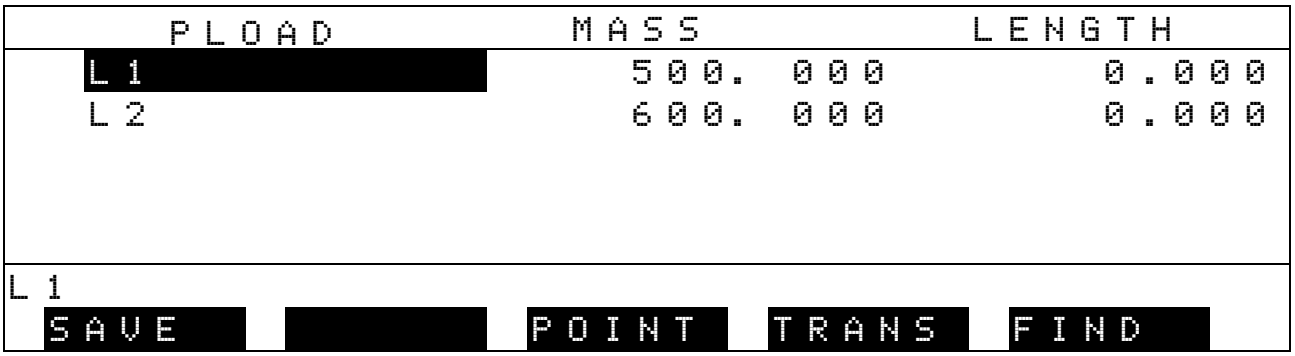
<Opening the PAYLOAD-type Data Edit Screen from the POINT-type Data List Screen or POINT-type Data Details Screen>

1 Press the NEXT key until "PYLD" is displayed in the function menu on the POINT-type data list screen or POINT-type data details screen.



2 Press the PYLD (F5) key.

This opens the PAYLOAD-type data list screen.



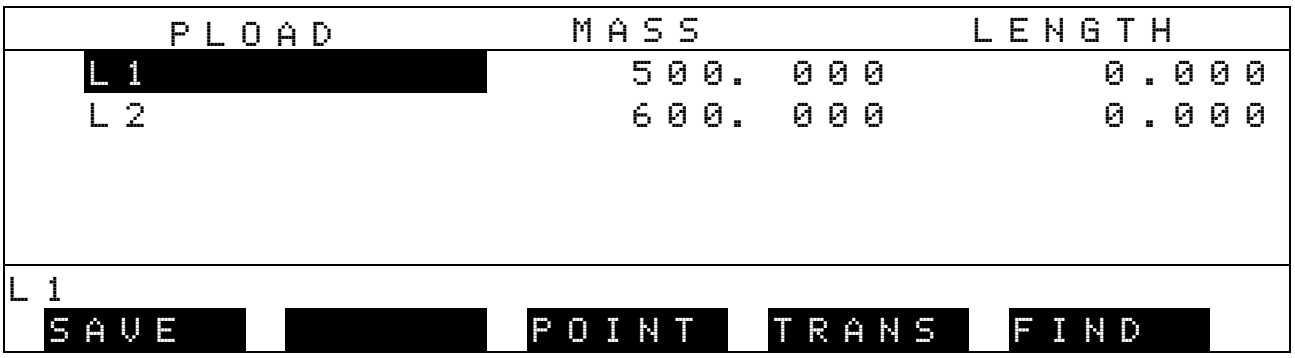
<Opening the PAYLOAD-type Data Edit Screen from the TRANS-type Data List Screen>

1 Press the NEXT key until "PYLD" is displayed in the function menu on the TRANS-type data list screen.

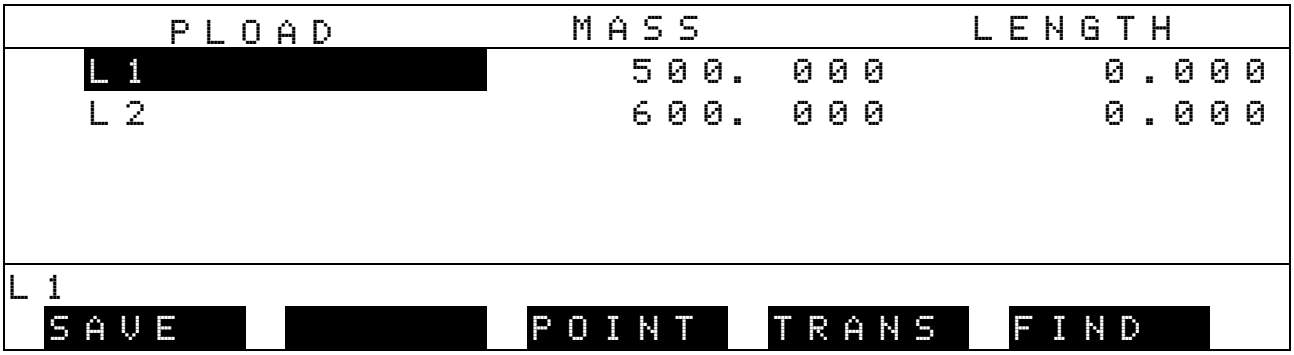


2 Press the PYLD (F3) key.

This opens the PAYLOAD-type data list screen.



PAYLOAD-type Data Edit Screen



The valid keys in the PAYLOAD-type data edit screen and their actions are shown in the following table.

Key	Description
↑ key	This moves the cursor up by one line.
↓ key	This moves the cursor down by one line.
ALT key + ↑ key	This moves the cursor to the previous page.
ALT key + ↓ key	This moves the cursor to the next page.
INS key	This adds data to the line after the cursor.
ALT key + BS key	This removes the data in the cursor line.
NEXT key or ALT key + NEXT key	This switches the function menu.
EXE key	This edits the cursor position.

The function menus in the PAYLOAD-type data edit screen and their actions are shown in the following table.

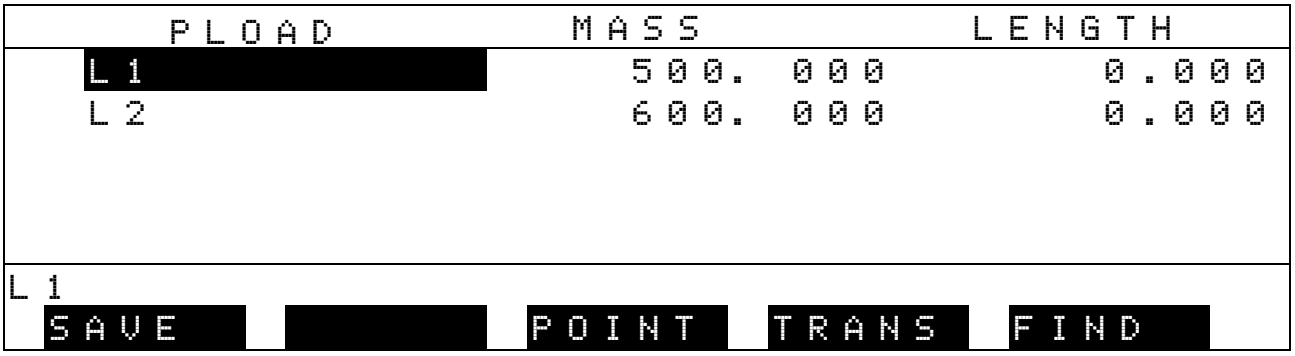
Displayed menu name	Description	Section number for detailed explanation
SAVE	This saves and exits the file.	3.3
NOSAV	This discards the file.	3.3
PRG	This opens the program section edit screen.	3.1.1
POINT	This opens the POINT-type data list screen.	3.2.2
TRANS	This opens the TRANS-type data list screen.	3.2.8
FIND	This searches for data.	3.2.17
CUT	This cuts data.	3.2.15
PASTE	This pastes data.	3.2.16

3.2.13 Editing PAYLOAD-type Data

<Editing Items in PAYLOAD-type Data>

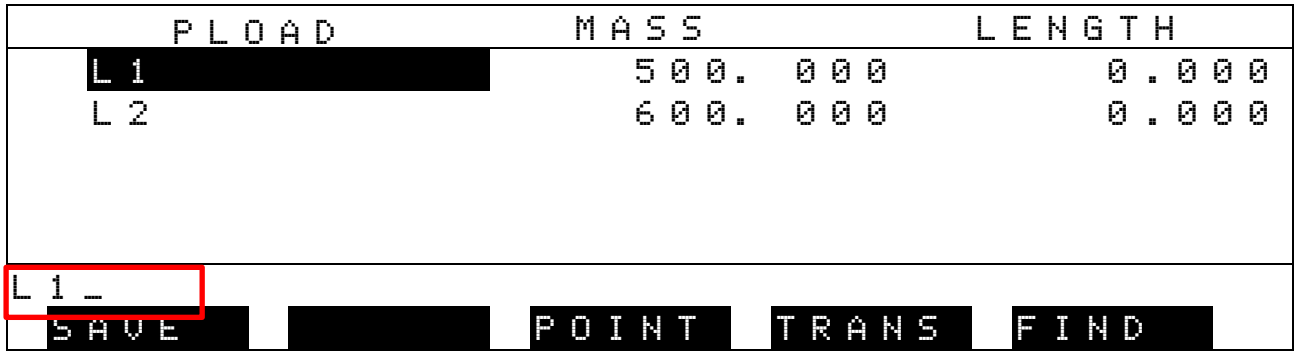
1 Move the cursor to the position of the item to be edited.

The content for the item is displayed in the command line.



2 Press the EXE key.

The cursor is displayed in the command line, and the content can be edited.



3 Edit the content in the command line, and press the EXE key.

This saves the edited changes.

MEMO

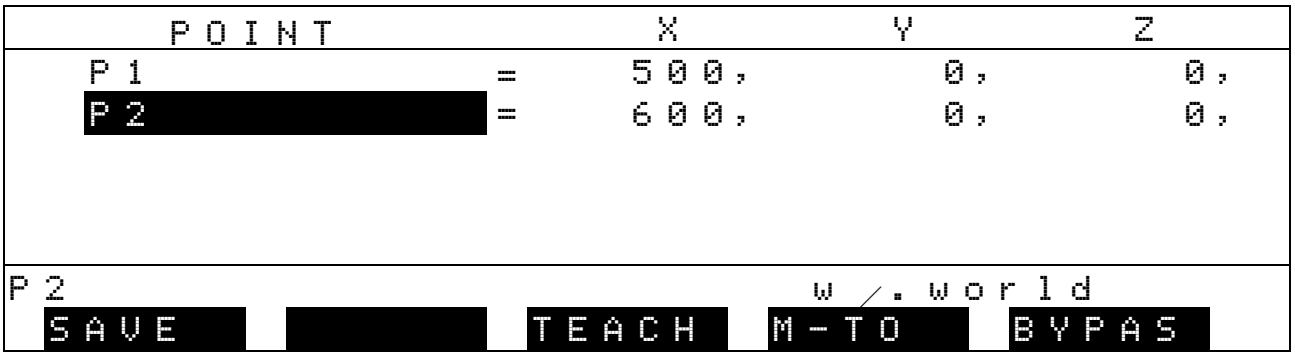
- Character strings that are set for the PAYLOAD-type data name must have 10 or less alphanumeric characters and begin with a letter.
  If a character string exceeds 10 characters or includes symbols, the editing changes are not accepted, and the message *"ERR: ILLEGAL NAME" is displayed.*
- A value in the range of -9999.999 to 9999.999 can be set for the value of the PAYLOAD-type data.
  If a value outside the range or an invalid character is entered, the editing changes are not accepted, and the message "ERR: ILLEGAL DATA" is displayed.
- *If the entered data name is already being used, the editing changes are not accepted, and the message "ERR: DUPLICATE NAME" is displayed.*

3.2.14 Adding Data

This describes how to add POINT-type data, TRANS-type data, and PAYLOAD-type data.

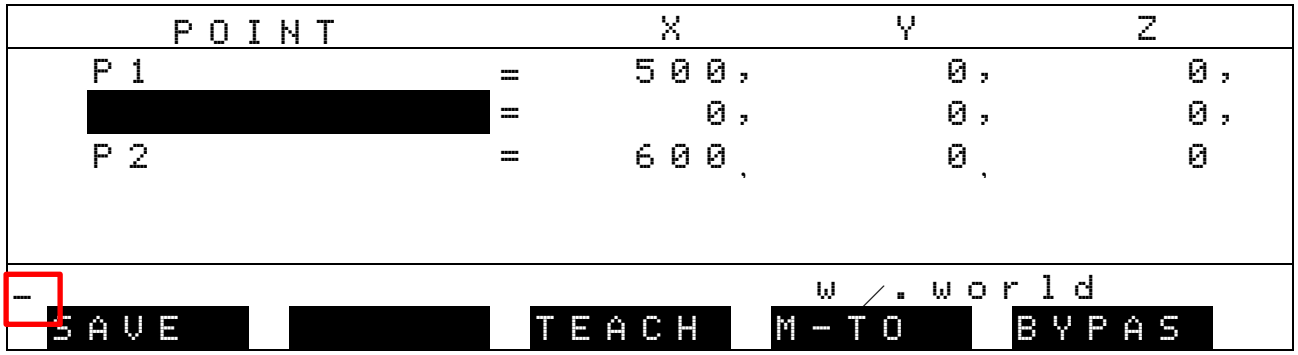
<Procedure for Adding New Data to Various Data>

1 Move the cursor to the position where the new line will be added from the POINT-type data list screen (or TRANS-type data list screen or PAYLOAD-type data edit screen). (The cursor can be at any position in the POINT-type data details screen or the TRANS-type data edit screen.)



2 Press the INS key.

A new line is added under the cursor, and the cursor is displayed in the command line.



3 Enter the name of the new data, and press the EXE key.

Data with the entered name is generated.

MEMO

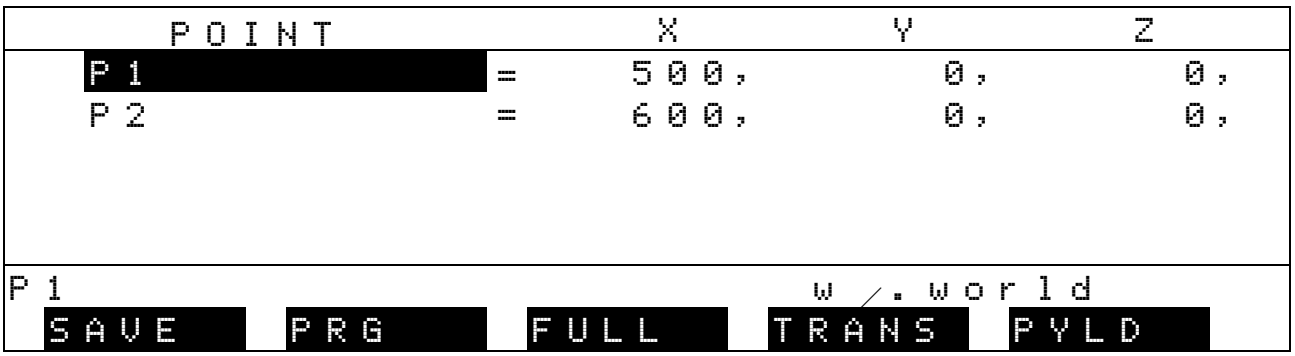
- Character strings that are set for the data name must have 10 or less alphanumeric characters and begin with a letter.
  If a character string exceeds 10 characters or includes symbols, the editing changes are not accepted, and the message *"ERR: ILLEGAL NAME" is displayed.*
- *If the entered data name is already being used, the editing changes are not accepted, and the message "ERR: DUPLICATE NAME" is displayed.*

3.2.15 Cutting the Data (CUT)

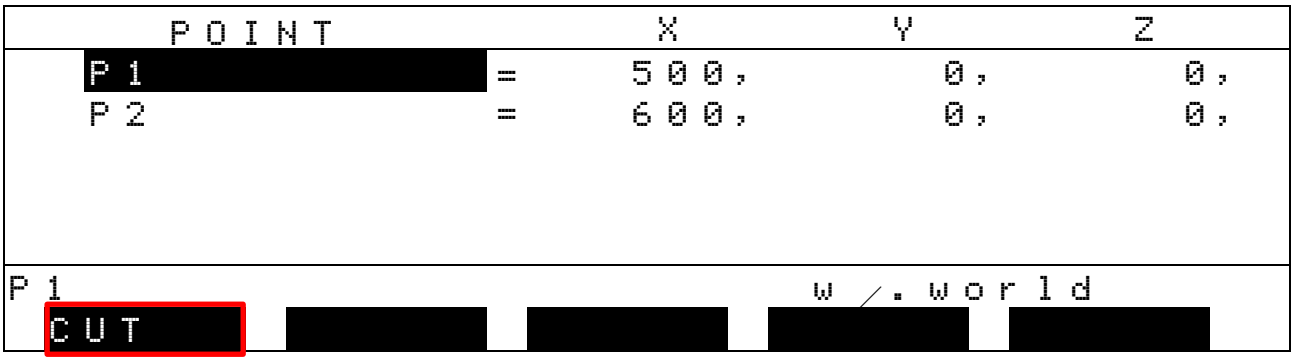
This describes how to cut data in the data section edit screen.

<Data Cutting Procedure>

1 Move the cursor to the line where the data will be cut from the POINT-type data list screen (or TRANS-type data list screen or PAYLOAD-type data edit screen).



2 Press the NEXT key until "CUT" is displayed in the function menu.



MEMO

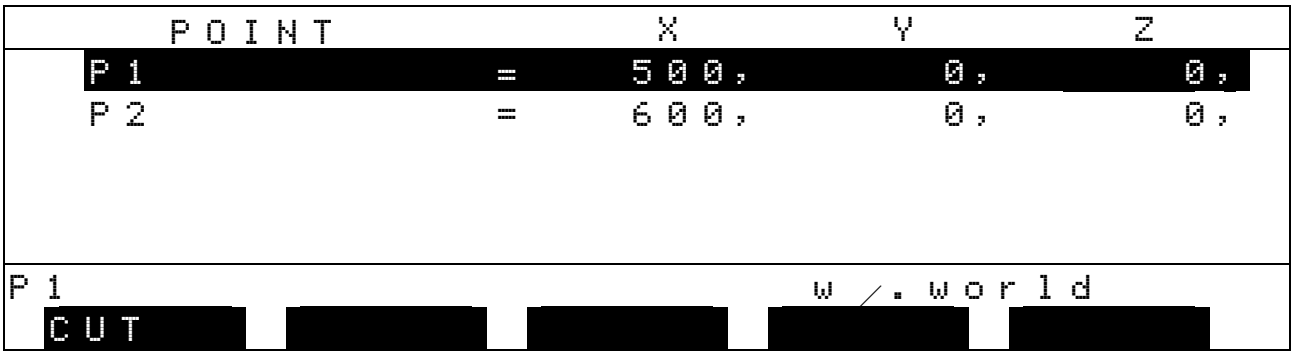
The position of the function key indicating "CUT" varies depending on the "BYPASS-SW" setting for [U011].

When [U011] "BYPASS-SW""" is 0: Position corresponding to F5 key

When [U011] "BYPASS-SW""" is 1 or 2: Position corresponding to F1 key

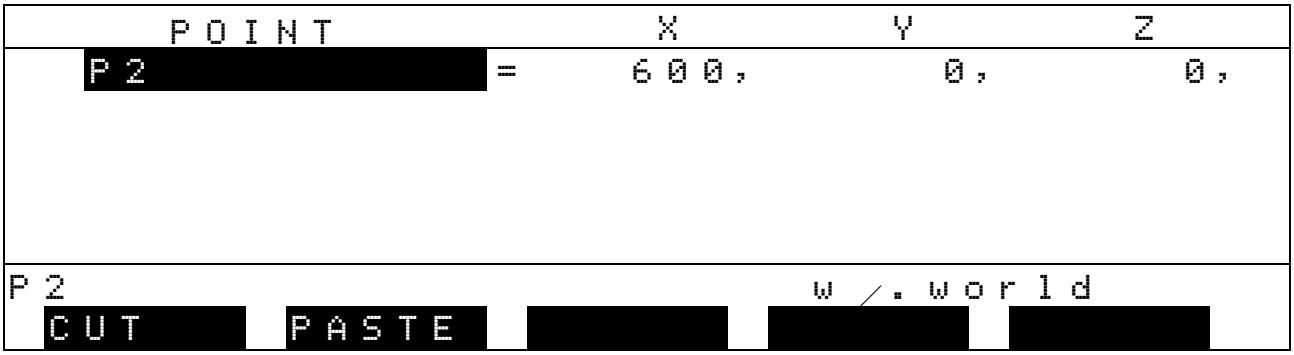
3 Press the CUT (F1 or F5) key.

The cursor extends to the entire line, and the function menu is hidden except for "CUT".



4 Press the CUT (F1 or F5) key again.

This cuts the line where the cursor is located.



MEMO

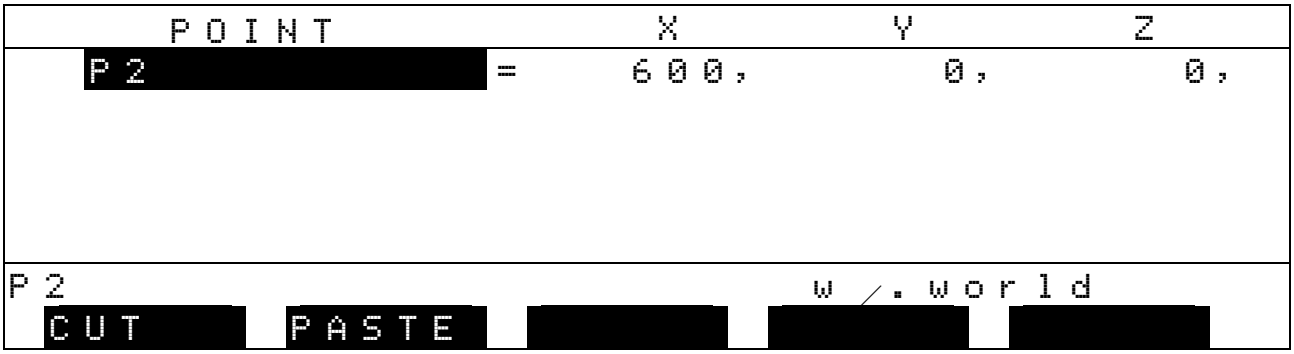
- *Only one type of data in the data section can be cut.*
  Note that, if data is cut while there is already data that was cut, the data that was cut first is lost.
  (Example: If the TRANS-type data is cut after POINT-type data was cut, the POINT-type data is lost.)
- *When editing of the data section is exited, the data that was cut is discarded.*

3.2.16 Pasting Data (PASTE)

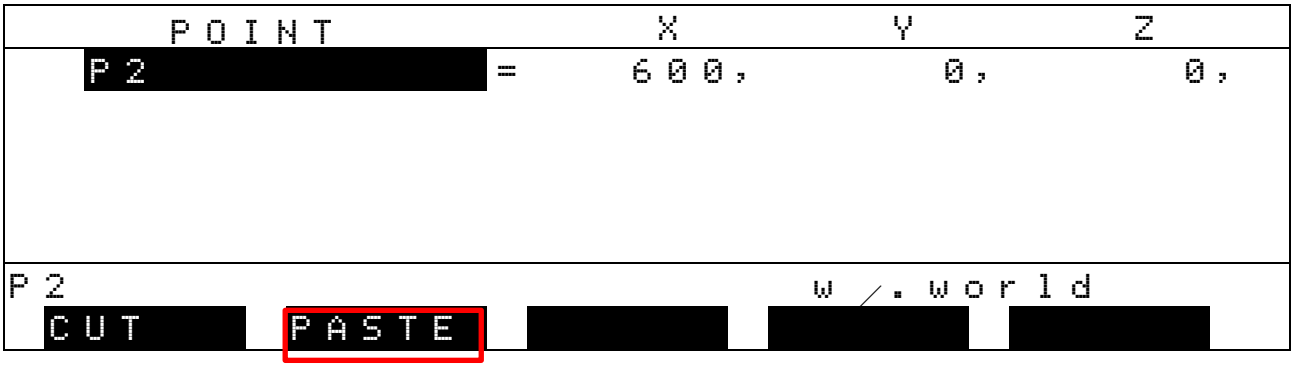
This describes how to paste data that was cut in the data section edit screen.

<Data Pasting Procedure>

1 Move the cursor to the line where the data will be pasted from the POINT-type data list screen (or TRANS-type data list screen or PAYLOAD-type data edit screen).



2 Press the NEXT key until "PASTE" is displayed in the function menu.



MEMO

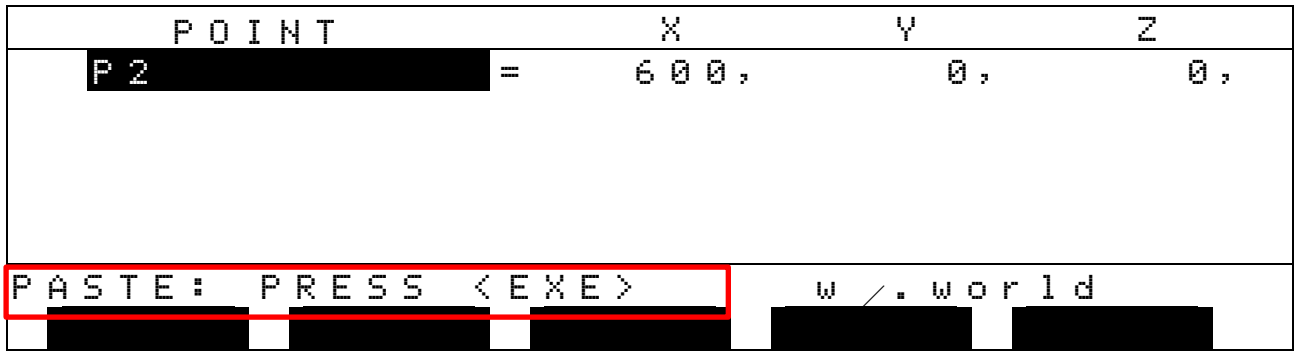
- *"PASTE" appears only when data that can be pasted is available.*
- *The position of the function key indicating "PASTE" varies depending on the "BYPASS-SW" setting for [U011].*

  When [U011] "BYPASS-SW""" is 0: Position corresponding to F1 key

  When [U011] "BYPASS-SW""" is 1 or 2: Position corresponding to F5 key

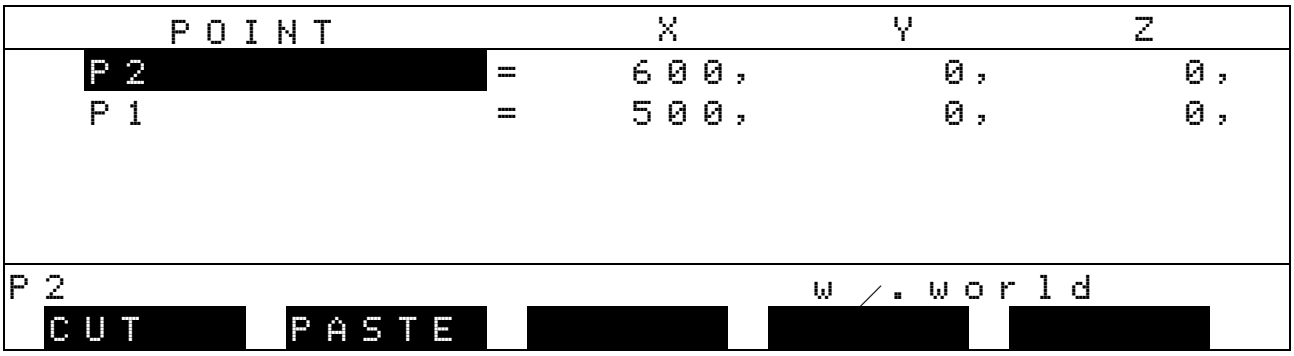
3 Press the PASTE (F1 or F2) key.

"PASTE:PRESS <EXE>" appears in the command line.



4 Press the PASTE (F1 or F2) key again.

Data is pasted to the line below the line where the cursor is located.



MEMO

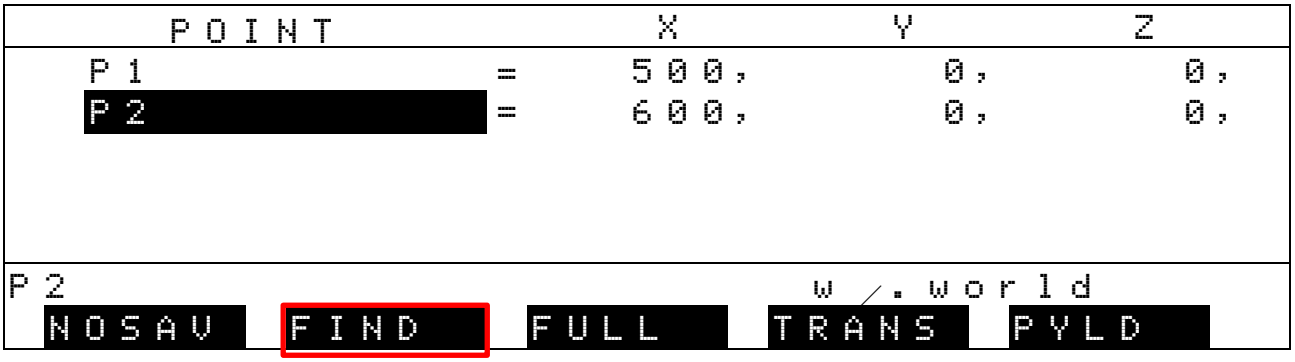
- *Data is not pasted if there is data with the same name as the data that was cut.*
- *The same type of data only can be pasted.*

3.2.17 Searching for Data Names (FIND)

This describes how to find a selected data name in the data section.

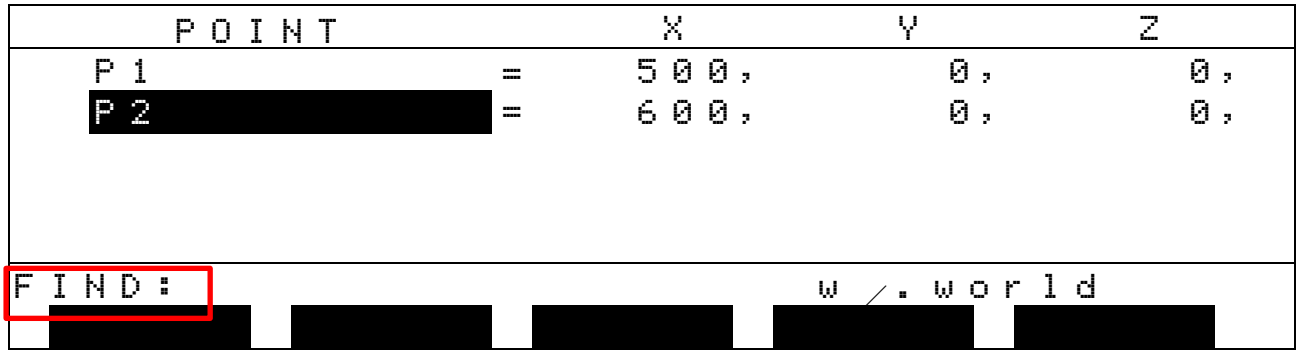
<Data Name Search Procedure>

1 Press the NEXT key until "FIND" is displayed in the function menu on the POINT-type data list screen (or POINT-type data details screen, TRANS-type data list screen, TRANS-type data edit screen, or PAYLOAD-type data edit screen).



2 Press the function key assigned to "FIND".

"FIND:" appears in the command line.



3 Enter the name of the data that will be searched for, and press the EXE key.

The cursor moves to the location of the data with the entered name.

If the entered data name is used in POINT-type data, the POINT-type data list screen is opened.

If the entered data name is used in TRANS-type data, the TRANS-type data list screen is opened.

If the entered data name is used in PAYLOAD-type data, the PAYLOAD-type data edit screen is opened.

MEMO

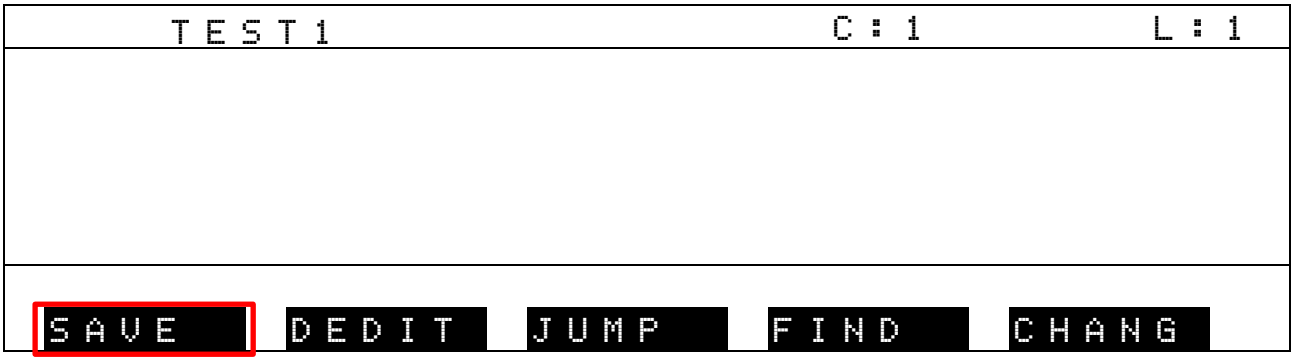
- *If no matching character string is found, "Not found." appears in the command line.*
- *If the entered name is used in POINT-type data, the TRANS-type data where the POINT-type data is linked is registered to work coordinates, and the POINT-type data list screen is opened.*
- *If the data name that was searched for in the POINT-type data details screen is POINT-type data, the POINT-type data details screen is opened.*
  If the data name that was searched for in the TRANS-type data edit screen is TRANS-type data, the TRANS-type data edit screen is opened.

3.3 Exiting of Saving and Editing of Files

This describes how to save the edited file and exit the editing process.

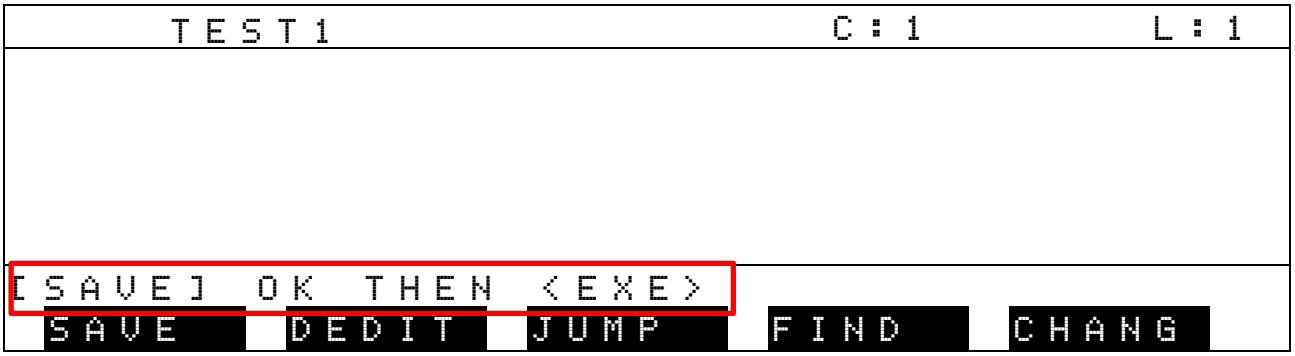
<File Saving and Exiting Procedure>

1 Press the NEXT key until "SAVE" is displayed in the function menu on the program section edit screen or the data section edit screen.



2 Press the SAVE (F1) key.

"[SAVE] OK THEN <EXE>" appears in the command line.



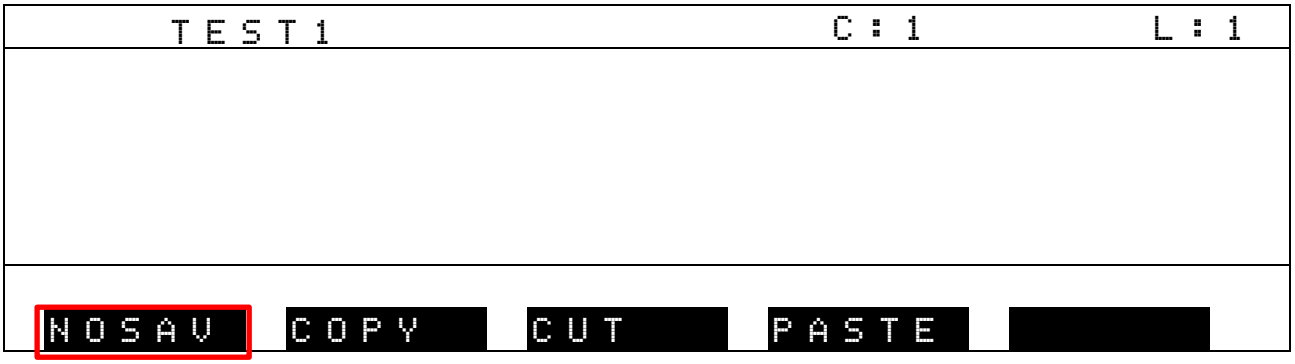
3 Press the EXE key.

This saves the file, exits editing, and returns to the basic screen.

> *MEMO*
>
> *If a file registered in the execution program is edited and saved, the program is recompiled.*

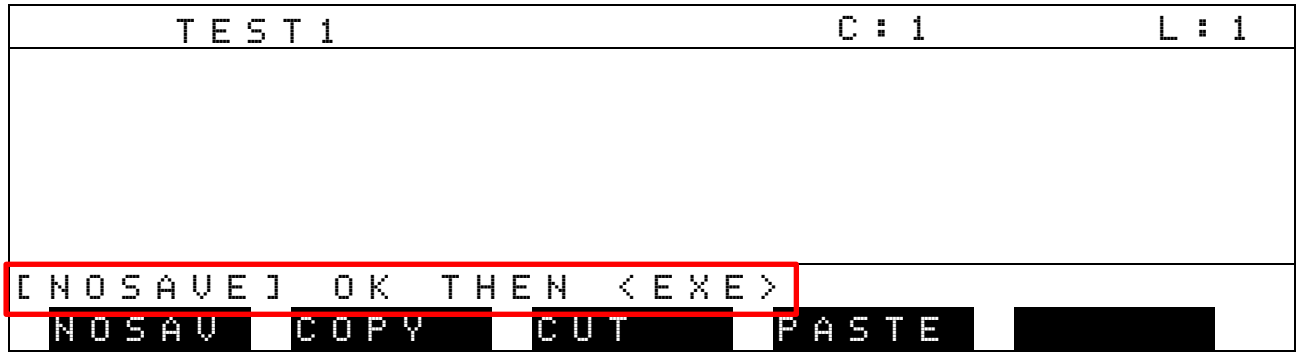
<Discarding File and Exiting Procedure>

1 Press the NEXT key until "NOSAV" is displayed in the function menu on the program section edit screen or the data section edit screen.



2 Press the NOSAV (F1) key.

"[NOSAVE] OK THEN <EXE>" appears in the command line.



3 Press the EXE key.

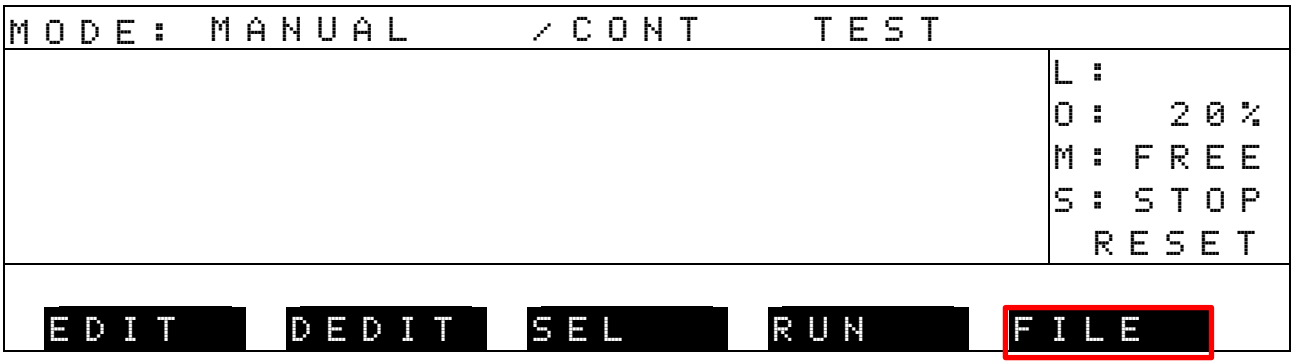
This discards the file and returns to the basic screen.

3.4 File Operations

This describes how to use the TP1000 to perform file operations (copy, delete, rename).

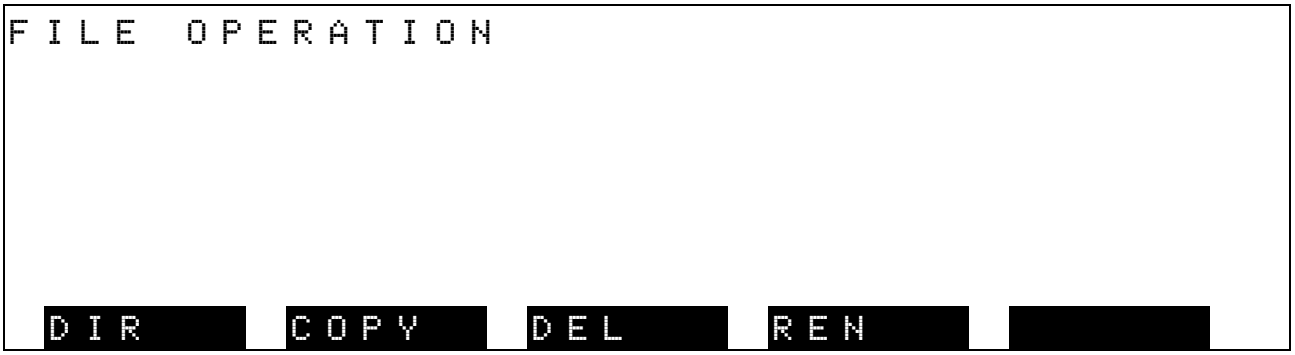
3.4.1 Opening the File Operation Screen

1 Press the NEXT key until "FILE" appears in the function menu on the basic screen.

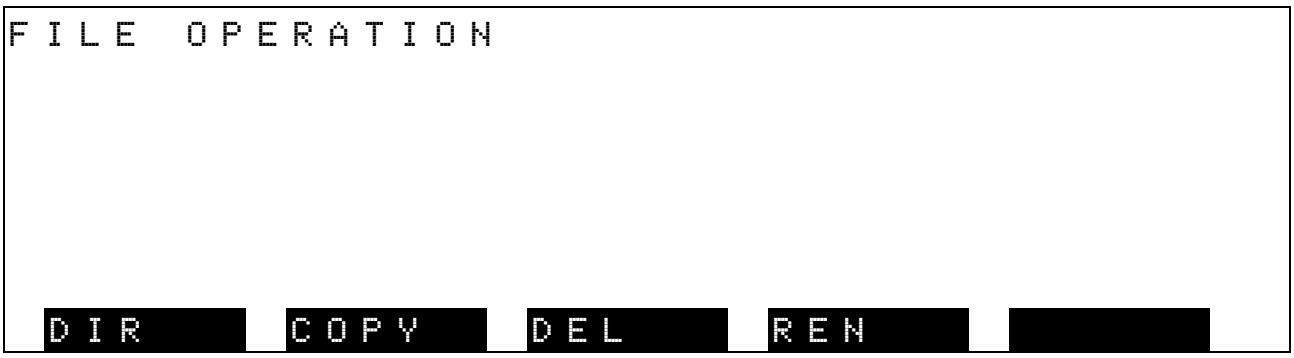


2 Press the FILE (F5) key.

The file operation screen is opened.



File Operation Screen



The valid keys in the file operation screen and their actions are shown in the following table.

Key	Description
ESC key	This returns to the basic screen.
DIR (F1) key	This displays the file list.
COPY (F2) key	This copies a file.
DEL (F3) key	This deletes a file.
REN (F4) key	This renames a file.

The function menus in the file operation screen and their actions are shown in the following table.

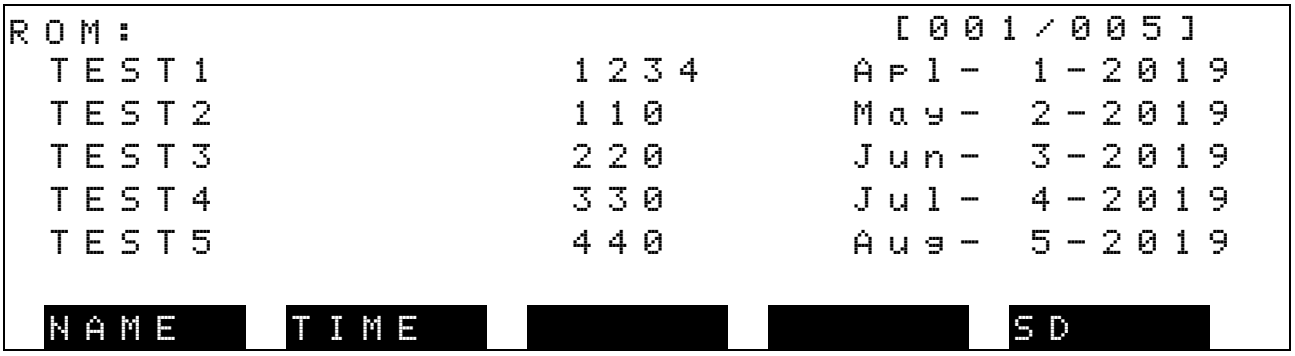
Displayed menu name	Description	Section number for detailed explanation
DIR	This displays the file list.	3.4.2
COPY	This copies a file.	3.4.3
DEL	This deletes a file.	3.4.4
REN	This renames a file.	3.4.5

3.4.2 Displaying the File List (DIR)

<File List Display Procedure>

1 Press the DIR (F1) key in the file operation screen.

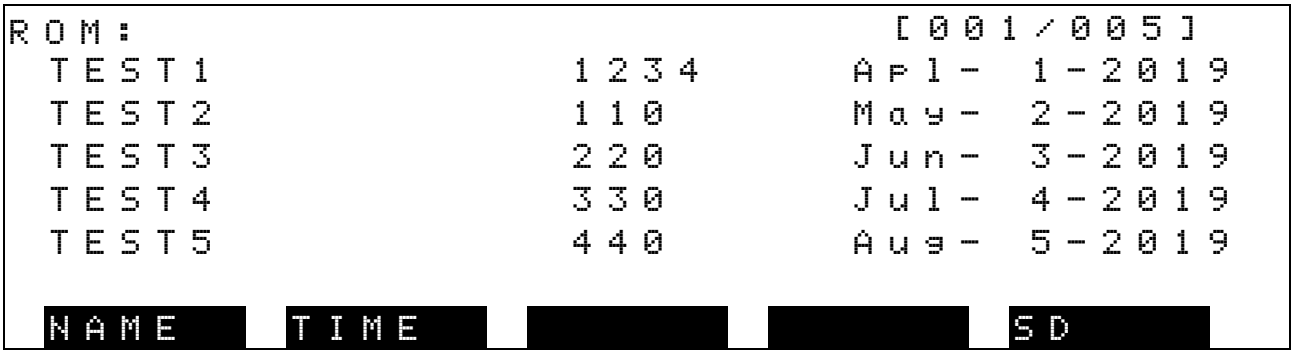
A list of the files appears.



MEMO

When the file list screen is displayed, a list of the files in the ROM (inside the controller) is shown.

File List Screen



The valid keys in the file list screen and their actions are shown in the following table.

Key	Description
↑ key	This shows the file above the current position.
↓ key	This shows the file below the current position.
ALT key + ↑ key or ALT key + NEXT key	This shows the files on the previous page.
ALT key + ↓ key or NEXT key	This shows the files on the next page.
ESC key	This returns to the file operation screen.

The function menus in the file list screen and their actions are shown in the following table.

Displayed menu name	Description	Section number for detailed explanation
NAME	This is used to sort files by name.	
TIME	This is used to sort files by date and time created.	
ROM	This switches the displayed drive to ROM.	3.4.2
SD	This switches the displayed drive to SD.	3.4.2

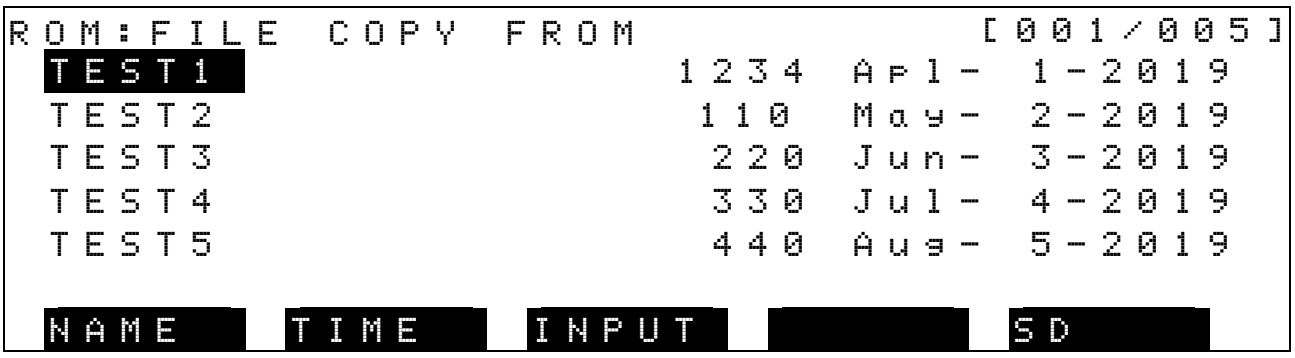
3.4.3 Copying Files (COPY)

This describes how to copy files.

<Procedure for Copying File to Same Drive>

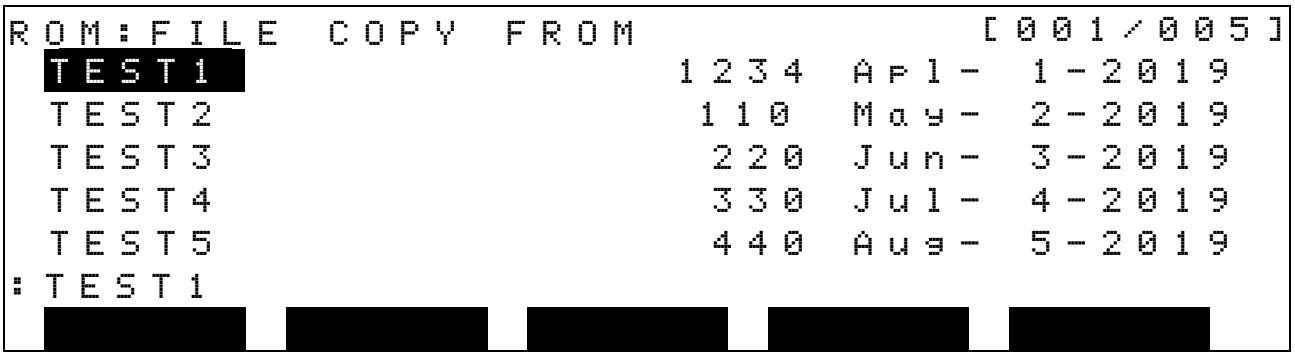
1 Press the COPY (F2) key in the file operation screen.

The operation file selection screen is opened.



2 Move the cursor to the location of the file to be copied, and press the EXE key.

The file name appears in the command line.

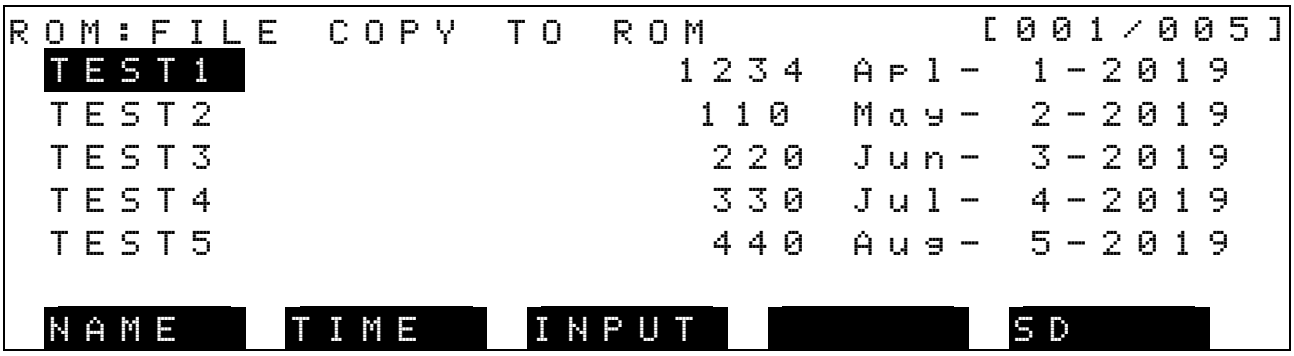


MEMO

The same operation can be performed by pressing the INPUT (F3) key and entering the file name to be copied to the command line.

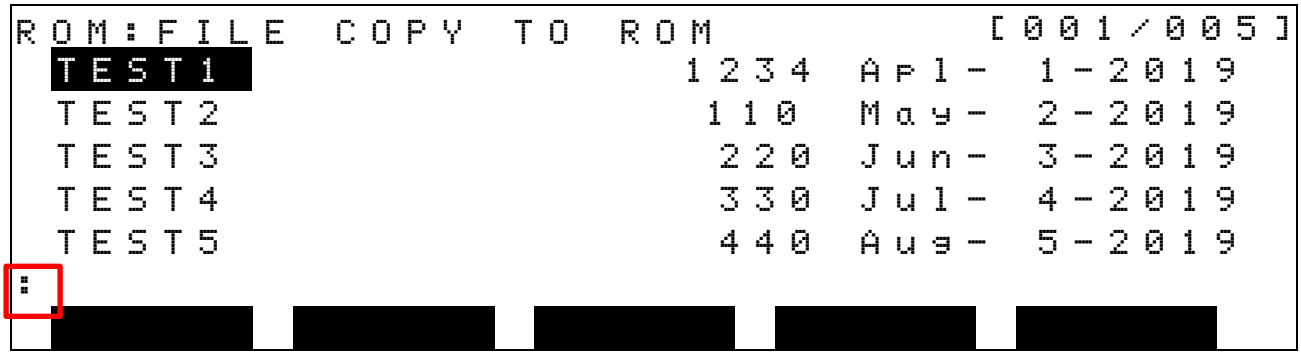
3 Press the EXE key.

The operation file selection screen is opened again.



4 Press the INPUT (F3) key.

The symbol ":" is displayed in the command line, and the content can be edited.

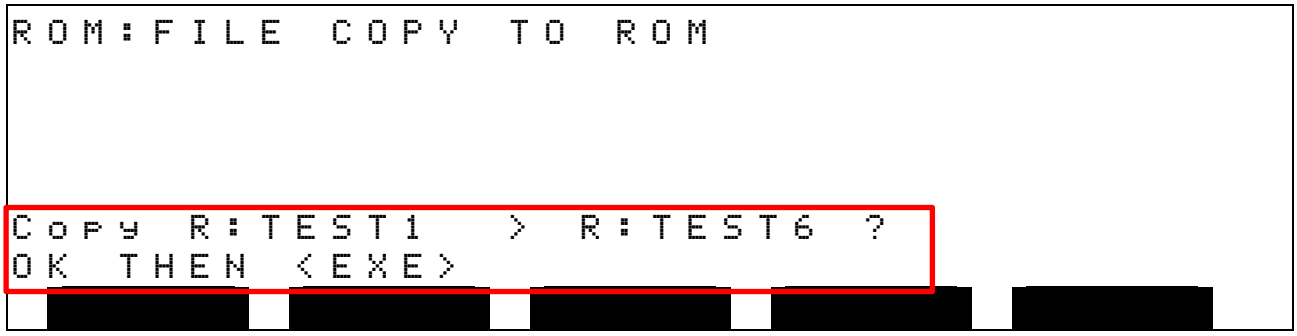


MEMO

When the cursor is aligned with a selected file name and the EXE key is pressed, the file name appears in the command line.

5 Enter the file name for the copy target, and click [EXE].

A confirmation message appears.

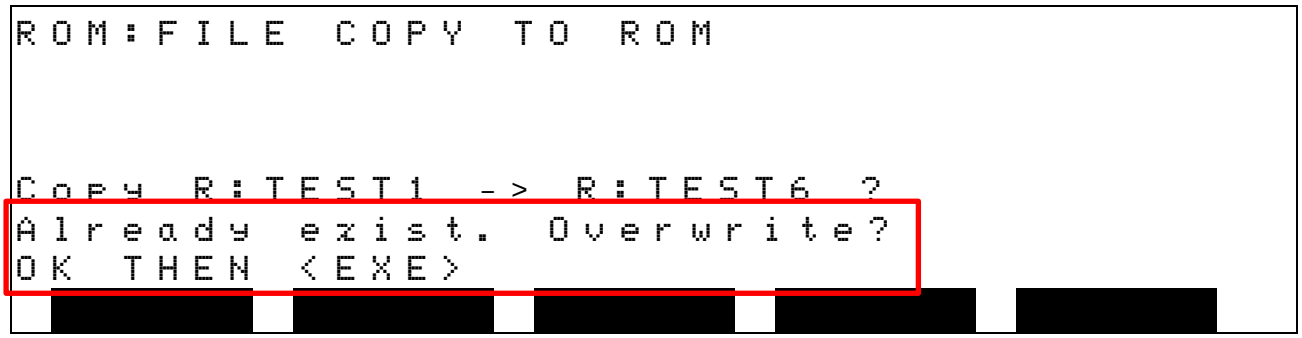


6 Press the EXE key.

If a file with the same name as the copy target file does not exist, the file copy operation is performed. When the copy operation is finished, "Completed." and "HIT ANY KEY" appear.



If a file with the same name as the copy target file already exists, a confirmation message appears asking whether it is OK to overwrite the file at the copy target.



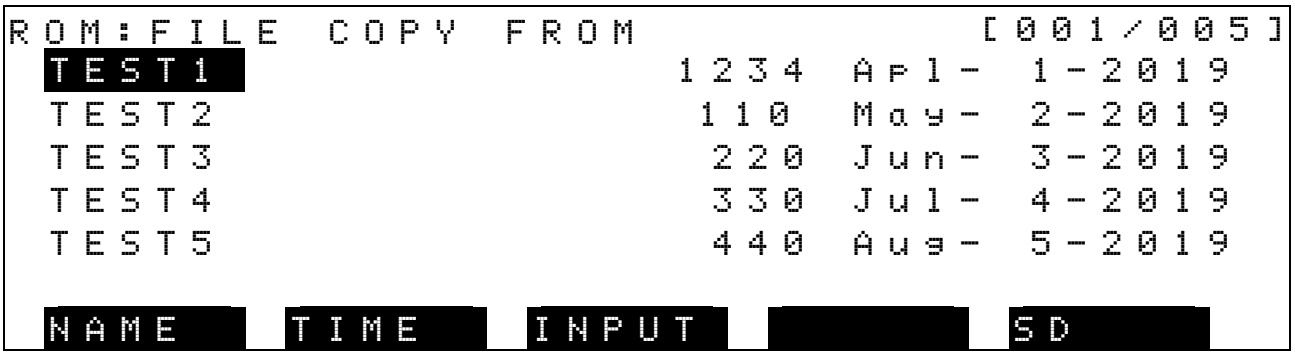
7 If a file with the same name as the copy target file already exists, press the EXE key again.

The file is copied. When the copy operation is finished, "Completed." and "HIT ANY KEY" appear.



8 Press any key.

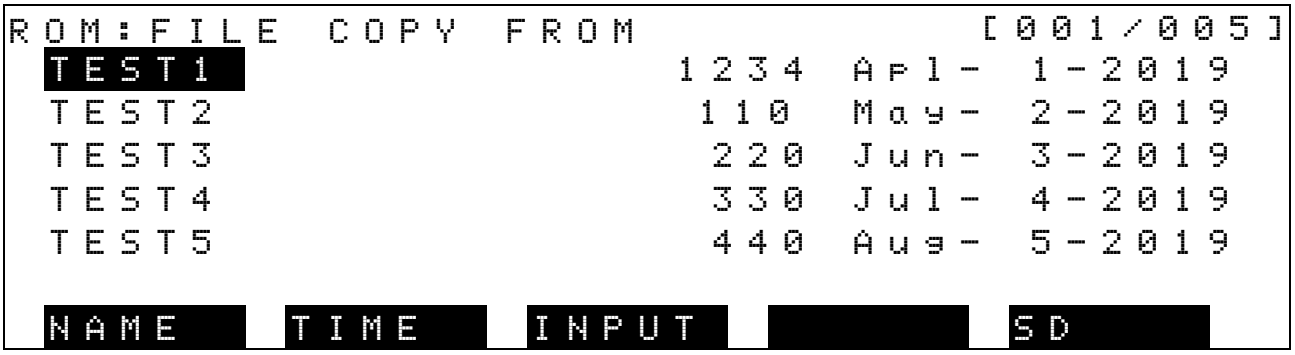
This returns to the operation file selection screen.



<Procedure for Copying File to Different Drive>

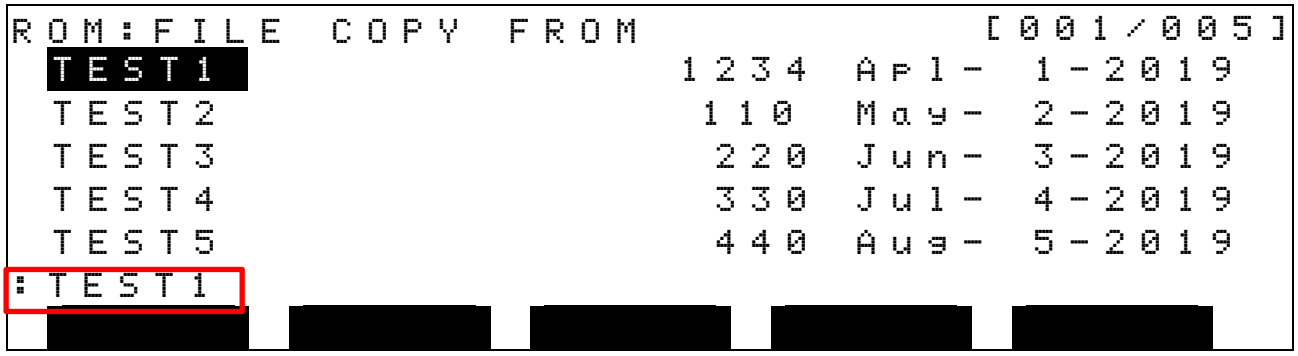
1 Press the COPY (F2) key in the file operation screen.

The operation file selection screen is opened.



2 Move the cursor to the location of the file to be copied, and press the EXE key.

The file name appears in the command line.

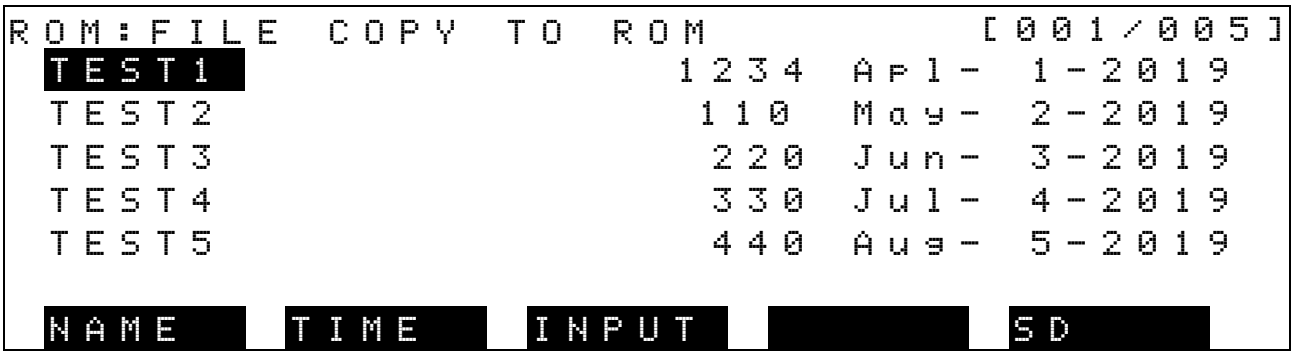


MEMO

The same operation can be performed by pressing the INPUT (F3) key and entering the file name to be copied to the command line.

3 Press the EXE key.

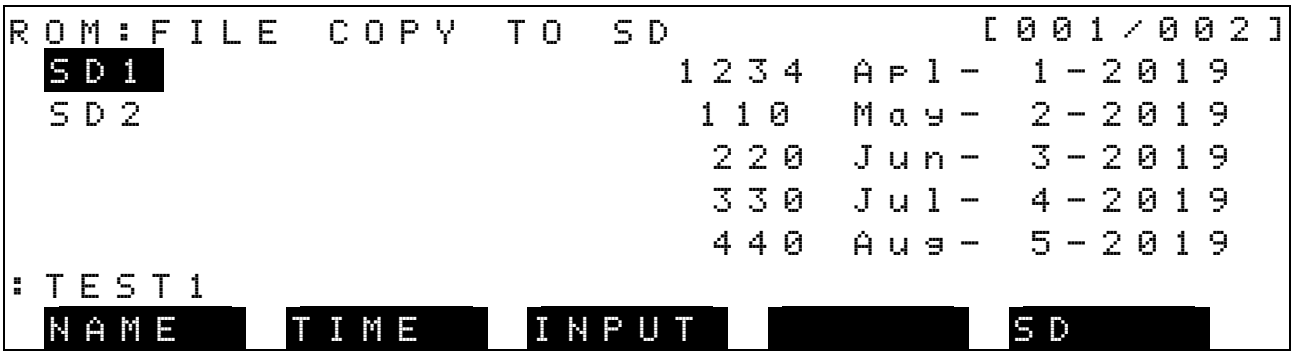
The operation file selection screen is opened again.



4 Press the SD/ROM (F5) key.

The operation file selection screen in the SD card (ROM) is opened.

The name of the file at the copy source appears in the command line.

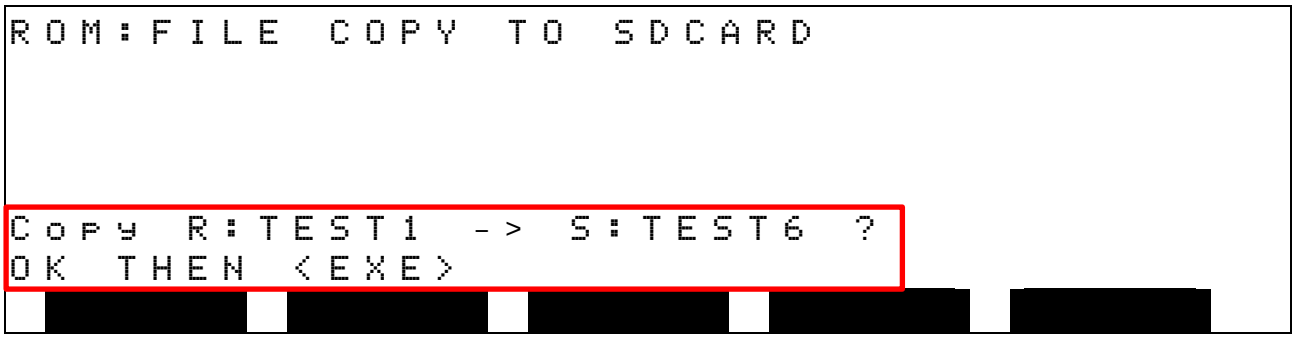


MEMO

The name of the file at the copy target can be edited in the command line.

5 Enter the file name for the copy target, and press the EXE key.

A confirmation message appears.

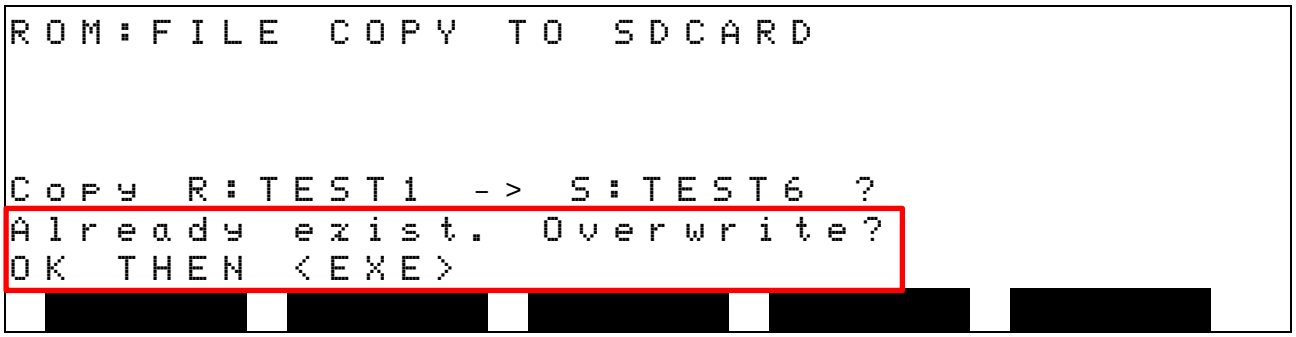


6 Press the EXE key.

If a file with the same name as the copy target file does not exist, the file copy operation is performed. When the copy operation is finished, "Completed." and "HIT ANY KEY" appear.

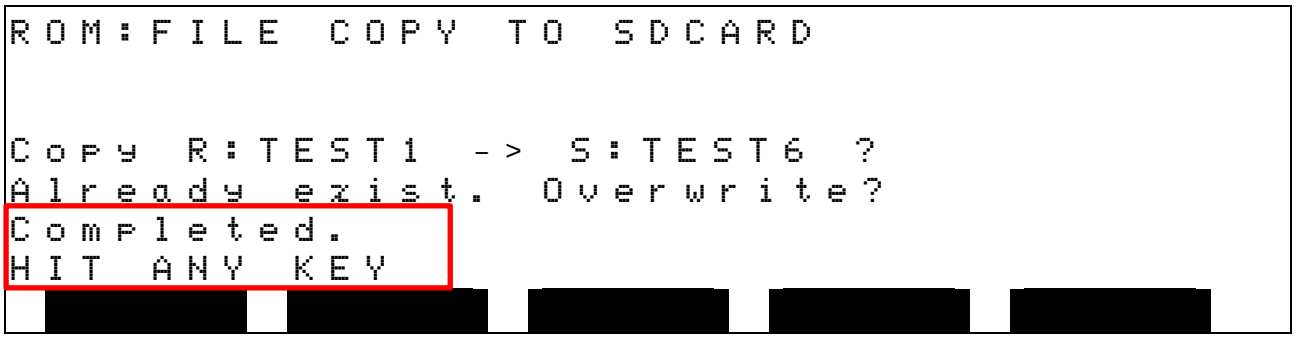


If a file with the same name as the copy target file already exists, a confirmation message appears asking whether it is OK to overwrite the file at the copy target.



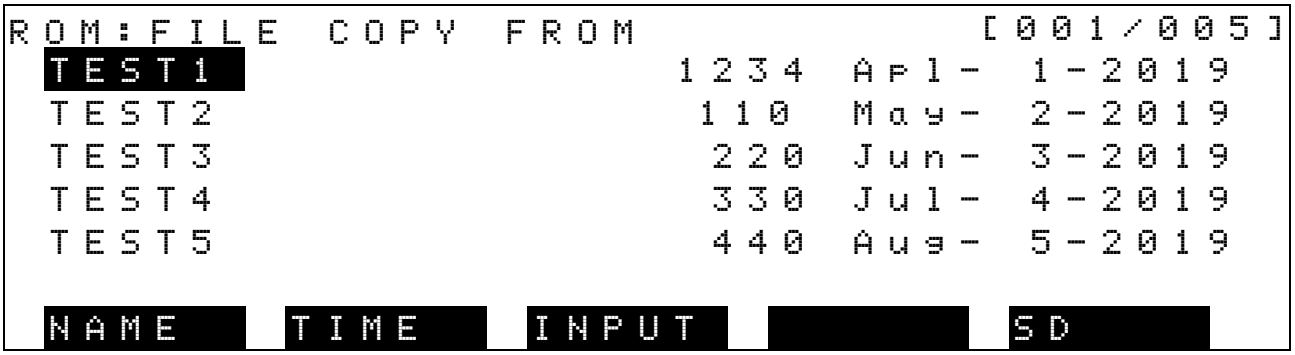
7 If a file with the same name as the copy target file already exists, press the EXE key again.

The file is copied. When the copy operation is finished, "Completed." and "HIT ANY KEY" appear.

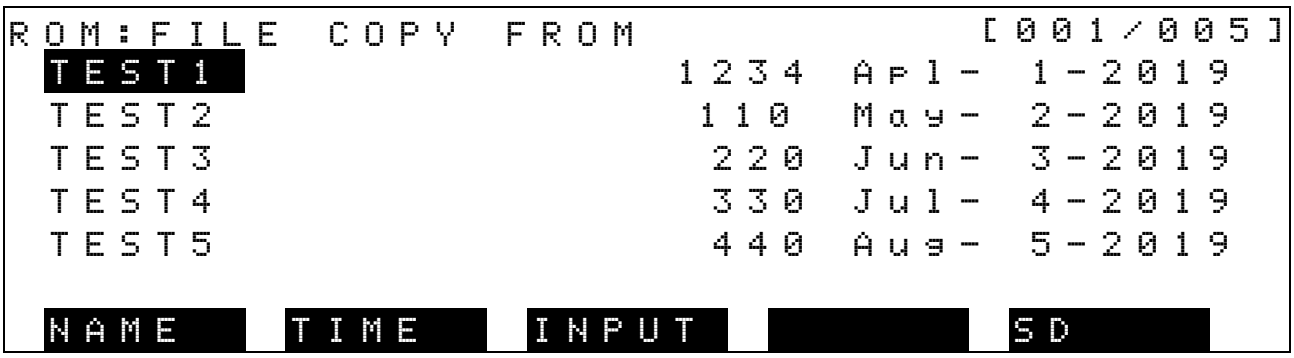


8 Press any key.

This returns to the operation file selection screen.



Operation File Selection Screen



The valid keys in the operation file selection screen and their actions are shown in the following table.

Key	Description
↑ key	This moves the cursor up by one line.
↓ key	This moves the cursor down by one line.
ALT key + ↑ key or ALT key + NEXT key	This moves the cursor to the previous page.
ALT key + ↓ key or NEXT key	This moves the cursor to the next page.
ESC key	This returns to the previous screen.
NAME (F1) key	This is used to sort files by name.
TIME (F2) key	This is used to sort files by date and time created.
INPUT/NEW (F3) key	The ":" is displayed in the command line to indicate that the command line can be edited.
SD/ROM (F5) key	This switches the displayed drive.

The function menus in the file list screen and their actions are shown in the following table.

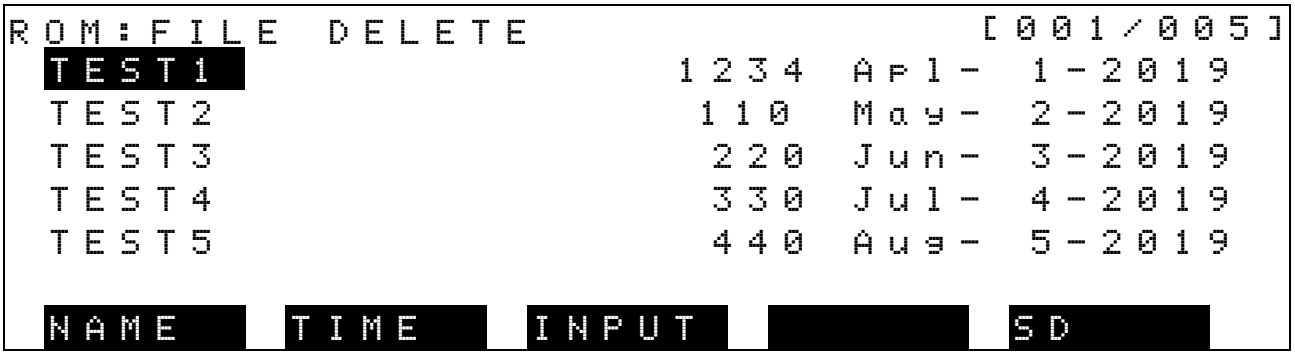
Displayed menu name	Description	Section number for detailed explanation
NAME	This is used to sort files by name.	
TIME	This is used to sort files by date and time created.	
INPUT	The ":" is displayed in the command line to indicate that the command line can be edited.	
NEW	The ":" is displayed in the command line to indicate that the command line can be edited.	
ROM	This switches the displayed drive to ROM.	3.4.2
SD	This switches the displayed drive to SD.	3.4.2

3.4.4 Deleting Files (DEL)

<File Delete Procedure>

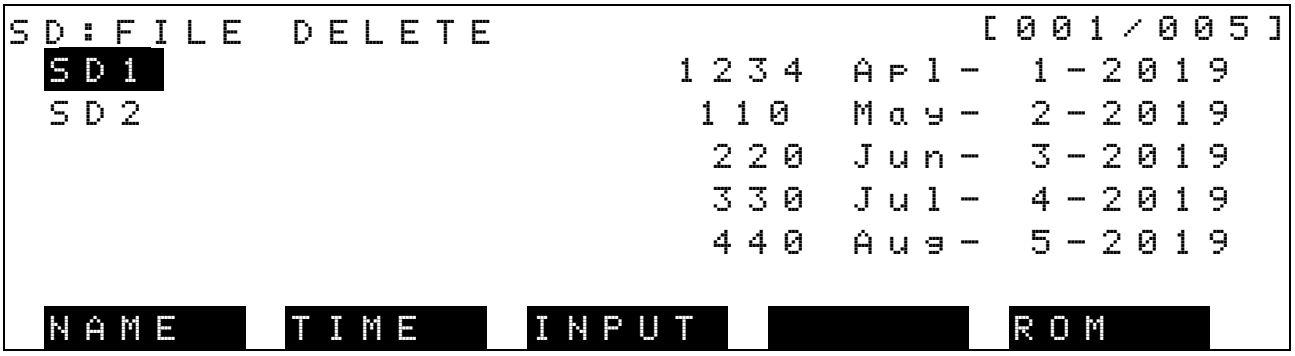
1 Press the DEL (F3) key in the file operation screen.

The operation file selection screen is opened.



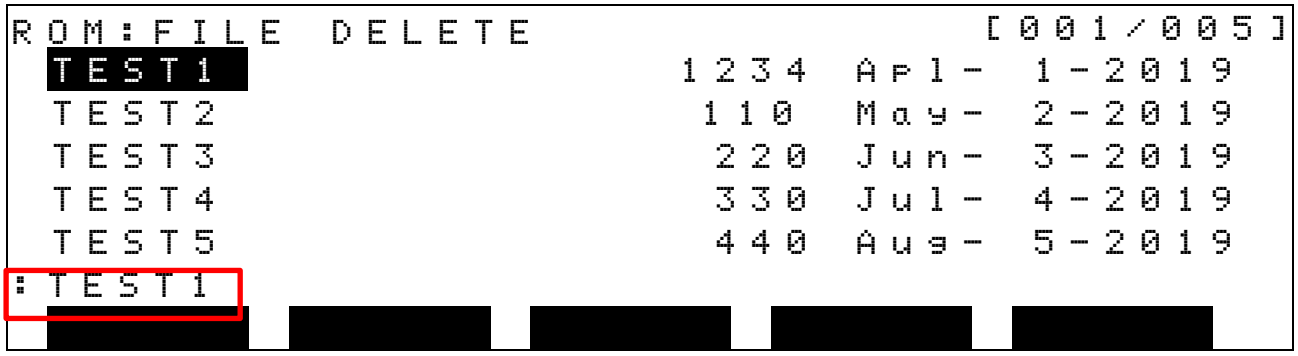
2 To delete a file in the SD card, press the SD (F5) key.

The files in the SD card are displayed in the operation file selection screen.



3 Move the cursor to the location of the file to be deleted, and press the EXE key.

The file name appears in the command line.

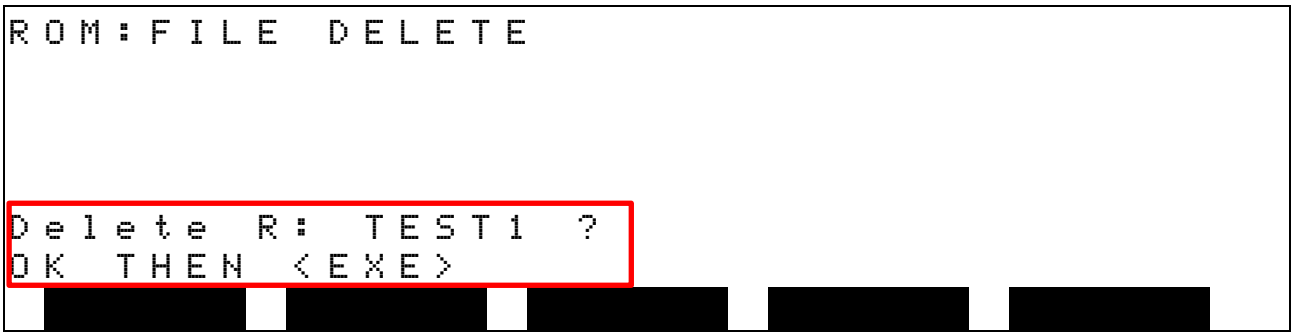


MEMO

The same operation can be performed by pressing the INPUT (F3) key and entering the file name to be copied to the command line.

4 Press the EXE key.

A confirmation message appears.



5 Press the EXE key.

The file is deleted. When the copy operation is finished, "Completed." and "HIT ANY KEY" appear.

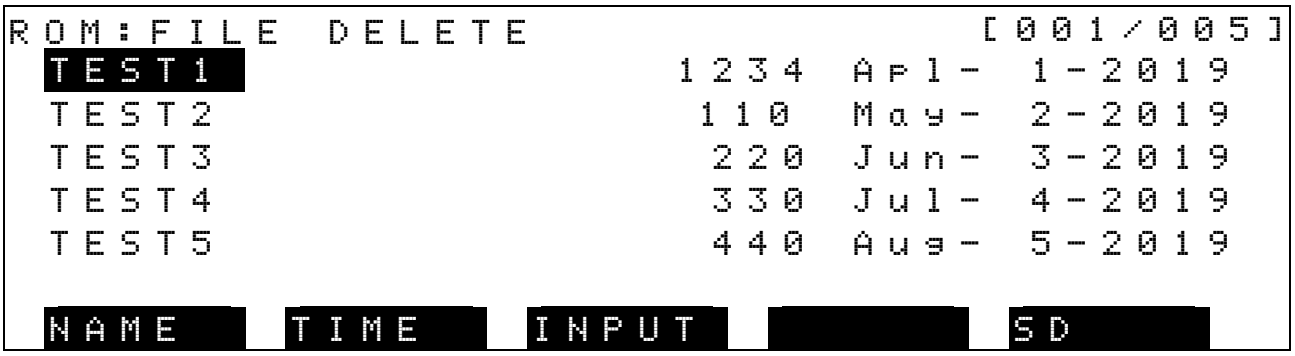


MEMO

*If the selected file is registered to an execution program, "*** File already selected" (where *** is the file name) is displayed, and the file is not deleted.*

6 Press any key.

This returns to the operation file selection screen.

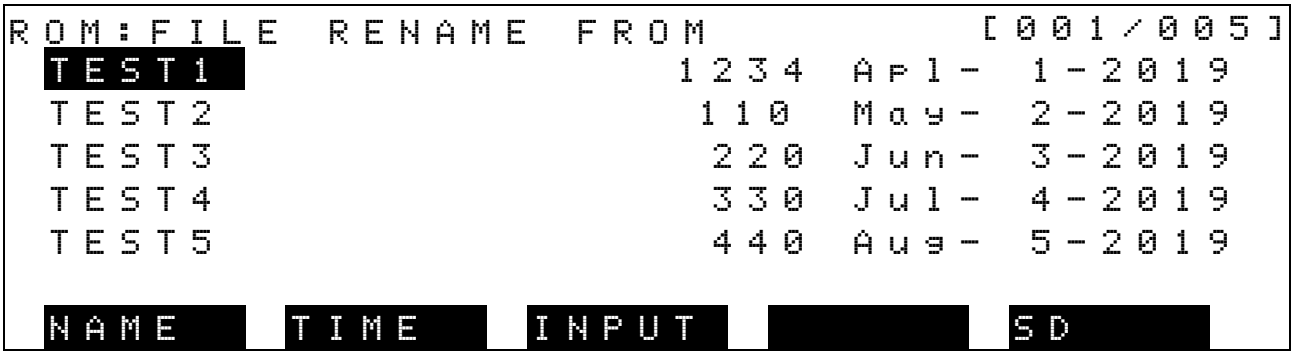


3.4.5 Renaming Files (REN)

<File Rename Procedure>

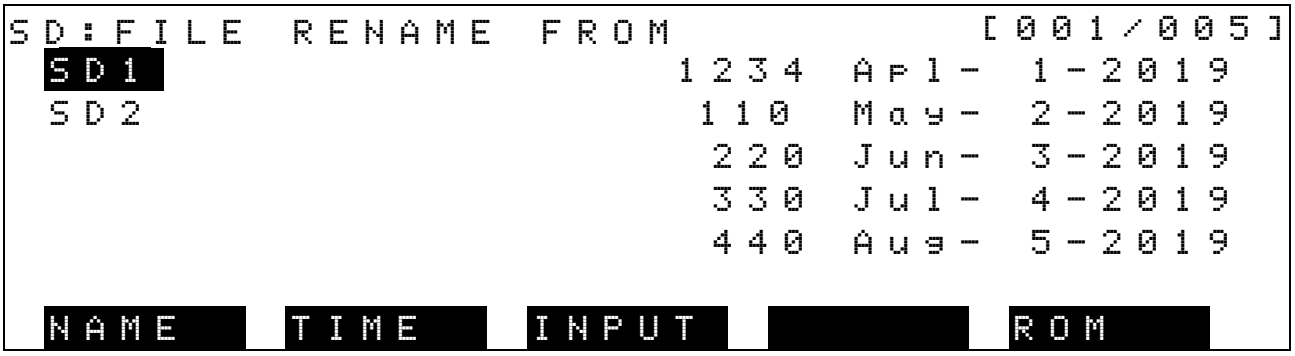
1 Press the REN (F4) key in the file operation screen.

The operation file selection screen is opened.



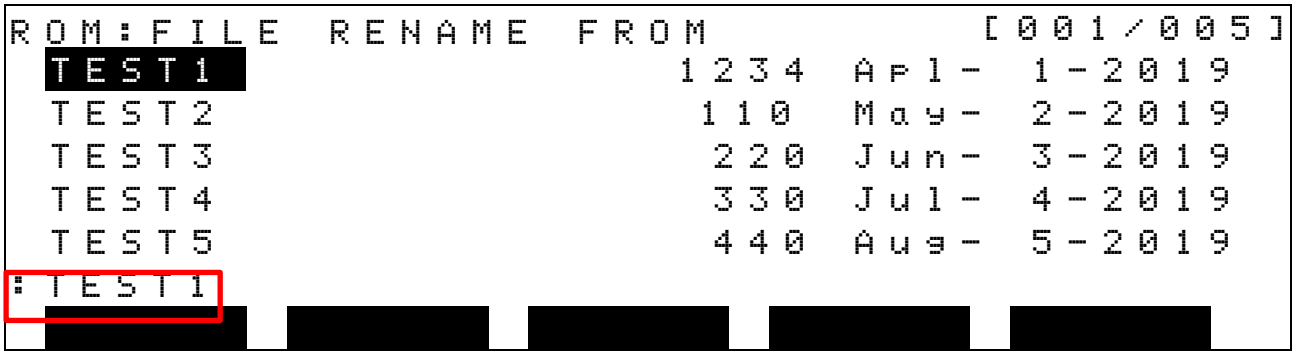
2 To rename a file in the SD card, press the SD (F5) key.

The files in the SD card are displayed in the operation file selection screen.



3 Move the cursor to the location of the file to be renamed, and press the EXE key.

The file name appears in the command line.

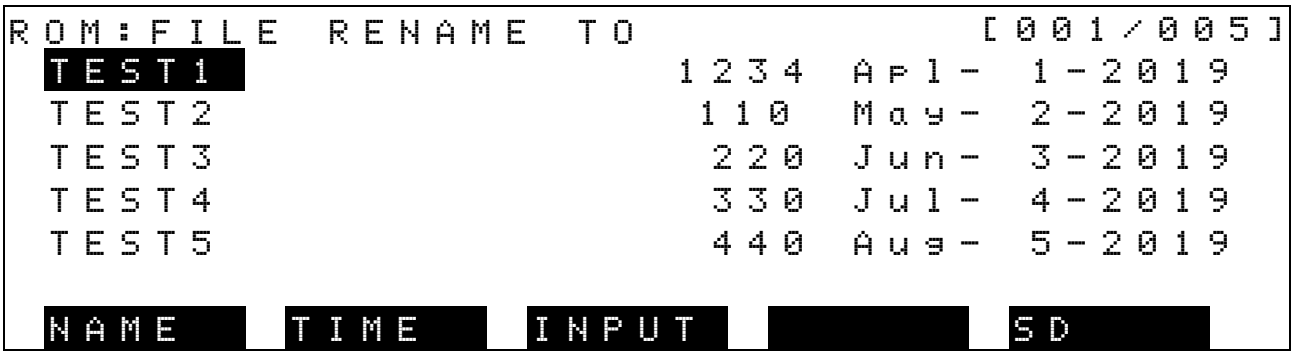


MEMO

The same operation can be performed by pressing the INPUT (F3) key and entering the file name to be copied to the command line.

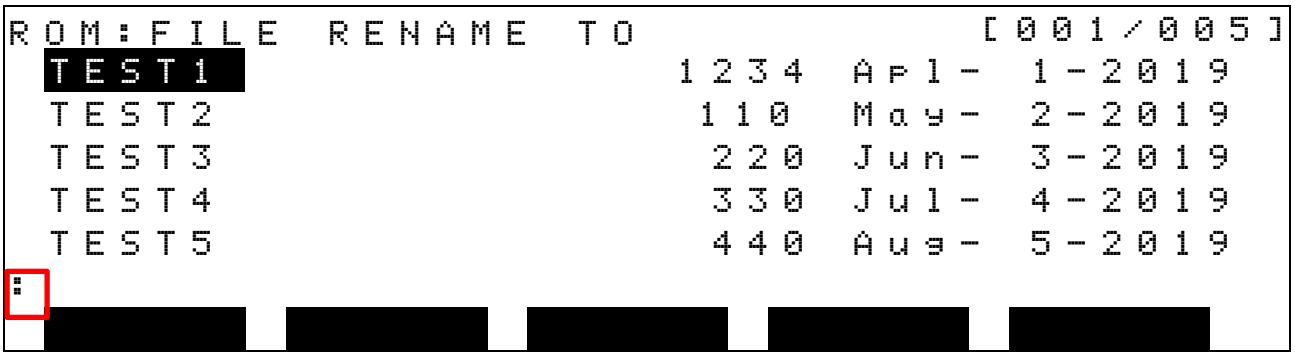
4 Press the EXE key.

The operation file selection screen is opened again.



5 Press the INPUT (F3) key.

The symbol ":" is displayed in the command line, and the content can be edited.



MEMO

When the cursor is aligned with a selected file name and the EXE key is pressed, the file name appears in the command line.

6 Enter the new name for the file, and click [EXE].

A confirmation message appears.

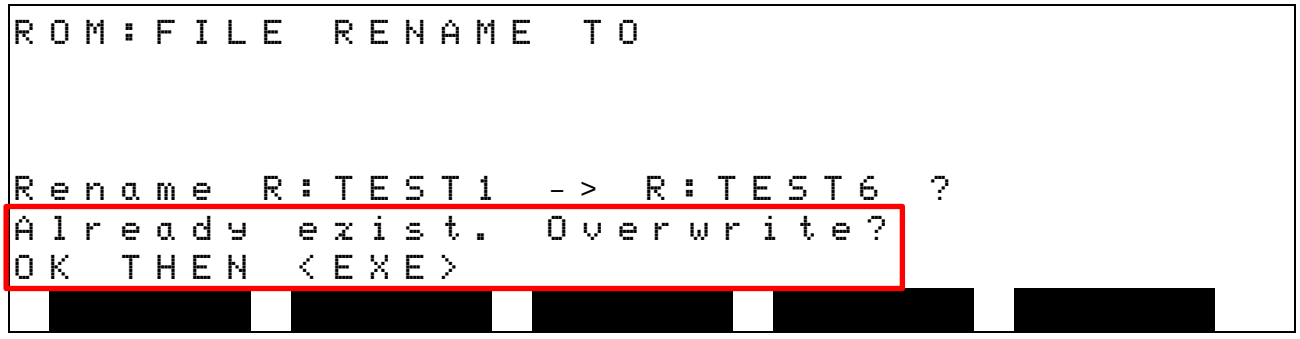


7 Press the EXE key.

If a file with the same name as the renamed file does not exist, the file rename operation is performed. When the rename operation is finished, "Completed." and "HIT ANY KEY" appear.

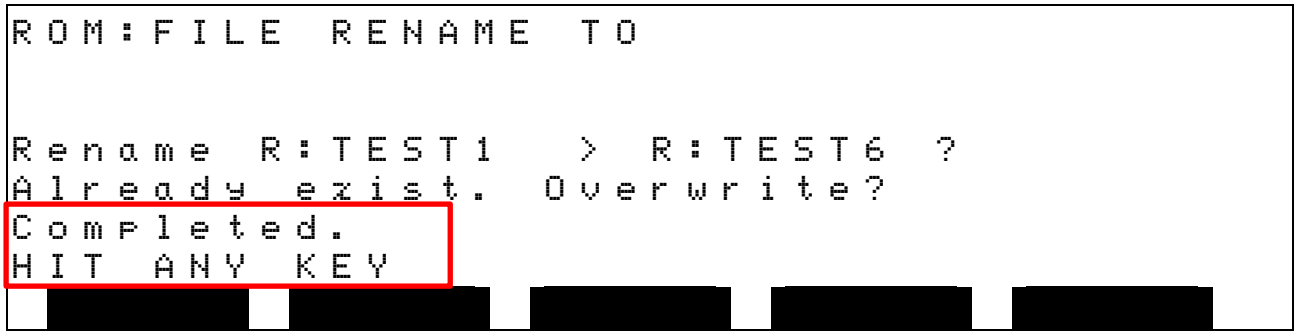


If a file with the same name as the renamed file already exists, a confirmation message appears asking whether it is OK to overwrite the file.



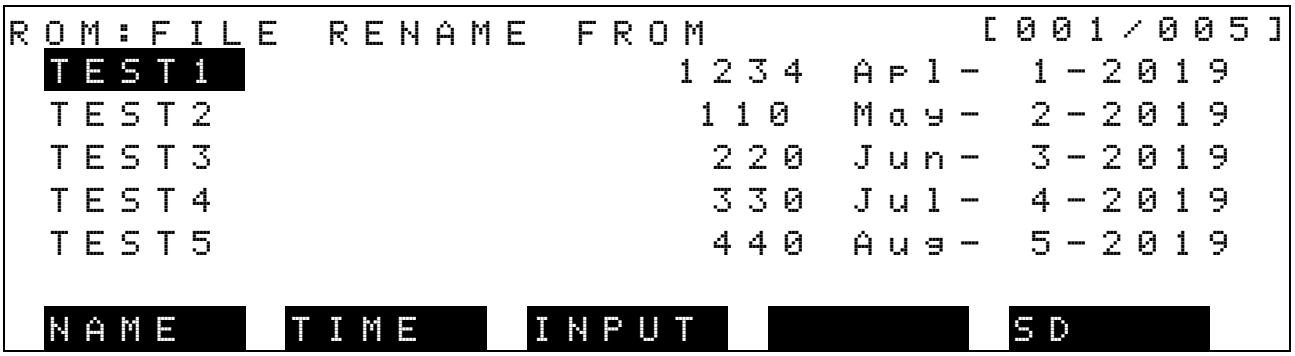
8 If a file with the same name as the renamed file already exists, press the EXE key again.

The original file is deleted, and the file is renamed. When the rename operation is finished, "Completed." and "HIT ANY KEY" appear.



9 Press any key.

This returns to the operation file selection screen.



4. Operations in AUTO Mode

This chapter describes the operations performed in AUTO mode.
AUTO mode is used when operating the robot.
During AUTO mode, the TP1000 can be used to perform the "Override Setting", "Execution Program Monitor" and "Global Variable Display and Overwriting" operations only.
During AUTO mode, operations where the robot moves, such as servo ON or program execution, and file operations cannot be performed from the TP1000.

> *MEMO*
> *During AUTO mode, operations where the robot moves, such as servo ON or program execution, can be performed by turning communication commands and system signals (interface relay) on and off.*
> *For details on the communication commands, see the separate "STA1164 Communication Manual".*
> *For details on system signals (interface relay), see the separate "STA1167 Interface Manual"and "STA1168 TCmini (Internal PLC) Manual".*

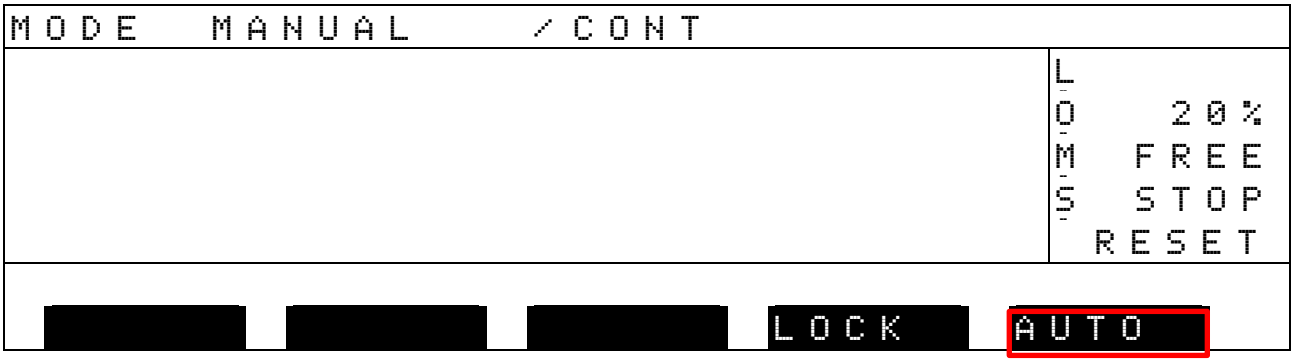
Use the following procedure to switch from MANUAL mode to AUTO mode.

4.1 Master Mode Switching

4.1.1 Master Mode Switching

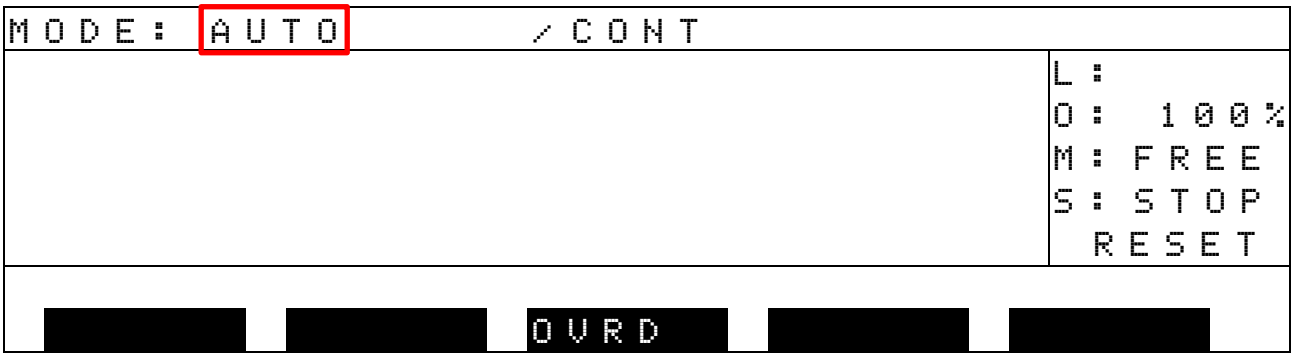
<Switching to AUTO Mode>

1 Press the NEXT key until "AUTO" appears in the function menu on the basic screen.



2 Press the AUTO (F5) key.

The controller master mode is switched, and the mode indicator at the top of the basic screen appears as "AUTO".



MEMO

Once master mode switching lock is enabled, the message "Master Mode is Locked" is displayed. If necessary, disable the master mode switching lock by referring to the next section, "Master Mode Switching Lock".

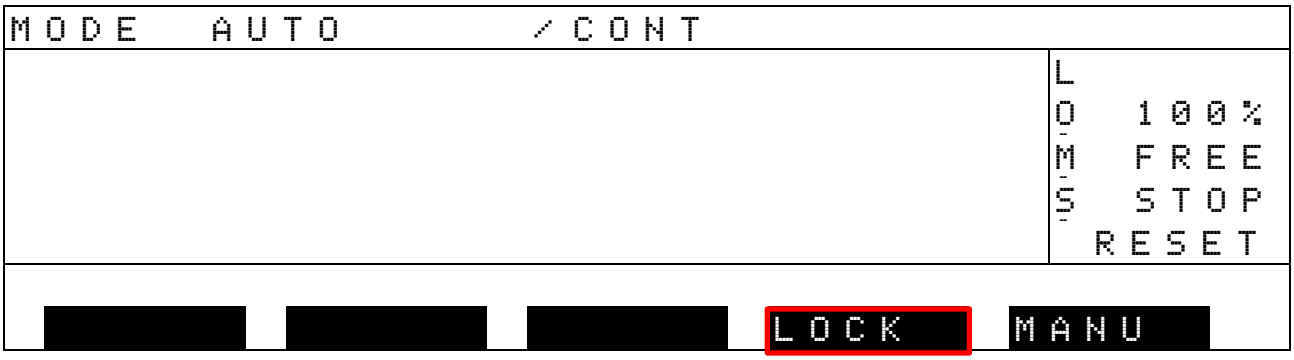
4.1.2 Master Mode Switching Lock

Master mode switching lock can be enabled to prevent switching of the master mode by a third party to a mode unintended by the operator. This section describes how to enable and disable the switching lock for master mode.

> *MEMO*
>
> *Master mode switching lock is disabled when the controller power is turned off.*

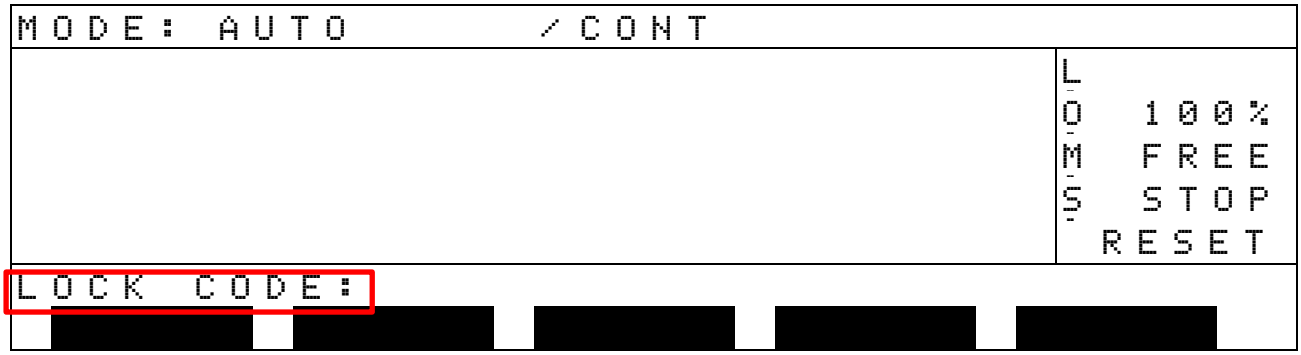
<Master Mode Switching Lock Enable Procedure>

1 Press the NEXT key until "LOCK" appears in the function menu on the basic screen.



2 Press the LOCK (F4) key.

The command line changes as shown below.



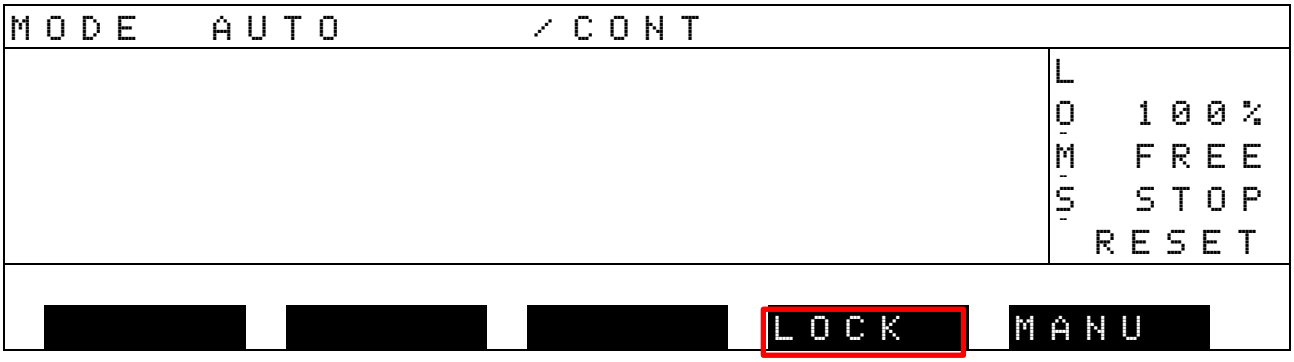
3 Enter the desired lock code, and press the EXE key.

The master mode switching lock is enabled.

> *MEMO*
>
> *Any character string from 1 to 4 characters can be set as the lock code.*

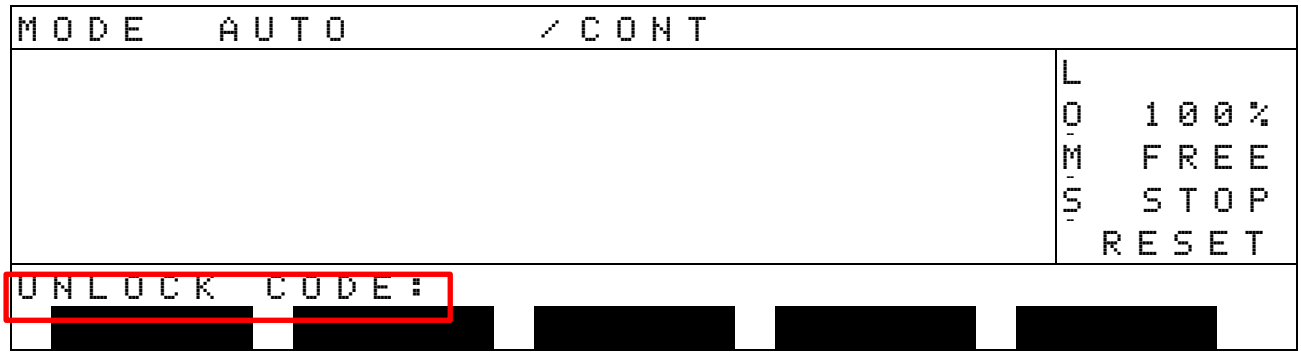
<Master Mode Switching Lock Disable Procedure>

1 Press the NEXT key until "LOCK" appears in the function menu on the basic screen.



2 Press the "LOCK (F4)" key.

The command line changes as shown below.



3 Enter the lock code that was entered when setting, and press the EXE key.

The master mode switching lock is disabled.

> *MEMO*
>
> *If the entered lock code is incorrect, the message "Lock code is missmatch" is displayed.*

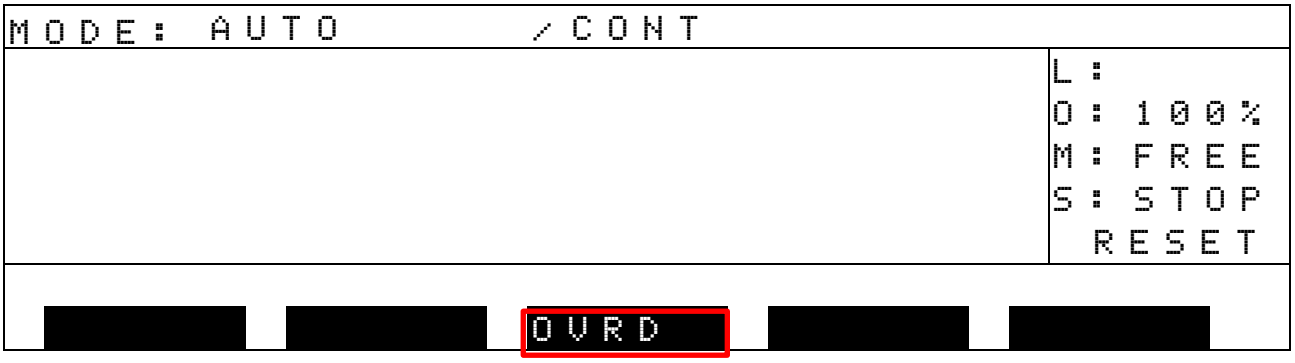
4.2 Override Setting

This describes how to set the override.

> *MEMO*
>
> *• The override can be set even while a program is running.*

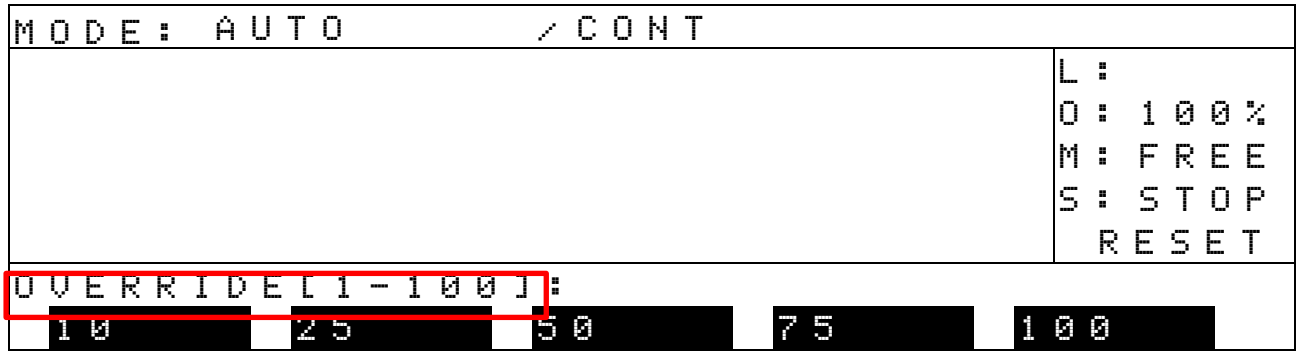
<Override Setting Procedure>

1 Press the NEXT key until "OVRD" appears in the function menu on the basic screen.



2 Press the OVRD (F3) key.

"OVERRIDE[1-100]:" is displayed in the command line, and this switches to the function menu.



3 Enter the value to be set for the override into the command line, and press the EXE key.

A check is conducted that the entered value is within the proper range, and if it is in the range, the override is overwritten by the entered value.

If the value is outside the proper range, "CAN NOT EXECUTE" appears.

MEMO

• When the function key is pressed when entering to the command line, the number appearing on the function key is entered to the command line.

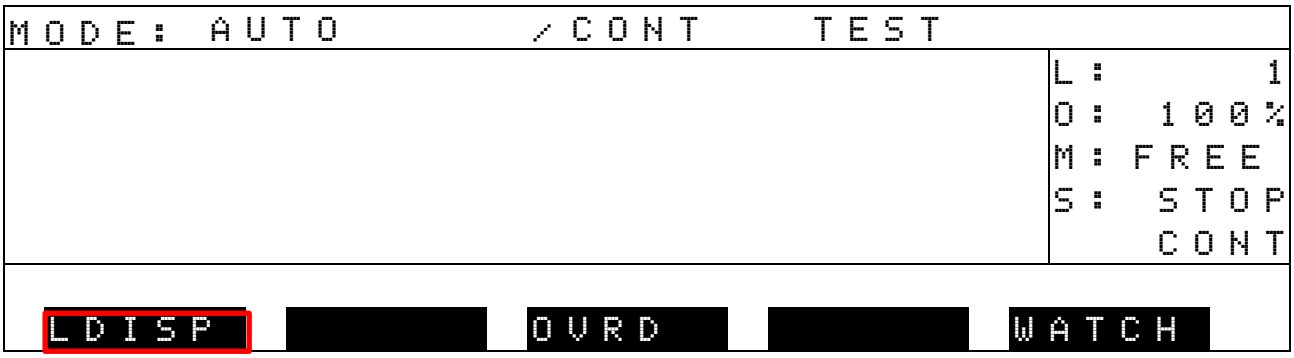
4.3 Monitoring of Execution Programs

This describes how to monitor the content of the execution program.

The content of the execution program is displayed on the LDISP screen.

<LDISP Screen Opening Procedure>

1 Press the NEXT key until "LDISP" appears in the function menu on the basic screen.

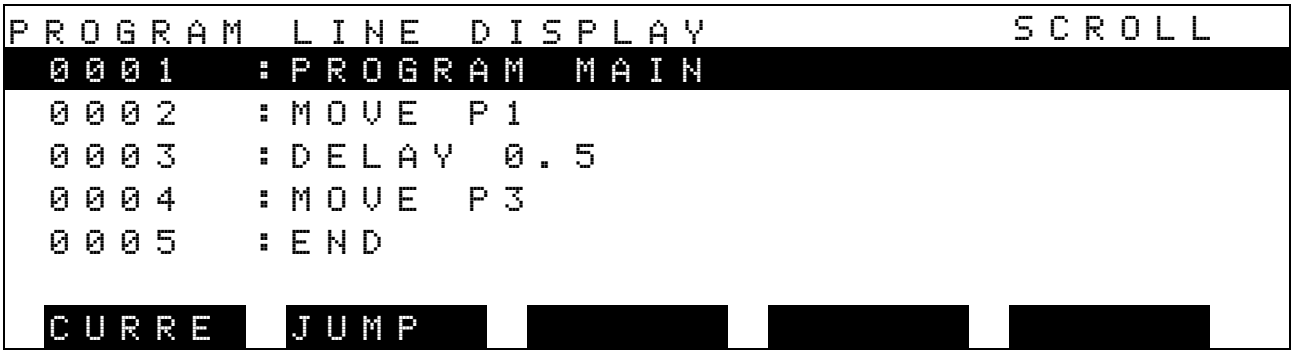


MEMO

"LDISP" does not appear in the function menu unless the execution program is selected.

2 Press the LDISP (F1) key.

The LDISP screen is opened.



The execution program content can be displayed on the LDISP screen using one of two methods.

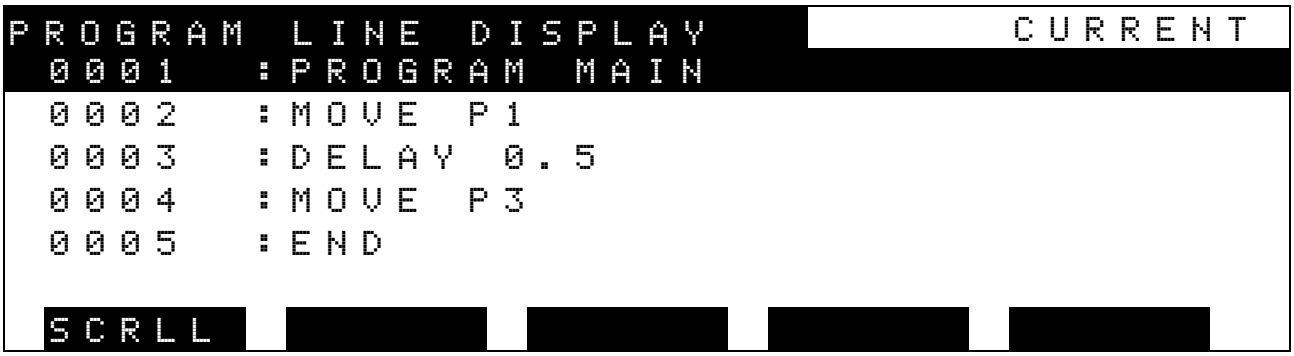
- Displaying in synchronization with the currently-displayed line (Synchronous Display (CURRENT))
- Displaying a selected line (Asynchronous Display (SCROLL))

MEMO

• You can switch between synchronous display and asynchronous display by pressing the SCRLL/CURRE (F1) key.

Execution Program Synchronous Display Screen

In synchronous display, the screen is scrolled based on the line currently being executed.
The currently-displayed execution line appears highlighted as shown in the figure below.



The valid keys in the execution program synchronous display screen and their actions are shown in the following table.

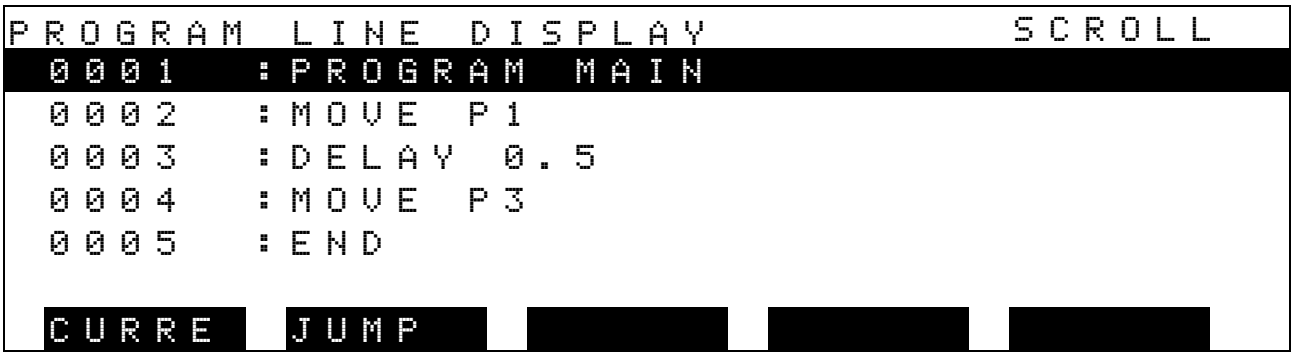
Key	Description
ESC key	This returns to the basic screen.

The function menus in the execution program synchronous display screen and their actions are shown in the following table.

Displayed menu name	Description	Section number for detailed explanation
SCRLL	This opens the asynchronous display screen.	This chapter
BREAK	This aborts execution of a program.	4.5.2
STOP	This stops execution of the program.	4.5.1

Execution Program Asynchronous Display Screen

In asynchronous display, the cursor can be moved and the screen can be scrolled.
The line where the cursor is located appears highlighted as shown in the figure below.



The valid keys in the execution program asynchronous display screen and their actions are shown in the following table.

Key	Description
↑ key	This moves the cursor up by one line.
↓ key	This moves the cursor down by one line.
ESC key	This returns to the basic screen.

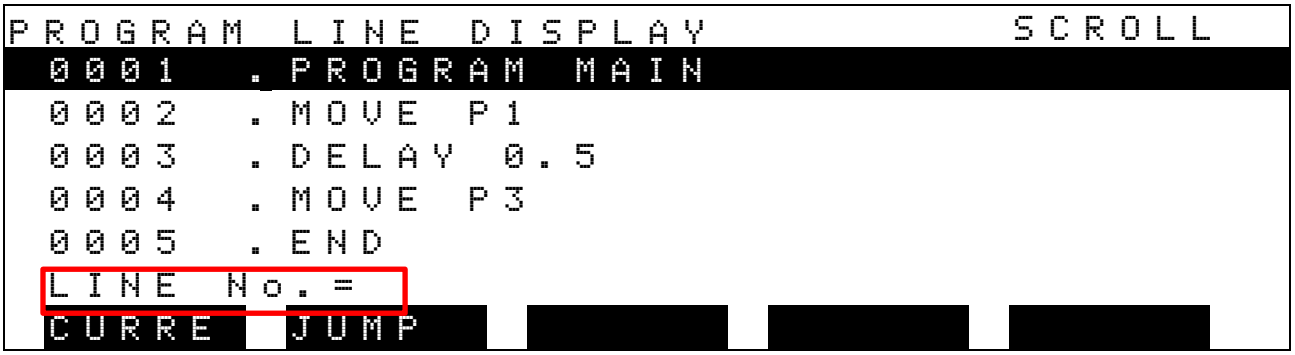
The function menus in the execution program asynchronous display screen and their actions are shown in the following table.

Displayed menu name	Description	Section number for detailed explanation
SCRLL	This opens the asynchronous display screen.	This chapter
JUMP	This jumps to a specified line.	This chapter
BREAK	This aborts execution of a program.	4.5.2
STOP	This stops execution of the program.	4.5.1

<Jumping to a Selected Line>

1 Press the JUMP (F2) key.

"LINE NO.=" appears in the command line.



2 Enter the line number to be displayed in the command line, and press the EXE key.

The cursor moves to the entered line.

MEMO

If a non-numerical value is entered, this moves to the first line.

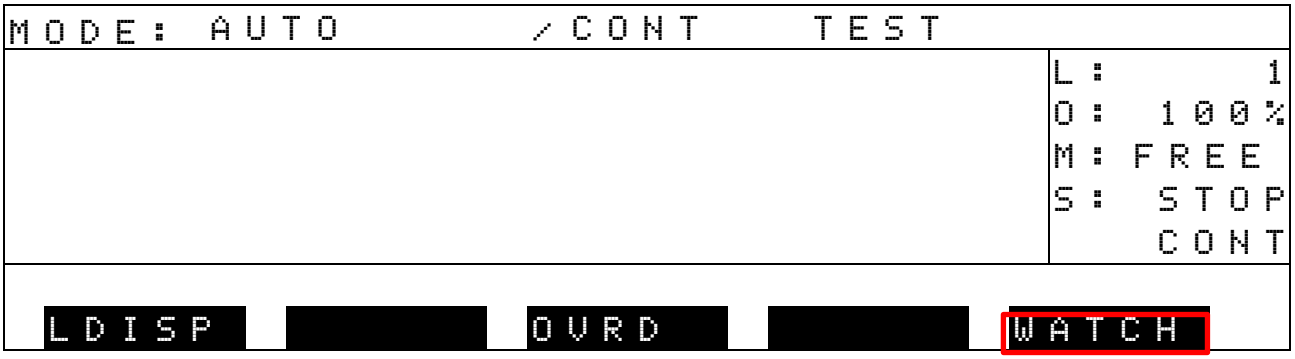
4.4 Displaying and Overwriting Global Variables (WATCH)

This describes how to display and overwrite the global variables in the execution program.

A global variable is a variable defined in the area enclosed by GLOBAL-END or DATA-END in a program. Global variables can be displayed and overwritten on the global variable monitor screen.

<Global Variable Monitor Screen Opening Procedure>

1 Press the NEXT key until "WATCH" appears in the function menu on the basic screen.



MEMO

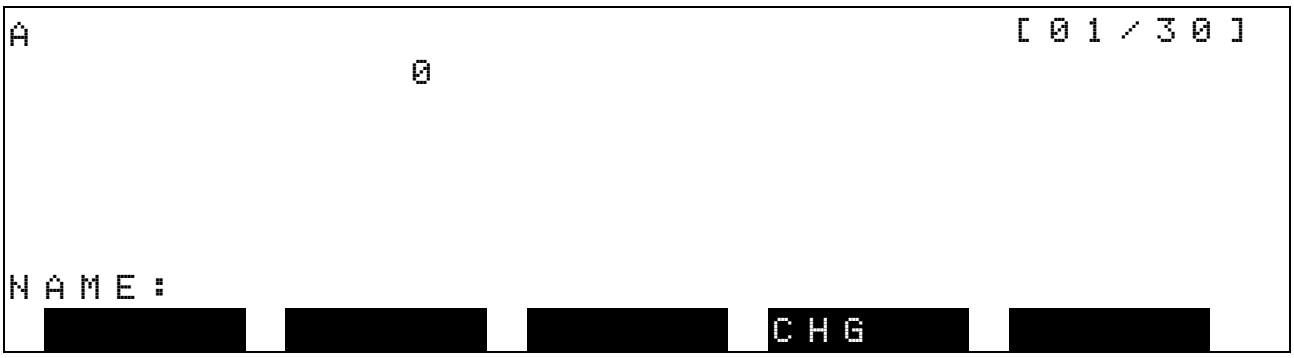
"WATCH" does not appear in the function menu unless the execution program is selected.

2 Press the WATCH (F5) key.

The global variable monitor screen is opened.



Global Variable Monitor Screen



The valid keys in the global variable monitor screen and their actions are shown in the following table.

Key	Description
ESC key	This returns to the basic screen.
↑ key	This displays the previous variable in the watchlist.
↓ key	This displays the next variable in the watchlist.
ALT key + ↑ key or NEXT key	This displays the variable that is five variables before the current variable in the watchlist.
ALT key + ↓ key or ALT key + NEXT key	This displays the variable that is five variables after the current variable in the watchlist.
ALT key + BS key	This removes the displayed variable from the watchlist.

The function menus in the global variable monitor screen and their actions are shown in the following table.

Displayed menu name	Description	Section number for detailed explanation
CHG	This overwrites the value of the displayed variable.	2.2.10

<Registering Variables to Watchlist>

1 Use the character input keys to enter the name of the variable to be registered.

For array variables, include the element number when entering.

> *MEMO*
>
> *A maximum of 25 variable names can be registered.*

2 Press the EXE key.

The entered variable name is registered to the watchlist, and the current value appears on the WATCH screen.

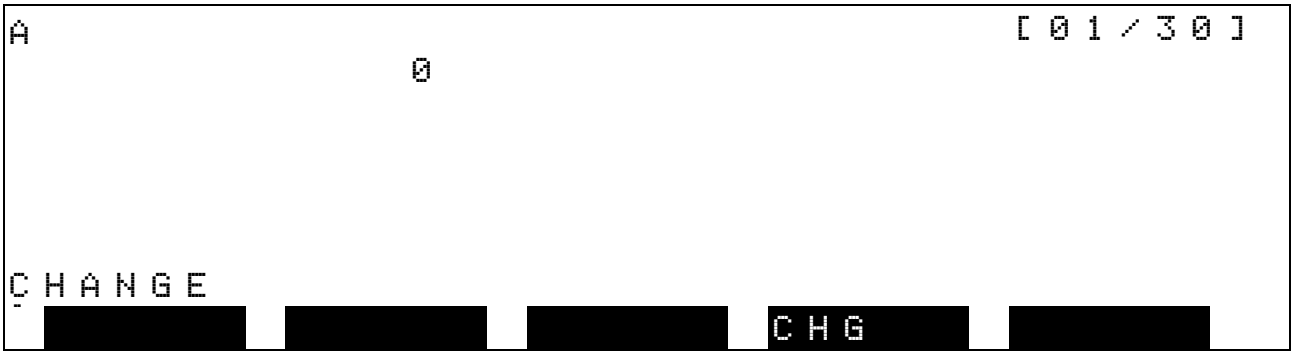
> *MEMO*
>
> *• A maximum of 30 variables can be registered to the watchlist.*
>
> *• If there are no global variables for the entered name, "----.---""""appears.*

<Overwriting Variable Values>

1 This shows the variable whose value will be overwritten.

2 Press the CHG (F4) key.

"CHANGE:" appears in the command line.



3 Move the cursor to the element that will be overwritten, and use the character input keys to enter the value.

4 Press the EXE key.

The value of the displayed variable is overwritten by the value in the command line.

MEMO

- *To prevent the robot from running out of control due to mistaken input, for position-type and coordinate-type variables, overwriting is not allowed to element values that exceed ±3 mm (or deg) from their current value.*
- *The overwritten value is a temporary change, and the value is reset by performing a program reset. To permanently apply the overwritten value, consider either editing the file or using a memory variable.*
  For details on memory variables, see the "MEMORY" section in the Language Manual.

⚠ DANGER



MANDATORY

- Before overwriting a target global variable, confirm the effect that the variable will have on the program.
  If a global variable is overwritten, the robot could perform an unexpected motion, and a collision could occur.

4.5 Program Execution Stopping and Aborting

This describes how to stop program execution.

Program execution can be stopped by a "stop"or "abort" operation.

Each procedure is described in detail below.

Stop: Stops program execution once execution of the current command is completed.

Abort: Stops program execution immediately. In this state, the robot decelerates to a stop.

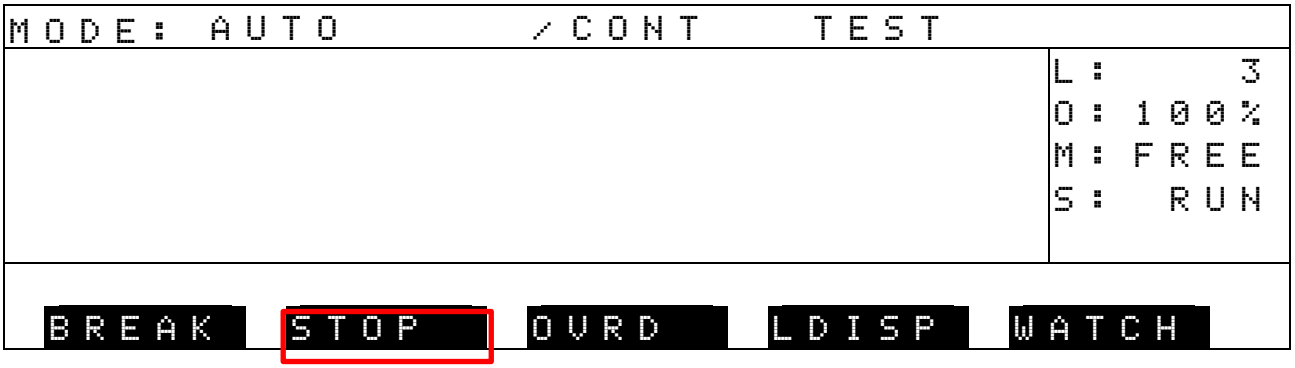
> *MEMO*
>
> *During AUTO mode, program execution cannot be resumed from the TP1000.*
>
> *To resume the program, use a communication command or system signal.*
>
> *For details on the communication commands, see the separate "STA1164 Communication Manual".*
>
> *For details on system signals (interface relay), see the separate "STA1167 Interface Manual"and "STA1168 TCmini (Internal PLC) Manual".*

4.5.1 Program Execution Stop (STOP)

This describes how to stop program execution.

<Program Execution Stop Procedure>

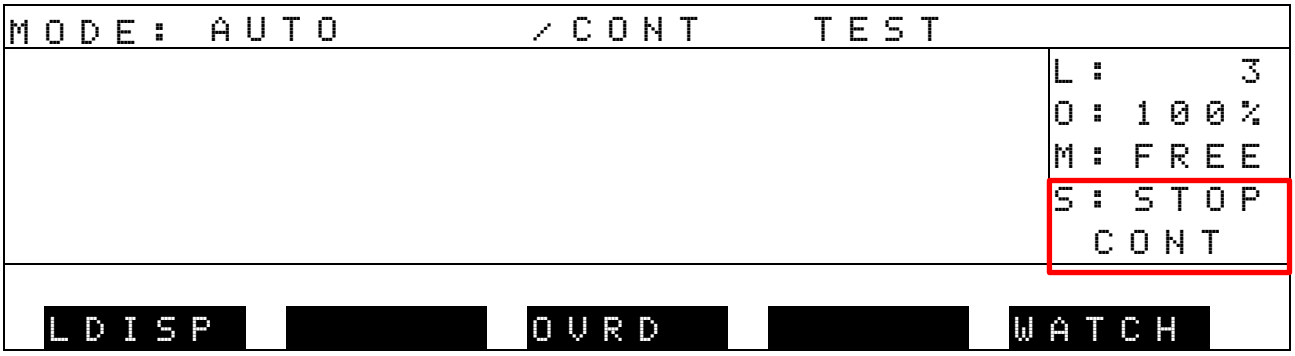
1 Press the NEXT key until "STOP" appears in the function menu on the basic screen.



2 Press the STOP (F2) key.

This stops execution of the program.

When program execution is stopped, "STOP CONT" appears in the status indicator.



MEMO

Program execution can be stopped by performing the following operations.

• Setting the STOP signal to OFF during program execution

(For details, refer to the "STA1167 Interface Manual".)

• Sending the communication command "SOP1" to the HOST port of the controller

(For details, refer to the "STA1164 Communication Manual".)

• Executing the STOP command in SCOL2 language

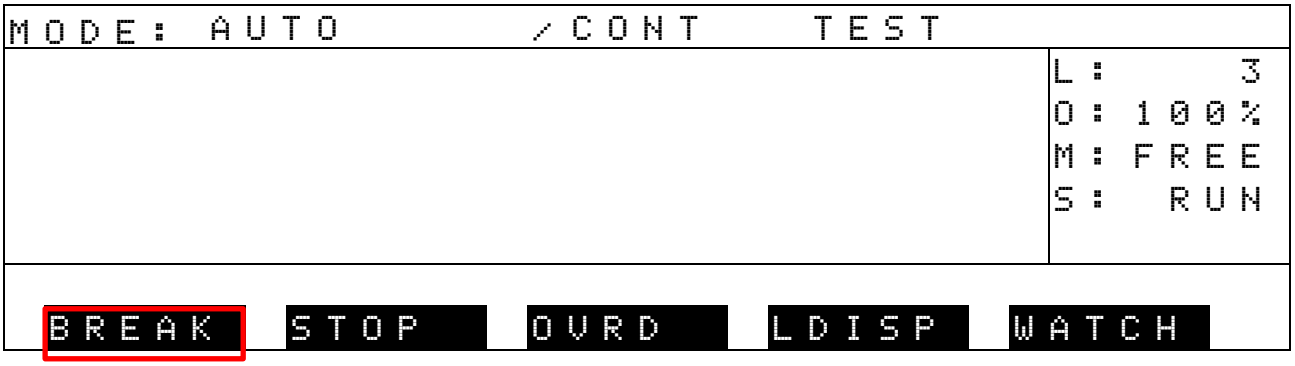
(For details, refer to "STA1163 Robot Language Manual (SCOL2)".)

4.5.2 Program Execution Abort (BREAK)

This describes how to abort program execution.

<Program Execution Abort Procedure>

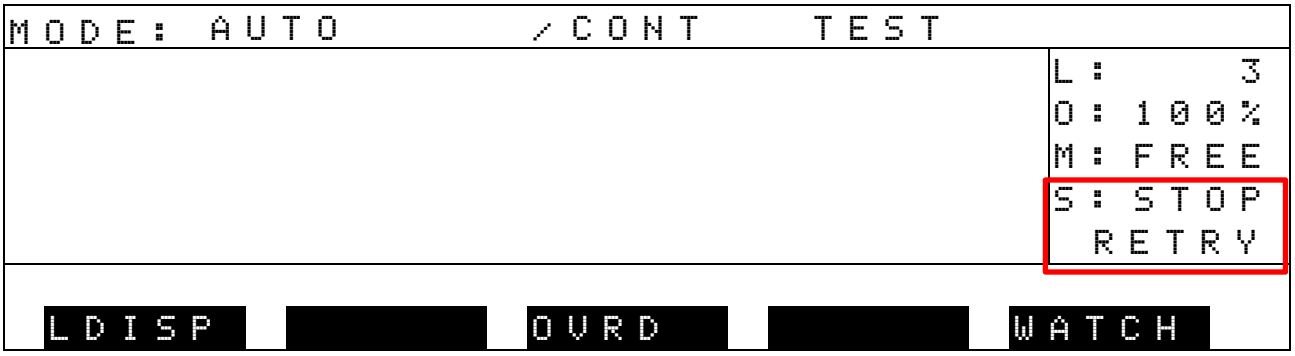
1 Press the NEXT key until "BREAK" appears in the function menu on the basic screen.



2 Press the BREAK (F1) key.

This aborts execution of a program.

When program execution is aborted, "STOP RETRY" appears in the status indicator.



MEMO

After aborting, when program execution is performed, program execution is performed again from the line where execution was aborted.

MEMO

In addition to the above operation, the following operations can be used to abort program execution.

• Setting the BREAK signal to OFF during program execution

(For details, refer to the "STA1167 Interface Manual".)

• Sending the communication command "BRK1" to the HOST port of the controller

(For details, refer to the "STA1164 Communication Manual".)

4.6 Program Execution Feedhold (FEEDHOLD)

This describes how to apply or remove a feedhold on program execution.

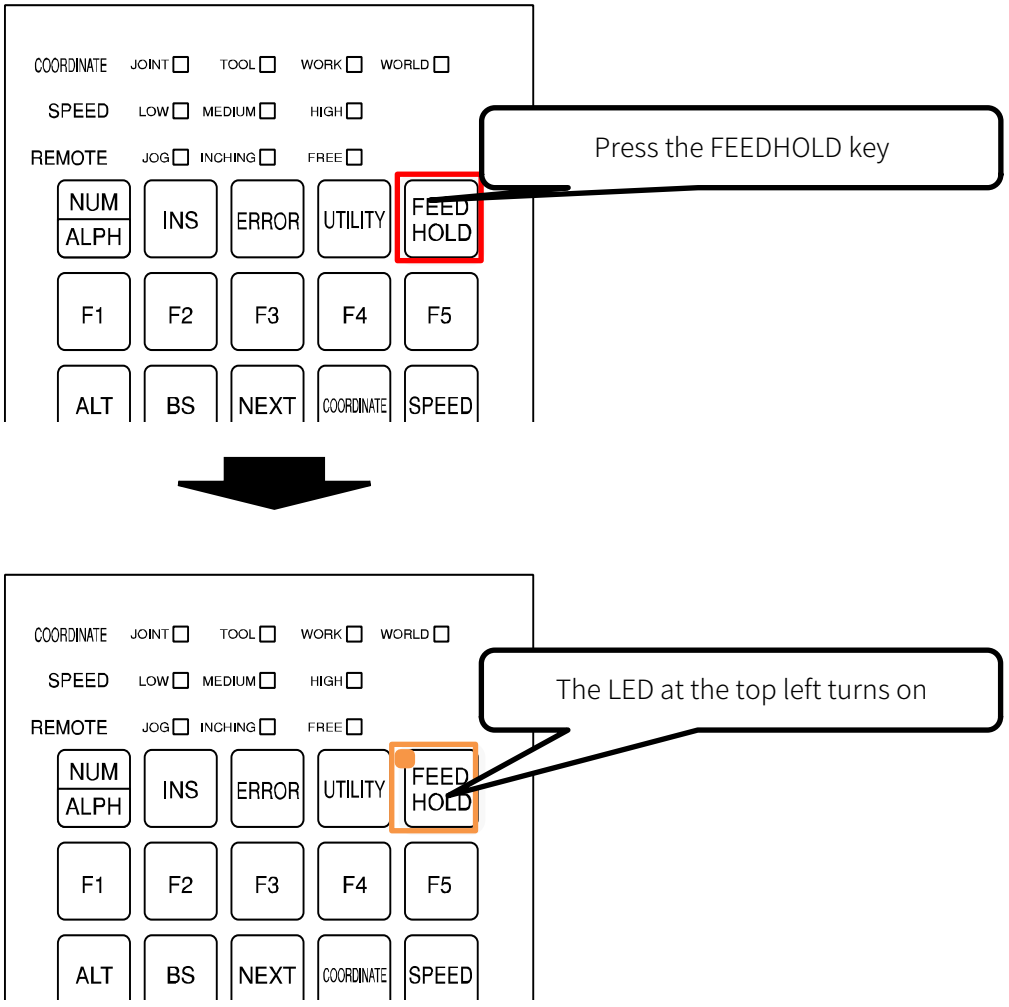
<Applying a Feedhold to Program Execution>

1 Press the FEEDHOLD key on the key sheet.

A feedhold is applied to program execution.

The LED at the top left of the FEEDHOLD key on the key sheet turns on.

While a feedhold is applied to program execution, "RUN" appears in the status indicator.



MEMO

Program execution can still be stopped or aborted during FEEDHOLD.

If this happens, then program execution is immediately stopped.

MEMO

In addition to the above operations, a feedhold can also be applied by performing the following operations.

• Enabling [P006] and setting FEEDHOLD (register G12C) to OFF during program execution

(For details, refer to the "STA1166 Parameter Manual"and "STA1168 TCmini (Internal PLC) Manual".)

• Sending the communication command "FON1" to the HOST port of the controller

(For details, refer to the "STA1164 Communication Manual".)

5. Utility Screen

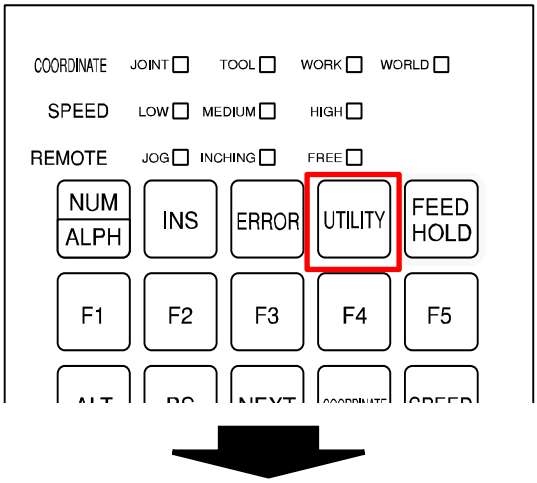
This chapter describes the utility screen.

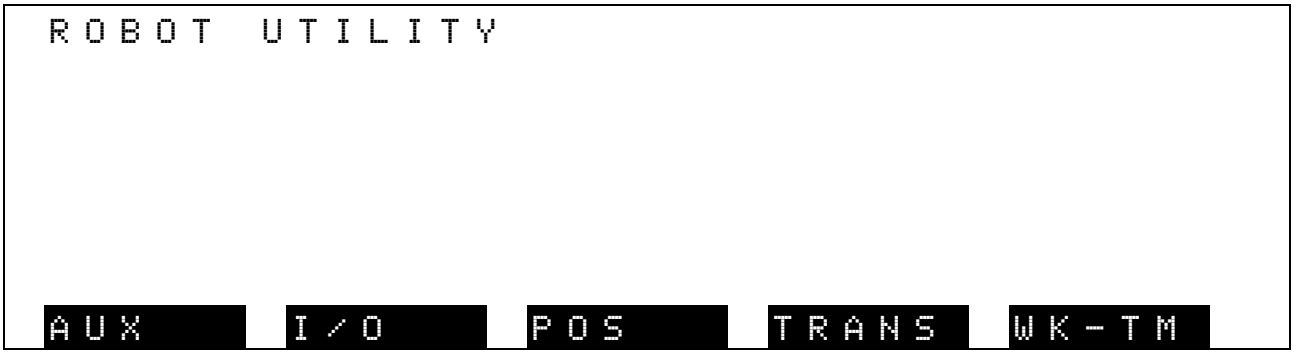
In the utility screen, external input/output signals can be displayed and used, the current position can be displayed, work/tool coordinates can be selected, and other operations can be performed.
The utility screen can be displayed in either MANUAL or AUTO modes.

<Utility Screen Opening Procedure>

1 Press the UTILITY key.

The utility menu screen is opened.

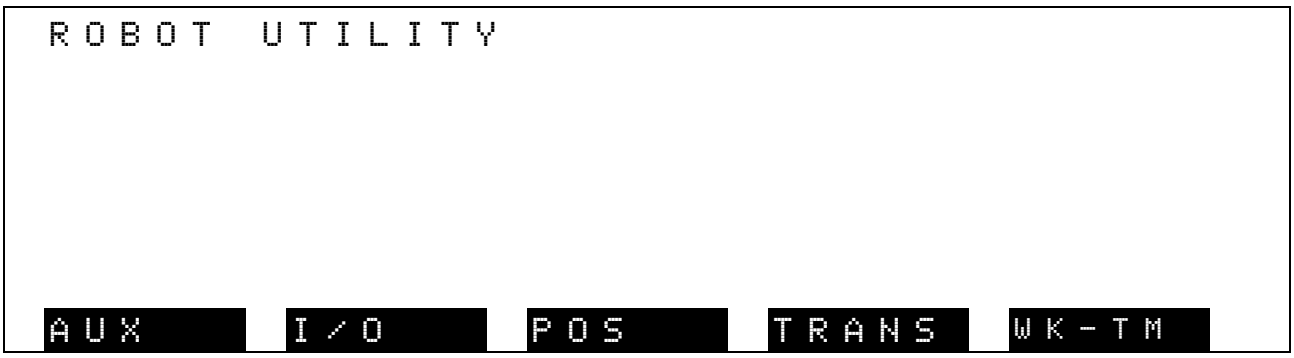




MEMO

During command line editing and in certain screens, the utility screen is not opened even when the UTILITY key is pressed.

Utility Menu Screen



The valid keys in the utility menu screen and their actions are shown in the following table.

Key	Description
ESC key	This returns to the basic screen.
NEXT key	This switches the function menu.
ALT key + NEXT key	This switches the function menu.

The function menus in the utility menu screen and their actions are shown in the following table.

Displayed menu name	Description	Section number for detailed explanation
***	This opens the corresponding screen.	5.1

5.1 Screens

The utility screens include the following screen types.

Table 5.1. Utility Screens

Screen Name	Description	Is Screen Displayable?	
		MANUAL	AUTO
AUX	This screen enables auxiliary signal output operation.	Yes	Yes
I/O	This screen shows the I/O signal status. It enables output signal ON/OFF switching operation.	Yes	Yes
POS	This screen shows the current position of the robot.	Yes	Yes
TRANS	This screen shows the coordinate-type data of the execution program.	Yes*1	Yes
WK-TM	This screen shows and sets the running time of the robot.	Yes	Yes
REORG	This sets the origin.	Yes*2	Yes*2
J-LIM	This screen enables setting of the software limit for each axis.	Yes	Yes
DATE	This screen shows and sets the date.	Yes	Yes
MOTOR	This screen shows the motor status.	Yes	Yes
ENC	This screen shows the encoder information.	Yes	Yes
PLC-M	This screen shows the TCmini status.	Yes	Yes
ZEROP	This screen enables setting of the origin.	Yes*2	Yes*2
VER	This screen shows the controller version.	Yes	Yes
ETHER	This screen shows the Ethernet communication status.	Yes	Yes
PAYLD	This screen enables setting of the payload for manual guidance.	Yes	Yes
FCTRY	This opens the factory setting screen.	Yes	Yes
BCKUP	This screen enables backing up and restoring of files in the controller.	Yes*2	Yes*2

*1 The screen cannot be opened during file editing.

*2 The screen cannot be opened in the servo ON state.

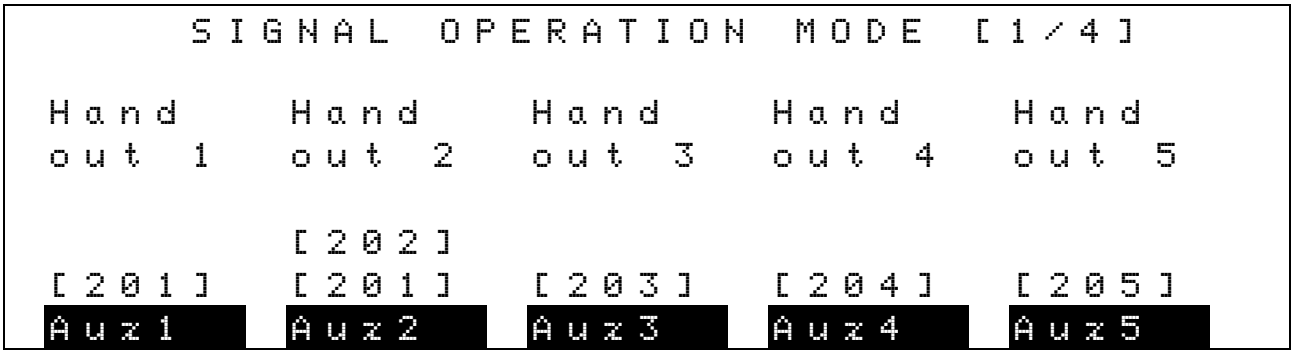
5.2 Screen Details

AUX

This screen enables display of the auxiliary signal status and on/off operation.

AUX Screen

This screen shows the status of the registered auxiliary signal.



The valid keys in the AUX screen and their actions are shown in the following table.

Key	Description
ESC key	This returns to the utility menu screen.
↑ key or ALT key + ↑ key or ALT key + NEXT key	This shows the previous page. If located on the first page, this shows the last page.
↓ key or ALT key + ↓ key or NEXT key	This shows the next page. If located on the last page, this shows the first page.

The function menus in the AUX screen and their actions are shown in the following table.

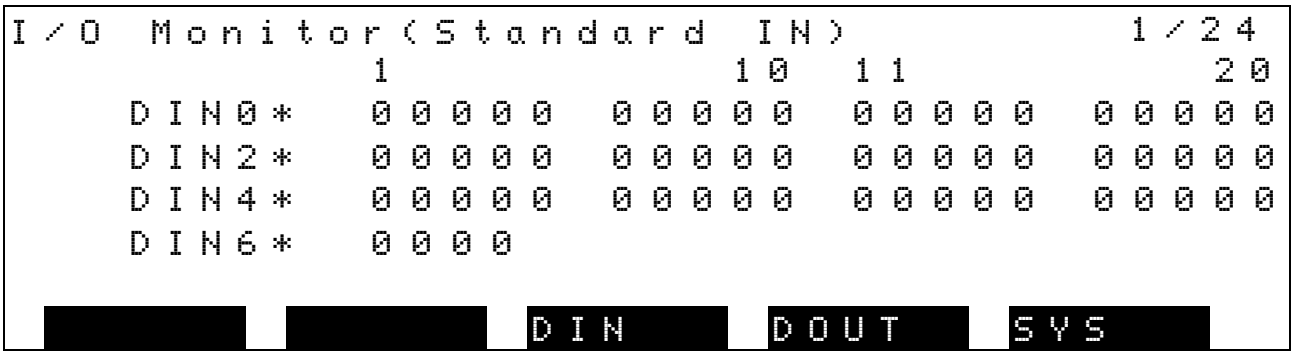
Displayed menu name	Description
AUXn	This switches the corresponding auxiliary signal on and off.

I/O

This shows the status of the external input/output signal.
This screen enables turning of the external output signal on and off.
When the I/O (F2) key is pressed in the utility menu screen, the I/O (input signal) screen is opened.
When the target signal is on, it appears as "1". When the target signal is off, it appears as "0".

I/O (DIN) Screen

This screen shows the input signal status.



The valid keys in the I/O (DIN) screen and their actions are shown in the following table.

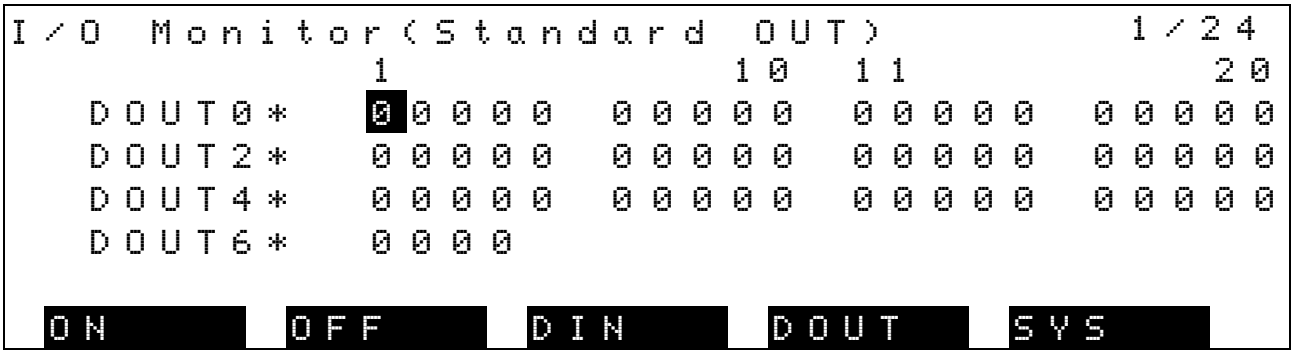
Key	Description
ESC key	This returns to the utility menu screen.
ALT key + ↑ key or ALT key + NEXT key	This shows the previous page. If located on the first page, this shows the last page.
ALT key + ↓ key or NEXT key	This shows the next page. If located on the last page, this shows the first page.

The function menus in the I/O (DIN) screen and their actions are shown in the following table.

Displayed menu name	Description
DIN	Disabled
DOUT	This switches to the I/O (DOUT) screen.
SYS	This switches to the screen that was displayed last among the I/O(HAND), I/O(SYSTEM IN), and I/O(SYSTEM OUT) screens.

I/O (DOUT) Screen

This screen shows the output signal status.



The valid keys in the I/O (DOUT) screen and their actions are shown in the following table.

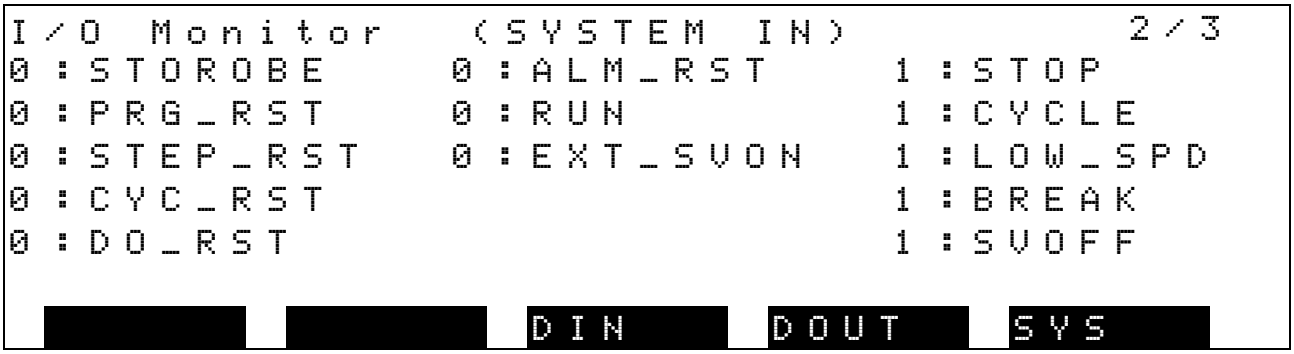
Key	Description
ESC key	This returns to the utility menu screen.
↑ key	This moves the cursor up.
↓ key	This moves the cursor down.
← key	This moves the cursor to the left.
→ key	This moves the cursor to the right.
ALT key + ↑ key or ALT key + NEXT key	This shows the previous page. If located on the first page, this shows the last page.
ALT key + ↓ key or NEXT key	This shows the next page. If located on the last page, this shows the first page.

The function menus in the I/O (DOUT) screen and their actions are shown in the following table.

Displayed menu name	Description
ON	The output signal at the cursor position is set to ON.
OFF	The output signal at the cursor position is set to OFF.
DIN	This switches to the I/O (DIN) screen.
DOUT	This switches to the I/O (DOUT) screen.
SYS	This switches to the screen that was displayed last among the I/O (HAND), I/O (SYSTEM IN), and I/O (SYSTEM OUT) screens.

I/O (SYSTEM IN) Screen

This screen shows the system input signal status.



The valid keys in the I/O (SYSTEM IN) screen and their actions are shown in the following table.

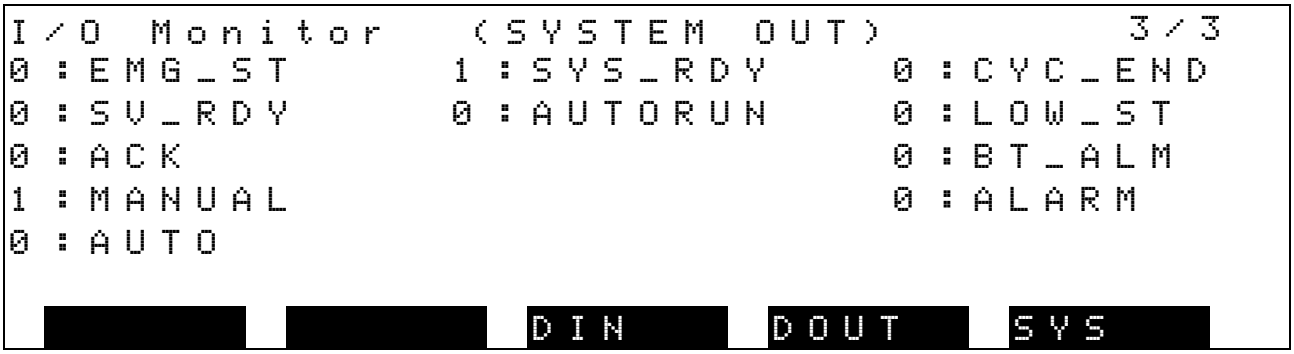
Key	Description
ESC key	This returns to the utility menu screen.
ALT key + ↑ key or ALT key + NEXT key	This switches to the I/O (HAND) screen.
ALT key + ↓ key or NEXT key	This switches to the I/O (SYSTEM OUT) screen.

The function menus in the I/O (SYSTEM IN) screen and their actions are shown in the following table.

Displayed menu name	Description
DIN	This switches to the I/O (DIN) screen.
DOUT	This switches to the I/O (DOUT) screen.
SYS	Disabled.

I/O (SYSTEM OUT) Screen

This screen shows the system output signal status.



The valid keys in the I/O (SYSTEM OUT) screen and their actions are shown in the following table.

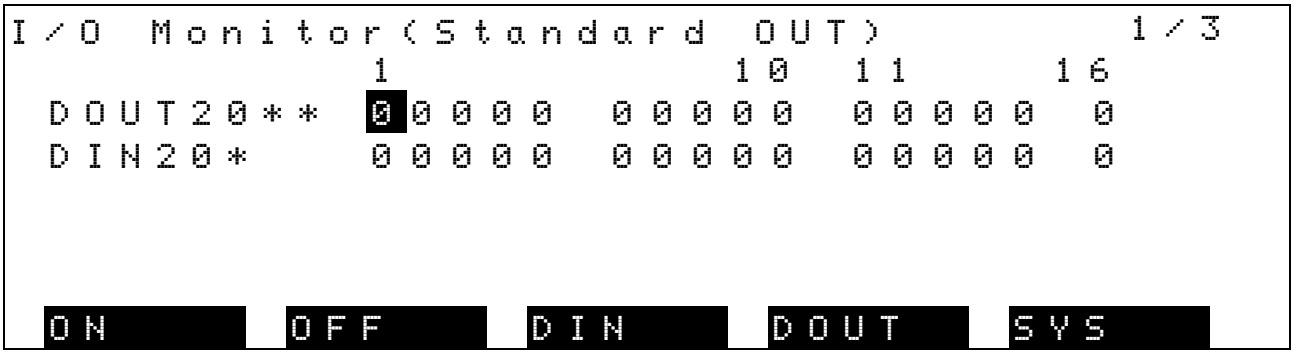
Key	Description
ESC key	This returns to the utility menu screen.
ALT key + ↑ key or ALT key + NEXT key	This switches to the I/O (SYSTEM IN) screen.
ALT key + ↓ key or NEXT key	This switches to the I/O (HAND) screen.

The function menus in the I/O (SYSTEM OUT) screen and their actions are shown in the following table.

Displayed menu name	Description
DIN	This switches to the I/O (DIN) screen.
DOUT	This switches to the I/O (DOUT) screen.
SYS	Disabled.

I/O (HAND) Screen

This screen shows the hand I/O signal status.



The valid keys in the I/O (SYSTEM IN) screen and their actions are shown in the following table.

Key	Description
ESC key	This returns to the utility menu screen.
← key	This moves the cursor to the left.
→ key	This moves the cursor to the right.
ALT key + ↑ key or ALT key + NEXT key	This switches to the I/O (HAND) screen.
ALT key + ↓ key or NEXT key	This switches to the I/O (SYSTEM OUT) screen.

The function menus in the I/O (SYSTEM IN) screen and their actions are shown in the following table.

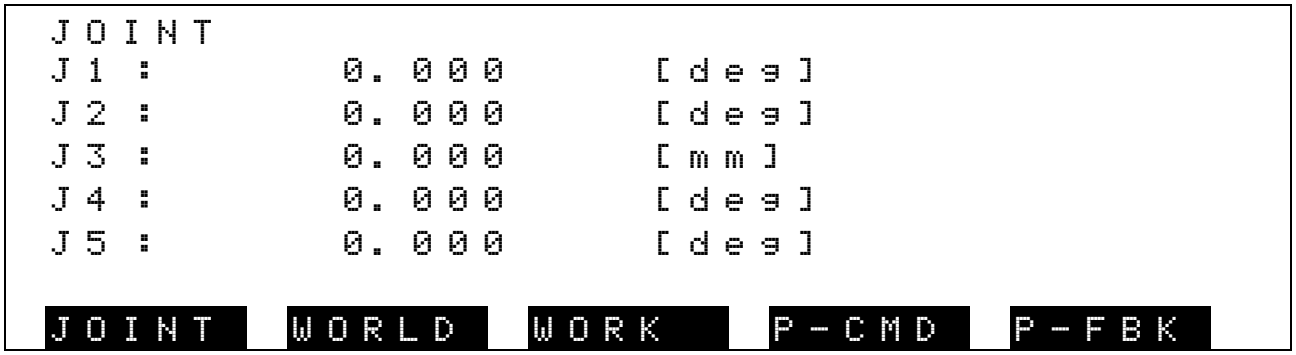
Displayed menu name	Description
ON	The output signal at the cursor position is set to ON.
OFF	The output signal at the cursor position is set to OFF.
DIN	This switches to the I/O (DIN) screen.
DOUT	This switches to the I/O (DOUT) screen.
SYS	Disabled.

POS

This screen shows the current position of the robot and the current angles of the joint axes.

POS (JOINT) Screen

This screen shows the current angles of the joint axes for the robot.



The valid keys in the POS (JOINT) screen and their actions are shown in the following table.

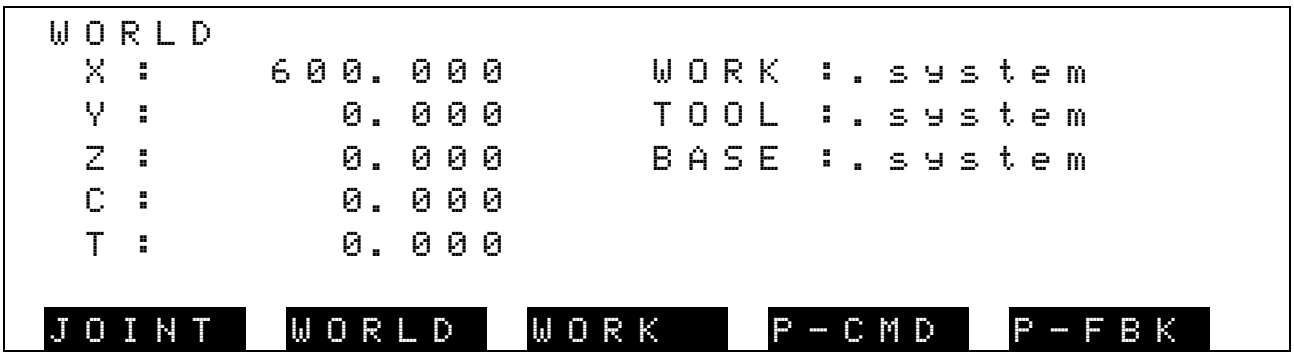
Key	Description
ESC key	This returns to the utility menu screen.

The function menus in the POS (JOINT) screen and their actions are shown in the following table.

Displayed menu name	Description
JOINT	Disabled.
WORLD	This switches to the POS (WORLD) screen.
WORK	This switches to the POS (WORK) screen.
P-CMD	This switches to the POS (P-CMD) screen.
P-FBK	This switches to the POS (P-FBK) screen.

POS (WORLD) Screen

This screen shows the current position on the world coordinate system of the robot and the names of the current work coordinates, tool coordinates, and base coordinates.



The valid keys in the POS (WORLD) screen and their actions are shown in the following table.

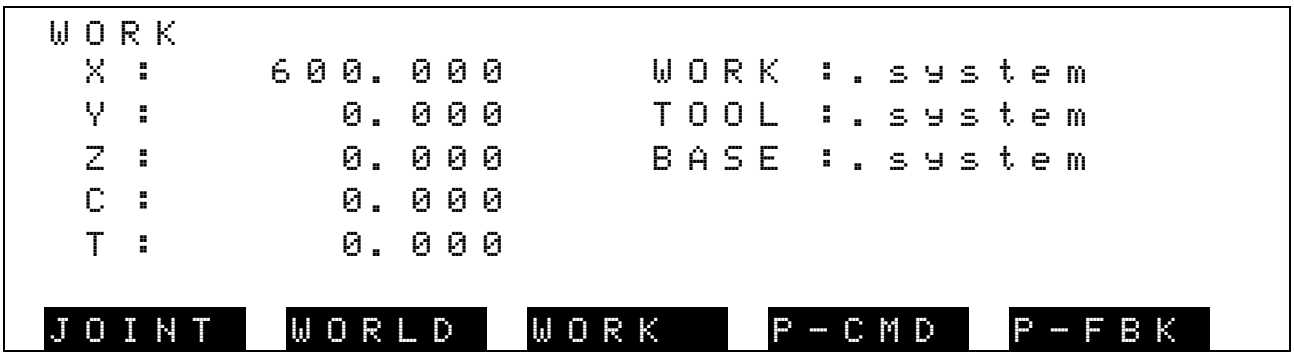
Key	Description
ESC key	This returns to the utility menu screen.

The function menus in the POS (WORLD) screen and their actions are shown in the following table.

Displayed menu name	Description
JOINT	This switches to the POS (JOINT) screen.
WORLD	Disabled.
WORK	This switches to the POS (WORK) screen.
P-CMD	This switches to the POS (P-CMD) screen.
P-FBK	This switches to the POS (P-FBK) screen.

POS (WORK) Screen

This screen shows the current position on the work coordinate system of the robot and the names of the current work coordinates, tool coordinates, and base coordinates.



The valid keys in the POS (WORK) screen and their actions are shown in the following table.

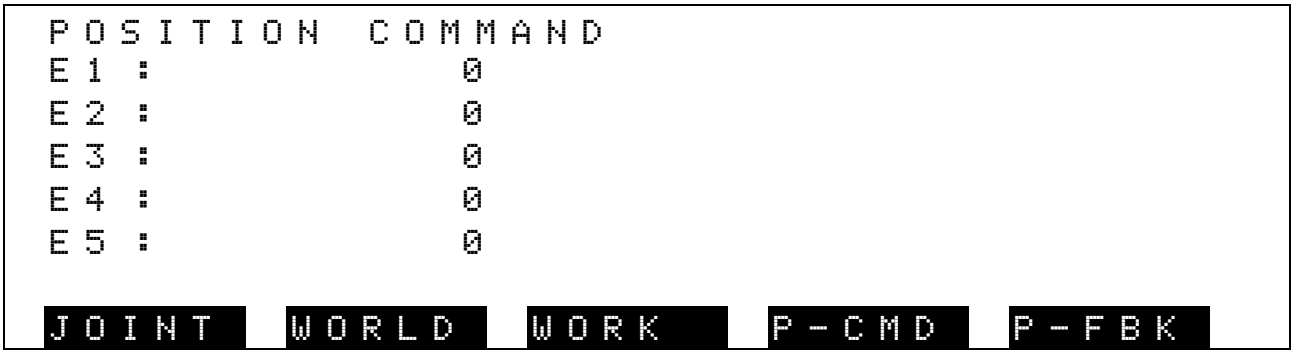
Key	Description
ESC key	This returns to the utility menu screen.

The function menus in the POS (WORK) screen and their actions are shown in the following table.

Displayed menu name	Description
JOINT	This switches to the POS (JOINT) screen.
WORLD	This switches to the POS (WORLD) screen.
WORK	Disabled.
P-CMD	This switches to the POS (P-CMD) screen.
P-FBK	This switches to the POS (P-FBK) screen.

POS (P-CMD) Screen

This screen shows the pulse command values of the joint axes for the robot.



The valid keys in the POS (P-CMD) screen and their actions are shown in the following table.

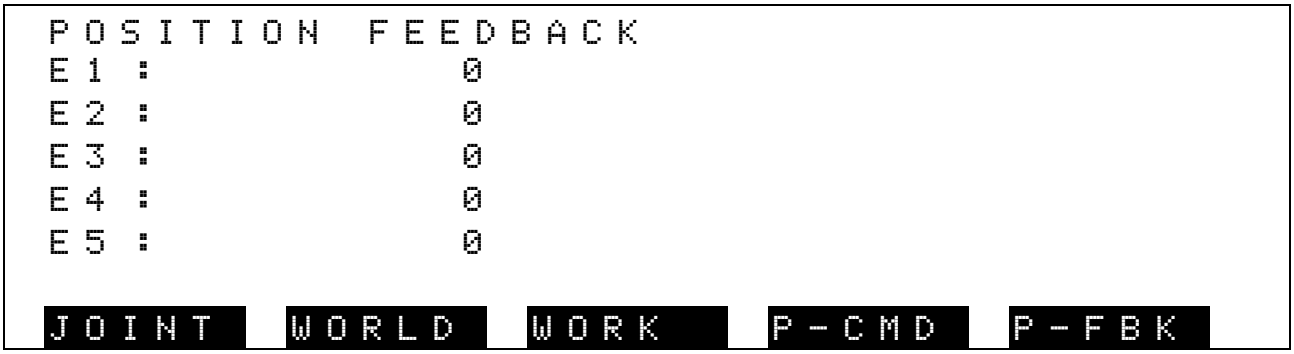
Key	Description
ESC key	This returns to the utility menu screen.

The function menus in the POS (P-CMD) screen and their actions are shown in the following table.

Displayed menu name	Description
JOINT	This switches to the POS (JOINT) screen.
WORLD	This switches to the POS (WORLD) screen.
WORK	This switches to the POS (WORK) screen.
P-CMD	Disabled.
P-FBK	This switches to the POS (P-FBK) screen.

POS (P-FBK) Screen

This screen shows the current pulses of the joint axes for the robot.



The valid keys in the POS (P-FBK) screen and their actions are shown in the following table.

Key	Description
ESC key	This returns to the utility menu screen.

The function menus in the POS (P-FBK) screen and their actions are shown in the following table.

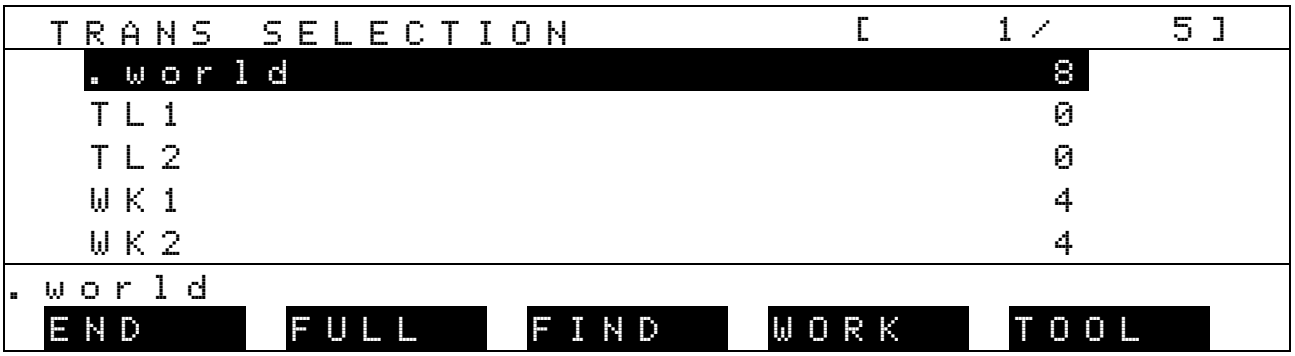
Displayed menu name	Description
JOINT	This switches to the POS (JOINT) screen.
WORLD	This switches to the POS (WORLD) screen.
WORK	This switches to the POS (WORK) screen.
P-CMD	This switches to the POS (P-CMD) screen.
P-FBK	Disabled.

TRANS

This screen shows the coordinate-type data of the execution program.
TRANS-type data can be registered to tool coordinates/work coordinates.
The TRANS screen cannot be opened unless the execution program is registered.

TRANS (DIR) Screen

This screen shows the TRANS-type data of the execution program.



The valid keys in the TRANS (DIR) screen and their actions are shown in the following table.

Key	Description
ESC key	This returns to the utility menu screen.
↑ key	This moves the cursor up by one line.
↓ key	This moves the cursor down by one line.
ALT key + ↑ key	This moves the cursor to the previous page.
ALT key + ↓ key	This moves the cursor to the next page.

The function menus in the TRANS (DIR) screen and their actions are shown in the following table.

Displayed menu name	Description
END	This returns to the utility menu screen.
FULL	This switches to the TRANS (FULL) screen.
FIND	This searches for TRANS-type data.
WORK	This registers the TRANS-type data in the cursor line to the work coordinates.
TOOL	This registers the TRANS-type data in the cursor line to the tool coordinates.

MEMO

- *A "t" is displayed at the start of the line for the TRANS-type data registered to the tool coordinates.*
- *A "w" is displayed at the start of the line for the TRANS-type data registered to the work coordinates.*

TRANS (FULL) Screen

Details of the TRANS-type data in the cursor line are shown in the TRANS (DIR) screen.



The valid keys in the TRANS (FULL) screen and their actions are shown in the following table.

Key	Description
↑ key or ALT key + ↑ key	This moves the cursor in the TRANS (DIR) screen up by one line. The displayed coordinate-type data is changed.
↓ key or ALT key + ↓ key	This moves the cursor in the TRANS (DIR) screen down by one line. The displayed coordinate-type data is changed.
ESC key	This returns to the utility menu screen.

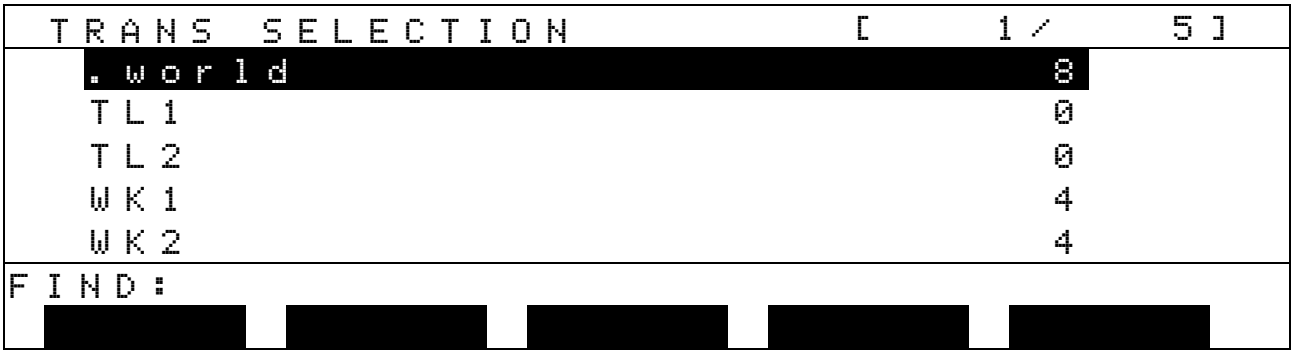
The function menus in the TRANS (FULL) screen and their actions are shown in the following table.

Displayed menu name	Description
END	This returns to the utility menu screen.
DIR	This switches to the TRANS (DIR) screen.
FIND	This searches for TRANS-type data.
WORK	This registers the TRANS-type data in the cursor line to the work coordinates.
TOOL	This registers the TRANS-type data in the cursor line to the tool coordinates.

<TRANS-type Data Search Procedure>

1 Press the FIND (F3) key in either the TRANS (DIR) screen or TRANS (FULL) screen.

"FIND:" appears in the command line.



2 Enter the name of the data that will be searched for, and press the EXE key.

The cursor moves to the location of the data with the entered name.

MEMO

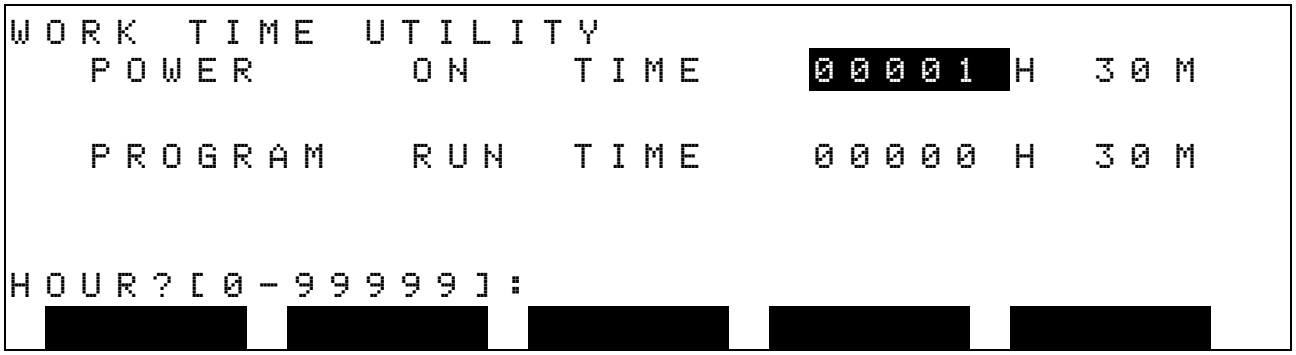
- *If no matching character string is found, "Not found." appears in the command line.*
- *If the search was performed in the TRANS (DIR) screen, the TRANS (DIR) screen is opened. If the search was performed in the TRANS (FULL) screen, the TRANS (FULL) screen is opened.*

WK-TM

This screen shows the controller power-on time and the program operation time.
This enables changing of the controller power-on time and program operation time.

WK-TM Screen

This screen shows the controller power-on time and program run time.



The valid keys in the WK-TM screen and their actions are shown in the following table.

Key	Description
↑ key	This moves the cursor up by one line.
↓ key	This moves the cursor down by one line.
← key	This moves the cursor by one column to the left.
→ key	This moves the cursor by one column to the right.
EXE key	This changes the item at the cursor position to the value in the command line.
ESC key	This returns to the utility menu screen.

<Procedure for Changing the Controller Power-on Time and Program Operation Time>

1 Move the cursor to the position of the item to be changed.

2 Enter the value to be changed into the command line.

3 Press the EXE key.

The value of the item at the cursor position is changed to the value in the command line.

MEMO

The allowable ranges of values are 0 to 99999 for the hours and 0 to 59 for the minutes.

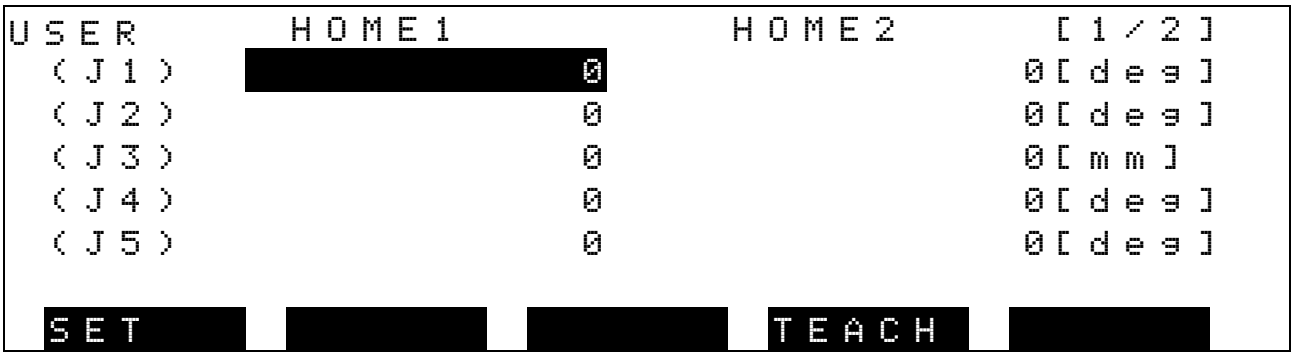
The value is not updated if a value outside the range is entered and the EXE key is pressed.

REORG

This screen enables setting of the origin and registering of the origin return position for each joint axis.

REORG Screen

This shows the registered origin return position.



The valid keys in the REORG screen and their actions are shown in the following table.

Key	Description
↑ key	This moves the cursor up by one line.
↓ key	This moves the cursor down by one line.
← key	This moves the cursor by one column to the left.
→ key	This moves the cursor by one column to the right.
ALT key + ↑ key or ALT key + NEXT key	This switches to the previous page.
ALT key + ↑ key or NEXT key	This switches to the next page.
EXE key	Applies the origin setting and the origin return position registration.
ESC key	This returns to the utility menu screen.

The function menus in the REORG screen and their actions are shown in the following table.

Displayed menu name	Description
SET	This sets the origin.
TEACH	This registers the origin return position. *This cannot be used when the cursor is in the HOME3 or HOME4 column.

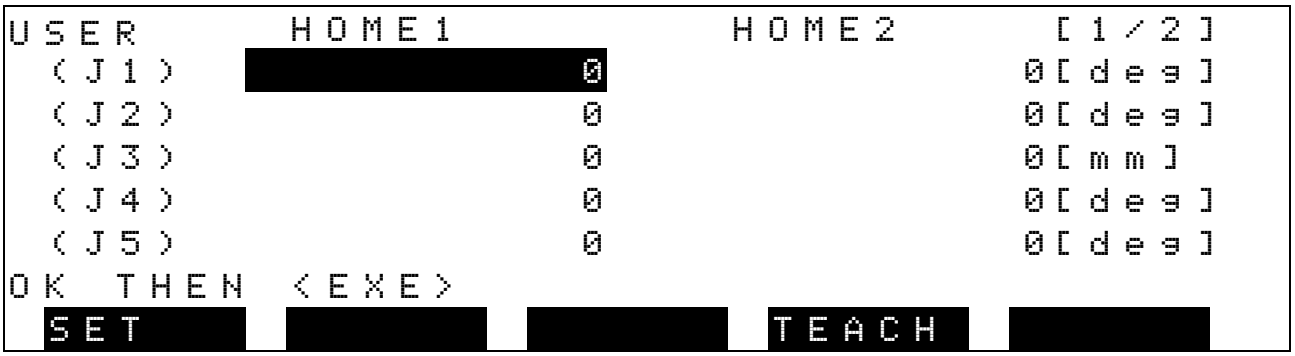
<Origin Setting Procedure>

1 Move the cursor to the position of the joint axis where the origin will be set.

2 Use your hand to move the robot to the predetermined origin return position.

3 Press the SET (F1) key.

"OK THEN <EXE>" appears in the command line.



4 Press the EXE key.

The motor angle of the joint axis at the cursor position is overwritten by the value at the cursor position.

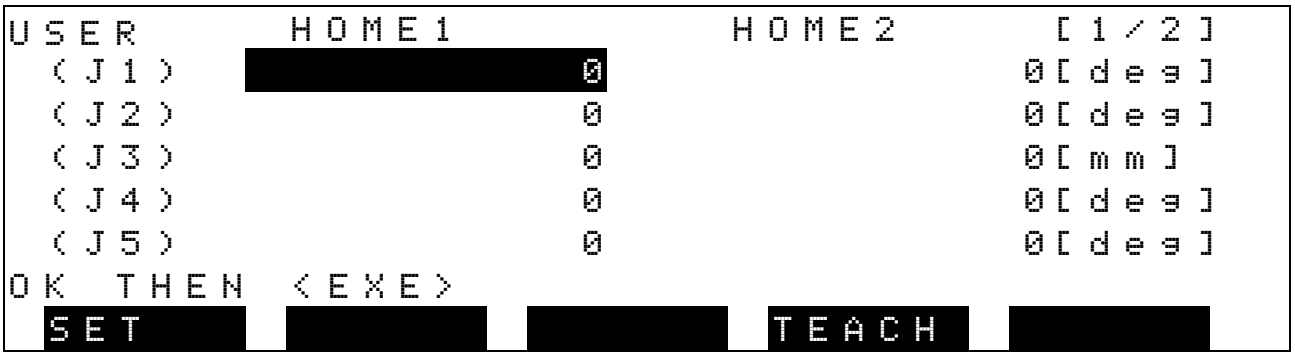
<Origin Return Position Registration Procedure>

1 Move the cursor to the position of the joint axis where the origin return position will be registered.

2 Use your hand to move the robot to the predetermined origin return position.

3 Press the TEACH (F4) key.

"OK THEN <EXE>" appears in the command line.



4 Press the EXE key.

The motor angle of the current joint axis is registered at the cursor position.

MEMO

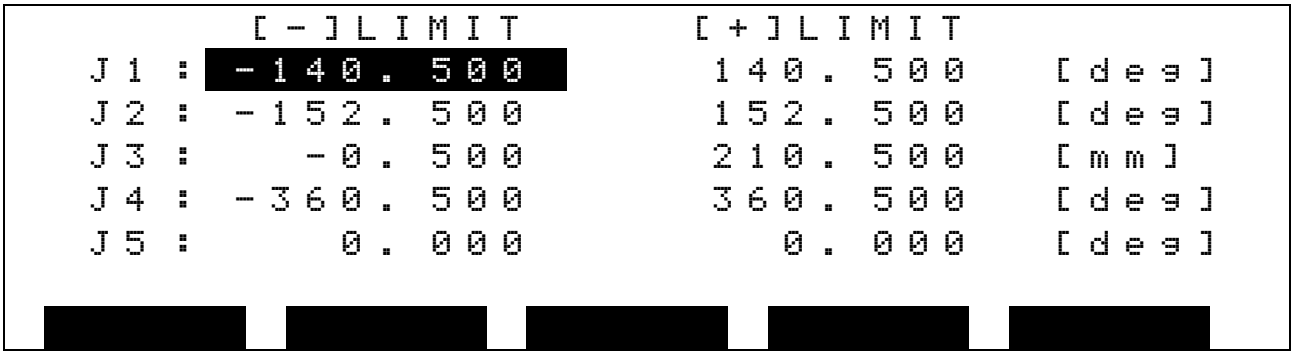
- *For SYSTEM (HOME3, HOME4), the origin return value at factory shipping is registered. The customer cannot overwrite the registration.*
- *Servo ON cannot be set while the REORG screen is open.*

J-LIM

This screen enables setting of the software limit for each joint axis.

J-LIM Screen

This screen shows the current software limit setting values.



The valid keys in the J-LIM screen and their actions are shown in the following table.

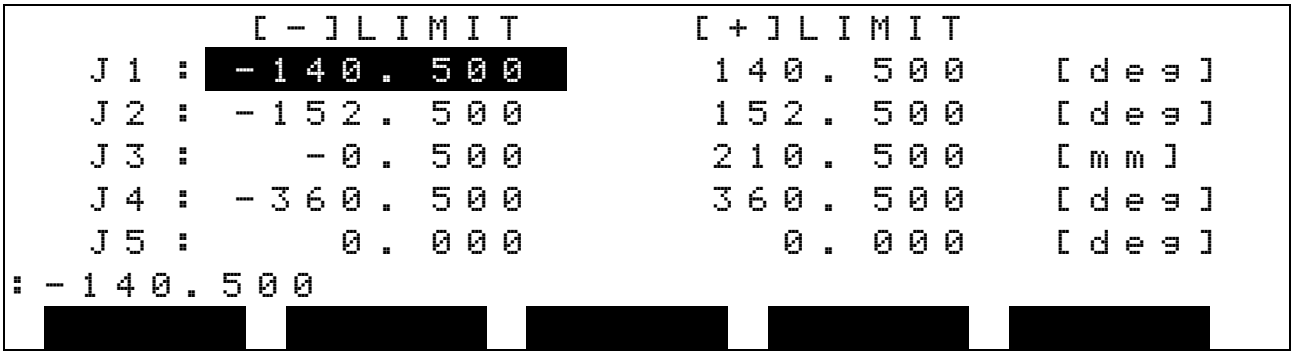
Key	Description
↑ key	This moves the cursor up by one line.
↓ key	This moves the cursor down by one line.
← key	This moves the cursor by one column to the left.
→ key	This moves the cursor by one column to the right.
EXE key	This edits the value at the cursor position.
ESC key	This returns to the utility menu screen.

<Software Limit Setting Procedure>

1 Move the cursor to the position of the item where the software limit will be changed.

2 Press the EXE key.

The current setting value is displayed in the command line, and the content can be edited.



3 Change the command line value to the value for the software limit setting.

> *MEMO*
>
> *If the ESC key is pressed while editing the command line, all command line values are erased.*
>
> *If the ESC key is pressed while nothing is entered into the command line, the editing process is canceled.*

4 Press the EXE key.

The software limit setting value is changed to the command line value.

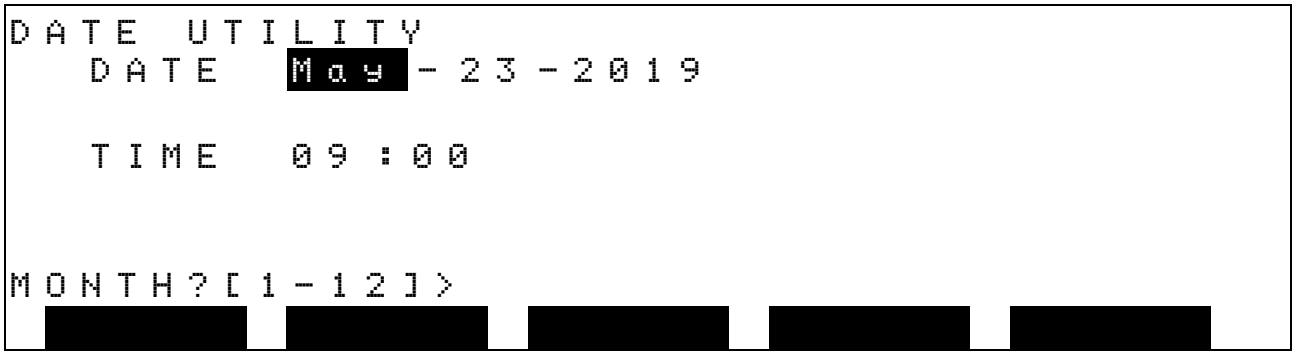
> *MEMO*
>
> *The range of values that can be set for the software limit is a range specified by the manufacturer for each joint axis.*
>
> *If a value outside the range is entered and the EXE key is pressed, "Data Error" appears.*

DATE

The date setting for the controller is displayed.
The date setting for the controller can be changed.

DATE Screen

This screen shows the controller date and time.



The valid keys in the DATE screen and their actions are shown in the following table.

Key	Description
↑ key	This moves the cursor up by one line.
↓ key	This moves the cursor down by one line.
← key	This moves the cursor by one column to the left.
→ key	This moves the cursor by one column to the right.
EXE key	This changes the item at the cursor position to the value in the command line. This changes to the value in the command line.
ESC key	This returns to the utility menu screen.

<Procedure for Changing the Controller Date>

1 Move the cursor to the position of the item to be changed.

2 Enter the value to be changed into the command line.

3 Press the EXE key.

The value of the item at the cursor position is changed to the value in the command line.

MEMO

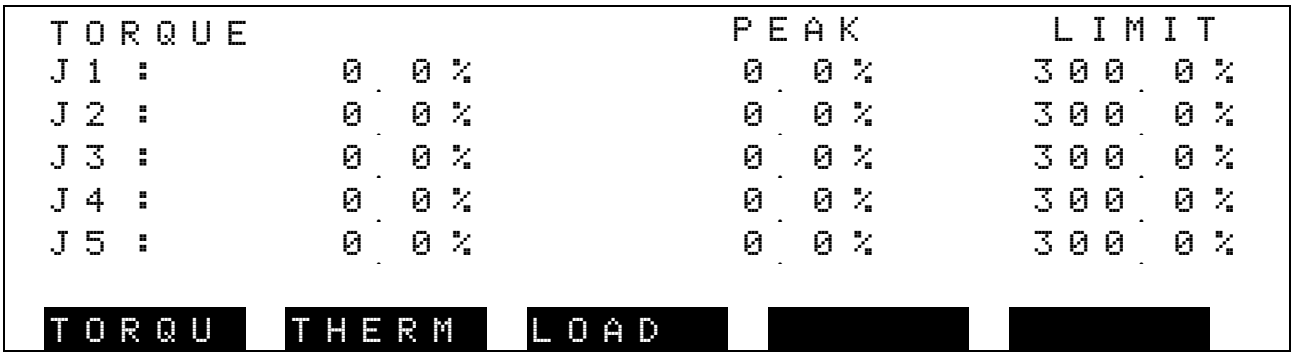
- *The ranges of allowable values are 0 to 99 for the year, 1 to 12 for the month, 1 to 31 for the day, 0 to 23 for the hour, and 0 to 59 for the minutes.*
  If a value outside the range is entered, and the EXE key is pressed, the value is not updated, and the message "Input Error" appears.
- *If a date within the allowable setting range is entered, but the date is not found within the setting month, and the EXE key is pressed, the date is changed to the final day of that month.*
- *If the day is outside the setting range after changing the year and month, the day is automatically changed to the final date of that month.*

MOTOR

This shows the information about each joint axis.

MOTOR (TORQU) Screen

This screen shows the current value, maximum value, and limit value of the motor torque for each joint axis.



The valid keys in the MOTOR (TORQU) screen and their actions are shown in the following table.

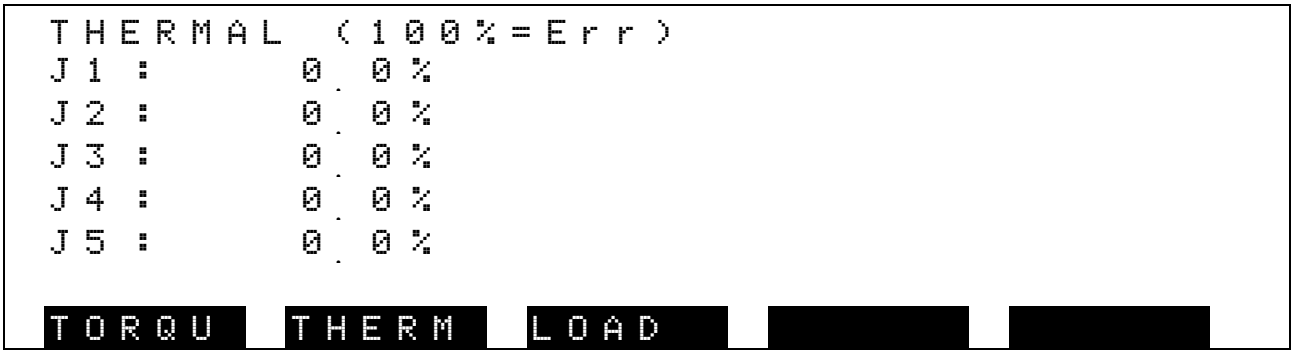
Key	Description
ESC key	This returns to the utility menu screen.

The function menus in the MOTOR (TORQU) screen and their actions are shown in the following table.

Displayed menu name	Description
TORQU	Disabled.
THERM	This switches to the MOTOR (THERM) screen.
LOAD	This switches to the MOTOR (LOAD) screen.

MOTOR (THERM) Screen

This screen shows the thermal values for each joint axis.



The valid keys in the MOTOR (THERM) screen and their actions are shown in the following table.

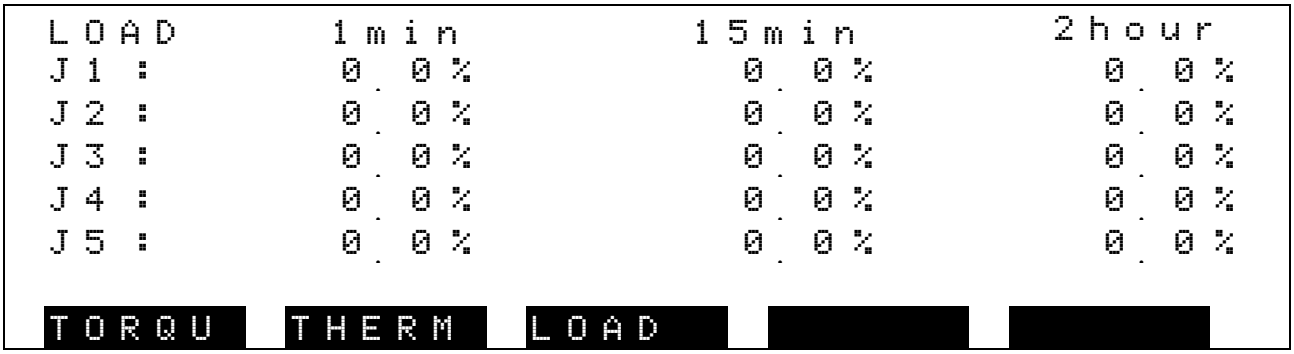
Key	Description
ESC key	This returns to the utility menu screen.

The function menus in the MOTOR (THERM) screen and their actions are shown in the following table.

Displayed menu name	Description
TORQU	This switches to the MOTOR (TORQU) screen.
THERM	Disabled.
LOAD	This switches to the MOTOR (LOAD) screen.

MOTOR (LOAD) Screen

This screen shows the load factors (1 minute, 15 minutes, 2 hours) for each joint axis.



The valid keys in the MOTOR (LOAD) screen and their actions are shown in the following table.

Key	Description
ESC key	This returns to the utility menu screen.

The function menus in the MOTOR (LOAD) screen and their actions are shown in the following table.

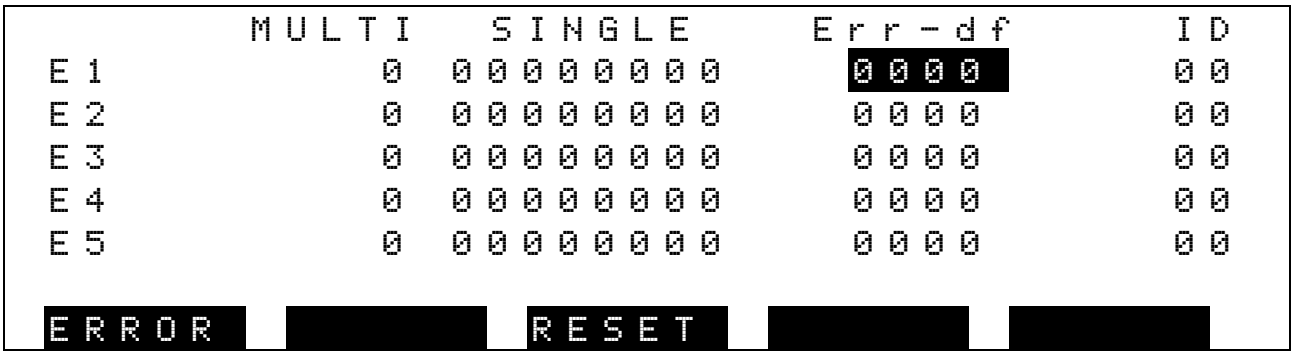
Displayed menu name	Description
TORQU	This switches to the MOTOR (TORQU) screen.
THERM	This switches to the MOTOR (THERM) screen.
LOAD	Disabled.

ENC

This screen shows the encoder information for each joint axis.

ENC (STAT) Screen

This screen shows the encoder status for each joint axis.



The valid keys in the ENC (STAT) screen and their actions are shown in the following table.

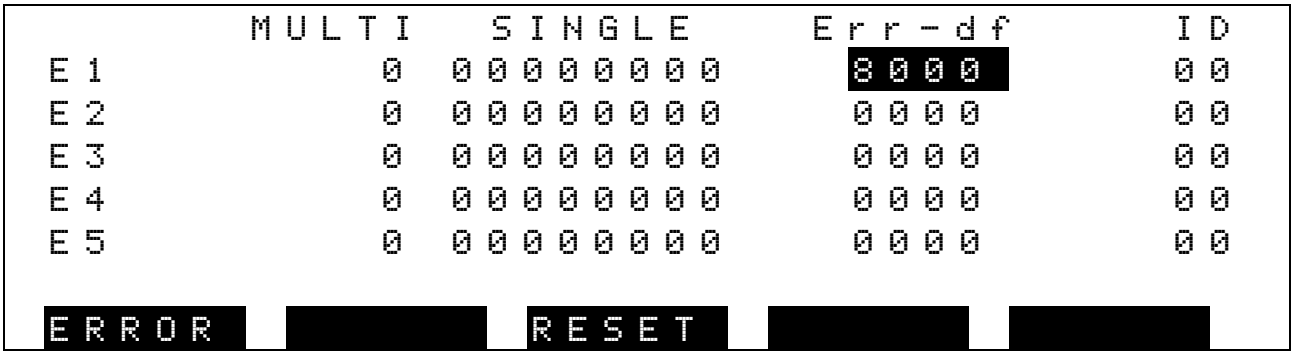
Key	Description
↑ key	This moves the cursor up by one line.
↓ key	This moves the cursor down by one line.
ESC key	This returns to the utility menu screen.

The function menus in the ENC (STAT) screen and their actions are shown in the following table.

Displayed menu name	Description
ERROR	This switches to the ENC (ERROR) screen.
RESET	This is used to perform the reset operation.

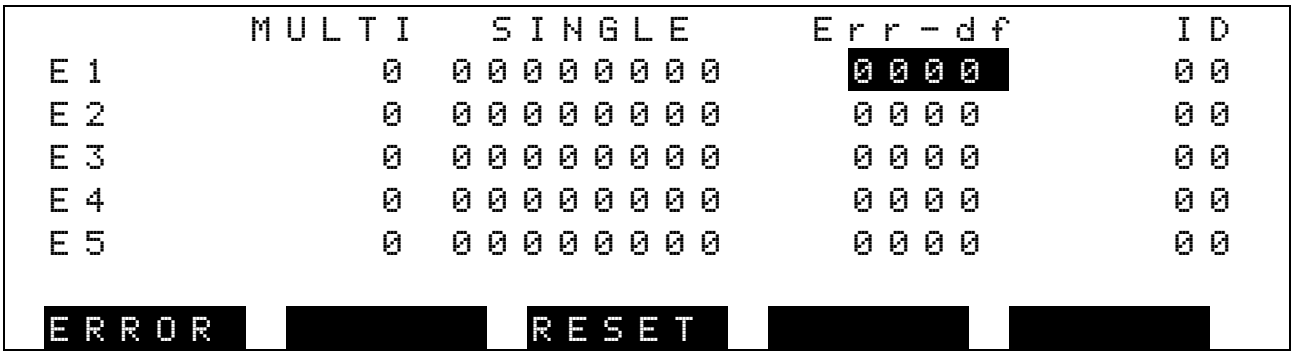
<Encoder Error Reset Procedure>

1 Move the cursor to the line of the joint axis where the encoder error will be reset.



2 Press the RESET (F3) key.

The encoder error for the joint axis in the cursor line is reset.



MEMO

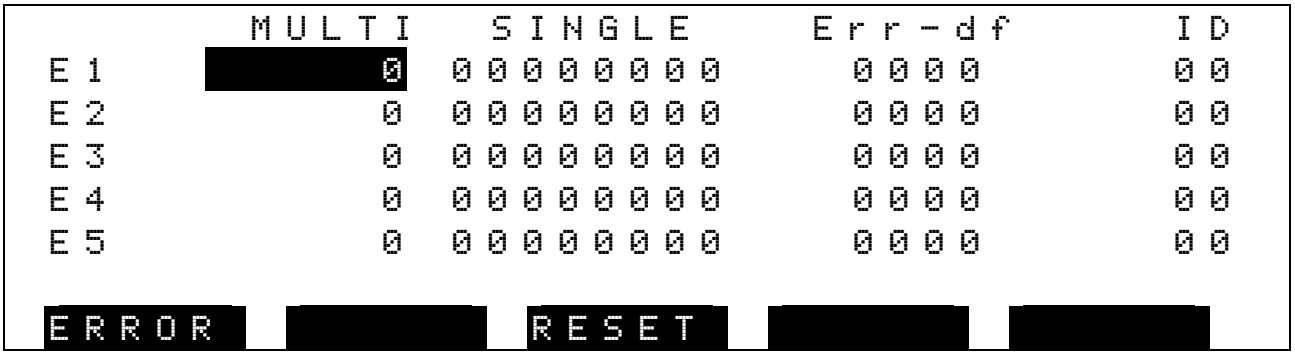
Errors are not reset unless the cause of the error was removed.

<Multi-rotational Information Reset Procedure>

1 Press the ALT key and 0 key simultaneously in the ENC (STAT) screen.

This enables movement of the cursor key to the MULTI column.

2 Move the cursor to the MULTI column for the line of the joint axis where the multi-rotational information will be reset.



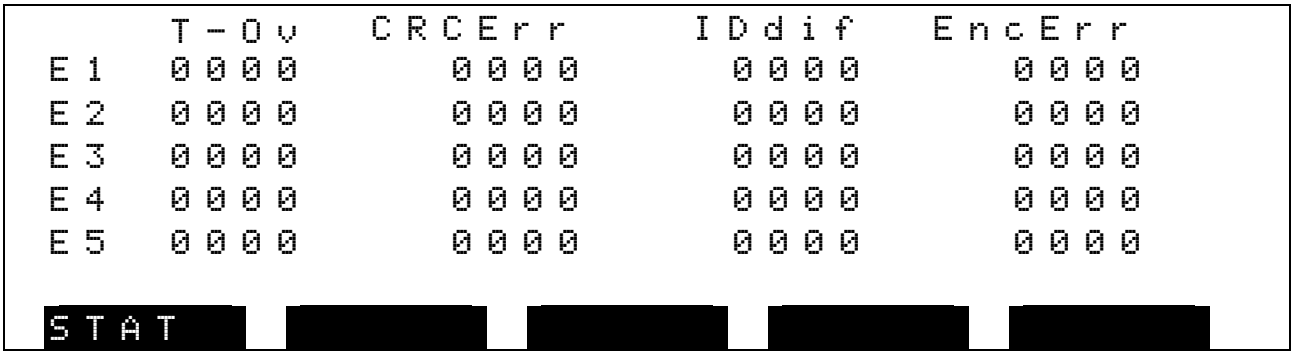
3 Press the RESET (F3) key.

The multi-rotational information for the joint axis in the cursor line is reset.

> *MEMO*
>
> *Errors are not reset unless the cause of the error was removed.*

ENC (ERROR) Screen

This shows the error information for each joint axis.



The valid keys in the ENC (STAT) screen and their actions are shown in the following table.

Key	Description
ESC key	This returns to the utility menu screen.

The function menus in the ENC (STAT) screen and their actions are shown in the following table.

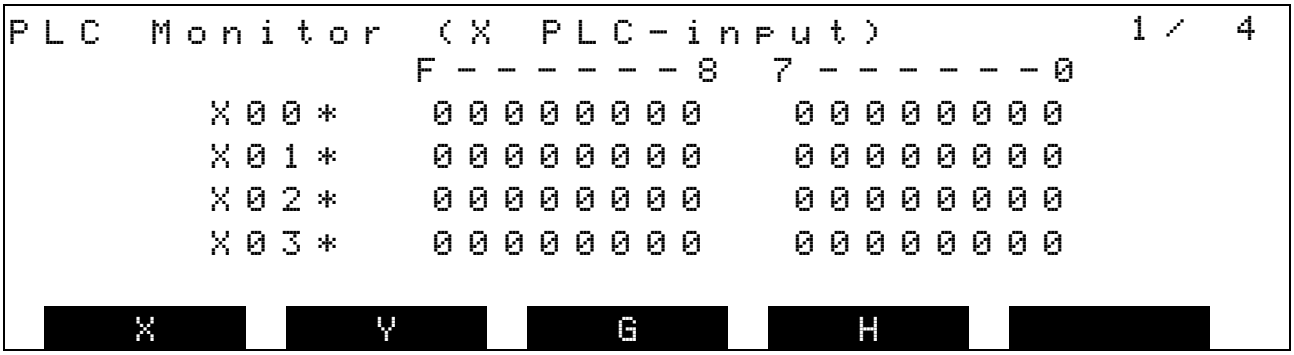
Displayed menu name	Description
STAT	This switches to the ENC (STAT) screen.

PLC-M

This shows the status of the relays for the TCmini (internal PLC).

PLC-M Screen

This screen shows the status of the relays for the TCmini.



The valid keys in the PLC-M screen and their actions are shown in the following table.

Key	Description
↑ key or ALT key + ↑ key or ALT key + NEXT key	This shows the previous page.
↓ key or ALT key + ↓ key or NEXT key	This shows the next page.
ESC key	This returns to the utility menu screen.

The function menus in the PLC-M screen and their actions are shown in the following table.

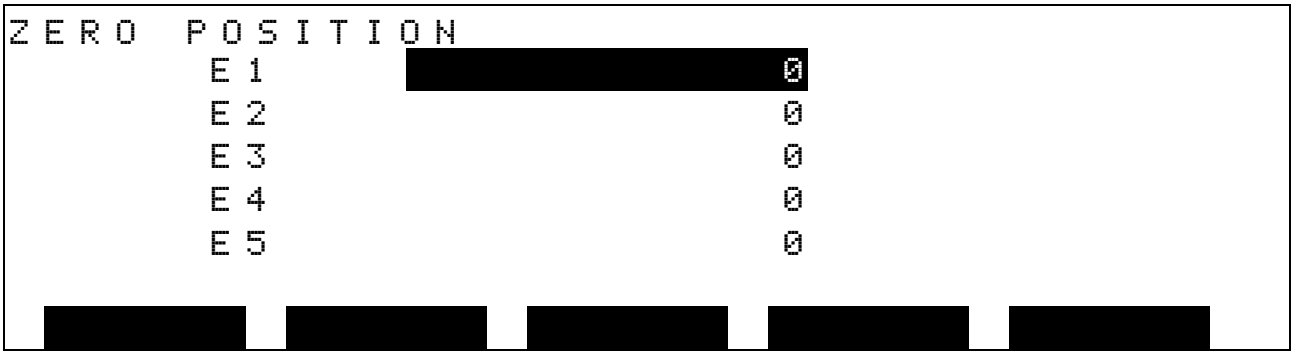
Displayed menu name	Description
X	This shows the status of the X register.
Y	This shows the status of the Y register.
G	This shows the status of the G register.
H	This shows the status of the H register.

ZEROP

The origin can be set for each joint axis.

ZEROP Screen

This screen shows the current pulses of the joint axes.



The valid keys in the ZEROP screen and their actions are shown in the following table.

Key	Description
↑ key	This moves the cursor up by one line.
↓ key	This moves the cursor down by one line.
ESC key	This returns to the utility menu screen.

MEMO

Servo ON cannot be set from the ZEROP screen.

<Joint Axis Origin Setting Procedure>

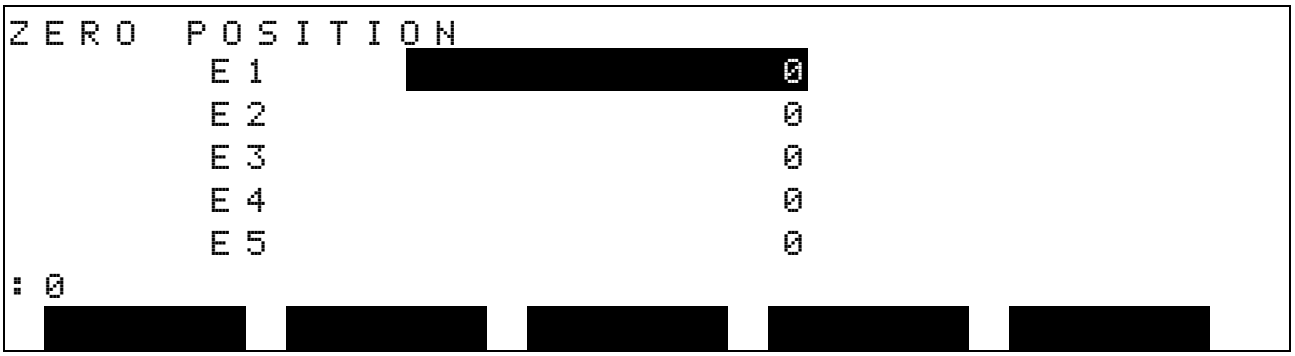
1 Press the ALT key and 0 key simultaneously.

This enables movement of the cursor key.

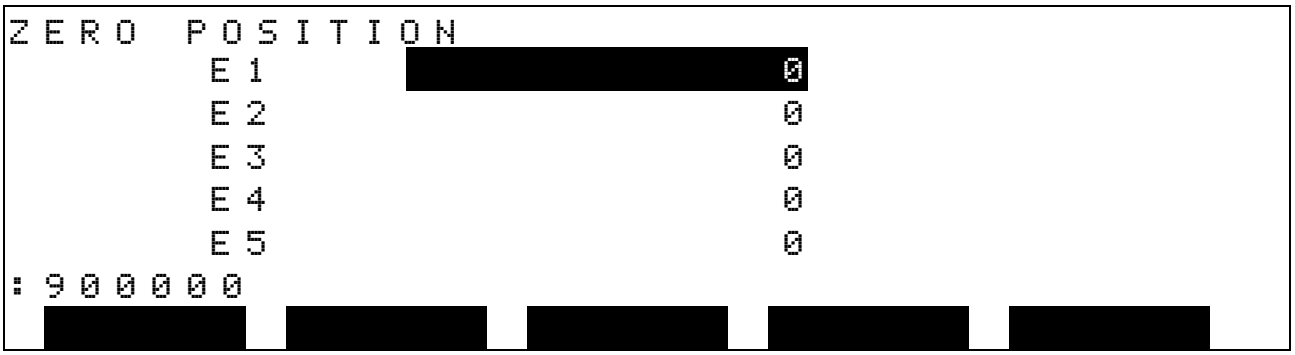
2 Move the cursor to the position of the joint axis where the origin will be set.

3 Press the EXE key.

The current value is displayed in the command line, and the content can be edited.

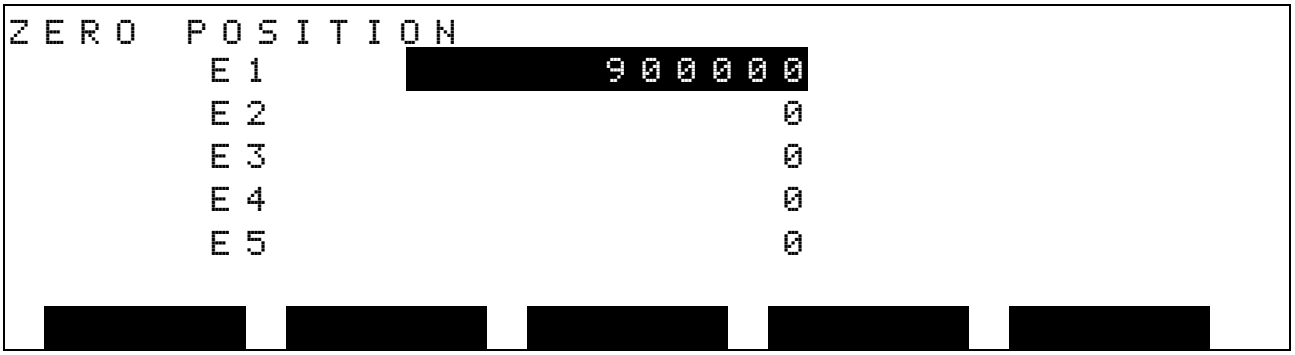


4 Change the command line value to the setting value.



5 Press the EXE key.

The setting value is changed to the command line value.

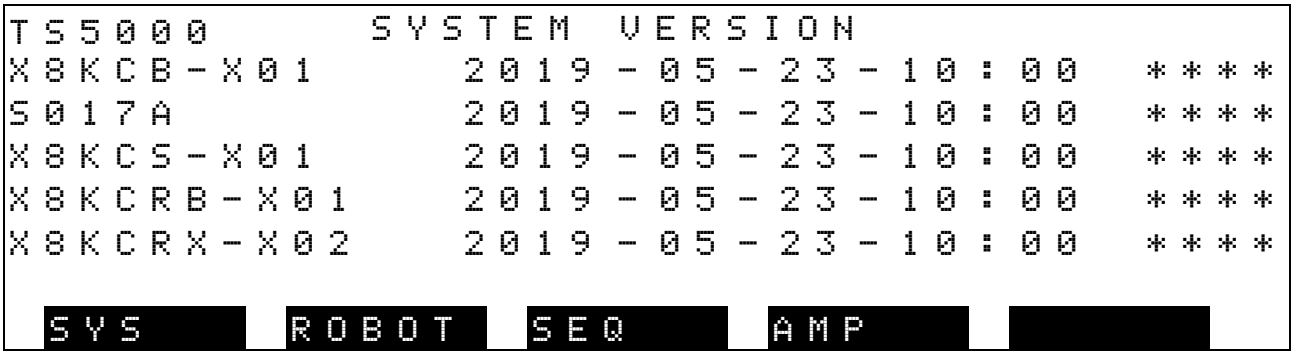


VER

This screen shows the system versions of the robot and controller.

VER (SYS) Screen

This screen shows the controller system version.



The valid keys in the VER (SYS) screen and their actions are shown in the following table.

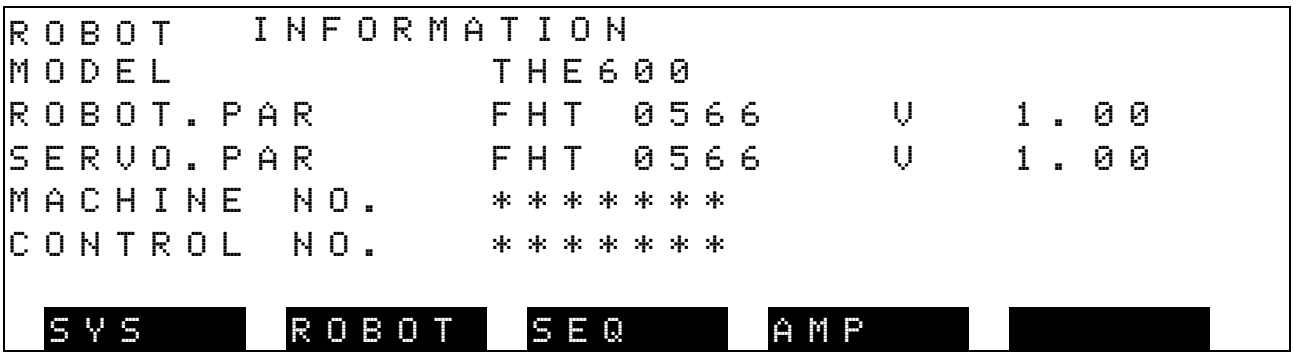
Key	Description
ESC key	This returns to the utility menu screen.

The function menus in the VER (SYS) screen and their actions are shown in the following table.

Displayed menu name	Description
SYS	Disabled.
ROBOT	This switches to the VER (ROBOT) screen.
SEQ	This switches to the VER (SEQ) screen.
AMP	This switches to the VER (AMP) screen.

VER (ROBOT) Screen

This screen shows the robot system and parameter versions and the robot and controller serial numbers.



The valid keys in the VER (ROBOT) screen and their actions are shown in the following table.

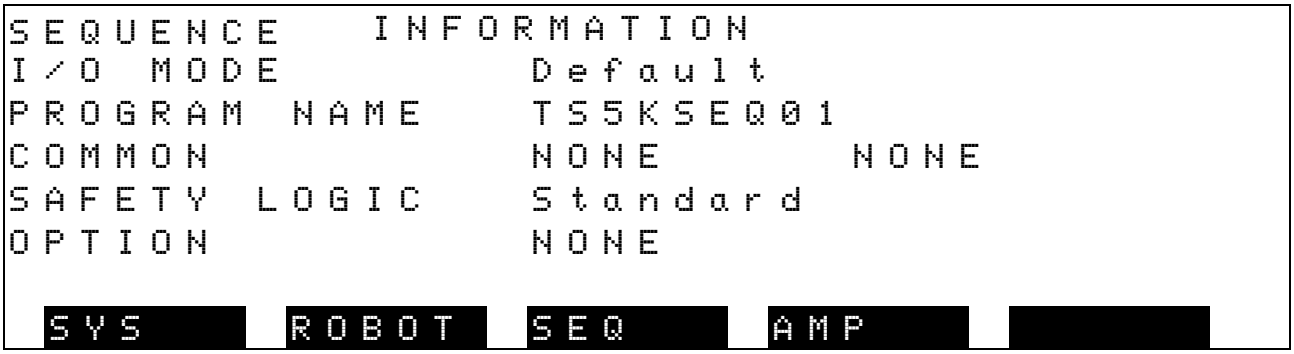
Key	Description
ESC key	This returns to the utility menu screen.

The function menus in the VER (ROBOT) screen and their actions are shown in the following table.

Displayed menu name	Description
SYS	This switches to the VER (SYS) screen.
ROBOT	Disabled.
SEQ	This switches to the VER (SEQ) screen.
AMP	This switches to the VER (AMP) screen.

VER (SEQ) Screen

This screen shows the TCmini (internal PLC) information.



The valid keys in the VER (SEQ) screen and their actions are shown in the following table.

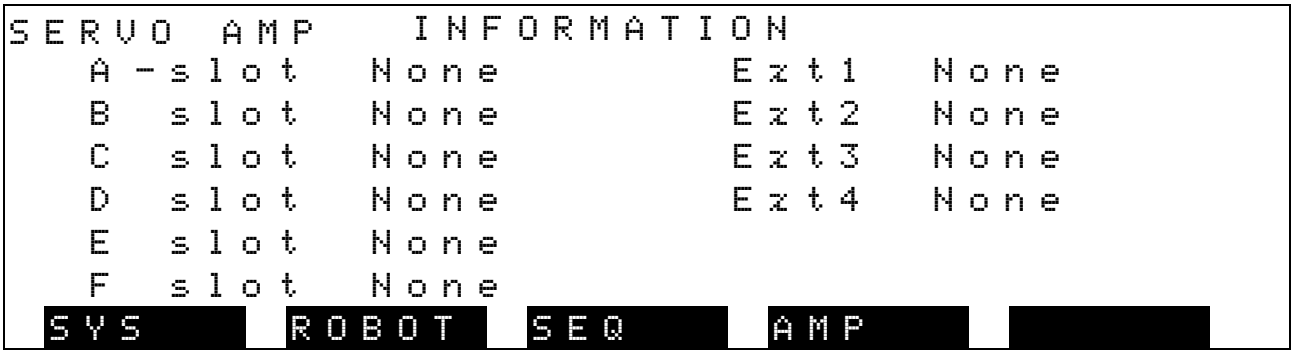
Key	Description
ESC key	This returns to the utility menu screen.

The function menus in the VER (SEQ) screen and their actions are shown in the following table.

Displayed menu name	Description
SYS	This switches to the VER (SYS) screen.
ROBOT	This switches to the VER (ROBOT) screen.
SEQ	Disabled.
AMP	This switches to the VER (AMP) screen.

VER (AMP) Screen

This screen shows the controller servo amplifier information.



The valid keys in the VER (AMP) screen and their actions are shown in the following table.

Key	Description
ESC key	This returns to the utility menu screen.

The function menus in the VER (AMP) screen and their actions are shown in the following table.

Displayed menu name	Description
SYS	This switches to the VER (SYS) screen.
ROBOT	This switches to the VER (ROBOT) screen.
SEQ	This switches to the VER (SEQ) screen.
AMP	Disabled.

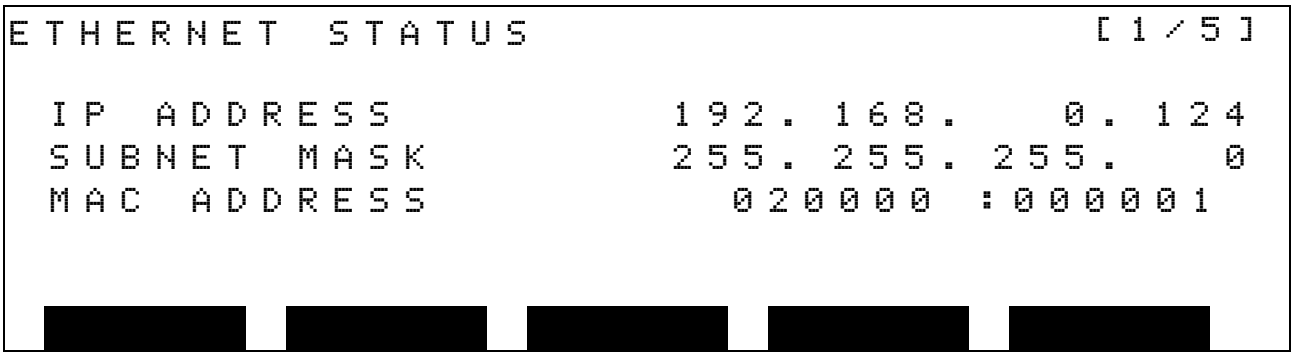
ETHER

This shows the controller Ethernet information and the status of each Ethernet port.

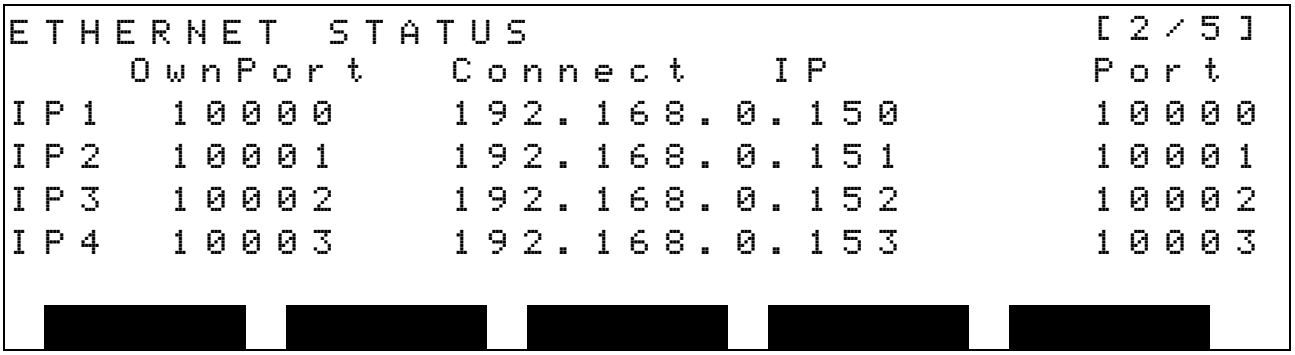
ETHER Screen

The controller IP address, subnet mask, default gateway, and non-protocol communication port status are displayed.

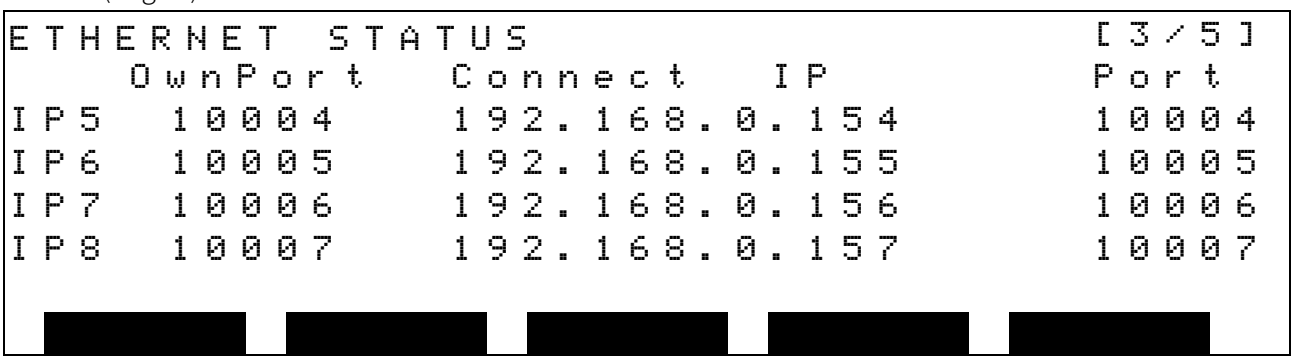
(Page 1)



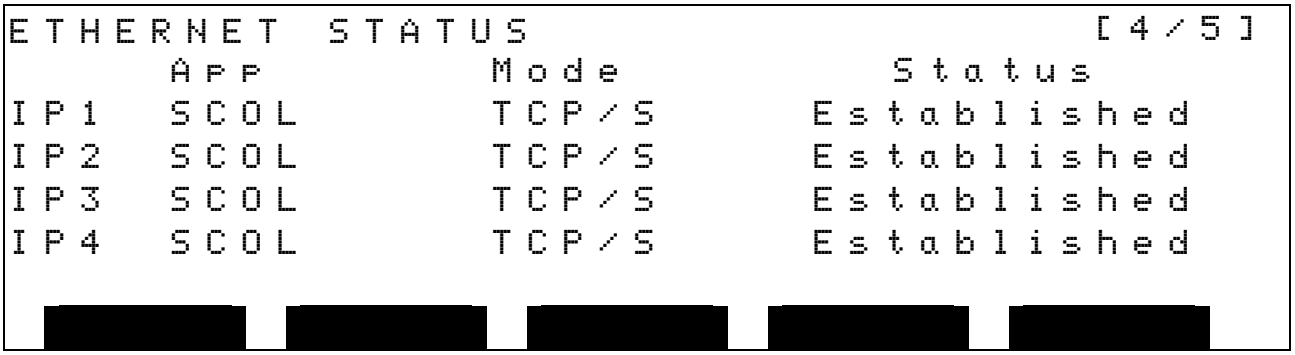
(Page 2)



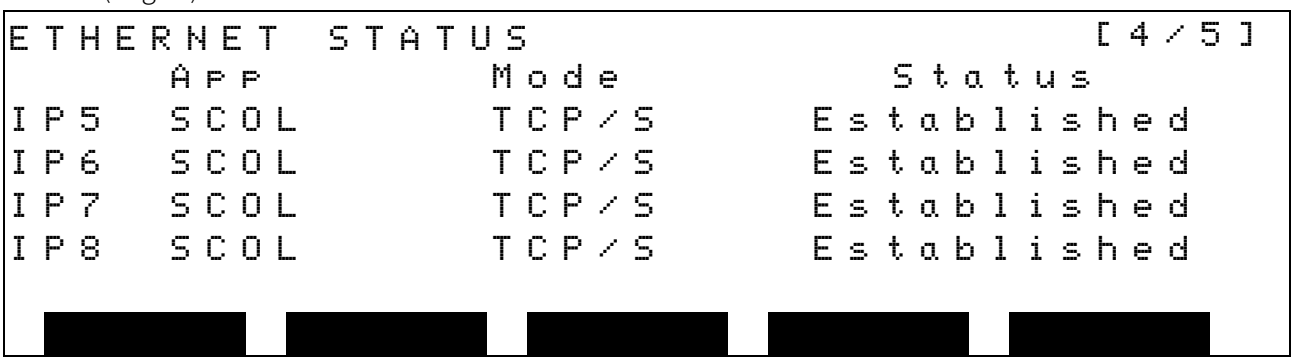
(Page 3)



(Page 4)



(Page 5)



The valid keys in the ETHER screen and their actions are shown in the following table.

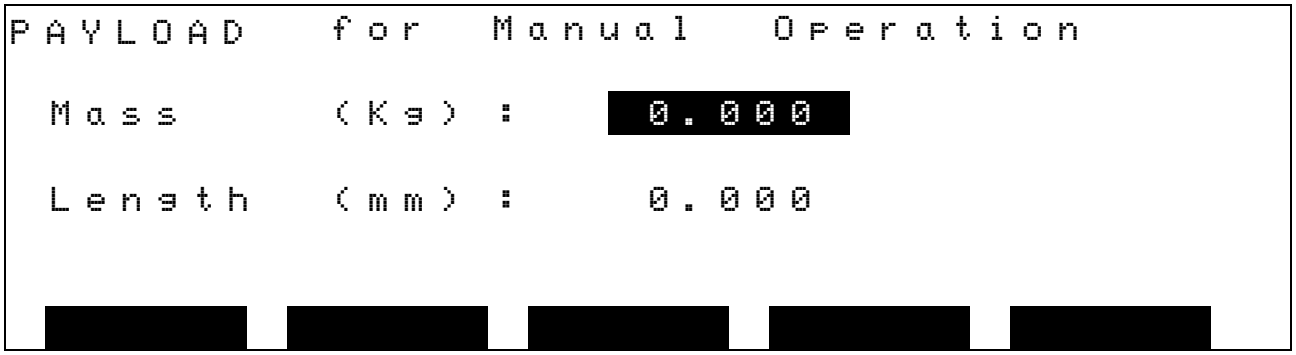
Key	Description
NEXT key	This shows the next page.
ALT key + NEXT key	This shows the previous page.
ESC key	This returns to the utility menu screen.

PAYLD

This screen enables setting of the payload for manual guidance.

PAYLD Screen

This screen shows the payload setting value for manual guidance.



The valid keys in the PYLD screen and their actions are shown in the following table.

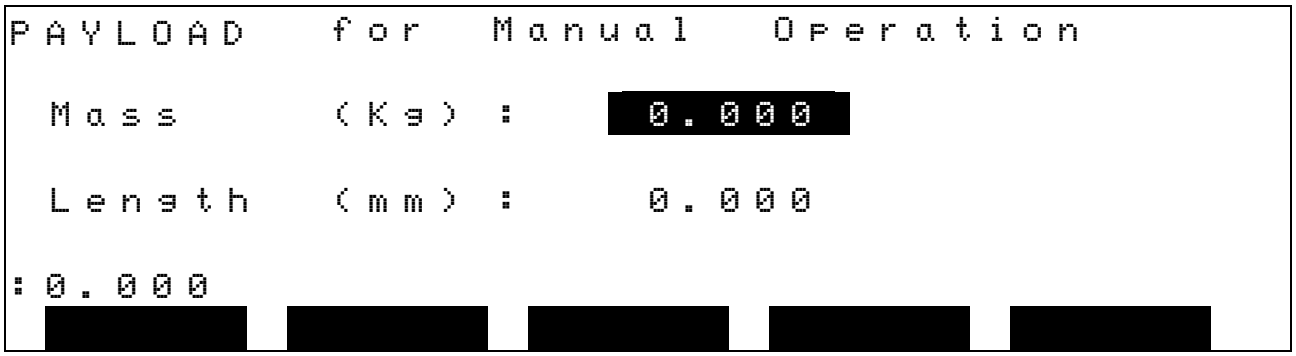
Key	Description
↑ key	This moves the cursor up by one line.
↓ key	This moves the cursor down by one line.
EXE key	This edits the value at the cursor position.
ESC key	This returns to the utility menu screen.

<Manual Guidance Load Setting Procedure>

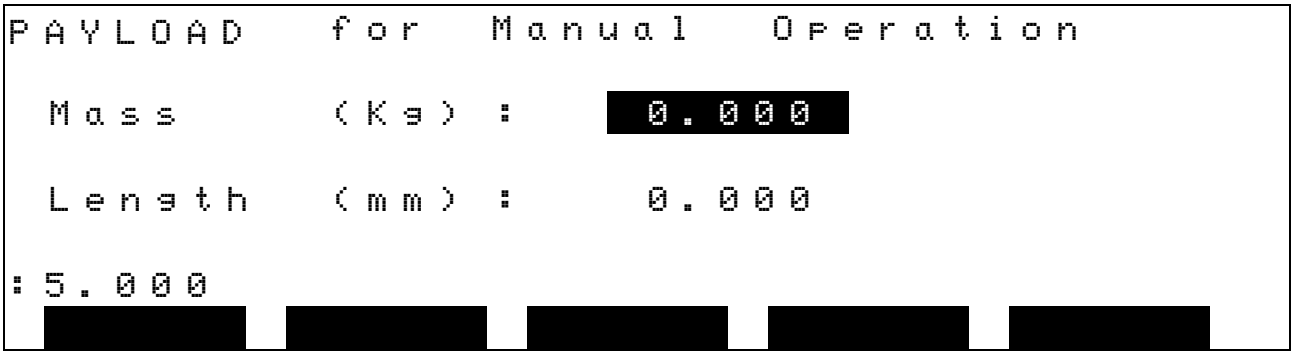
1 Move the cursor to the position of the item to be changed.

2 Press the EXE key.

The current setting value is displayed in the command line, and the content can be edited.



3 Change the command line value to the setting value.



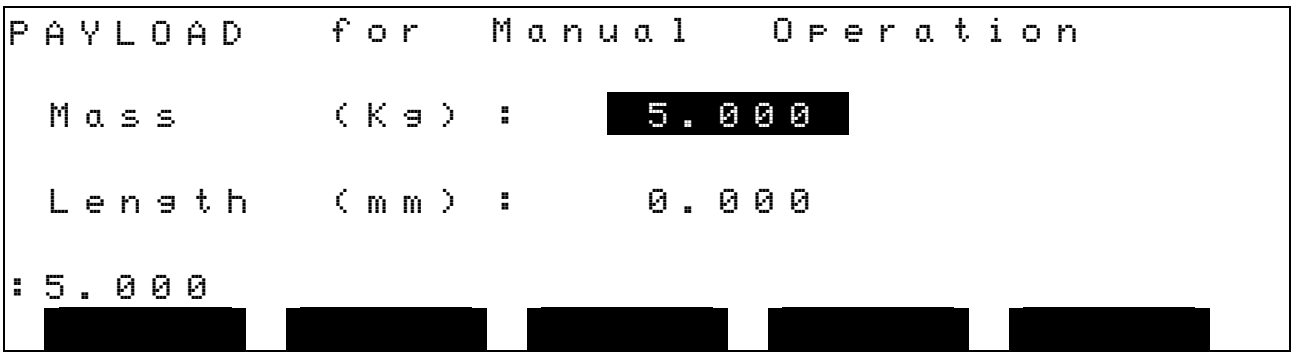
MEMO

If the ESC key is pressed while editing the command line, all command line values are erased.

If the ESC key is pressed while nothing is entered into the command line, the editing process is canceled.

4 Press the EXE key.

The setting value is changed to the command line value.



5 Press any key.

The displayed information in the command line is hidden.

MEMO

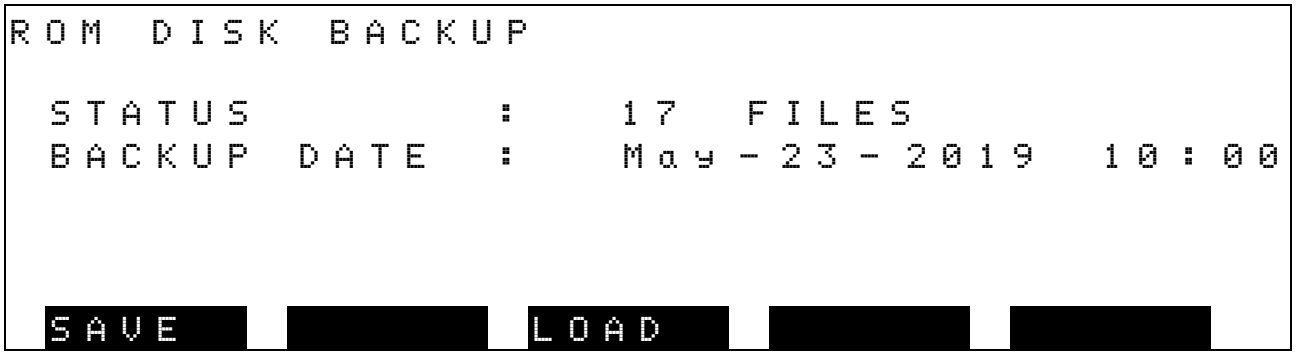
If manual guidance is performed while a negative value is set for the mass, the alarm "PAYLOAD ERROR" occurs.

BCKUP

This enables backing up of files in the controller and restoring of files from the backup.

BCKUP Screen

This screen shows the number of files backed up in the controller and the backup date and time.



The valid keys in the BCKUP screen and their actions are shown in the following table.

Key	Description
EXE key	Executes the backup operation or restoration operation from the backup.
ESC key	This returns to the utility menu screen.

The function menus in the BCKUP screen and their actions are shown in the following table.

Displayed menu name	Description
SAVE	This performs the backup operation.
LOAD	This performs the restoration operation from the backup.

MEMO

Servo ON cannot be set from the BCKUP screen.

<Controller File Backup Procedure>

1 Press the ALT key and 0 key simultaneously.

2 Press the SAVE (F1) key.

"Push EXE Then ROM -> BACKUP" appears on the screen.

3 Press the EXE key.

This backs up the files in the ROM area of the current controller to the backup area of the controller.

MEMO

- *The files at factory shipping are already backed up to the controller.*

  Be aware that, when a new backup is performed, the backup at factory shipping is overwritten.

NOTICE



PROHIBITED

- Do not turn off the controller during the backup process. Doing so could corrupt the files in the controller.

<Procedure for Restoring from Controller File Backup>

1 Press the ALT key and 0 key simultaneously.

2 Press the LOAD (F3) key.

"Push EXE Then BACKUP -> ROM" appears on the screen.

3 Press the EXE key.

This restores the files in the backup area of the controller to the controller ROM area.

NOTICE	
 **PROHIBITED**	• Do not turn off the controller during the backup restore process. Doing so could corrupt the files in the controller.

6. Alarm Screen

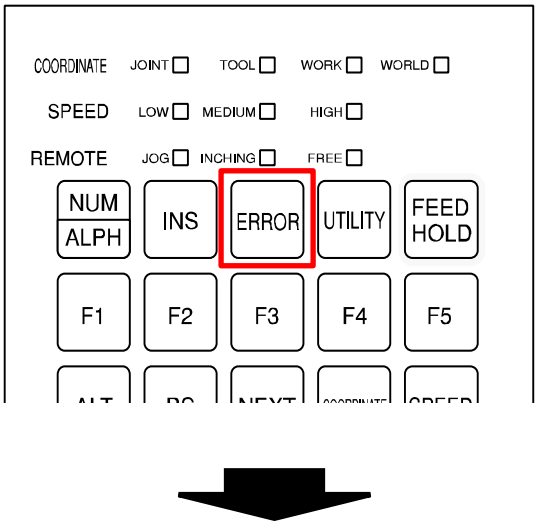
This chapter describes the alarm screen.

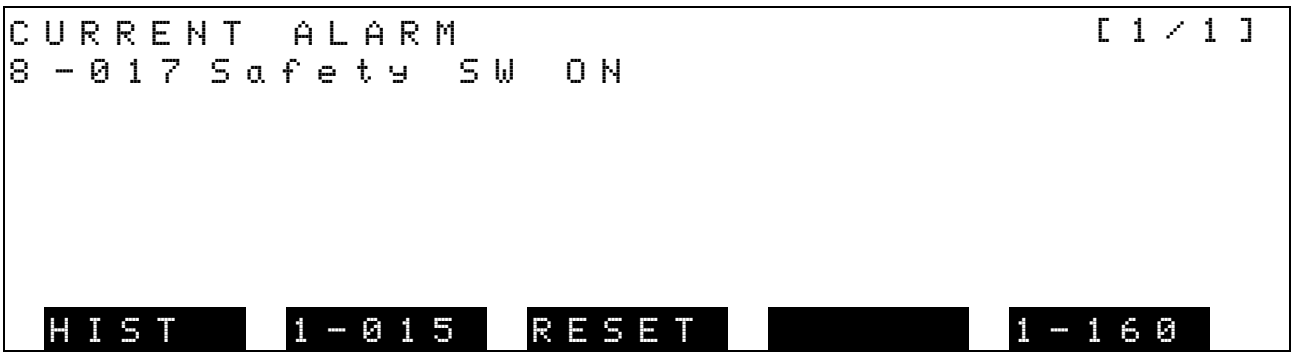
For details on the alarms and their causes, see the separate STA1165 Alarm Manual.

<Alarm Screen Opening Procedure>

1 Press the ERROR key.

The alarm screen (current alarm screen) is opened.





6.1 Current Alarms

6.1.1 Opening the Current Alarm

<Procedure for Opening the Current Alarm Screen from the Alarm Screen>

1 Press the CURRE (F1) key.

The CURRENT ALARM screen is opened.

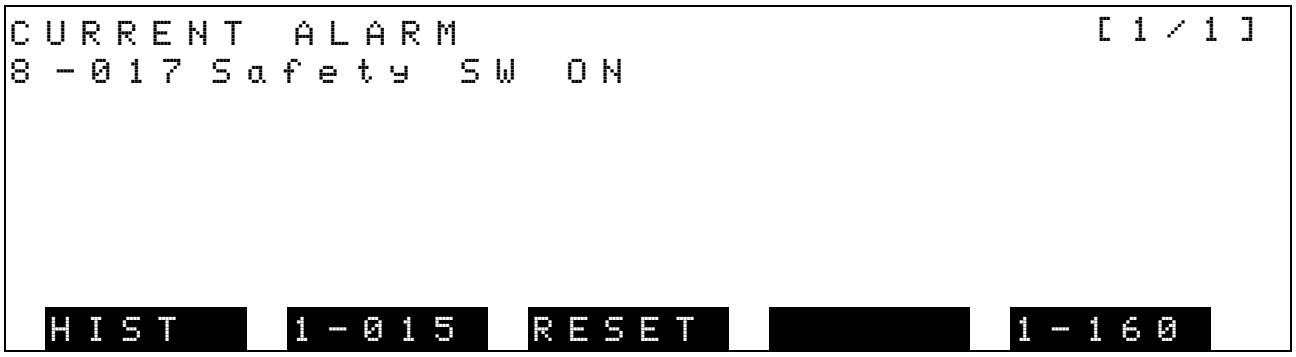
<Procedure for Opening the Current Alarm Screen from a Non-alarm Screen>

1 Press the ERROR key.

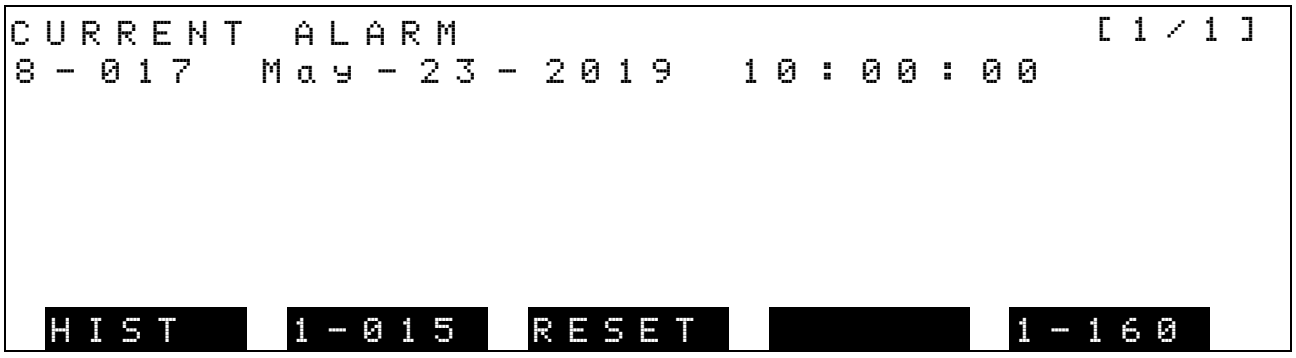
The CURRENT ALARM screen is opened.

Current Alarm Screen

(When an alarm message is displayed)



(When alarm occurrence time is displayed)



The valid keys in the current alarm screen and their actions are shown in the following table.

Key	Description
NEXT key	This shows the next page.
ALT key + NEXT key	This shows the previous page.
← key	This shows the alarm message.
→ key	This shows the time that the alarm occurred.
ESC key	This hides the alarm screen.

The function menus in the current alarm screen and their actions are shown in the following table.

Displayed menu name	Description
HIST	This shows the alarm history screen.
1-015	This shows the parameter error details screen.
RESET	This resets the alarm.
1-160	This shows the compile error details screen.

6.1.2 Clearing Current Alarms

<Current Alarm Clear Procedure>

1 Press the RESET (F3) key in the current alarm screen.

The current alarms are cleared.

MEMO

If the cause of the alarm was not removed before clearing the alarm, the alarm will occur again even after the alarm is cleared.

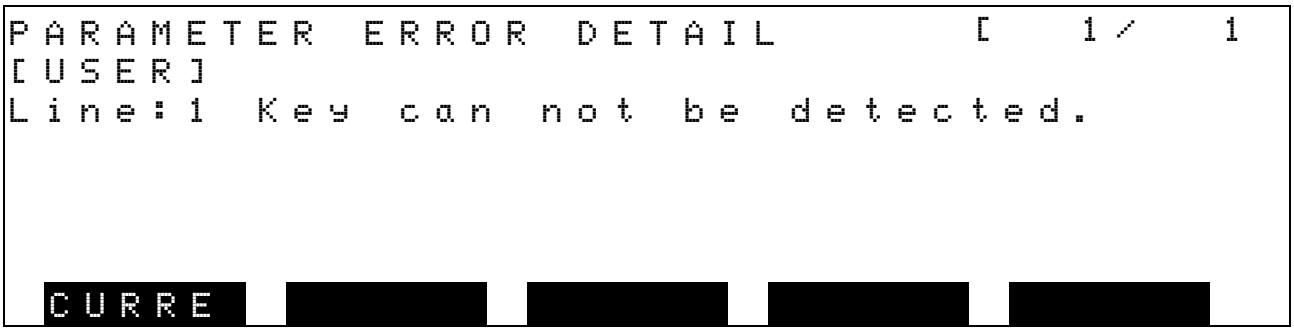
6.1.3 Opening Details of "1-015 Parameter error"

<Parameter Error Details Screen Opening Procedure>

1 Press the 1-015 (F2) key in the current alarm screen.

The parameter error details screen is opened.

Parameter Error Details Screen



The valid keys in the parameter error details screen and their actions are shown in the following table.

Key	Description
↑ key or ALT key + ↑ key	This shows the previous page. If located on the first page, this shows the last page.
↓ key or ALT key + ↓ key or NEXT key	This shows the next page. If located on the last page, this shows the first page.
ESC key	This hides the alarm screen.

The function menus in the parameter error details screen and their actions are shown in the following table.

Displayed menu name	Description
CURRE	This shows the current alarm screen.

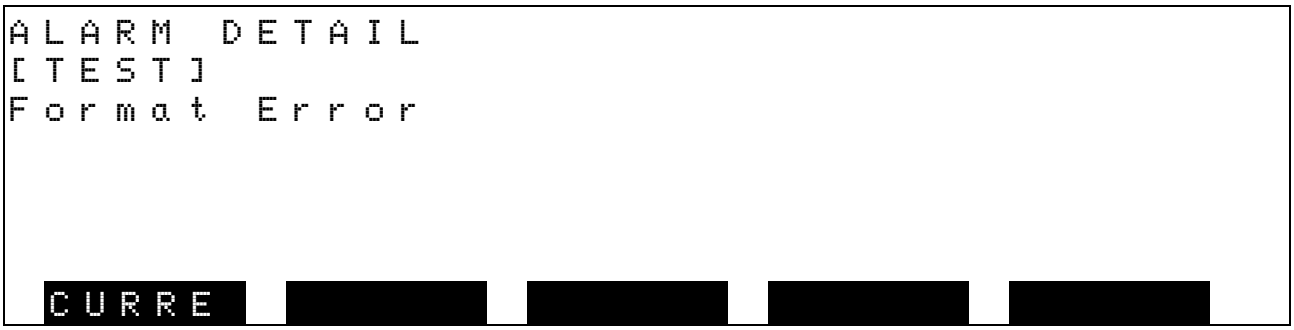
6.1.4 Opening Details of "1-160 Compile error"

<Compile Error Details Screen Opening Procedure>

1 Press the 1-160 (F5) key in the current alarm screen.

The compile error details screen is opened.

Compile Error Details Screen



The valid keys in the compile error details screen and their actions are shown in the following table.

Key	Description
ESC key	This hides the alarm screen.

The function menus in the compile error details screen and their actions are shown in the following table.

Displayed menu name	Description
CURRE	This shows the current alarm screen.

6.2 Alarm History

6.2.1 Displaying the Error History

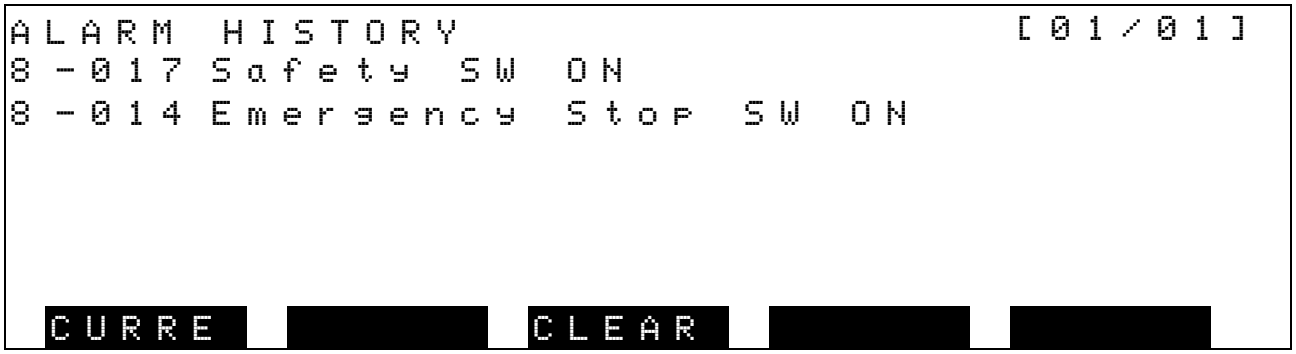
<Alarm History Screen Opening Procedure>

1 Press the HIST (F1) key in the current alarm screen.

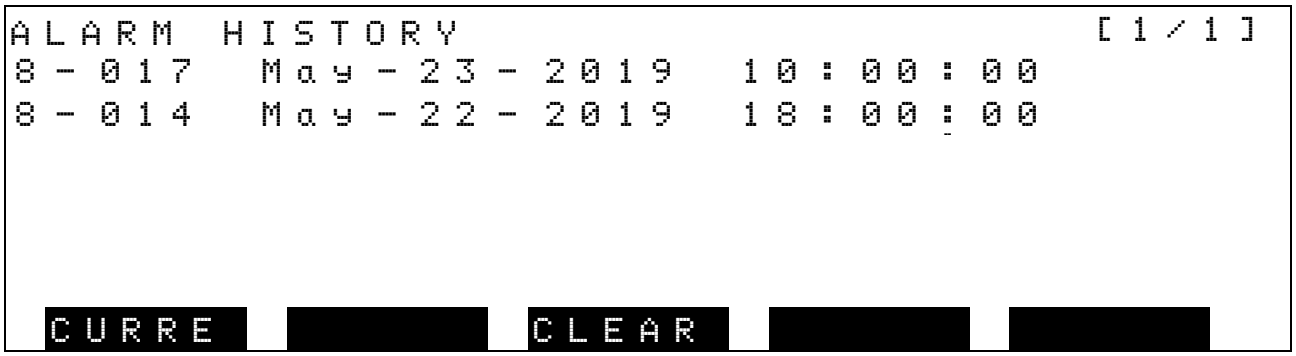
The ALARM HISTORY screen is opened.

Alarm History Screen

(When an alarm message is displayed)



(When alarm occurrence time is displayed)



The valid keys in the alarm history screen and their actions are shown in the following table.

Key	Description
↑ key	This shows the previous page.
↓ key or NEXT key	This shows the next page.
← key	This shows the alarm message.
→ key	This shows the time that the alarm occurred.
ESC key	This hides the alarm screen.

The function menus in the alarm history screen and their actions are shown in the following table.

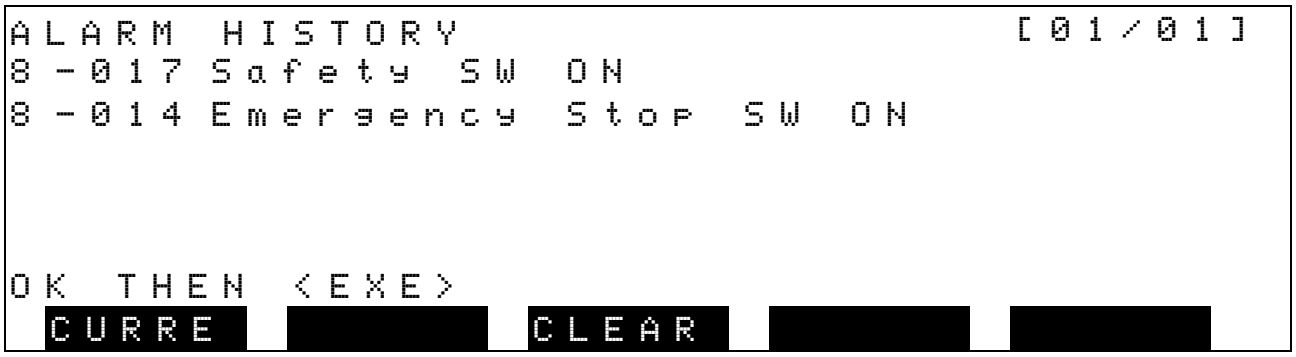
Displayed menu name	Description
CURRE	This shows the current alarm screen.
CLEAR	This clears the alarm history.

6.2.2 Clearing the Alarm History

<Alarm History Clear Procedure>

1 Press the CLEAR (F3) key in the alarm history screen.

"OK THEN <EXE>" appears in the command line.



MEMO

If the ESC key is pressed while "OK THEN <EXE>" is displayed, this returns to the status before the CLEAR (F3) key was pressed.

2 Press the EXE key.

The alarm history is cleared.

